

M O D U L A R S Y S T E M

Class 9 **GEOMETRY**

?



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Digital Assembly

Zambak Typesetting & Design

Page Design

Serdar YILDIRIM

Aydın ÇETİN

Proofreader

Zoe Barnett

Publisher

Zambak Yayıncılık ve

Eğitim Gereçleri A.Ş.

Printed by

Çağlayan A.Ş.

Gaziemir / zmir, August 2012

Tel: +90-232 252 22 85

+90-232 522 20 96 / 97

ISBN: 978-605-112-467-4

Printed in Turkey

DISTRIBUTION

Zambak Yayıncılık ve

Eğitim Gereçleri A.Ş.

Mahmutbey Merkez Mah.

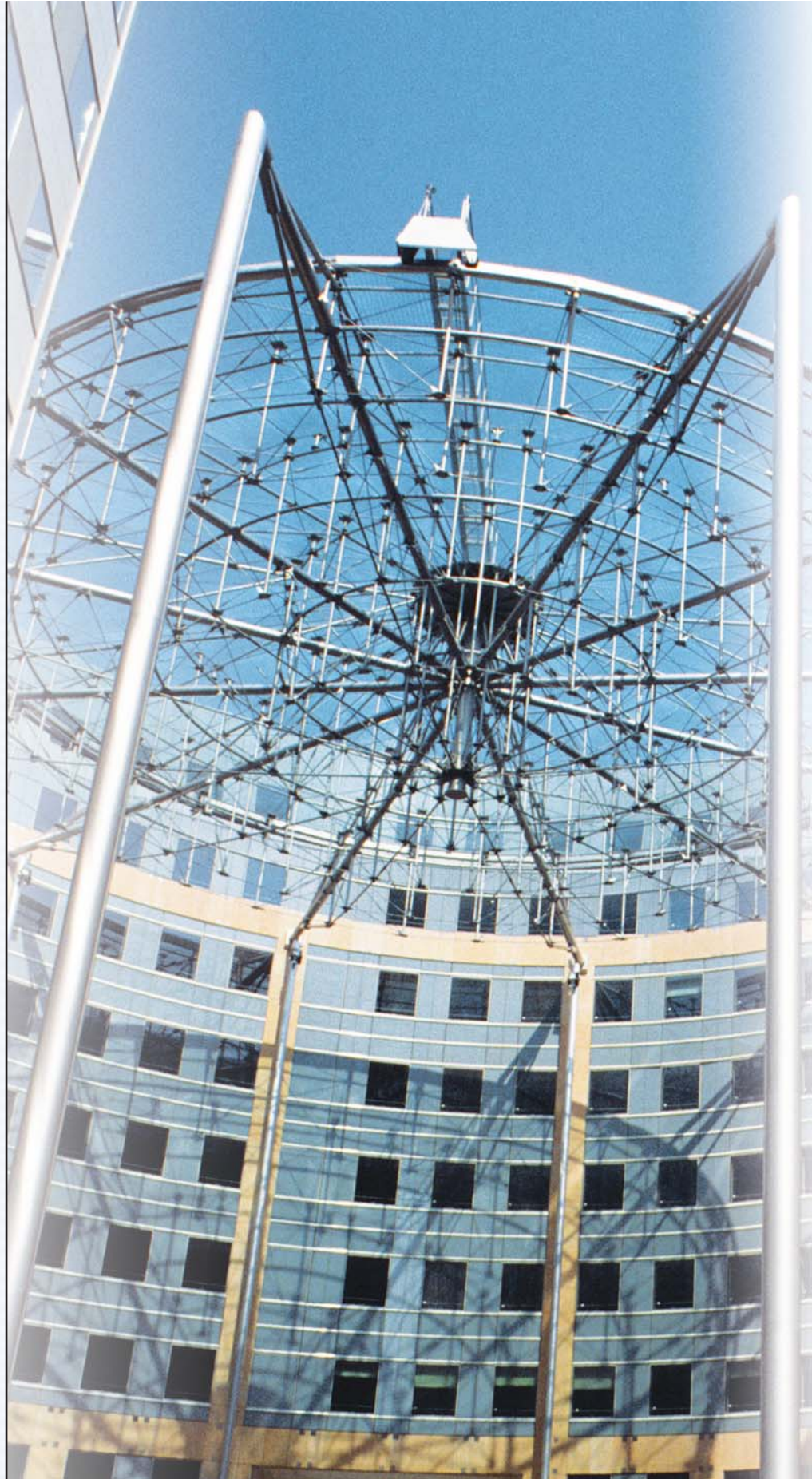
Souksu Cad. No. 31 Tek-er Merkezi

Baclar / İSTANBUL

Tel.: +90-212 604 21 00

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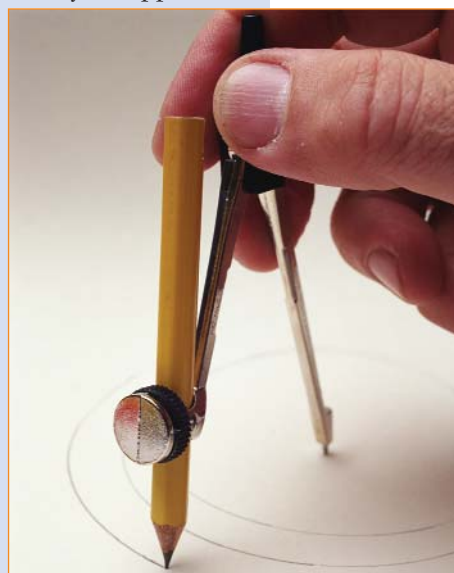


PREFACE

To the Teacher,

Analytic Analysis of Lines and Circles is designed to provide students with the analytic geometry background needed for further college-level geometry courses. Analytic geometry can be defined as algebraic analysis applied to geometrical concepts and figures, or the use of geometrical concepts and figures to illustrate algebraic forms.

Analytic geometry has many applications in different branches of science and makes it easier to solve a wide variety of problems. The goal of this text is to help students develop the skills necessary for solving analytic geometry problems, and then help students apply these skills. By the end of the book, students will have a good understanding of the analytic approach to solving problems. In addition, we have provided many systematic explanations throughout the text that will help instructors to reach the goals that they have set for their students. As always, we have taken particular care to create a book that students can read, understand, and enjoy, and that will help students gain confidence in their ability to use analytic geometry.



To the Student,

This book consists of two chapters, which cover analytical analysis of lines and circles respectively. Each chapter begins with basic definitions, theorems, and explanations which are necessary for understanding the subsequent chapter material. In addition, each chapter is divided into subsections so that students can follow the material easily.

Every subsection includes self-test **Check Yourself** problem sections followed by basic examples illustrating the relevant definition, theorem, rule, or property. Teachers should encourage their students to solve Check Yourself problems themselves because these problems are fundamental to understanding and learning the related subjects or sections. The answers to most Check Yourself problems are given directly after the problems, so that students have immediate feedback on their progress. Answers to some Check Yourself problems are not included in the answer key, as they are basic problems which are covered in detail in the preceding text or examples.

Giving answers to such problems would effectively make the problems redundant, so we have chosen to omit them, and leave students to find the basic answers themselves.

At the end of every section there are exercises categorized according to the structure and subject matter of the section. **Exercises** are graded in order,

from easy (at the beginning) to difficult (at the end). Exercises which involve more ability and effort are denoted by one or two stars. In addition, exercises which deal with more than one subject are included in a separate bank of mixed problems at the end of the section. This organization allows the instructor to deal

with only part of a section if necessary and to easily determine which exercises are appropriate to assign.

Every chapter ends with three important sections.

The **Chapter Summary** is a list of important concepts and formulas covered in the chapter that students can use easily to get direct information whenever needed.

A **Concept Check** section contains questions about the main concepts of the subjects covered, especially about the definitions, theorems or derived formulas.

Finally, a **Chapter Review Test** section consists of three tests, each with sixteen carefully-selected problems. The first test covers primitive and basic problems. The second and third tests include more complex problems. These tests help students assess their ability in understanding the coverage of the chapter.

The answers to the exercises and the tests are given at the end of the book so that students can compare their solution with the correct answer.

Each chapter also includes some subjects which are denoted as **optional**. These subjects complement the topic and give some additional information. However, completion of optional sections is left to the discretion of the teacher, who can take into account regional curriculum requirements.

EXERCISES 1.1

A. Analytic Analysis of Points

1. Plot the following points in the coordinate plane.

- a. $A(2, 3)$ b. $B(-3, 1)$ c. $C(-3, 2)$
d. $D(5, -3)$ e. $E(0, -4)$ f. $F(-3, 0)$

CHAPTER SUMMARY

- There is a one-to-one correspondence between the points in a plane and the Cartesian coordinates. The point A can be represented by two components, the abscissa and the ordinate, $A(x, y)$.

Concept Check

- What is the coordinate plane?
- How can a point be represented in the coordinate plane?
- Define the concept of line. Find examples from the coordinate plane.

CHAPTER REVIEW TEST 1A

1. What is the length of the median passing through the vertex A of a triangle ABC with vertices $A(4, 7)$, $B(-1, 2)$, and $C(3, 4)$?

- A) 5 B) 6 C) 7 D) 8 E) 9

G. BUNCH OF LINES (OPTIONAL)

Definition

bunch of lines

In the coordinate plane, a set of the lines passing through a point is called a **bunch of lines**.

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VECTORS IN THE PLANE



A. BASIC VECTOR CONCEPTS

Some of the quantities we measure in our daily lives are completely determined by their magnitudes, for example, length, mass, area, temperature, and energy. When we speak of a length of 3 cm or an area of 5 cm², we only need one number to describe each of these quantities. We call such quantities **scalar quantities**.

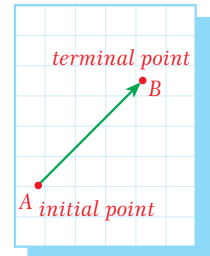
On the other hand, to describe a force, we need to record its direction as well as its size. For example, to describe the velocity of a moving object, we must specify both the speed and the direction of travel. Quantities such as displacement, velocity, acceleration, and other forces that have magnitude as well as direction are called **vector quantities**. We usually show a vector quantity as an arrow that points in the direction of the action, with length that shows the magnitude of the action in terms of a suitable unit. The way to represent such quantities mathematically is through the use of **vectors**.

1. Directed Line Segment

When we move from Antalya to Berlin by bus, we have two quantities: the direction from Antalya to Berlin, and the length of the displacement between these cities.



We can sketch a line segment AB as shown in the figure with starting point A and finishing point B to represent the movement from Antalya to Berlin. The line segment AB with an arrow has direction and length. The arrow head specifies the direction, and the length of the arrow specifies the magnitude, at a suitable scale. A and B are the endpoints of the segment. Point A is called the **initial point** and point B is called the **terminal point** of the line segment. The resulting segment AB is called a **directed line segment**.



Definition

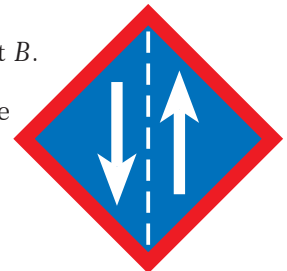
directed line segment

A line segment with direction is called a **directed line segment**.

We write $\overrightarrow{AB}$ to denote a directed line segment from point A to point B .

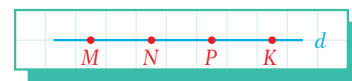
Directed line segments are used in daily life. For example, some traffic signs for drivers use directed line segments.

In technology we also use directed line segments.



EXAMPLE

1 Points M, N, P and K on line d are given. Write all the directed line segments with endpoints M, N, P , or K .



Solution The directed line segments with endpoints M, N, P , or K are $\overrightarrow{MN}, \overrightarrow{MP}, \overrightarrow{MK}, \overrightarrow{NP}, \overrightarrow{NK}, \overrightarrow{NM}, \overrightarrow{PK}, \overrightarrow{PN}, \overrightarrow{PM}, \overrightarrow{KP}, \overrightarrow{KN},$ and $\overrightarrow{KM}$.

Notice that $\overrightarrow{MN}$ is not the same as $\overrightarrow{NM}$, and $\overrightarrow{MP}$ is not the same as $\overrightarrow{PM}$. This is because the line segments have direction. Pairs such as $\overrightarrow{MN}$ and $\overrightarrow{NM}$ have the same magnitude but opposite direction.

2. Definition of a Vector

Definition

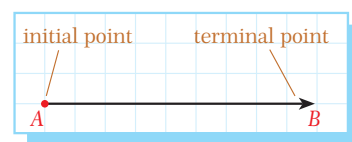
vector

A directed line segment in the plane is called a **vector**.

The length of the directed line segment is the length of the vector.

The direction of the directed line segment is the direction of the vector.

We write $\overrightarrow{AB}$ to mean a vector with initial point A and terminal point B . Alternatively, we can name a vector with a lower-case letter such as $\vec{u}$ or $\vec{p}$.



For example, consider a line segment $\overrightarrow{AB}$ with length 2 cm.

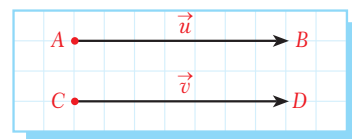
We can say the length of vector $\overrightarrow{AB}$ is 2 cm, and write $|\overrightarrow{AB}| = 2$ cm.

3. Equal Vectors

Definition

equal vectors

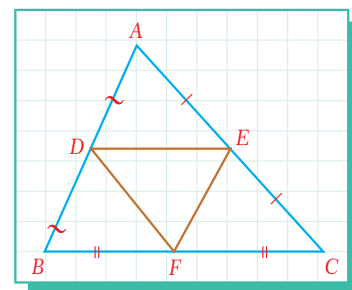
Two vectors that have the same direction and length are called **equal vectors**. We show that two vectors $\vec{u}$ and $\vec{v}$ are equal by writing $\vec{u} = \vec{v}$.



EXAMPLE

2 In the figure, D, E , and F are the midpoints of AB, AC and BC respectively, and $DE \parallel BC, EF \parallel AB, DF \parallel AC$. Name all the equal vectors.

Solution In triangle ABC , $|DE| = |BF| = |FC|$
 $|EF| = |AD| = |DB|$
 $|DF| = |AE| = |EC|$.
 So $\overrightarrow{DE} = \overrightarrow{BF} = \overrightarrow{FC}$ $\overrightarrow{ED} = \overrightarrow{FB} = \overrightarrow{CF}$
 $\overrightarrow{EF} = \overrightarrow{AD} = \overrightarrow{DB}$ and $\overrightarrow{FE} = \overrightarrow{DA} = \overrightarrow{BD}$
 $\overrightarrow{DF} = \overrightarrow{AE} = \overrightarrow{EC}$ $\overrightarrow{FD} = \overrightarrow{EA} = \overrightarrow{CE}$.



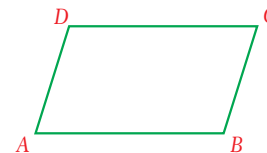
Check Yourself

$ABCD$ is a parallelogram in the figure.

- How many pairs of equivalent directed line segments are there?
- How many pairs of equal vectors are there?

Answers

- 4
- 4

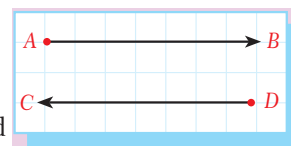


Definition

opposite vectors

Two vectors are called **opposite vectors** if and only if their magnitudes (lengths) are the same but their directions are opposite.

For example, in the figure, $\vec{AB}$ and $\vec{BA}$ are opposite vectors. $\vec{CD}$ and $\vec{DC}$ are also opposite vectors. We can write $\vec{AB} = -\vec{BA}$ and $\vec{CD} = -\vec{DC}$.



Definition

zero vector

A vector whose initial and terminal points are the same is called a **zero vector**.

We write a zero vector as $\vec{0}$.

A zero vector has no direction and no size.

B. VECTOR OPERATIONS

1. Addition of Vectors

Let $\vec{PQ}$ and $\vec{QR}$ be two vectors in a plane. $\vec{PQ} + \vec{QR}$ denotes the **sum** of the vectors $\vec{PQ}$ and $\vec{QR}$. There are two ways to find the sum of two or more vectors.

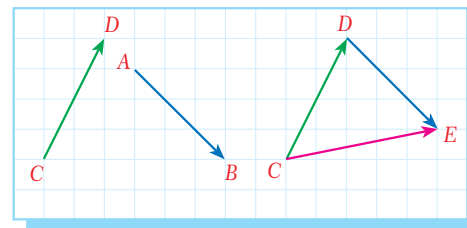
a. The Polygon Method

Imagine we want to add n vectors together. Using the polygon method, we draw the first vector. Then we place the initial point of the second vector at the terminal point of the first vector, the initial point of the third vector at the terminal point of the second vector, and so on until we place the initial point of the n^{th} vector at the terminal point of the $(n - 1)^{\text{th}}$ vector. The sum is the vector whose initial point is the initial point of the first vector and whose terminal point is the terminal point of the last vector.

Let us look at an example.

Let $\vec{AB}$ and $\vec{CD}$ be two vectors in a plane, as in the diagram. We place the initial point of $\vec{AB}$ at the terminal point of $\vec{CD}$ to make $\vec{DE}$ ($\vec{AB} = \vec{DE}$).

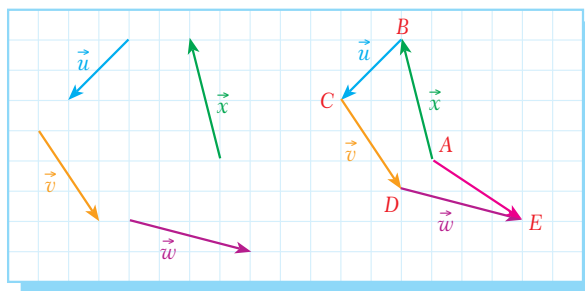
Using the polygon method,
 $\vec{CD} + \vec{AB} = \vec{CD} + \vec{DE} = \vec{CE}$.



Now look at an example of adding more than two vectors using the polygon method.

As shown in the figure,

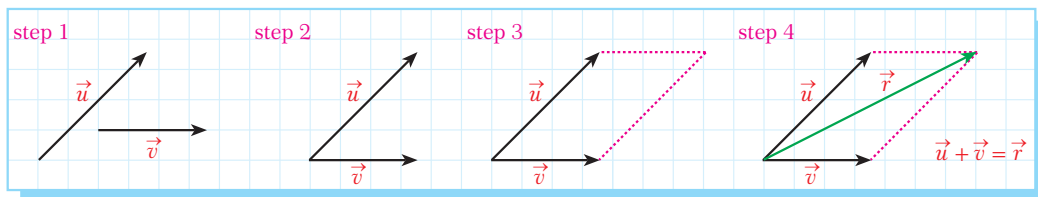
$$\vec{u} + \vec{v} + \vec{w} + \vec{x} = \vec{AE}.$$



b. The Parallelogram Method

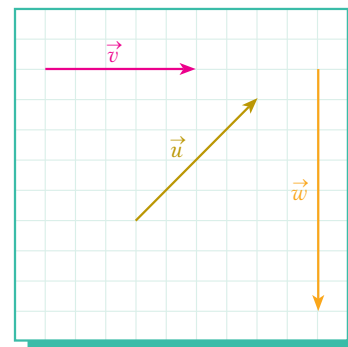
To add one vector to another using the parallelogram method, we draw the first vector, and then we draw the second vector with its initial point at the initial point of the first vector. We make a parallelogram by drawing two additional sides, each passing through the terminal point of one of the vectors and parallel to the other vector. We find the sum by drawing a vector along the diagonal from the common initial point to the intersection of the two lines.

Look at the example of adding $\vec{u}$ and $\vec{v}$ using the parallelogram method:

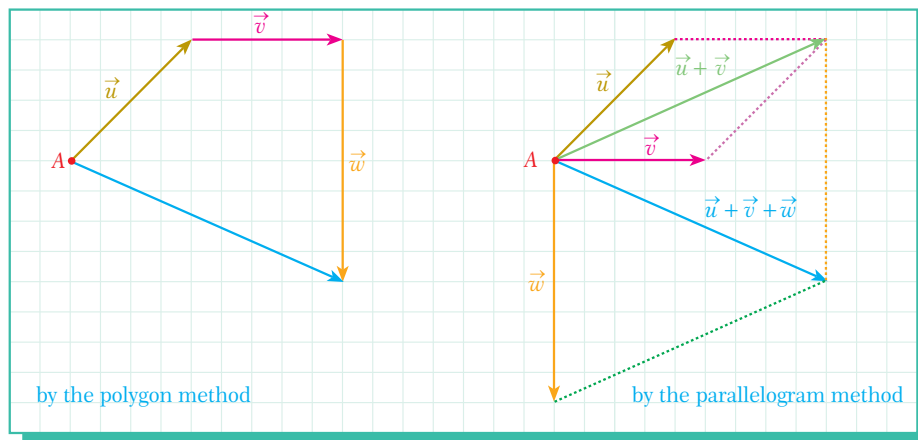


EXAMPLE

3 Find $\vec{u} + \vec{v} + \vec{w}$ in the figure on the right.



Solution Let us choose A as a fixed point. We can use the polygon method or the parallelogram method to add the given vectors $\vec{u} + \vec{v} + \vec{w}$.



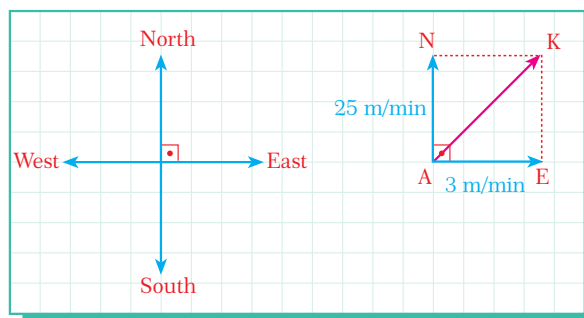
EXAMPLE

- 4** The velocity of a boat is 25 m/min north and the velocity of a river current is 3 m/min east. Draw a scale diagram to show the velocities as vectors and find the sum.

Solution First we choose a starting point A and write $\vec{AN}$ = velocity of the boat due north.
 $\vec{AE}$ = velocity of the current due east.
 $\vec{AN}$ and $\vec{AE}$ are perpendicular, and $\vec{AK}$ is the sum of $\vec{AN}$ and $\vec{AE}$:

$$|\vec{AK}| = \sqrt{25^2 + 3^2} = \sqrt{634}.$$

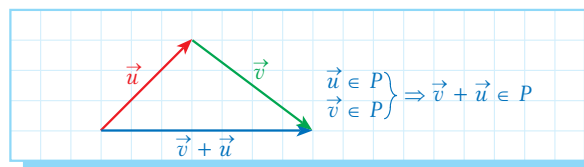
This is the sum of the vectors.



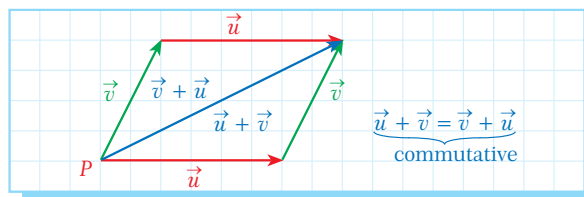
c. Properties of Vector Addition

Let $\vec{u}$, $\vec{v}$, and $\vec{w}$ be three vectors in a plane P .

1. The sum of any two vectors in P is also a vector in P (closure property).

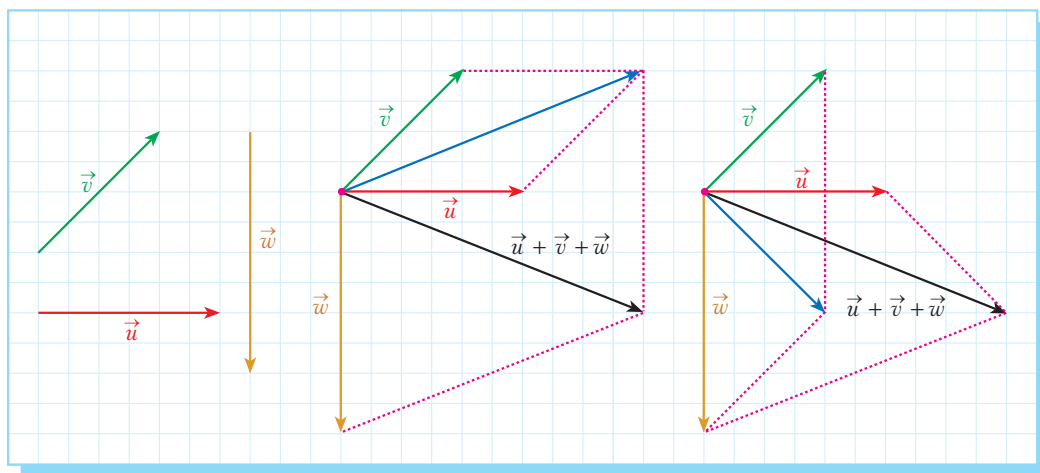


2. The sum of any two vectors in P is commutative (commutative property).

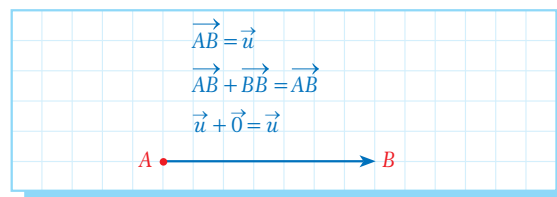


3. The sum of any three vectors in P is associative (associative property).

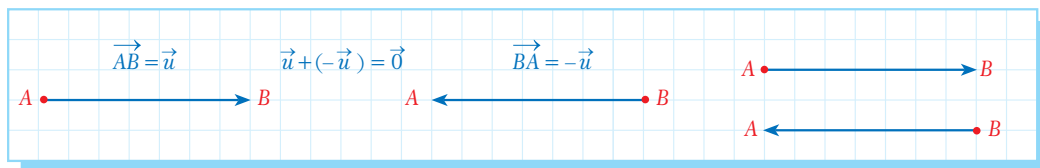
$$\vec{u} + (\vec{v} + \vec{w}) = (\vec{u} + \vec{v}) + \vec{w}$$



4. The sum of the zero vector and a vector in P is the vector itself (identity element).



5. The additive inverse of any vector $\vec{u}$ is $-\vec{u}$: $\vec{u} + (-\vec{u}) = \vec{0}$ (additive inverse).

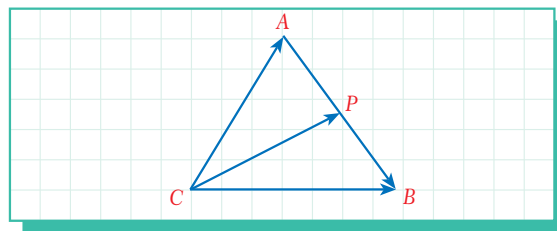


EXAMPLE

In a triangle ABC , P is the midpoint of $\overline{AB}$. Express $\overrightarrow{CP}$ in terms of $\overrightarrow{CA}$ and $\overrightarrow{CB}$.

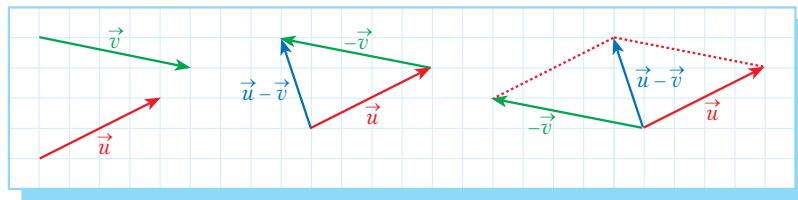
Solution

$$\begin{aligned} \overrightarrow{CP} &= \overrightarrow{CA} + \overrightarrow{AP} \\ + \quad \overrightarrow{CP} &= \overrightarrow{CB} + \overrightarrow{BP} \\ \hline 2 \cdot \overrightarrow{CP} &= \overrightarrow{CA} + \overrightarrow{CB} + \underbrace{\overrightarrow{AP} + \overrightarrow{BP}}_{\vec{0}} \\ \overrightarrow{CP} &= \frac{1}{2} \cdot (\overrightarrow{CA} + \overrightarrow{CB}) \end{aligned}$$



2. Subtraction of Vectors

Since subtraction is the inverse of addition, we can find the difference of two vectors $\vec{u}$ and $\vec{v}$ by adding the vectors $\vec{u}$ and $-\vec{v}$ (opposite of $\vec{v}$) using either the parallelogram method ($\vec{u} - \vec{v} = \vec{u} + (-\vec{v})$) or the polygon method.

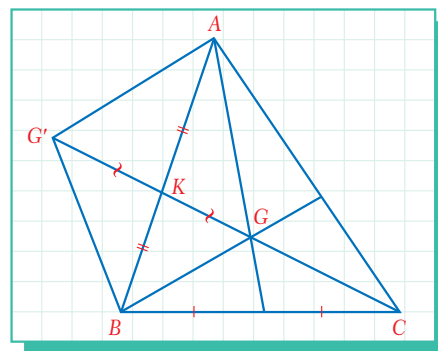


In a triangle ABC , G is the centroid. Find $\vec{GA} + \vec{GB} + \vec{GC}$.

Solution Let us label a point G' on the extension of CG which satisfies $|CG| = |GG'|$. Since G is the centroid of $\triangle ABC$, $|CG| = 2 \cdot |GK|$. Therefore $|GG'| = 2 \cdot |GK|$, which means that K is the midpoint of GG' . We conclude that $AG'BG$ is a parallelogram because K is the midpoint of both diagonals AB and GG' . So we have $\vec{AG'} = \vec{GB}$ which gives us $\vec{GA} + \vec{GB} = \vec{GG'}$.

On the other hand, we have $\vec{CG} = \vec{GG'} = -\vec{GC}$. Using this result in $\vec{GA} + \vec{GB} = \vec{GG'}$, we get $\vec{GA} + \vec{GB} = -\vec{GC}$ which gives us

$$\vec{GA} + \vec{GB} + \vec{GC} = \vec{0}.$$

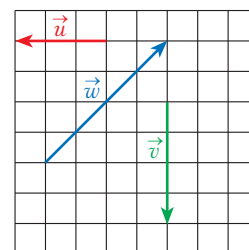


Check Yourself

- Find the following using the vectors in the figure.
 - $\vec{v} - \vec{u}$
 - $\vec{u} + \vec{w}$
 - $\vec{w} + \vec{v} - \vec{u}$
- In a triangle ABC , $D \in [BC]$ and $|BD| = 2 \cdot |DC|$. Express $\vec{AD}$ in terms of $\vec{AB}$ and $\vec{AC}$.

Answers

- use the polygon method
- $\vec{AD} = \frac{2}{3}\vec{AC} + \frac{1}{3}\vec{AB}$



3. Multiplication of a Vector by a Scalar

Multiplying a vector by a scalar makes the vector longer or shorter depending on the value of the scalar. If the scalar is greater than 1 or less than -1 , multiplying makes a longer vector. If the scalar is between -1 and 1 and non-zero, it makes a shorter vector.

If the scalar is positive, multiplying does not change the direction.

If the scalar is negative, multiplying will make the vector's direction opposite.

For a real number a and a vector $\vec{u}$,

1. if $a > 0$ then vector $a \cdot \vec{u}$ has the same direction as $\vec{u}$ and the length $|a \cdot \vec{u}| = a \cdot |\vec{u}|$.
2. if $a < 0$ then vector $a \cdot \vec{u}$ has the opposite direction to $\vec{u}$ and the length $|a \cdot \vec{u}| = |a| \cdot |\vec{u}|$.
3. if $a = 0$ then $a \cdot \vec{u} = \vec{0}$.

EXAMPLE

5

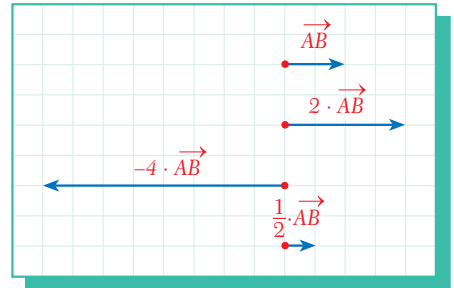
Using $\vec{AB}$ as shown in the figure, draw vector diagrams to show $2 \cdot \vec{AB}$, $-4 \cdot \vec{AB}$, and $\frac{1}{2} \cdot \vec{AB}$.



Solution

Since 2 and $\frac{1}{2}$ are positive, $2 \cdot \vec{AB}$ and $\frac{1}{2} \cdot \vec{AB}$ have the same direction as $\vec{AB}$. However, $2 \cdot \vec{AB}$ is twice as long as $\vec{AB}$ and $\frac{1}{2} \cdot \vec{AB}$ is half as long.

On the other hand, $-4 \cdot \vec{AB}$ has opposite direction to $\vec{AB}$ (since -4 is a negative scalar) and it is four times as long as $\vec{AB}$.



a. Properties of the Multiplication of a Vector by a Scalar

For any vectors $\vec{u}$, $\vec{v}$, and $\vec{w}$ and real numbers a and b , the following properties are satisfied.

1. $a \cdot \vec{u}$ is a vector in the plane
2. $(a \cdot b) \cdot \vec{u} = a \cdot (b \cdot \vec{u})$
3. $(a + b) \cdot \vec{u} = a \cdot \vec{u} + b \cdot \vec{u}$
4. $a \cdot (\vec{u} + \vec{v}) = a \cdot \vec{u} + a \cdot \vec{v}$
5. $1 \cdot \vec{u} = \vec{u}$
6. $a \cdot \vec{0} = \vec{0}$

EXAMPLE

Points A, B, C , and M are on the same line. M is between C and B . $\vec{AB} = 3\vec{AC}$. Express the vector $\vec{MC}$ in terms of vectors $\vec{MA}$ and $\vec{MB}$.

Solution $\vec{AB} = 3\vec{AC}$ so $\vec{CB} = 2\vec{AC}$ (1)

$$\vec{MA} + \vec{AC} = \vec{MC} \quad (2)$$

$$\vec{CM} + \vec{MB} = 2\vec{AC} \quad (3)$$

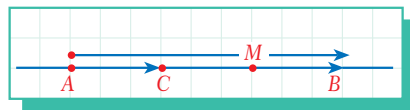
$$\vec{AC} = -\frac{1}{2}\vec{MC} + \frac{1}{2}\vec{MB} \quad (4)$$

$$\vec{MA} - \frac{1}{2}\vec{MC} + \frac{1}{2}\vec{MB} = \vec{MC} \text{ by (2) and (4).}$$

$$\vec{MA} + \frac{1}{2}\vec{MB} = \frac{1}{2}\vec{MC} + \vec{MC}$$

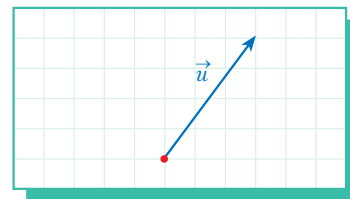
$$\vec{MA} + \frac{1}{2}\vec{MB} = \frac{3}{2}\vec{MC}$$

$$\text{So } \vec{MC} = \frac{2}{3}\vec{MA} + \frac{1}{3}\vec{MB}.$$



Check Yourself

- Multiply the vector $\vec{u}$ by the scalars $-2, 3, 0.5$ and $-\frac{1}{3}$ and draw a vector diagram to show them.
- Points A, B, C , and M are on the same line. M is between A and C . $\vec{AB} = 2\vec{AC}$. Express the vector $\vec{MC}$ in terms of the vectors $\vec{MA}$ and $\vec{MB}$.



Answers

2. $\vec{MC} = \frac{1}{2}(\vec{MA} + \vec{MB})$

C. PARALLEL VECTORS

1. Parallel Vectors

Definition

parallel vectors

Let $\vec{a}$ and $\vec{b}$ be two vectors. $\vec{a}$ and $\vec{b}$ are called **parallel vectors** if and only if $\vec{a} = k\vec{b}$ where $k \neq 0$ and $k \in \mathbb{R}$. We write $\vec{a} \parallel \vec{b}$ to show that two vectors are parallel.

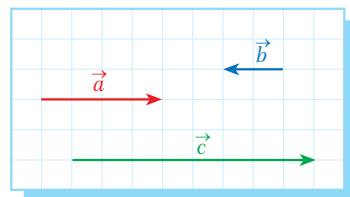


For example, in the diagram, $|\vec{a}| = 2 \text{ cm}$,

$|\vec{b}| = 1 \text{ cm}$ and $|\vec{c}| = 4 \text{ cm}$.

We can express vector $\vec{a}$ as $\vec{a} = \frac{1}{2}\vec{c}$ and $\vec{a} = -2\vec{b}$.

Therefore the vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$ are parallel, i.e. $\vec{a} \parallel \vec{b} \parallel \vec{c}$.



EXAMPLE

7

In a triangle ABC , D and E are the midpoints of sides AB and AC respectively.

Show that $\overrightarrow{DE} \parallel \overrightarrow{BC}$.

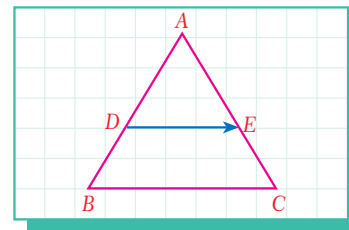
Solution

$\overrightarrow{BA} + \overrightarrow{AC} = \overrightarrow{BC}$ and $\overrightarrow{DA} + \overrightarrow{AE} = \overrightarrow{DE}$ by the addition of vectors.

$$\overrightarrow{DA} = \frac{1}{2} \cdot \overrightarrow{BA}, \quad \overrightarrow{AE} = \frac{1}{2} \cdot \overrightarrow{AC}$$

$$\overrightarrow{DE} = \frac{1}{2} \cdot \overrightarrow{BA} + \frac{1}{2} \cdot \overrightarrow{AC} = \frac{1}{2} \cdot (\overrightarrow{BA} + \overrightarrow{AC}) = \frac{1}{2} \cdot \overrightarrow{BC}$$

Now $\overrightarrow{DE} = \frac{1}{2} \cdot \overrightarrow{BC}$, so $\overrightarrow{DE} \parallel \overrightarrow{BC}$ by the definition of parallel vectors.



EXAMPLE

8

In a quadrilateral $ABCD$, points E and F are the midpoints of side AB and diagonal AC , respectively. Show that $\overrightarrow{EF} \parallel \overrightarrow{BC}$.

Solution

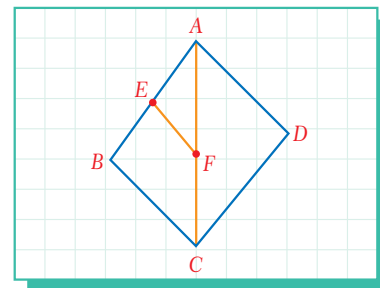
$\overrightarrow{BA} + \overrightarrow{AC} = \overrightarrow{BC}$ (1) and $\overrightarrow{EA} + \overrightarrow{AF} = \overrightarrow{EF}$ (2).

$$\overrightarrow{EA} = \frac{1}{2} \cdot \overrightarrow{BA}$$

$$+ \overrightarrow{AF} = \frac{1}{2} \cdot \overrightarrow{AC}$$

$$\overrightarrow{EA} + \overrightarrow{AF} = \frac{1}{2} \cdot (\overrightarrow{BA} + \overrightarrow{AC})$$

$\overrightarrow{EF} = \frac{1}{2} \cdot \overrightarrow{BC}$ by (1) and (2). Therefore, $\overrightarrow{EF} \parallel \overrightarrow{BC}$.



2. Non-Parallel Vectors

By the definition of parallel vectors we can conclude that if $\vec{a}$ and $\vec{b}$ are non-zero, non-parallel vectors, then $h \cdot \vec{a} = k \cdot \vec{b}$ when $h = k = 0$. Look at the proof:

Suppose that $h \neq k \neq 0$, $\vec{a} = \frac{k}{h} \cdot \vec{b}$.

Then $\vec{a} \parallel \vec{b}$. This is a contradiction, since $\vec{a}$ and $\vec{b}$ are non-parallel. As a result, $h = k = 0$.

EXAMPLE

9

Prove that the diagonals of a parallelogram intersect at their midpoints by using vectors.

Solution

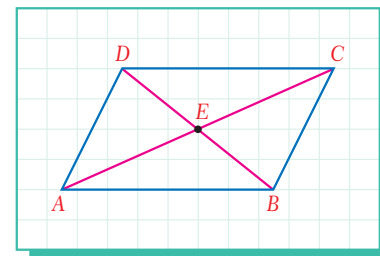
Look at the diagram. Let $\overrightarrow{AB} = \vec{a}$ and $\overrightarrow{BC} = \vec{b}$, so

$$\overrightarrow{AC} = \vec{a} + \vec{b}$$

$$\overrightarrow{DB} = \vec{a} - \vec{b}$$

$$\overrightarrow{AE} = m \cdot (\vec{a} + \vec{b})$$

$$\overrightarrow{EB} = n \cdot (\vec{a} - \vec{b})$$



$$\begin{aligned}\vec{AE} + \vec{EB} &= m \cdot (\vec{a} + \vec{b}) + n \cdot (\vec{a} - \vec{b}) \\ \vec{a} &= m \cdot (\vec{a} + \vec{b}) + n \cdot (\vec{a} - \vec{b}) \\ (m + n - 1) \cdot \vec{a} &= \vec{b} \cdot (n - m).\end{aligned}$$

Since $\vec{a}$ and $\vec{b}$ are non-zero and non-parallel, we have $(m + n - 1) = (n - m) = 0$. Therefore $m + n = 1$ and $m = n$, and so $m = n = \frac{1}{2}$.

EXAMPLE

10

In a triangle ABC , D , E and F are the midpoints of sides BC , BA and AC respectively. Find the sum $\vec{AD} + \vec{BF} + \vec{CE}$.

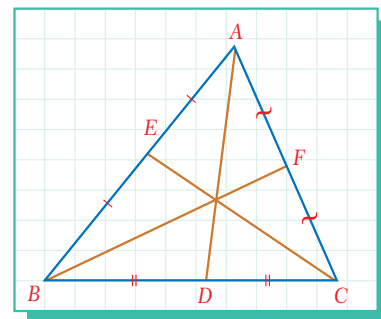
Solution Let us draw the triangle ABC as in the figure.

If D , E and F are midpoints then

$$\begin{aligned}\vec{AD} &= \frac{1}{2} \cdot (\vec{AB} + \vec{AC}) \quad \text{by the result of Example 5.} \\ \vec{BF} &= \frac{1}{2} \cdot (\vec{BA} + \vec{BC}) \\ + \vec{CE} &= \frac{1}{2} \cdot (\vec{CA} + \vec{CB})\end{aligned}$$

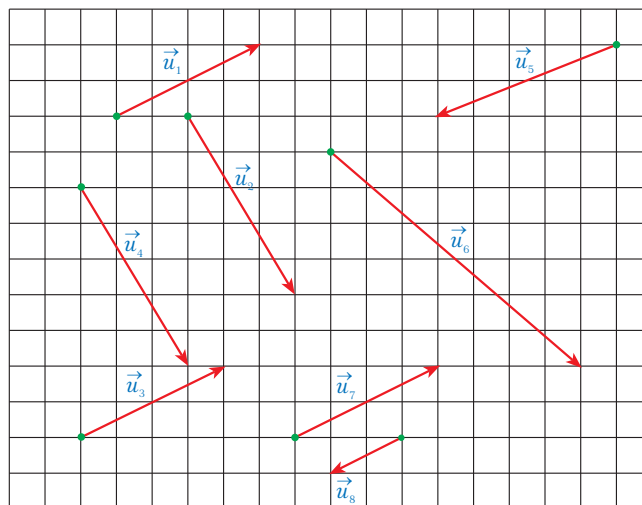
$$\vec{AD} + \vec{BF} + \vec{CE} = \frac{1}{2} \cdot \underbrace{(\vec{AB} + \vec{BA})}_{\vec{0}} + \underbrace{(\vec{AC} + \vec{CA})}_{\vec{0}} + \underbrace{(\vec{BC} + \vec{CB})}_{\vec{0}}$$

Therefore, $\vec{AD} + \vec{BF} + \vec{CE} = \vec{0}$.



Check Yourself

1. Name all the pairs of parallel vectors in the figure.
2. Add the vector pairs $\vec{u}_1$ and $\vec{u}_5$, $\vec{u}_2$ and $\vec{u}_6$, and $\vec{u}_3$ and $\vec{u}_7$.
3. Find the additive inverse of $\vec{u}_7$ and $\vec{u}_1$.
4. Subtract $\vec{u}_8$ from $\vec{u}_4$.
5. In a quadrilateral $ABCD$, P and K are the midpoints of sides AB and DC respectively. Express $\vec{PK}$ in terms of $\vec{DA}$ and $\vec{CB}$.



Answers

1. look at the directions and lengths
2. use the polygon method
3. use the polygon method
4. use the polygon method
5. $\vec{PK} = -\frac{1}{2} \cdot (\vec{DA} + \vec{CB})$

Traffic Signs



Project: Describe some other areas in which we use directed line segments (for example: flowcharts, keyboards, ...).

EXERCISES 1

A. Basic Vector Concepts

1. Draw any two parallel directed line segments with the same length but opposite direction. Express one of the line segments in terms of the other.
2. How many equal directed line segments can we find on two parallel lines?
3. Make a scale diagram showing the vectors in each statement and find their sum.
 - a. A 6 km trip east is followed by a 3 km trip southeast.
 - b. The velocity of a swimmer is 5 m/min west and the velocity of a river current is 2 m/min north.

B. Vector Operations

4. Using the vectors given on the right, sketch the following vectors.

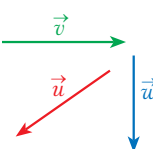
a. $\vec{u} + \vec{v}$

b. $\vec{w} - (\vec{u} + \vec{v})$

c. $-\vec{w} - \vec{v} - \vec{u}$

d. $\vec{u} + (\vec{w} + \vec{v})$

e. $\vec{u} - 3\vec{v} + 2\vec{w}$



5. In a plane, $[AB]$ is given. Point K is the midpoint of $[AB]$ and point O is any point in the same plane. Express $\vec{OK}$ in terms of $\vec{OA}$ and $\vec{OB}$.
6. In a triangle ABC , points D and E lie on $[BC]$ and $|BD| = |DE| = |EC|$. Express the vector $\vec{AD} + \vec{AE}$ in terms of $\vec{AB}$ and $\vec{AC}$.

7. Show that $(\vec{AB} + \vec{BC}) + \vec{CD} = \vec{AB} + (\vec{BC} + \vec{CD})$ by using the parallelogram method.

8. Point O is in the plane of a triangle ABC . Point G is the centroid of triangle ABC . Show that $\vec{OA} + \vec{OB} + \vec{OC} = 3\vec{OG}$.

9. Consider any two points A and B in a plane. For any point P in the same plane, the symmetry of point P with respect to point A is Q and the symmetry of point Q with respect to point B is point R . Show that $\vec{PR}$ is always $2\vec{AB}$. (Hint: let A , B , and C be collinear. If $|AB| = |BC|$ then A is the symmetry of C with respect to B .)

C. Parallel Vectors

10. In a triangle ABC , G is the point of intersection of the medians and P is the midpoint of $\vec{BG}$. Show that $\vec{PA} + \vec{PC} = 4\vec{PG}$.

11. $ABCD$ is a quadrilateral and M, N, P, Q are the midpoints of AB, BC, CD , and DA respectively. Show that $MNPQ$ is a parallelogram.

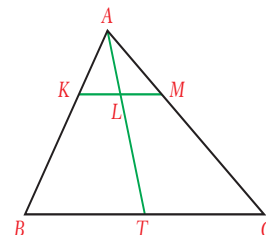
12. In a six-sided polygon $ABCDEF$, $\vec{AB} = \vec{ED}$, $\vec{BC} = \vec{FE}$, and $\vec{CD} = \vec{AF}$. Show that $FBCE$ is a parallelogram.

13. A trapezoid is a four-sided figure with only two parallel sides. A line segment which joins the midpoints of the non-parallel sides is called the *median* of the trapezoid. Prove that the median of a trapezoid is parallel to the two parallel sides, and has magnitude equal to half of their sum.

14. In a triangle ABC , $|BD| = |DE| = |EC|$, and $E, D \in [BC]$. If $|\vec{AD} + \vec{AE}| = 9$ cm, find $|\vec{AB} + \vec{AC}|$.

15. In a quadrilateral $ABCD$, E and F are the midpoints of the diagonals AC and BD respectively. Show that $\vec{AB} + \vec{AD} + \vec{CB} + \vec{CD} = 4\vec{EF}$.

16. In the figure,
 $\star$ T is the midpoint of BC ,
 ABC is a triangle,
 $2 \cdot |AK| = |KB|$, and
 $2 \cdot |AM| = |MC|$.
 Use vectors to show that
 $|\vec{AL}| = \frac{1}{2} \cdot |\vec{LT}|$.



17. Show that the centroid of a triangle divides a median in the ratio 1:2 using vectors.
 $\star\star$

18. In a parallelogram $ABCD$, K is the midpoint of side DC , and the intersection point of diagonals AC and BK is T . Show that $\vec{AC} = 3\vec{TC}$.
 $\star\star$

ANALYSIS OF VECTORS ANALYTICALLY

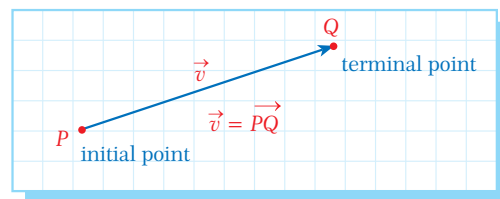
We have studied vectors geometrically. Now let us look at a method for describing vectors analytically.

We will begin this section by looking at some important axioms.

A. BASIC CONCEPTS OF VECTORS IN THE ANALYTIC PLANE

1. Axioms

1. For each pair of points P and Q there exists a unique vector $\vec{v}$ such that $\vec{PQ} = \vec{v}$.
2. For each point P and vector $\vec{v}$, there is a unique point Q such that $\vec{v} = \vec{PQ}$.



Conclusion

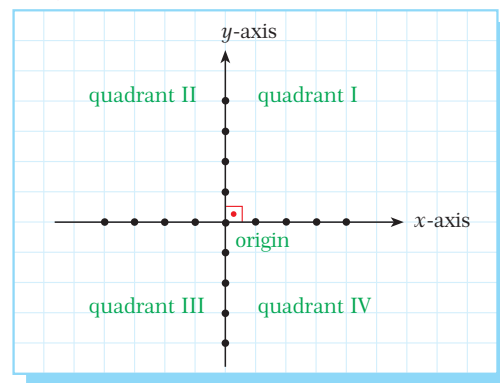
1. Two points in a plane determine two opposite vectors.
2. In a plane, if one point is fixed as an initial point then all the other points in the plane can be chosen as the terminal point of any vector.

2. The Rectangular Coordinate System

The **rectangular coordinate system** is formed by two perpendicular intersecting number lines, as shown in the diagram opposite.

1. The horizontal number line is called the **x -axis**.
2. The vertical number line is called the **y -axis**.

The **origin** is the point of intersection. At this intersection, both number lines are 0. The rectangular coordinate system is split into four **quadrants**, which are marked in the diagram with roman numerals.



Each point in the coordinate system is associated with a **pair of real numbers**. In an x, y system, the **x -coordinate** always comes first and the **y -coordinate** always comes second in the pair (x, y) . The first coordinate is called the **abscissa** of the point and the second coordinate is called the **ordinate** of the point.

EXAMPLE

11

Plot each pair of coordinates and name the quadrant or axis in which the point lies.

$A(2, 3)$, $B(-1, 2)$, $C(-3, -4)$, $D(2, 0)$, $E(0, 5)$

Solution

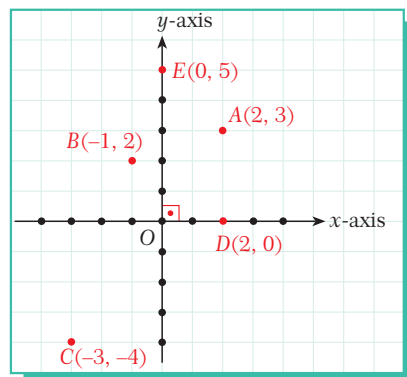
$A(2, 3)$ lies in quadrant I.

$B(-1, 2)$ lies in quadrant II.

$C(-3, -4)$ lies in quadrant III.

$D(2, 0)$ lies on the x -axis.

$E(0, 5)$ lies on the y -axis.



Check Yourself

Plot the pairs of real numbers and name the quadrant or axis in which the point lies.

$A(3, 2)$, $B(-2, 1)$, $C(-4, -3)$, $D(0, 2)$, $E(5, 0)$

3. Position Vector

Definition

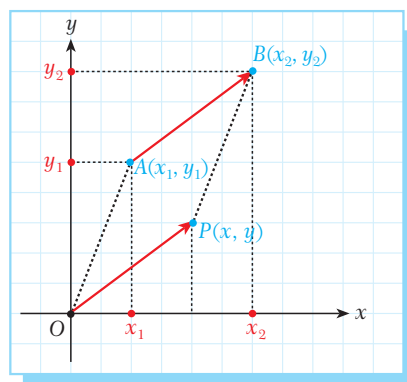
position vector

A vector $\vec{OP}$ whose initial point is at the origin of the rectangular coordinate plane and which is parallel to a vector $\vec{AB}$ is called the **position vector** of $\vec{AB}$ in the plane. In other words, if $\vec{OP}$ is the position vector of $\vec{AB}$, then $\vec{OP} \parallel \vec{AB}$, $|\vec{OP}| = |\vec{AB}|$, and $\vec{OP} = \vec{AB}$.

From this definition we can conclude the following:

1. For every vector in the plane there exists a position vector $\vec{OP}$ which is determined by a pair (x, y) , that is, $\vec{OP} = (x, y)$.
2. If $\vec{OP} = \vec{AB}$, then $OPBA$ is a parallelogram.

Look at the diagram. We can calculate that $x = x_2 - x_1$ and $y = y_2 - y_1$. Therefore the vector $\vec{AB}$ determined by the points $A(x_1, y_1)$ and $B(x_2, y_2)$ has position vector $\vec{OP} = \vec{AB} = \vec{OB} - \vec{OA} = (x, y) = (x_2 - x_1, y_2 - y_1)$.



EXAMPLE

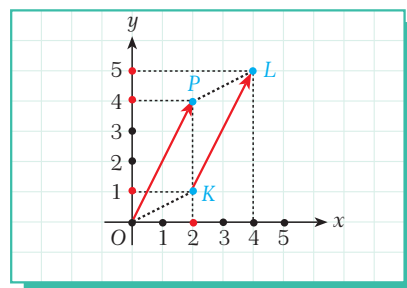
12

Find the position vector of $\vec{KL}$ with endpoints $K(2, 1)$ and $L(4, 5)$.

Solution

$$\vec{OP} = \vec{KL} = \vec{OL} - \vec{OK} = (4 - 2, 5 - 1)$$

$$\vec{OP} = (2, 4)$$



4. Components of a Vector

We have seen how to describe a vector in the plane with reference to its unique position vector.

We can also express a vector in the plane as the sum of two vectors, one of which is parallel to the x -axis and the other parallel to the y -axis. These two vectors are called the **components** of the original vector. The component parallel to the x -axis is called the **horizontal component** of the vector and the component parallel to y -axis is called the **vertical component**. Expressing a vector as the

sum of its components like this is called **resolving** the vector. For example, in the figure opposite, the vector $\vec{u}$ is the sum of the two components $\vec{u}_x$ and $\vec{u}_y$.

We can represent the vector $\vec{u}$ as an ordered pair of real numbers:

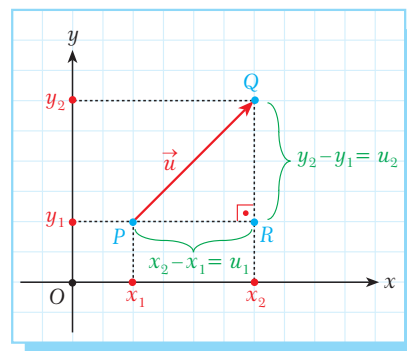
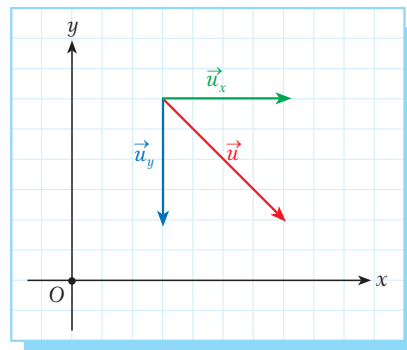
$$\vec{u} = (u_1, u_2) \text{ or } \vec{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix},$$

where u_1 is the horizontal scalar component of $\vec{u}$, and u_2 is the vertical scalar component of $\vec{u}$.

Now, $\vec{u}_x = (u_1, 0)$ and $\vec{u}_y = (0, u_2)$.

Look at the diagram opposite. By applying the Pythagorean theorem for triangle PQR in the figure, we can see that the length of the vector $\vec{u} = (u_1, u_2)$ is

$$|\vec{u}| = \sqrt{u_1^2 + u_2^2}.$$



EXAMPLE 13 Find the length of each vector.

a. $\vec{u} = (2, -3)$

b. $\vec{v} = (3, 0)$

c. $\vec{w} = \left(\frac{3}{5}, \frac{4}{5}\right)$

Solution a. $|\vec{u}| = \sqrt{2^2 + (-3)^2} = \sqrt{4 + 9} = \sqrt{13}$

b. $|\vec{v}| = \sqrt{3^2 + 0^2} = \sqrt{9} = 3$

c. $|\vec{w}| = \sqrt{\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2} = \sqrt{\frac{9}{25} + \frac{16}{25}} = \sqrt{\frac{9+16}{25}} = \sqrt{\frac{25}{25}} = 1$

EXAMPLE 14 Find the length of the vector $\vec{u}$ with initial point $(1, 2)$ and terminal point $(3, 6)$.

Solution $\vec{u} = (3 - 1, 6 - 2) = (2, 4)$

$$|\vec{u}| = \sqrt{2^2 + 4^2} = \sqrt{4 + 16} = \sqrt{20}$$

EXAMPLE

15

Find the possible values of a given $|\vec{AB}| = 2\sqrt{5}$ and the endpoints $A(a, 2)$ and $B(1, 4)$.

Solution

$$\begin{aligned} |\vec{AB}| &= \sqrt{(1-a)^2 + (4-2)^2} = \sqrt{(1-a)^2 + 4} = 2\sqrt{5} \\ (1-a)^2 + 4 &= 20 \\ (1-a)^2 &= 16 \\ 1-a &= 4 \quad \text{or} \quad 1-a = -4 \\ a &= -3 \quad \text{or} \quad a = 5 \end{aligned}$$

Check Yourself

- Find the position vector of $\vec{ML}$ with endpoints $M(3, 2)$ and $L(2, 3)$.
- Find the length of each vector.

a. $\vec{u} = (3, 1)$

b. $\vec{v} = (0, 3)$

c. $\vec{AB}$ with endpoints $A(2, 1)$ and $B(5, 4)$

Answers

1. $\vec{ML} = (-1, 1)$ 2. a. $\sqrt{10}$ b. 3 c. $3\sqrt{2}$

5. Equal Vectors

Definition

equal vectors

Two vectors are **equal** if and only if their corresponding scalar components are equal. In other words, the vectors $\vec{u} = (u_1, u_2)$ and $\vec{v} = (v_1, v_2)$ are equal if and only if $u_1 = v_1$ and $u_2 = v_2$.

EXAMPLE

16

$\vec{u}$ and $\vec{v}$ are equal vectors with $\vec{u} = (a + b, 3)$, $\vec{v} = (-1, b - a)$. Find a and b .

Solution

$\vec{u} = \vec{v}$, so $a + b = -1$ and $3 = b - a$.

$$\left. \begin{aligned} a + b &= -1 \\ b - a &= 3 \end{aligned} \right\}, \text{ so } b = 1 \text{ and } a = -2.$$

EXAMPLE

17

The vector $\vec{v} = (3, 7)$ has initial point $(2, 4)$. What is its terminal point?

Solution

Let the terminal point of $\vec{v}$ be (x, y) , so

$$(x - 2, y - 4) = (3, 7)$$

$$x - 2 = 3 \text{ and } y - 4 = 7 \text{ by the equality of vectors.}$$

$$\text{So } x = 5 \text{ and } y = 11.$$

EXAMPLE
18

The initial point and terminal point of $\overrightarrow{MN} = (2, 3)$ are $M(2, k)$ and $N(p, 1)$ respectively. Find $p + k$.

Solution

$$\begin{aligned}\overrightarrow{MN} &= (p - 2, 1 - k) = (2, 3) \\ p - 2 &= 2 \quad \text{and} \quad 1 - k = 3 \\ p &= 4 \quad \text{and} \quad k = -2 \\ \text{So } p + k &= 4 - 2 = 2.\end{aligned}$$

Check Yourself

1. $A(2, 1)$, $B(1, -3)$, $C(3, 2)$, and $D(a, b)$ are given. If $\overrightarrow{AC} = \overrightarrow{BD}$, find $|\overrightarrow{AD}|$.
2. Let $\vec{u} = (a, a + 1)$ and $\vec{v} = (3, b)$ such that $\vec{u} = \vec{v}$. Find a and b .

Answers

1. 3
2. $a = 3, b = 4$

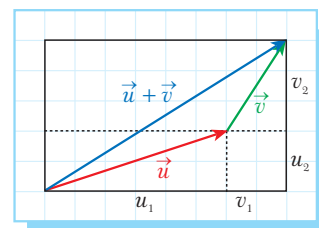
B. VECTOR OPERATIONS

1. Addition of Vectors

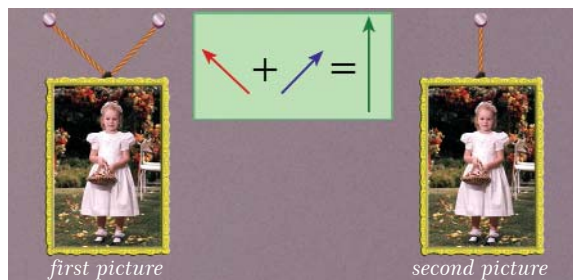
If $\vec{u} = (u_1, u_2)$ and $\vec{v} = (v_1, v_2)$, then

$$\vec{u} + \vec{v} = (u_1 + v_1, u_2 + v_2).$$

The figure opposite shows how the analytic definition of addition corresponds to the geometric one.



Two identical pictures are hung on a wall by pieces of string as shown in the figure. Each piece of string exerts a force upon the picture to support its weight. The sum of the forces of the first picture is equal to the force of the second picture. We can represent the forces as vectors. The box shows that the sum of the two vectors of the first picture equals the vector of the second picture.


EXAMPLE
19

$\vec{u} = (1, -2)$ and $\vec{v} = (-2, 1)$. Find $\vec{u} + \vec{v}$.

Solution

$$\vec{u} + \vec{v} = (1 - 2, -2 + 1) = (-1, -1).$$

EXAMPLE
20

$A(-3, 2)$ and $\overrightarrow{AB} = (5, -2)$, are given. Find the coordinates of B .

Solution

$$\begin{aligned}\text{Let us write } B(a, b). \text{ Now, } \overrightarrow{AB} &= (a - (-3), b - 2) = (a + 3, b - 2) \\ (a + 3, b - 2) &= (5, -2).\end{aligned}$$

$a + 3 = 5$ and $b - 2 = -2$ by the equality of vectors, so $a = 2$ and $b = 0$. Therefore, the coordinates of B are $(2, 0)$.

a. Properties of Vector Addition

Let $\vec{u} = (u_1, u_2)$, $\vec{v} = (v_1, v_2)$, and $\vec{w} = (w_1, w_2)$ be vectors in a plane. Then the following properties hold.

1. The sum of any two vectors in a plane is a vector. (closure property)
2. $\vec{u} + \vec{v} = \vec{v} + \vec{u}$ (commutative property)
3. $\vec{u} + (\vec{v} + \vec{w}) = (\vec{u} + \vec{v}) + \vec{w}$ (associative property)
4. $\vec{0}$ is the identity element:

$$\vec{u} + \vec{0} = (u_1 + 0, u_2 + 0) = (u_1, u_2) = \vec{u}.$$
5. $-\vec{u}$ is additive inverse of $\vec{u}$:

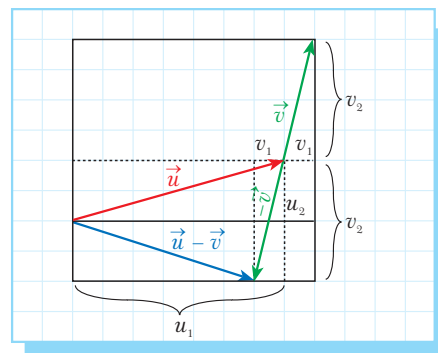
$$\vec{u} + (-\vec{u}) = (u_1 - u_1, u_2 - u_2).$$

2. Subtraction of Vectors

If $\vec{u} = (u_1, u_2)$ and $\vec{v} = (v_1, v_2)$ then

$$\vec{u} - \vec{v} = (u_1 - v_1, u_2 - v_2).$$

The figure opposite shows how the analytic definition of vector subtraction corresponds to the geometric one.



EXAMPLE

21

Subtract $\vec{v} = (3, 2)$ from $\vec{u} = (2, 4)$.

Solution

$$\vec{u} - \vec{v} = (2 - 3, 4 - 2) = (-1, 2)$$

a. Properties of Vector Subtraction

Let $\vec{u} = (u_1, u_2)$, $\vec{v} = (v_1, v_2)$, and $\vec{w} = (w_1, w_2)$ be vectors in a plane. Then the following properties hold.

1. The difference of any two vectors in a plane is a vector.
2. $\vec{u} - \vec{v} \neq \vec{v} - \vec{u}$. Therefore, vector subtraction is not commutative.
3. $\vec{u} - (\vec{v} - \vec{w}) \neq (\vec{u} - \vec{v}) - \vec{w}$. Therefore, vector subtraction is not associative.
4. $\vec{u} - \vec{0} \neq \vec{0} - \vec{u}$. Therefore there is no identity element for subtraction.

EXAMPLE

22

Given $\vec{w} = (1, -3)$, $K(3, 2)$ and $P(-1, 4)$, find $\vec{w} + \vec{KP}$ and $\vec{w} - \vec{KP}$.

Solution

$$\begin{aligned}\overrightarrow{KP} &= (-1 - 3, 4 - 2) = (-4, 2) \\ \vec{w} + \overrightarrow{KP} &= (1, -3) + (-4, 2) = (1 - 4, -3 + 2) = (-3, -1) \\ \vec{w} - \overrightarrow{KP} &= (1, -3) - (-4, 2) = (1 + 4, -3 - 2) = (5, -5)\end{aligned}$$

EXAMPLE

23

$M(1, 4)$, $N(3, 2m)$, $K(m, -1)$, and $|\overrightarrow{MN} + \overrightarrow{NK}| = 13$ are given. Find m .

Solution

$$\begin{aligned}\overrightarrow{MN} + \overrightarrow{NK} &= \overrightarrow{MK} \\ |\overrightarrow{MN} + \overrightarrow{NK}| &= |\overrightarrow{MK}| = 13 \\ \sqrt{(m-1)^2 + (-1-4)^2} &= 13 \\ (m-1)^2 + 25 &= 169 \\ (m-1)^2 &= 144 \\ m-1 &= 12 \text{ or } m-1 = -12 \\ m &= 13 \text{ or } m = -11\end{aligned}$$

3. Multiplication of a Vector by a Scalar

Let $\vec{v} = (v_1, v_2)$ and $c \in \mathbb{R}$, then $c \cdot \vec{v} = (c \cdot v_1, c \cdot v_2)$.

EXAMPLE

24

$\vec{u} = (-3, 2)$ and $\vec{v} = (2, -1)$ are given. Find $-2 \cdot \vec{u}$, $3 \cdot \vec{v}$, and $3 \cdot \vec{u} + 2 \cdot \vec{v}$.

Solution

$$\begin{aligned}-2 \cdot \vec{u} &= (-2 \cdot (-3), (-2) \cdot 2) = (6, -4) \\ 3 \cdot \vec{v} &= (3 \cdot 2, 3 \cdot (-1)) = (6, -3) \\ 3 \cdot \vec{u} + 2 \cdot \vec{v} &= 3 \cdot (-3, 2) + 2 \cdot (2, -1) = (-9, 6) + (4, -2) = (-5, 4)\end{aligned}$$

a. Properties of the Multiplication of a Vector by a Scalar

Let $\vec{u} = (u_1, u_2)$, $\vec{v} = (v_1, v_2)$ and $c, d \in \mathbb{R}$. Then the following properties hold:

1. $c \cdot (\vec{u} + \vec{v}) = c \cdot \vec{u} + c \cdot \vec{v}$. Look at the proof:

$$\begin{aligned}c \cdot (\vec{u} + \vec{v}) &= c \cdot (u_1 + v_1, u_2 + v_2) \\ &= (c \cdot u_1 + c \cdot v_1, c \cdot u_2 + c \cdot v_2) \\ &= c \cdot (u_1, u_2) + c \cdot (v_1, v_2) \\ &= c \cdot \vec{u} + c \cdot \vec{v}.\end{aligned}$$

2. $(c + d) \cdot \vec{u} = c \cdot \vec{u} + d \cdot \vec{u}$

3. $(c \cdot d) \cdot \vec{u} = c \cdot (d \cdot \vec{u}) = d \cdot (c \cdot \vec{u})$

4. $1 \cdot \vec{u} = \vec{u}$

5. $0 \cdot \vec{u} = \vec{0}$

6. $c \cdot \vec{0} = \vec{0}$

7. $|c \cdot \vec{u}| = |c| \cdot |\vec{u}|$.

EXAMPLE

25

Find $|2\vec{u} - 3\vec{v}|$ given $\vec{u} = (-2, 3)$ and $\vec{v} = (1, -1)$.

Solution

$$2\vec{u} - 3\vec{v} = 2(-2, 3) - 3(1, -1) = (-4, 6) - (3, -3) = (-4 - 3, 6 + 3) = (-7, 9)$$

$$\text{Therefore, } |2\vec{u} - 3\vec{v}| = \sqrt{(-7)^2 + 9^2} = \sqrt{49 + 81} = \sqrt{130}.$$

Check Yourself

- Find $|\vec{w}|$ given $\vec{u} = (3, -2)$, $\vec{v} = (-1, 4)$, $\vec{w} = (a, b)$ and $\vec{v} - \vec{w} = 3\vec{u}$.
- Find $\vec{x}$ given $2\vec{x} + \vec{y} = (1, 2)$, and $\vec{x} - \vec{y} = (-4, 4)$.

Answers

- $|\vec{w}| = 10\sqrt{2}$
- $\vec{x} = (-1, 2)$

EXAMPLE

26

Find each vector, given $\vec{u} = (-3, 1)$ and $\vec{v} = (2, -2)$.

a. $\vec{u} - 3\vec{v}$

b. $3\vec{u} + 2\vec{v}$

c. $4\vec{u} + \vec{v}$

Solution

a. $\vec{u} - 3\vec{v} = (-3, 1) - 3(2, -2) = (-3, 1) + (-6, 6) = (-3 - 6, 1 + 6) = (-9, 7)$

b. $3\vec{u} + 2\vec{v} = 3(-3, 1) + 2(2, -2) = (-9, 3) + (4, -4) = (-9 + 4, 3 - 4) = (-5, -1)$

c. $4\vec{u} + \vec{v} = 4(-3, 1) + (2, -2) = (-12, 4) + (2, -2) = (-12 + 2, 4 - 2) = (-10, 2)$

4. Standard Base Vectors

Definition

unit vector

A vector of length 1 is called a **unit vector**.

For example, the vector $\vec{w} = \left(\frac{3}{5}, \frac{4}{5}\right)$ is a unit vector.

For any non-zero vector $\vec{u} = (u_1, u_2)$, $\frac{\vec{u}}{|\vec{u}|}$ is a unit vector because $\frac{1}{|\vec{u}|} = \frac{1}{\sqrt{u_1^2 + u_2^2}}$ and

$$\frac{\vec{u}}{|\vec{u}|} = \frac{1}{|\vec{u}|} \cdot \vec{u} = \left(\frac{u_1}{\sqrt{u_1^2 + u_2^2}}, \frac{u_2}{\sqrt{u_1^2 + u_2^2}} \right).$$

$$\text{So } \left| \frac{\vec{u}}{|\vec{u}|} \right| = \sqrt{\left(\frac{u_1}{\sqrt{u_1^2 + u_2^2}} \right)^2 + \left(\frac{u_2}{\sqrt{u_1^2 + u_2^2}} \right)^2} = \sqrt{\frac{u_1^2}{u_1^2 + u_2^2} + \frac{u_2^2}{u_1^2 + u_2^2}} = \sqrt{\frac{u_1^2 + u_2^2}{u_1^2 + u_2^2}} = 1.$$

We sometimes use $\frac{\vec{u}}{|\vec{u}|}$ to find the direction of $\vec{u}$.

There are two important unit vectors, $\vec{i}$ and $\vec{j}$, defined as $\vec{i} = (1, 0)$ and $\vec{j} = (0, 1)$.

These vectors are special because we can use them to express any vector.

We call these vectors **standard base vectors**.

For example, let us express the vector $\vec{v} = (v_1, v_2)$ in terms of $\vec{i}$ and $\vec{j}$:
 $\vec{v} = (v_1, v_2) = v_1 \cdot \vec{i} + v_2 \cdot \vec{j}$.

In this expression,

$v_1 \cdot \vec{i}$ is the horizontal component,

$v_2 \cdot \vec{j}$ is the vertical component,

v_1 is the horizontal scalar component, and

v_2 is the vertical scalar component.

We can prove the proposition above by using algebraic operations on vectors and the properties of real numbers:

$$\begin{aligned}\vec{v} = (v_1, v_2) &= (1 \cdot v_1 + 0, 0 + 1 \cdot v_2) = (1 \cdot v_1, 0) + (0, 1 \cdot v_2) \\ &= v_1 \cdot (1, 0) + v_2 \cdot (0, 1) \\ &= v_1 \cdot \vec{i} + v_2 \cdot \vec{j}.\end{aligned}$$

EXAMPLE

27

- a. Write the vector $\vec{u} = (5, -8)$ in terms of $\vec{i}$ and $\vec{j}$.
 b. If $\vec{u} = 3 \cdot \vec{i} + 2 \cdot \vec{j}$ and $\vec{v} = -\vec{i} + 6 \cdot \vec{j}$, find $2 \cdot \vec{u} + 3 \cdot \vec{v}$.

Solution

a. $\vec{u} = 5 \cdot \vec{i} + (-8) \cdot \vec{j} = 5\vec{i} - 8\vec{j}$.

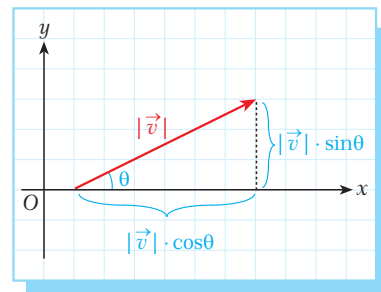
- b. The properties of addition and scalar multiplication of vectors show that we can manipulate vectors in the same way we manipulate algebraic expressions. Therefore,

$$\begin{aligned}2 \cdot \vec{u} + 3 \cdot \vec{v} &= 2 \cdot (3 \cdot \vec{i} + 2 \cdot \vec{j}) + 3 \cdot (-\vec{i} + 6 \cdot \vec{j}) \\ &= (6 \cdot \vec{i} + 4 \cdot \vec{j}) + (-3 \cdot \vec{i} + 18 \cdot \vec{j}) \\ &= (6 - 3) \cdot \vec{i} + (4 + 18) \cdot \vec{j} \\ &= 3 \cdot \vec{i} + 22 \cdot \vec{j}.\end{aligned}$$

Let $\vec{v}$ be a vector in the plane with its initial point at the origin. Let θ be the positive angle between the positive x -axis and $\vec{v}$ (see the figure). If we know the length and direction of $\vec{v}$, then we can resolve the vector into horizontal and vertical components in terms of θ :

$\vec{v}$ has length $|\vec{v}|$, and $\vec{v} = (v_1, v_2) = v_1 \cdot \vec{i} + v_2 \cdot \vec{j}$.

So $v_1 = |\vec{v}| \cdot \cos \theta$ and $v_2 = |\vec{v}| \cdot \sin \theta$.

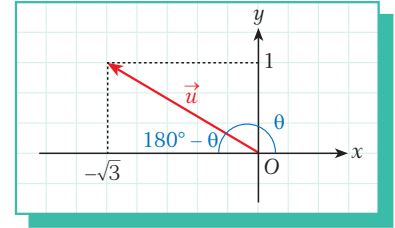


EXAMPLE

28

- a. $|\vec{v}| = 4$ cm and the angle between $\vec{v}$ and the positive x -axis is 60° . Find the horizontal and vertical components of $\vec{v}$ and express $\vec{v}$ in terms of $\vec{i}$ and $\vec{j}$.
 b. Find the angle θ between the vector $\vec{u} = -\sqrt{3} \cdot \vec{i} + \vec{j}$ and the positive x -axis.

- Solution** a. We have $\vec{v} = (v_1, v_2)$, where the scalar components are given by $v_1 = 4 \cdot \cos 60^\circ = 2$ and $v_2 = 4 \cdot \sin 60^\circ = 2\sqrt{3}$. Therefore, the horizontal component is $2 \cdot \vec{i}$ and the vertical component is $2\sqrt{3} \cdot \vec{j}$. Therefore, $\vec{v} = 2 \cdot \vec{i} + 2\sqrt{3} \cdot \vec{j}$.
- b. From the figure we see that θ has the property that
- $$\tan(180^\circ - \theta) = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}.$$
- Thus $180 - \theta = 30^\circ$, and so $\theta = 150^\circ$.



Check Yourself

- Express $\vec{v} = (3, 2)$ in terms of standard base vectors.
- Resolve the vector $\vec{u}$ with length 3 cm into its components if the angle between the positive x -axis and $\vec{u}$ is 120° .
- Find the angle θ between the vector $\vec{v} = -3 \cdot \vec{i} - 3\sqrt{3} \cdot \vec{j}$ and the positive x -axis.

Answers

- $\vec{v} = 3 \cdot \vec{i} + 2 \cdot \vec{j}$
- $\vec{u}_x = (-\frac{3}{2}, 0)$, $\vec{u}_y = (0, \frac{3\sqrt{3}}{2})$
- $\theta = 240^\circ$

C. PARALLEL VECTORS

We know from the geometrical analysis of vectors that two non-zero vectors are parallel if and only if multiplying one of them by a suitable scalar equals the other, that is,

for any $c \neq 0$, $\vec{u} \neq 0$, and $\vec{v} \neq 0$, $\vec{u} \parallel \vec{v}$ if and only if $\vec{u} = c \cdot \vec{v}$.

It follows that if $\vec{u} = (u_1, u_2)$ and $\vec{v} = (v_1, v_2)$, then $(u_1, u_2) = (c \cdot v_1, c \cdot v_2)$.

So $\vec{u} \parallel \vec{v}$ if and only if $\frac{u_1}{v_1} = \frac{u_2}{v_2} = c$.

EXAMPLE 29 Show that $\vec{u} = (2, 1)$ and $\vec{v} = (1, \frac{1}{2})$ are parallel.

Solution $\frac{2}{1} = \frac{1}{\frac{1}{2}} = 2$, so they are parallel.

EXAMPLE 30 Find the relation between x and y given $A(3, -1)$, $B(2, 3)$, $C(5, -4)$, $D(x, y)$, and $\vec{CD} \parallel \vec{AB}$.

Solution $\vec{AB} = (2 - 3, 3 - (-1)) = (-1, 4)$
 $\vec{CD} = (x - 5, y + 4)$
 $\vec{CD} \parallel \vec{AB}$ so $\frac{x-5}{-1} = \frac{y+4}{4}$.
 $4x - 20 = -y - 4$, so $4x + y - 16 = 0$.

EXAMPLE 31 A triangle ABC has vertices $A(3, 2)$, $B(-3, -2)$, and $C(2, -3)$. E and F are the midpoints of sides AB and AC respectively. Find the coordinates of E and F .

Solution Let us write $E(x, y)$ and $F(m, n)$.

Remember!
 If two parallel vectors $\vec{a}$ and $\vec{b}$ have at least one point in common, then $\vec{a}$ and $\vec{b}$ lie on the same straight line (they are collinear).
 For instance, if $\vec{AB} = k \cdot \vec{BC}$ for some $k \in \mathbb{R} \setminus \{0\}$, then A , B , and C are collinear.

$$\begin{array}{ll} \vec{BE} = \vec{EA} & \text{and} \quad \vec{CF} = \vec{FA} \\ \vec{BE} = (x + 3, y + 2) & \vec{CF} = (m - 2, n + 3) \\ \vec{EA} = (3 - x, 2 - y) & \vec{FA} = (3 - m, 2 - n) \\ x + 3 = 3 - x & m - 2 = 3 - m \\ 2x = 0 & 2m = 5 \\ x = 0 & m = \frac{5}{2} \\ y + 2 = 2 - y & n + 3 = 2 - n \\ 2y = 0 & 2n = -1 \\ y = 0 & n = -\frac{1}{2} \end{array}$$

Therefore the coordinates are $E(0, 0)$ and $F(\frac{5}{2}, -\frac{1}{2})$.

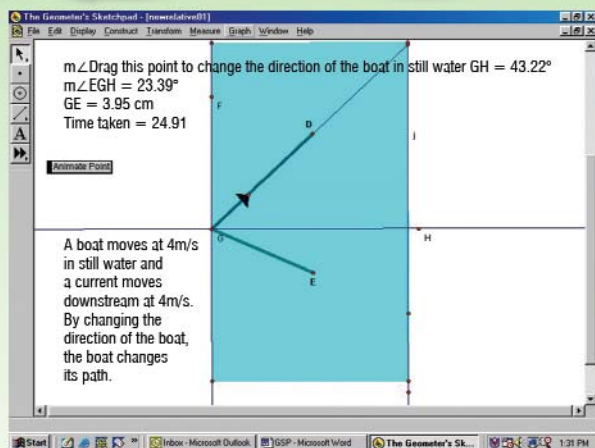
Check Yourself

1. Show that $\vec{u} = (-2, -3)$ and $\vec{v} = (4, 6)$ are parallel.
2. Find the value of k if $\vec{u} = (1, k)$ and $\vec{v} = (3, 6)$ are parallel.

Answers

1. check the scalar components 2. $k = 2$

Computer Applications



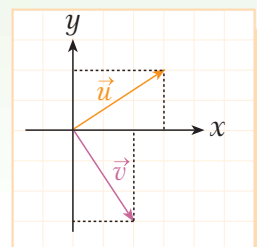
The screen opposite shows a simple problem illustrated using The Geometer's Sketchpad. Suppose a current flows at a certain velocity w downstream. A boat moves at a constant speed v . Which direction the boat take in order to reach the other side of the river in the shortest possible time?

The Geometer's Sketchpad calculates the time as we move the point H on the screen to set the direction of the boat. Using the Sketchpad we can move H to find the shortest possible time in the problem.

We can use computer applications such as The Geometer's Sketchpad, Cabri Geometry, or Javascript sketchpad to sketch vectors and solve problems. We can use an application to a problem, and then change certain values to see their effect. We can also use a computer application to add and subtract vectors, and to multiply a vector by a scalar.

Project:

Use The Geometer's Sketchpad, Cabri Geometry, or Javascript sketchpad to sketch the vectors $2\vec{u}$, $-\vec{v}$, $\vec{u} + \vec{v}$, $\vec{u} - 3\vec{v}$ and $\vec{PQ}$, $\vec{QP}$ if $P(3, -4)$ and $Q(4, -3)$.



EXERCISES 1.2

A. Basic Concepts of Vectors in the Analytic Plane

- Plot the points $A(-1, 1)$, $B(2, -1)$, $C(3, 1)$, and $D(-3, -1)$ in the plane.

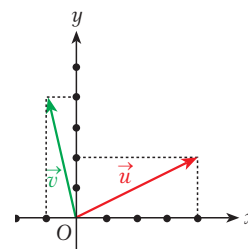
- Sketch the position vector of the vector with the given endpoints.

a. $\vec{AB}$	b. $\vec{CD}$	c. $\vec{EF}$
$A(-2, -3)$	$C(1, -5)$	$E(0, -3)$
$B(4, -1)$	$D(0, 2)$	$F(-4, 2)$

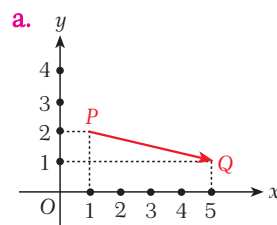
- Find the length of $\vec{MN}$ given $M(1, 4)$ and $N(-2, -1)$.

- The figure shows the vectors $\vec{u}$ and $\vec{v}$. Sketch the following vectors.

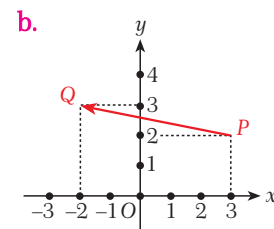
a. $2\vec{v}$	b. $-\vec{u}$
c. $\vec{u} + \vec{v}$	d. $\vec{u} - 2\vec{v}$
e. $2\vec{u} + \vec{v}$	



- Describe the vector with initial point P and terminal point Q.



c. $P(3, 2)$, $Q(8, 9)$



d. $P(-1, 3)$, $Q(1, 0)$

6. Find the vectors $\vec{a}$ and $\vec{b}$ if $2\vec{a} - 3\vec{b} = (4, 2)$ and $\vec{a} + \vec{b} = (2, 1)$.

B. Vector Operations

7. Find $\vec{u} + \vec{v}$, $\vec{u} - \vec{v}$, $2\vec{u}$, $2\vec{u} - \vec{v}$, $\vec{u} + 3\vec{v}$, and $-4\vec{u} + 3\vec{v}$ for the given vectors $\vec{u}$ and $\vec{v}$.

a. $\vec{u} = (2, 6)$, $\vec{v} = (1, 3)$

b. $\vec{u} = (-2, 3)$, $\vec{v} = (8, -2)$

c. $\vec{u} = (1, 0)$, $\vec{v} = (0, -2)$

d. $\vec{u} = \vec{i}$, $\vec{v} = \vec{j}$

e. $\vec{u} = -3\vec{i} + \vec{j}$, $\vec{v} = \vec{i} - \vec{j}$

f. $\vec{u} = 7\vec{i} + 5\vec{j}$, $\vec{v} = \vec{j} - \vec{i}$

8. Find $|\vec{u}|$, $|\vec{v}|$, $|2\vec{v}|$, $|\frac{1}{3}\vec{v}|$, $|\vec{u} + \vec{v}|$, and $|\vec{u} - \vec{v}|$ for the given vectors $\vec{u}$ and $\vec{v}$.

a. $\vec{u} = 3\vec{i} + \vec{j}$, $\vec{v} = -\vec{i} + 2\vec{j}$

b. $\vec{u} = 2\vec{i} - \vec{j}$, $\vec{v} = -\vec{i} - \vec{j}$

c. $\vec{u} = (2, 3)$, $\vec{v} = (0, 1)$

d. $\vec{u} = (3, 4)$, $\vec{v} = (2, 5)$

9. Find the horizontal and vertical scalar components of the vector with the given length and angle with the positive x -axis. Express the vector in terms of standard base vectors.

a. $|\vec{v}| = 20$, $\theta = 30^\circ$

b. $|\vec{v}| = 30$, $\theta = 120^\circ$

c. $|\vec{v}| = 1$, $\theta = 225^\circ$

d. $|\vec{u}| = 80$, $\theta = 135^\circ$

e. $|\vec{v}| = 4$, $\theta = 10^\circ$

f. $|\vec{u}| = \sqrt{3}$, $\theta = 300^\circ$

10. Given $\vec{AB} = -7\vec{i} + 2\vec{j}$ and $B(3, 11)$, find the coordinates of point A.

11. Given $\vec{AB} = 5\vec{i} + 6\vec{j}$ and $B(-4, 8)$, find the coordinates of point A.

12. $\vec{u} = 3\vec{i} + 4\vec{j}$ and $\vec{v} = 4\vec{i} + \vec{j}$ are given. Which vector is the longest?

13. Given $\vec{u} = 3\vec{i} + 4\vec{j}$, calculate $|\frac{1}{2}\vec{u} - \vec{i}|$.

C. Parallel Vectors

14. Show that $\vec{u} = (a, b)$ and $\vec{v} = (2a, 2b)$ are parallel.

15. $\vec{u} \parallel \vec{v}$, $\vec{u} = (1, k - 3)$, and $\vec{v} = (k, k - 4)$ are given. Find the value of k .

16. In a triangle ABC , the vertices are $A(-2, 3)$, $B(0, 1)$ and $C(4, 1)$. Points $D(-1, 2)$ and $E(1, 2)$ are on the sides AB and AC respectively. Show that $\vec{DE} \parallel \vec{BC}$.

17. In a triangle ABC , the vertices are $A(1, 3)$, $B(2, 1)$, and $C(3, 2)$. $E(\frac{3}{2}, 2)$ is on the side AB . Find the coordinates of F if F is on AC and $\vec{EF} \parallel \vec{BC}$.

THE DOT PRODUCT OF TWO VECTORS

We have seen how to add and subtract vectors, and how to multiply vectors by a scalar.

In this section we will introduce another operation on vectors, called the **dot product**.

The dot product is sometimes called the scalar product because the resulting product is a number and not a vector. It has applications in mathematics, as well as in engineering and physics.

A. DOT PRODUCT

Definition

dot product (scalar product, or Euclidean inner product)

Let $\vec{u} = (u_1, u_2)$ and $\vec{v} = (v_1, v_2)$ be two vectors in the plane.

The **dot product** of $\vec{u}$ and $\vec{v}$, denoted by $\vec{u} \cdot \vec{v}$, is defined by

$$\vec{u} \cdot \vec{v} = u_1 \cdot v_1 + u_2 \cdot v_2.$$

Thus, to find the dot product of two vectors we multiply the corresponding scalar components and then add them together.

EXAMPLE

32

Find the dot product of the given vectors.

a. $\vec{u} = (-2, 3)$ and $\vec{v} = (5, 4)$

b. $\vec{u} = -\vec{i} + \vec{j}$ and $\vec{v} = 5\vec{i} + \vec{j}$

Solution a. $\vec{u} \cdot \vec{v} = -2 \cdot 5 + 3 \cdot 4 = -10 + 12 = 2$

b. $\vec{u} \cdot \vec{v} = -1 \cdot 5 + 1 \cdot 1 = -5 + 1 = -4$

1. Properties of the Dot Product

The definition of the dot product gives us the following properties.

1. $\vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$ (commutative property)

2. $\vec{u} \cdot (\vec{v} + \vec{w}) = \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w}$ (associative property)

3. $c \cdot (\vec{u} \cdot \vec{v}) = (c \cdot \vec{u}) \cdot \vec{v}$

4. $\vec{u} \cdot \vec{u} = |\vec{u}|^2$

5. $\vec{u} \cdot \vec{v} \geq 0$, and $\vec{u} \cdot \vec{u} = 0$ if and only if $\vec{u} = \vec{0}$.

Proof

1. $\vec{u} \cdot \vec{v} = u_1 \cdot v_1 + u_2 \cdot v_2 = v_1 \cdot u_1 + v_2 \cdot u_2 = \vec{v} \cdot \vec{u}$, by the commutative property of real numbers.

2. $\vec{u} \cdot (\vec{v} + \vec{w}) = (u_1, u_2) \cdot (v_1 + w_1, v_2 + w_2) = u_1 \cdot (v_1 + w_1) + u_2 \cdot (v_2 + w_2)$
 $= u_1 \cdot v_1 + u_1 \cdot w_1 + u_2 \cdot v_2 + u_2 \cdot w_2$
 $= u_1 \cdot v_1 + u_2 \cdot v_2 + u_1 \cdot w_1 + u_2 \cdot w_2$
 $= \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w}$

3. $c \cdot (\vec{u} \cdot \vec{v}) = c \cdot (u_1 \cdot v_1 + u_2 \cdot v_2) = c \cdot u_1 \cdot v_1 + c \cdot u_2 \cdot v_2 = (c \cdot \vec{u}) \cdot \vec{v}$
4. $\vec{u} \cdot \vec{u} = u_1 \cdot u_1 + u_2 \cdot u_2 = u_1^2 + u_2^2 = |\vec{u}|^2$, so $\sqrt{\vec{u} \cdot \vec{u}} = |\vec{u}|$.
5. This proof is left as an exercise for you.

EXAMPLE

33

Find the length of the vector $\vec{u} = (2, -2)$ by using the dot product.

Solution $\vec{u} \cdot \vec{u} = 2 \cdot 2 + (-2) \cdot (-2) = 4 + 4 = 8$. Therefore, $|\vec{u}| = \sqrt{8}$.

Check Yourself

1. Find the dot product of $\vec{u} = 3 \cdot \vec{j}$ and $\vec{v} = \vec{i} + \vec{j}$.
2. Find the quantity $(\vec{u} + \vec{v})(\vec{u} - 3 \cdot \vec{v})$ given $\vec{u} = \vec{j}$, $\vec{v} = \vec{i}$.

Answers

1. 3
2. -2

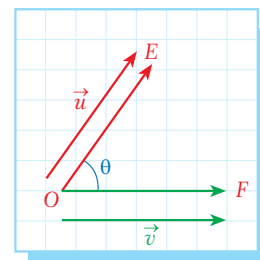
B. ANGLE BETWEEN TWO VECTORS

1. Angle Between Two Vectors

Definition

angle between two vectors

Let $\vec{u} = \overrightarrow{OE}$ and $\vec{v} = \overrightarrow{OF}$ be two non-zero vectors. The angle $\angle EOF$ is called the **angle between $\vec{u}$ and $\vec{v}$** . We use θ to represent the smaller angle between $\vec{u}$ and $\vec{v}$ when their initial points coincide.



Theorem

dot product theorem

Let θ be the angle measure between two non-zero vectors $\vec{u}$ and $\vec{v}$. Then

$$\vec{u} \cdot \vec{v} = |\vec{u}| \cdot |\vec{v}| \cdot \cos \theta.$$

Proof

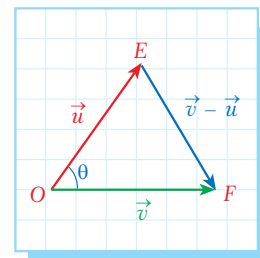
The proof is a nice application of the law of cosines.

As we know, $|\vec{v} - \vec{u}|^2 = (\vec{v} - \vec{u}) \cdot (\vec{v} - \vec{u}) = \vec{v}^2 + \vec{u}^2 - 2 \cdot \vec{v} \cdot \vec{u}$ (1).

Applying the cosine law to triangle EOF in the figure,

$$|\vec{v} - \vec{u}|^2 = |\vec{u}|^2 + |\vec{v}|^2 - 2 \cdot |\vec{u}| \cdot |\vec{v}| \cdot \cos \theta \quad (2) \quad (0 < \theta < \pi)$$

$$(\vec{v} - \vec{u}) \cdot (\vec{v} - \vec{u}) = |\vec{u}|^2 + |\vec{v}|^2 - 2 \cdot |\vec{u}| \cdot |\vec{v}| \cos \theta.$$



cosine law:
 $a^2 = b^2 + c^2 - 2bc \cdot \cos \angle A$

By (1) and (2),

$$\vec{v}^2 - 2 \cdot \vec{v} \cdot \vec{u} + \vec{u}^2 = |\vec{u}|^2 + |\vec{v}|^2 - 2 \cdot |\vec{u}| \cdot |\vec{v}| \cos \theta$$

$$2 \cdot \vec{v} \cdot \vec{u} = 2 \cdot |\vec{u}| \cdot |\vec{v}| \cos \theta.$$

Therefore, $\vec{u} \cdot \vec{v} = |\vec{u}| \cdot |\vec{v}| \cdot \cos \theta.$

EXAMPLE

34

Given $\vec{u} = (2, -2)$, $\vec{v} = (5, 8)$ and $\vec{w} = (-4, 3)$, find the following.

- a. $\vec{u} \cdot \vec{v}$ b. $(\vec{u} \cdot \vec{v}) \cdot \vec{w}$ c. $\vec{u} \cdot (2 \cdot \vec{v})$ d. $|\vec{w}|^2$

Solution

- a. By the definition of the dot product we have $\vec{u} \cdot \vec{v} = 2 \cdot 5 + (-2) \cdot 8 = -6$.
 b. Using the result from (a) we have $(\vec{u} \cdot \vec{v}) \cdot \vec{w} = -6 \cdot \vec{w} = -6 \cdot (-4, 3) = (24, -18)$.
 c. By property 3 of the dot product we have $\vec{u} \cdot (2 \cdot \vec{v}) = 2 \cdot (\vec{u} \cdot \vec{v}) = 2 \cdot (-6) = -12$.
 d. By property 4 of the dot product we have $|\vec{w}|^2 = \vec{w} \cdot \vec{w} = (-4)^2 + 3^2 = 25$.

One of the most important uses of the dot product is to find the angle between two vectors if the scalar components of the vectors are given. We simply calculate the angle by solving the equation given by the dot product theorem for $\cos \theta$. Let us state this important result clearly.

FINDING THE ANGLE BETWEEN TWO NON-ZERO VECTORS

If θ is the angle measure between two non-zero vectors $\vec{u}$ and $\vec{v}$ then $\cos \theta = \frac{\vec{u} \cdot \vec{v}}{|\vec{u}| \cdot |\vec{v}|}$.

EXAMPLE

35

- a. Find the cosine of the angle between the vectors $\vec{u} = (2, 5)$ and $\vec{v} = (5, 2)$.
 b. Find the angle between the vectors $\vec{u} = (1, \sqrt{2})$ and $\vec{v} = (2, 2\sqrt{2})$.

Solution

- a. $\vec{u} \cdot \vec{v} = 2 \cdot 5 + 5 \cdot 2 = 20$
 $|\vec{u}| = \sqrt{2^2 + 5^2} = \sqrt{29}$
 $|\vec{v}| = \sqrt{5^2 + 2^2} = \sqrt{29}$
 $\cos \theta = \frac{20}{\sqrt{29} \cdot \sqrt{29}}$ by the formula, and so $\cos \theta = \frac{20}{29}$.
 b. $\vec{u} \cdot \vec{v} = 1 \cdot 2 + \sqrt{2} \cdot 2\sqrt{2} = 2 + 4 = 6$
 $|\vec{u}| = \sqrt{1^2 + 2} = \sqrt{3}$
 $|\vec{v}| = \sqrt{4 + 8} = \sqrt{12}$
 $\cos \theta = \frac{6}{\sqrt{3} \cdot \sqrt{12}} = \frac{6}{\sqrt{36}} = 1$

Therefore, $\theta = 0^\circ$.

Check Yourself

1. Find the dot product of $\vec{u} = (0, 3)$ and $\vec{v} = (-1, -2)$.
2. Find the length of $\vec{u} = (1, 3)$ using the dot product.
3. Find $(\vec{u} \cdot \vec{v}) \cdot \vec{w}$ given $\vec{u} = (0, 3)$, $\vec{v} = (1, 2)$, and $\vec{w} = (-2, -1)$.
4. Find the angle between $\vec{u} = (2, 1)$ and $\vec{v} = (1, 3)$.

Answers

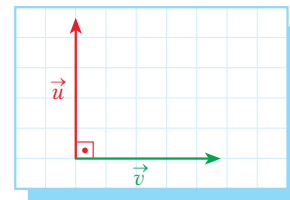
1. -6 2. $\sqrt{10}$ 3. $(-12, -6)$ 4. $\theta = 45^\circ$

2. Perpendicular and Parallel Vectors

Definition

perpendicular (orthogonal) vectors

Two non-zero vectors $\vec{u}$ and $\vec{v}$ are **perpendicular** if and only if the angle measure between them is 90° .



By the dot product theorem, $\vec{u} \cdot \vec{v} = |\vec{u}| \cdot |\vec{v}| \cdot \cos \theta$. We can conclude that for two non-zero vectors $\vec{u}$ and $\vec{v}$, $\vec{u} \cdot \vec{v}$ is zero if and only if θ equals 90° .

Theorem

Two non-zero vectors $\vec{u}$ and $\vec{v}$ are perpendicular if and only if $\vec{u} \cdot \vec{v} = 0$.

EXAMPLE

36

Are the vectors $\vec{u} = (6, -2)$ and $\vec{v} = (1, 3)$ perpendicular?

Solution

$$\vec{u} \cdot \vec{v} = 6 \cdot 1 + (-2) \cdot 3 = 0$$

So the vectors are perpendicular.

EXAMPLE

37

Are the vectors $\vec{u} = (3, 5)$ and $\vec{v} = (2, -6)$ perpendicular?

Solution

$$\vec{u} \cdot \vec{v} = 3 \cdot 2 + 5 \cdot (-6) = 6 - 30 = -24. \text{ So } \vec{u} \text{ and } \vec{v} \text{ are not perpendicular.}$$

From the definition of parallel vectors we know that $\vec{u} \parallel \vec{v}$ if and only if $\vec{u} = k \cdot \vec{v}$. We can write,

$$|\vec{u}| = |k| \cdot |\vec{v}| \quad (1)$$

$$\vec{u} \cdot \vec{v} = k \cdot \vec{v} \cdot \vec{v} = k \cdot |\vec{v}|^2 \quad (2)$$

$$\vec{u} \cdot \vec{v} = |\vec{u}| \cdot |\vec{v}| \cdot \cos \theta \quad (3).$$

From (1), (2) and (3),

$$|k| \cdot |\vec{v}| \cdot |\vec{v}| \cdot \cos \theta = k \cdot |\vec{v}|^2$$

$$\cos \theta = \frac{k}{|k|}$$

$$\cos \theta = 1 \text{ or } \cos \theta = -1$$

$$\theta = 0 \text{ or } \theta = 180.$$

From these results we can conclude that two vectors are parallel if and only if the angle measure between them is 0° or 180° .

Conclusion

Let θ be the angle measure between nonzero vectors $\vec{u}$ and $\vec{v}$.

Then $\vec{u} \parallel \vec{v}$ if and only if $\vec{u} \cdot \vec{v} = |\vec{u}| \cdot |\vec{v}|$ or $\vec{u} \cdot \vec{v} = -|\vec{u}| \cdot |\vec{v}|$.

EXAMPLE

38

Determine two vectors in the plane which are perpendicular to $\vec{u} = (4, 2)$.

Solution

Let $\vec{v} = (v_1, v_2)$ be perpendicular to $\vec{u}$.

$$\vec{u} \cdot \vec{v} = 4 \cdot v_1 + 2 \cdot v_2 = 0$$

$$v_2 = -2 \cdot v_1$$

If $v_1 = t \in \mathbb{R}$, $v_2 = -2t$, then $\vec{v} = (t, -2t)$.

Let $t = 1$ or 3 :

$\vec{v} = (1, -2)$ and $\vec{v} = (3, -6)$ are both perpendicular to $\vec{u}$.

EXAMPLE

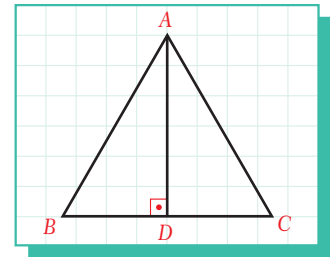
39

In an equilateral triangle ABC , D is the midpoint of BC . Find $\vec{AD} \cdot \vec{BC}$.

Solution

In an equilateral triangle, the median is also the altitude, as show in the diagram.

So $AD \perp BC$ and $\vec{AD} \cdot \vec{BC} = 0$.



EXAMPLE

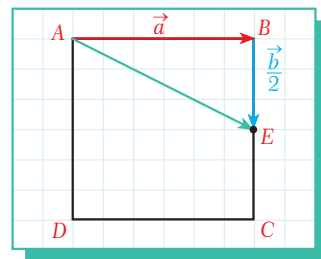
40

In a square $ABCD$, E is the midpoint of side BC and $|BC| = 4$ cm. Find $\vec{AE} \cdot \vec{AB}$.

Solution $\vec{AB} = \vec{a}$
 $\vec{AE} = \vec{a} + \frac{\vec{b}}{2}$

$$\vec{AE} \cdot \vec{AB} = \vec{a} \cdot \left(\vec{a} + \frac{\vec{b}}{2} \right) = \vec{a}^2 + \frac{\vec{a} \cdot \vec{b}}{2} = \vec{a}^2$$

Since $\vec{a} \perp \vec{b}$, $\vec{a}^2 = |\vec{a}|^2 = 4^2 = 16$.



EXAMPLE

41

Find the area of the triangle with vertices $A(2, 3)$, $B(0, 1)$, $C(3, 2)$.

Solution

$$A(\widehat{ABC}) = \frac{|\vec{AH}| \cdot |\vec{BC}|}{2}$$

$$\vec{AH} = (x_0 - 2, y_0 - 3)$$

$$\vec{BC} = (3, 1)$$

$$\vec{AH} \cdot \vec{BC} = 3 \cdot (x_0 - 2) + (y_0 - 3) = 0$$

$$3x_0 + y_0 - 6 - 3 = 0$$

$$3x_0 + y_0 = 9$$

$$\vec{BH} = k \vec{HC}$$

$$\vec{BH} = (x_0, y_0 - 1)$$

$$\vec{HC} = (3 - x_0, 2 - y_0)$$

$$\frac{x_0}{3 - x_0} = \frac{y_0 - 1}{2 - y_0}$$

$$2x_0 - x_0y_0 = 3y_0 - 3 - x_0y_0 + x_0$$

$$x_0 = 3y_0 - 3$$

$$x_0 - 3y_0 = -3$$

$$3x_0 + y_0 = 9 \quad (1)$$

$$x_0 - 3y_0 = -3 \quad (2)$$

Let us multiply (1) by 3. Then,

$$9x_0 + 3y_0 = 27$$

$$+ \quad x_0 - 3y_0 = -3$$

$$\hline 10x_0 = 24$$

$$x_0 = 2.4 \text{ and } y_0 = 1.8.$$

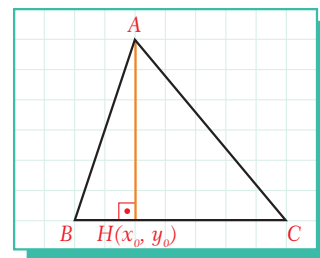
$$\vec{AH} = (0.4, -1.2)$$

$$|\vec{AH}| = \sqrt{0.16 + 1.44} = \sqrt{1.6} = 4 \cdot \frac{1}{\sqrt{10}}$$

$$\vec{BC} = (3, 1)$$

$$|\vec{BC}| = \sqrt{9 + 1} = \sqrt{10}$$

$$A(\widehat{ABC}) = \frac{1}{2} \cdot \frac{4}{\sqrt{10}} \cdot \sqrt{10} = 2$$



Theorem

triangle inequality

If $\vec{u}$ and $\vec{v}$ are vectors in the plane, then $|\vec{u} + \vec{v}| \leq |\vec{u}| + |\vec{v}|$. This property is called the **triangle inequality**.

Proof

By the property of the dot product, $|\vec{u} + \vec{v}|^2 = (\vec{u} + \vec{v}) \cdot (\vec{u} + \vec{v}) = \vec{u} \cdot \vec{u} + 2\vec{u} \cdot \vec{v} + \vec{v} \cdot \vec{v}$.

$$2\vec{u} \cdot \vec{v} \leq 2 \cdot |\vec{u}| \cdot |\vec{v}|$$

$$\{|\vec{u}|^2 + |\vec{v}|^2 + 2\vec{u} \cdot \vec{v}\} \leq \{|\vec{u}|^2 + |\vec{v}|^2 + 2|\vec{u}| \cdot |\vec{v}|\}$$

$|\vec{u} + \vec{v}|^2 \leq (|\vec{u}| + |\vec{v}|)^2$, since both $|\vec{u}| + |\vec{v}|$ and $|\vec{u} + \vec{v}|$ are non-negative.

Therefore, $|\vec{u} + \vec{v}| \leq |\vec{u}| + |\vec{v}|$.

Theorem

$\vec{u}$ and $\vec{v}$ are perpendicular in the plane if and only if $|\vec{u} + \vec{v}|^2 = |\vec{u}|^2 + |\vec{v}|^2$.

Proof

$$\begin{aligned} |\vec{u} + \vec{v}|^2 &= (\vec{u} + \vec{v}) \cdot (\vec{u} + \vec{v}) = \vec{u} \cdot \vec{u} + 2\vec{u} \cdot \vec{v} + \vec{v} \cdot \vec{v} = |\vec{u}|^2 + 2\vec{u} \cdot \vec{v} + |\vec{v}|^2 \\ &= |\vec{u}|^2 + |\vec{v}|^2, \text{ since } \vec{u} \cdot \vec{v} = 0. \end{aligned}$$

EXAMPLE

42

$|\vec{u}| = 3$, $|\vec{v}| = 4$, and the angle between $\vec{u}$ and $\vec{v}$ is 60° . Find $|2\vec{u} - 3\vec{v}|$.

Solution

$$\begin{aligned} |2\vec{u} - 3\vec{v}|^2 &= (2\vec{u} - 3\vec{v}) \cdot (2\vec{u} - 3\vec{v}) = 4\vec{u} \cdot \vec{u} + 9\vec{v} \cdot \vec{v} - 12\vec{u} \cdot \vec{v} \\ &= 4|\vec{u}|^2 + 9|\vec{v}|^2 - 12|\vec{u}| \cdot |\vec{v}| \cdot \cos\theta \\ &= 4 \cdot 3^2 + 9 \cdot 4^2 - 12 \cdot 3 \cdot 4 \cdot \frac{1}{2} \\ &= 36 + 144 - 72 \\ &= 108. \end{aligned}$$

Now $|2\vec{u} - 3\vec{v}|^2 = 108$, and so $|2\vec{u} - 3\vec{v}| = \sqrt{108} = 6\sqrt{3}$.

EXAMPLE

43

Find an equation for the line passing through $A(-1, 3)$ which is perpendicular to $\vec{n} = (3, 5)$.

Solution

Let us choose a variable point $B(x, y)$:

$$\vec{AB} = (x + 1, y - 3).$$

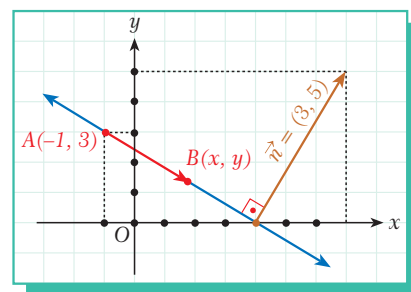
$\vec{AB}$ and $\vec{n}$ are perpendicular, so

$$\vec{AB} \cdot \vec{n} = 0.$$

$$3 \cdot (x + 1) + 5 \cdot (y - 3) = 0$$

$$3x + 3 + 5y - 15 = 0$$

$3x + 5y - 12 = 0$ is the required equation.



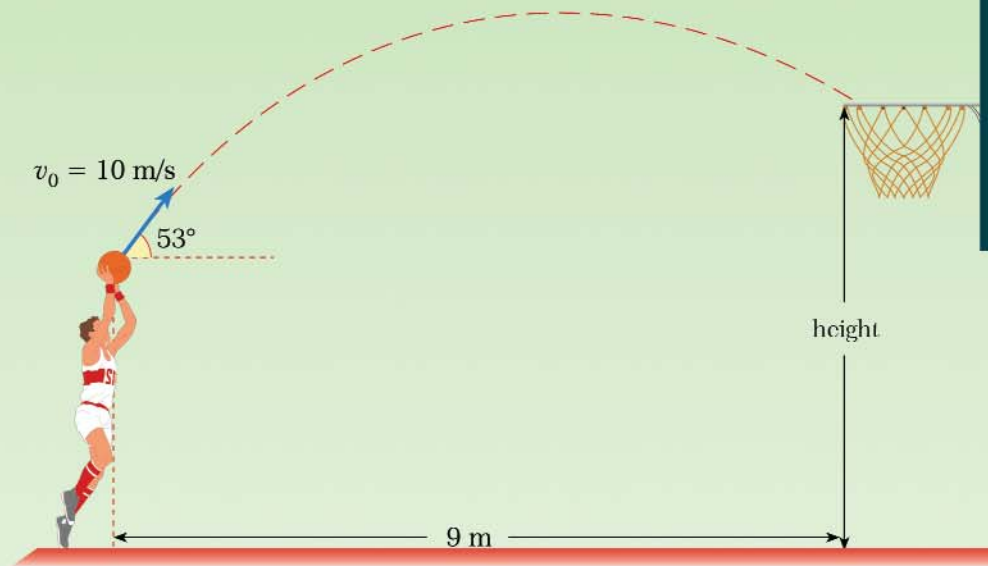
Check Yourself

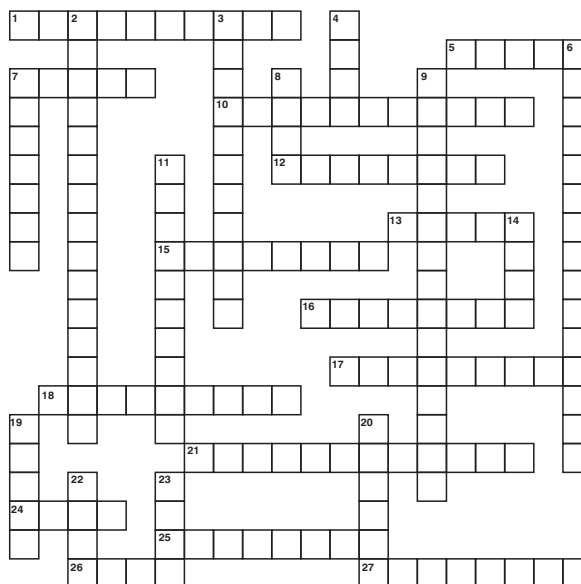
1. Find the value of a if $\vec{u} = (a, 1)$ is perpendicular to $\vec{v} = (3, 4)$.
2. In the right triangle ABC , AC is the hypotenuse, $BH \perp AC$, and H lies on AC .
Find $(\vec{HC} + \vec{CB}) \cdot (\vec{AB} + \vec{BH})$.
3. Find an equation for the line passing through $P(3, -1)$ which is perpendicular to $\vec{n} = (3, 1)$.

Answers

1. $a = -\frac{4}{3}$
2. 0
3. $3x + y - 8 = 0$

Projectile Motion in Sport





ACROSS

1. A vector with length zero.
5. An undefined concept in geometry which has no shape or size.
7. Being the main or most important part.
10. Being parallel.
12. The coordinate representing the position of a point along a line perpendicular to the y -axis in a plane.
13. A thin straight stick with a point, sometimes used to shoot things.
15. A mathematical statement that establishes the equality of two expressions.
16. Any of the four areas into which a plane is divided by the reference axes in a rectangular coordinate system, designated first, second, third, and fourth, counting counterclockwise from the area in which both coordinates are positive.
17. One of the two horizontal or vertical vectors whose sum is equal to a given vector.
18. In a triangle, the formula $\cos A = (b^2 + c^2 - a^2)/2bc$.
21. Vectors which have the same direction and length.
24. To arrange in sets of two.
25. The act or process of adding.
26. The direction 90° clockwise from north.
27. Either of two points marking the end of a line segment.

DOWN

2. A single vector that is the equivalent of a set of vectors.
3. The measure of heat of a body or environment.
4. The principal structural member of a ship.
6. The solution of an equation in which every variable is equal to zero.
7. An acute angle measured from due north or due south.
8. The size of a flat surface, calculated mathematically.
9. The property that states: if $a, b \in A$, then $a \cdot b \in A$ for an operation.
11. A point whose position is constant.
14. The direction 270° clockwise from due north and directly opposite east.
19. A picture such as a pie chart or bar graph, used to illustrate quantitative relationships.
20. To give the meaning of (a word or idea).
22. An undefined concept in geometry that describes a set of points along a path.
23. A relatively small, usually open craft used by fishermen.

EXERCISES 1.3

A. Dot Product

- Find the dot product of each pair of vectors.
 - $\vec{u} = (-2, 1)$ $\vec{v} = (3, 6)$
 - $\vec{u} = -7\vec{i} + 24\vec{j}$, $\vec{v} = 3\vec{i} - \vec{j}$
 - $\vec{u} = \sqrt{3}\vec{i} + \vec{j}$, $\vec{v} = 3\vec{i} - \vec{j}$
 - $\vec{u} = \vec{i} - \vec{j}$, $\vec{v} = 2\vec{i}$
 - $\vec{u} = -19\vec{j}$, $\vec{v} = 16\vec{j}$
- Find the following quantities given $\vec{u} = \vec{i} + 2\vec{j}$, $\vec{v} = -3\vec{i} + \vec{j}$, and $\vec{w} = 4\vec{i} + 3\vec{j}$.
 - $\vec{u} \cdot \vec{w} + \vec{u} \cdot \vec{w}$
 - $\vec{w} \cdot (\vec{u} + \vec{v})$
 - $(\vec{u} + \vec{w}) \cdot (\vec{u} - \vec{w})$
 - $2 \cdot (\vec{u} \cdot \vec{v})$
 - $(\vec{w} - \vec{v}) \cdot (\vec{w} + \vec{u})$
 - $(3\vec{u} - 2\vec{v}) \cdot (\vec{u} + 2\vec{v})$
- $\vec{u}$, $\vec{v}$ and $\vec{w}$ are unit vectors such that $\vec{u} + \vec{v} + \vec{w} = 0$. Find $\vec{u} \cdot \vec{v} + \vec{v} \cdot \vec{w} + \vec{w} \cdot \vec{u} = 0$.
- Evaluate the following quantities if the angle measure between $\vec{u}$ and $\vec{v}$ is 60° and $|\vec{u}| = 4$, $|\vec{v}| = 3$.
 - $\vec{u} \cdot \vec{v}$
 - $|\vec{u} + \vec{v}|^2$
 - $|\vec{u} - \vec{v}|^2$
 - $(3\vec{u} + 2\vec{v}) \cdot (\vec{u} + 2\vec{v})$
- Evaluate $|\vec{u} + \vec{v}|$ given $|\vec{u}| = 13$, $|\vec{v}| = 5$, and $|\vec{u} - \vec{v}| = 12$.

B. Angle Between Two Vectors

- Determine whether the given vectors are perpendicular or not.
 - $\vec{u} = (4, 6)$, $\vec{v} = (3, -2)$
 - $\vec{u} = (-5, 0)$, $\vec{v} = (0, 4)$
 - $\vec{u} = 3\vec{i}$, $\vec{v} = -\vec{j}$

- $\vec{u} = 2\vec{i} - 8\vec{j}$, $\vec{v} = 12\vec{i} + 3\vec{j}$
- $\vec{u} = 3\vec{i} - 4\vec{j}$, $\vec{v} = 8\vec{i} - 6\vec{j}$
- $\vec{u} = 4\vec{i}$, $\vec{v} = 3\vec{i}$

- Find the measure of the interior angles of the triangle ABC with vertices $A(1, 3)$, $B(1, 2)$, and $C(2, 2)$.
- Find two perpendicular vectors to each given vector.
 - $\vec{a} = 3\vec{i} - \vec{j}$
 - $\vec{b} = -8\vec{i} - 6\vec{j}$
 - $\vec{c} = \vec{i} + 2\vec{j}$
- For what values of t are $\vec{u} = 4\vec{i} - 5\vec{j} \cdot t$ and $\vec{v} = 3\vec{i} - 2\vec{j}$ perpendicular?
- $|\vec{u}| = 5$ and $|\vec{v}| = 3$ are given. For what values of t are $\vec{u} + t\vec{v}$ and $\vec{u} - t\vec{v}$ perpendicular?
- For what values of t do $\vec{a} = t\vec{i} + \vec{j}$ and $\vec{b} = \vec{i} + t\vec{j}$ have angle measure 120° between them?
- Show that the equality $|\vec{u} + \vec{v}| = |\vec{u} - \vec{v}|$ holds when $\vec{u}$ and $\vec{v}$ are orthogonal.
- Verify the equality $|\vec{u} + \vec{v}|^2 + |\vec{u} - \vec{v}|^2 = 2 \cdot (|\vec{u}|^2 + |\vec{v}|^2)$.

14. In a rhombus $ABCD$, one side measures 6 cm and E, F are the midpoints of sides AD and DC respectively. The angle measure between sides AD and DC is 120° . Find the quantity $\vec{BE} \cdot \vec{AF}$.
15. In a rectangle $ABCD$, $\vec{DC} = 3\vec{AD}$ and point E is on DC . Find the quantity $\vec{AE} \cdot \vec{BE}$ given $\vec{DE} = 2\vec{EC}$ and $|\vec{AD}| = 3$ cm.
16. Show that the inequality $|\vec{u} + \vec{v}| > |\vec{u} - \vec{v}|$ holds when the angle measure between $\vec{u}$ and $\vec{v}$ is less than 90° .
17. Find two unit vectors that make angles of 45° with $-\vec{i} + \vec{j}$.
18. Let $\vec{u}$ and $\vec{v}$ be vectors and let α be a scalar. Verify the given properties.
- $\vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$
 - $(\alpha \cdot \vec{u}) \cdot \vec{v} = \alpha \cdot (\vec{u} \cdot \vec{v}) = \vec{u} \cdot (\alpha \cdot \vec{v})$
19. Write the equation of the line passing through $A(-1, -1)$ which is perpendicular to $\vec{u} = (3, 4)$.
20. Show that the diagonals of a rhombus are perpendicular using vectors.
21. Find the area of a triangle with vertices $A(2, 2)$, $B(0, 2)$, and $C(1, 4)$.
22. Find the area of a rhombus with vertices $A(2, 0)$, $B(-3, 3)$, $C(-8, 0)$, and $D(-3, -3)$.
23. Find the area of a rectangle with vertices $A(3, 2)$, $B(9, 2)$, $C(9, 5)$, and $D(3, 5)$.
24. For any vectors $\vec{u}$, $\vec{v}$, and $\vec{w}$ prove that $(\vec{u} \cdot \vec{w}) \cdot \vec{v} - \vec{u} \cdot (\vec{v} \cdot \vec{w})$ is perpendicular to $\vec{w}$.
25. Show that the altitudes of an acute-angled triangle are concurrent using vectors.
26. Find the distance of $P(x_0, y_0)$ from the line $ax + by + c = 0$ using vectors.
27. Find the area of a parallelogram with vertices $A(2, -1)$, $B(6, 0)$, $C(8, 3)$, and $D(4, 2)$.

CHAPTER SUMMARY

- A line segment with direction is called a directed line segment.

- A vector in the plane is a directed line segment.

• Addition of Two Vectors Geometrically

For two vectors $\vec{PQ}$ and $\vec{QR}$ in a plane, $\vec{PQ} + \vec{QR}$ is the sum of $\vec{PQ}$ and $\vec{QR}$. There are two ways to add vectors geometrically: the polygon method and the parallelogram method.

• The Polygon Method

In this method we draw the first vector. Then we place the initial point of the second vector at the terminal point of the first, the initial point of the third vector at the terminal point of the second, and so on until we place the initial point of the n^{th} vector at the terminal point of the $(n - 1)^{\text{th}}$ vector. The vector whose initial point is the initial point of the first vector and whose terminal point is the terminal point of the last vector is the resultant (sum) vector.

• The Parallelogram Method

In this method we draw the first vector, and then draw the second vector with its initial point at the initial point of the first vector. We make a parallelogram by drawing two additional sides, each passing through the terminal point of one of the vectors and parallel to the other vector. The sum is drawn along the diagonal from the common initial point to the intersection of the two lines.

• Multiplication of a Vector by a Scalar

For a real number a and a vector $\vec{u}$:

1. If $a > 0$, then the vector $a \cdot \vec{u}$ has the same direction to $\vec{u}$ and the length $|a \cdot \vec{u}| = a \cdot |\vec{u}|$.
2. If $a < 0$, then the vector $a \cdot \vec{u}$ has the opposite direction to $\vec{u}$ and the length $|a \cdot \vec{u}| = |a| \cdot |\vec{u}|$.
3. If $a = 0$, then $a \cdot \vec{u} = \vec{0}$.

• Parallel Vectors

Let $\vec{a}$ and $\vec{b}$ be two vectors. $\vec{a}$ and $\vec{b}$ are parallel if and only if $\vec{a} = k \cdot \vec{b}$ where $k \neq 0$.

• Non-Parallel Vector

If $\vec{a}$ is not parallel to $\vec{b}$, then $h \cdot \vec{a} = k \cdot \vec{b}$ when $h = k = 0$.

• Components of a Vectors

A vector in a plane has two components, called the horizontal and vertical components of the vector.

Let $|\vec{a}| = (a_1, a_2)$ be a vector, then $|\vec{a}| = \sqrt{a_1^2 + a_2^2}$.

• Equal Vectors

Two vectors are equal if and only if their corresponding component vectors are equal.

In other words, for $\vec{a} = (a_1, a_2)$ and $\vec{b} = (b_1, b_2)$, then $\vec{a} = \vec{b}$ if and only if $a_1 = b_1$ and $a_2 = b_2$.

• Standard Base Vectors

$\vec{i} = (1, 0)$ and $\vec{j} = (0, 1)$ are called the standard base vectors.

• Linear Combination of Vectors

Let $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_k$ be vectors in a plane and let $c_1, c_2, \dots, c_k$ be scalars. $\vec{v} = c_1 \cdot \vec{u}_1 + c_2 \cdot \vec{u}_2 + \dots + c_k \cdot \vec{u}_k$ is called a linear combination of vectors.

• Dot Product

The dot product of $\vec{u} = (u_1, u_2)$ and let $\vec{v} = (v_1, v_2)$ is the scalar quantity $\vec{u} \cdot \vec{v} = u_1 \cdot v_1 + u_2 \cdot v_2$.

• Angle Between Two Vectors

Let θ be the angle between two non-zero vectors $\vec{u}$ and $\vec{v}$.

1. $\vec{u} \parallel \vec{v}$ if and only if $\vec{u} \cdot \vec{v} = |\vec{u}| \cdot |\vec{v}|$ or $\vec{u} \cdot \vec{v} = -|\vec{u}| \cdot |\vec{v}|$ because $\theta = 0^\circ$ or $\theta = 180^\circ$.
2. $\vec{u}$ and $\vec{v}$ are perpendicular if and only if $\vec{u} \cdot \vec{v} = 0$.

Concept Check

1. Can we use directed line segments in traffic? Give an example.
2. What is the difference between a vector and a scalar?
3. Is it possible to add a hundred vectors in a plane using the polygon method or the parallelogram method?
4. How do you change the direction of a vector using a real number?
5. a. Draw a diagram to show how to add two vectors.
b. Draw a diagram to show how to subtract two vectors.
6. Can you equalize two non-parallel vectors using multiplication by two real numbers?
7. In a plane, how many components do you need to resolve a vector?
8. How many standard base vectors are there in the plane?
9. What is the difference between the inner product and dot product of two vectors?
10. How do you use the dot product to find the angle measure between two vectors?
11. How do you use the dot product to determine whether two vectors are parallel or not?
12. How do you use the dot product to determine whether two vectors are perpendicular or not?

CHAPTER REVIEW TEST 1A

1. In triangle ABC , G is the centroid. Describe $\overrightarrow{AG} + \overrightarrow{GC} + \overrightarrow{CA}$.

A) $\vec{0}$ B) $\overrightarrow{AG}$ C) $\overrightarrow{BG}$ D) $2\overrightarrow{CA}$ E) $\overrightarrow{AC}$

2. In the figure, K, L, M, N, P, R are the midpoints of the sides $ABCDEF$ respectively. Describe $\overrightarrow{KB} + \overrightarrow{LC} + \overrightarrow{MD} + \overrightarrow{NE} + \overrightarrow{PF}$.

A) $\overrightarrow{AR}$ B) $\overrightarrow{RA}$ C) $\overrightarrow{FK}$ D) $\overrightarrow{FA}$ E) $\overrightarrow{PA}$

3. In the figure, $|AB| = |BC| = |CD|$ and the vectors $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are given. Which one of the following is the linear combination of $\vec{c}$ in terms of $\vec{a}$ and $\vec{d}$?

A) $\frac{\vec{a} + \vec{d}}{2}$ B) $\frac{2\vec{a} + \vec{d}}{2}$ C) $\frac{2\vec{a} + \vec{d}}{3}$
 D) $\frac{\vec{a} + 2\vec{d}}{3}$ E) $\frac{\vec{a} + 2\vec{d}}{2}$

4. $\overrightarrow{AD} = \frac{1}{4}\overrightarrow{AB}$ and $\overrightarrow{BE} = \frac{1}{2}\overrightarrow{BC}$ are given. Express $\overrightarrow{DE}$ in terms of $\overrightarrow{AB}$ and $\overrightarrow{BC}$.

A) $\frac{3}{4}\overrightarrow{AB} + \frac{1}{2}\overrightarrow{BC}$ B) $\frac{1}{4}\overrightarrow{AB} + \frac{1}{2}\overrightarrow{BC}$
 C) $\frac{3}{4}\overrightarrow{AB} - \frac{1}{2}\overrightarrow{BC}$ D) $\frac{1}{4}\overrightarrow{AB} - \frac{1}{2}\overrightarrow{BC}$
 E) $\frac{1}{2}\overrightarrow{BC} - \frac{3}{4}\overrightarrow{AB}$

5. $(-1, 4) + \vec{v} = (4, 5)$ is given. Find $\vec{v}$.

A) $(3, 9)$ B) $(-5, -1)$ C) $(5, 1)$
 D) $(1, 5)$ E) $(9, 3)$

6. Find the coordinates of B if $\overrightarrow{AB} = (7, 3)$ and $A(2, -1)$.

A) $(9, -4)$ B) $(9, -2)$ C) $(9, 4)$
 D) $(-9, -4)$ E) $(9, 2)$

7. $\vec{u} = -2\vec{i} + 3\vec{j}$ and $\vec{v} = \vec{i} - 2\vec{j}$ are given. Find $3\vec{u} - 2\vec{v}$.

A) $4\vec{i} + 5\vec{j}$ B) $8\vec{i} + 5\vec{j}$
 C) $-8\vec{i} + 13\vec{j}$ D) $-4\vec{i} + 13\vec{j}$
 E) $4\vec{i} + 13\vec{j}$

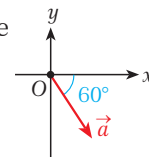
8. Find the unit vector with opposite direction to $\overrightarrow{AB}$ given $A(6, 3)$, and $B(2, 6)$.

A) $(\frac{4}{5}, \frac{3}{5})$ B) $(\frac{4}{5}, -\frac{3}{5})$ C) $(-\frac{4}{7}, \frac{3}{7})$
 D) $(\frac{4}{7}, -\frac{3}{7})$ E) $(-\frac{4}{5}, -\frac{3}{5})$

9. $\vec{u} = \vec{i} - 5\vec{j}$ and $\vec{v} = 2\vec{i} + 3\vec{j}$ are given. Find $3\vec{u} + 2\vec{v}$.

A) $3\vec{i} - 2\vec{j}$ B) $5\vec{i} - \vec{j}$ C) $7\vec{i} - 9\vec{j}$
 D) $2\vec{i} + 3\vec{j}$ E) $8\vec{i} - \vec{j}$

10. Describe the unit vector $\vec{a}$ in the figure.



A) $\frac{\sqrt{3}}{2}\vec{i} + \frac{1}{2}\vec{j}$ B) $-\frac{\sqrt{3}}{2}\vec{i} + \frac{1}{2}\vec{j}$
 C) $\frac{1}{2}\vec{i} - \frac{\sqrt{3}}{2}\vec{j}$ D) $\frac{1}{2}\vec{i} + \frac{\sqrt{3}}{2}\vec{j}$
 E) $\frac{\sqrt{3}}{2}\vec{i}$

11. For how many values of $m > 0$ are $\vec{a} = (2, 1 - m)$ and $\vec{b} = (m + 1, -4)$ parallel?

A) 1 B) 2 C) 3 D) 4 E) 5

12. Which one of the following is true for vectors $\vec{a} = (3, -6)$, $\vec{b} = (-6, 7)$ and $\vec{c} = (9, -13)$?

A) $\vec{a} = 4\vec{b} + 3\vec{c}$ B) $\vec{a} = 3\vec{b} + 4\vec{c}$

C) $\vec{a} = 4\vec{b} - 3\vec{c}$ D) $\vec{a} = 3\vec{b} - 4\vec{c}$

E) $\vec{a} = \vec{b} + \vec{c}$

13. Find the measure of the angle between the unit vectors $\vec{a}$ and $\vec{b}$ in degrees if their dot product is $\frac{1}{2}$.

A) 45° B) 60° C) 120° D) 135° E) 150°

14. $A(2, 5)$, $B(-1, 3)$, $C(m, 6)$, and $\vec{AB} \perp \vec{BC}$ are given. Find m .

A) 3 B) 2 C) 1 D) -2 E) -3

15. Which one of the following is perpendicular to $\vec{v} = (3, -4)$?

A) $(-\frac{4}{5}, -\frac{3}{5})$ B) $(4, -3)$ C) $(\frac{3}{5}, \frac{4}{3})$

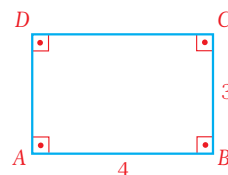
D) $(-\frac{3}{5}, -\frac{4}{5})$ E) $(-\frac{4}{5}, \frac{3}{5})$

16. ABCD is the rectangle in the figure.

Find $\vec{AB} \cdot \vec{AC}$ if

$|AB| = 4$ and

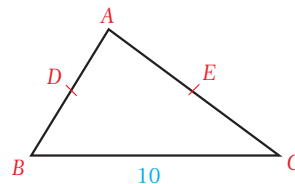
$|BC| = 3$.



A) 9 B) 12 C) 15 D) 16 E) 20

17. In the figure, D and E are the midpoints of the sides of triangle ABC.

Find $(\vec{AD} - \vec{AE}) \cdot \vec{BC}$.



A) 100 B) 150 C) 0 D) -50 E) -100

18. G is the centroid of a triangle with vertices $A(1, 1)$, $B(-4, 2)$, and $C(0, 6)$. Find the scalar product of $\vec{AB} \cdot \vec{BG}$.

A) -8 B) -14 C) 0 D) 1 E) 12

19. A triangle ABC has vertices $A(1, 1)$, $B(-4, 2)$, and $C(0, 6)$. Find the scalar component of $\vec{AB}$ along $\vec{BC}$.

A) $\frac{4}{\sqrt{2}}$ B) $-\frac{4}{\sqrt{2}}$ C) $\frac{3}{4}$ D) $-\frac{3}{4}$ E) $\frac{3}{2}$

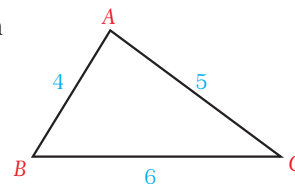
20. ABC is a triangle with

$|AB| = 4$,

$|AC| = 5$,

$|BC| = 6$. Find

$\vec{AC} \cdot (\vec{AB} + \vec{BC})$.



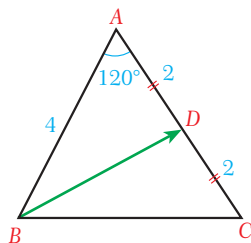
A) 20 B) 24 C) 25 D) 30 E) 50

CHAPTER REVIEW TEST 1B

1. $|\vec{a}| = 3$, $|\vec{b}| = 5$, and the angle between $\vec{a}$ and $\vec{b}$ is 60° . Find $|\vec{a} + \vec{b}|$.

A) 7 B) 8 C) 9 D) 10 E) 11

2. In the figure,
 $|AB| = |AC| = 4$,
 $m\angle A = 120^\circ$, and
 $|AD| = |DC|$.
 Find $\vec{BC} \cdot \vec{BD}$.



A) 48 B) 36 C) 30 D) 24 E) 18

3. Given $|\vec{a}| = 12$, $|\vec{b}| = 5$, and $|\vec{a} - \vec{b}| = 8$, find $\cos(\vec{a}, \vec{b})$.

A) $\frac{3}{8}$ B) $\frac{7}{8}$ C) $\frac{3}{4}$ D) $\frac{5}{13}$ E) $\frac{12}{13}$

4. In an equilateral triangle ABC , $D \in [AC]$, $|BC| = 6$, and $|CD| = 2$. Find $\vec{BC} \cdot \vec{BD}$.

A) 30 B) 24 C) 20 D) 18 E) 12

5. $|\vec{a}| = 7$, $|\vec{b}| = 10$, and $|\vec{a} + \vec{b}| = \sqrt{73}$ are given. Find $|\vec{a} - \vec{b}|$.

A) 3 B) 8 C) 13 D) 15 E) 17

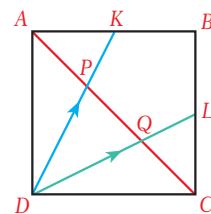
6. In the right triangle ABC , $m\angle A = 90^\circ$, $D \in [AC]$, and $|AB| = k$. Find $\vec{BA} \cdot \vec{BD}$.

A) k B) k^2 C) $\frac{k^2}{2}$ D) $\frac{k}{2}$ E) $2k^2$

7. What is the angle between $\vec{a} = (1, \sqrt{3})$ and $\vec{b} = (-4, -4\sqrt{3})$ in degrees?

A) 60° B) 120° C) 135° D) 150° E) 180°

8. A square $ABCD$ has side 3 cm, and K, L are the midpoints of sides AB and BC respectively. P and Q trisect AC as shown in the figure. Find $\vec{DP} \cdot \vec{PQ}$.



A) 0 B) 1 C) -1 D) -2 E) 3

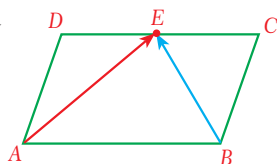
9. $\vec{a}$, $\vec{b}$, and $\vec{c}$ are three vectors such that $\vec{b} = \vec{a} - 2\vec{c}$, $\vec{b} \perp \vec{c}$, and $|\vec{a}| = 4|\vec{c}|$. Find the angle between $\vec{a}$ and $\vec{c}$ in degrees.

A) 30° B) 45° C) 60° D) 75° E) 90°

10. Which one of the following is false for two vectors $\vec{a}$, $\vec{b}$ and a scalar $k \in \mathbb{R}$?

A) If $\vec{a} \perp \vec{b}$, $\vec{a} \cdot \vec{b} = 0$.
 B) If $\vec{a} \parallel \vec{b}$, $\vec{b} \cdot \vec{a} = 0$.
 C) $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$
 D) $(k \cdot \vec{a} + k \cdot \vec{b}) = k \cdot (\vec{a} + \vec{b})$
 E) $(k \cdot \vec{a} \cdot k \cdot \vec{b}) = k^2 \cdot \vec{a} \cdot \vec{b}$

11. In the figure, $ABCD$ is a parallelogram and
 $|BC| = 1$,
 $|DC| = 2$,
 $|DE| = |EC|$. Find
 $(\vec{BC} + \vec{CE}) \cdot (\vec{AD} + \vec{DE})$.



A) 0 B) 1 C) 2 D) 3 E) 4

12. In an equilateral triangle ABC , D and E are the midpoints of $[AC]$ and $[BC]$ respectively. Find $\vec{DC} \cdot (\vec{BA} + \vec{AE})$.

A) $|\vec{AB}|^2$ B) $|\vec{EB}|^2$ C) $-\frac{1}{2} \cdot |\vec{DC}|^2$
D) $\frac{1}{2} \cdot |\vec{DC}|^2$ E) $\frac{1}{2} \cdot |\vec{EB}|^2$

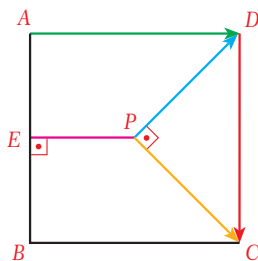
13. In an equilateral triangle ABC with side 1 cm, what is $\vec{BC} \cdot \vec{CA}$?

A) $\frac{1}{2}$ B) $-\frac{1}{2}$ C) $-\frac{\sqrt{3}}{2}$ D) -2 E) $\frac{\sqrt{3}}{2}$

14. For what values of k are the vectors $\vec{a} = (12k, 9)$ and $\vec{b} = (-4, 3)$ linearly dependent?

A) -3 B) -2 C) -1 D) -0 E) 1

15. In a square $ABCD$, the side is 4 cm,
 $[AB] \perp [EP]$, $[PD] \perp [PC]$,
and $|AE| = |EB|$.
Find $\vec{PE} \cdot (\vec{PC} + \vec{PD})$.

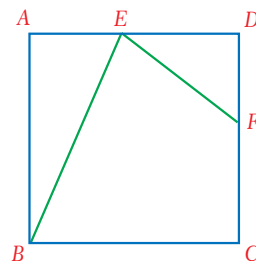


A) -16 B) -12 C) -10 D) -8 E) -6

16. In a triangle ABC , $\vec{AB} = (-4, 2a)$, $\vec{AC} = (a, -4)$ and the length of $\vec{BC}$ is 10 cm. Find a possible value of a .

A) 1 B) 2 C) 3 D) 4 E) 5

17. In a square $ABCD$ with side 2 cm,
 $|AE| = |ED|$ and
 $|DF| = |FC|$.
Find $\vec{EF} \cdot (\vec{EA} + \vec{AB})$.



A) 1 B) 2 C) -2 D) -1 E) 0

18. $\vec{a} + \vec{b} = (1, 3)$ and $2\vec{a} + \vec{b} = (4, 6)$ are given. Find the measure of the angle between $\vec{a}$ and $\vec{b}$.

A) 30° B) 45° C) 90° D) 135° E) 150°

19. In a triangle ABC , $\vec{AB} = (-2, -5)$ and $\vec{AC} = (2, -2)$. Find the length of $\vec{BC}$.

A) 1 B) 2 C) 3 D) 4 E) 5

20. $\vec{a} = (12, -5)$, and $\vec{b} = (3, 4)$ are given. Find $\sin(\angle \vec{a}, \vec{b})$.

A) $\frac{5}{13}$ B) $\frac{12}{13}$ C) $\frac{63}{65}$ D) $\frac{10}{13}$ E) $\frac{4}{13}$

A. THE CONCEPT OF CONGRUENCE

1. Congruent Figures and Polygons

The world around us is full of objects of various shapes and sizes. If we tried to compare some of these objects we could put them in three groups:

- ◆ objects which have a different shape and size,
- ◆ objects which are the same shape but a different size, and
- ◆ objects which are the same shape and size.

The tools in the picture at the right have different shape and size.

The pictures below show tools which have the same shape but different size. In geometry, figures like this are called **similar figures**. We will study similar figures in Chapter 3.



Factories often need to produce many parts with exactly the same size and shape.

The pictures below show objects which are the same size and shape.

In this section, we will study figures which have this property.



Definition

congruent figures

Figures that have the same size and shape are called **congruent figures**. We say 'A is congruent to B' (or 'B is congruent to A') if A and B are congruent figures.



Congruence in nature: the petals of this flower are congruent.

The pictures at the bottom of the previous page show some examples of congruent objects. The pictures below show two more examples. In these two examples there is only one piece left to fit in the puzzle. Therefore, without checking anything, we can say that each piece and its corresponding place are congruent.



2. Congruent Triangles

Definition

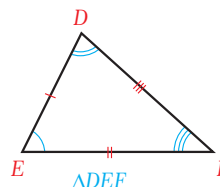
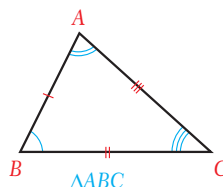
congruent triangles

Two triangles are congruent if and only if their corresponding sides and angles are congruent. We write $\triangle ABC \cong \triangle DEF$ to mean that $\triangle ABC$ and $\triangle DEF$ are congruent.

In the figure below, $\triangle ABC$ and $\triangle DEF$ are congruent because their corresponding parts are congruent. We can write this as follows:

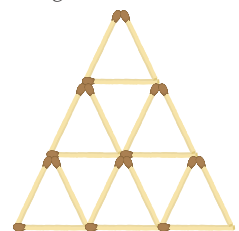
$$\begin{aligned} \angle A &\cong \angle D & AB &\cong DE \\ \angle B &\cong \angle E & \text{and} & BC &\cong EF \\ \angle C &\cong \angle F & AC &\cong DF. \end{aligned}$$

We can show this symbolically in a figure as follows:



Challenge!

Remove five toothpicks to make five congruent triangles.



EXAMPLE

1

Given that $\triangle MNP \cong \triangle STK$, state the congruent angles and sides in the two triangles without drawing them.

Solution

$$\begin{array}{lll} \angle M \cong \angle S & & MN \cong ST \\ \angle N \cong \angle T & \text{and} & NP \cong TK \\ \angle P \cong \angle K & & PM \cong KS. \end{array}$$

A short history of the $\cong$ symbol:

Gottfried Wilhelm

Leibniz

(1640-1716)

introduced $\cong$ for congruence in an unpublished manuscript in 1679.

In 1777,

Johann Friedrich

Häselor

(1372-1797)

used $\equiv$ (with the tilde reversed).

In 1824,

Carl Brandan

Mollweide

(1774-1825)

used the modern symbol $\cong$ for congruence in Euclid's *Elements*.

As we can see, the order of the vertices in congruent triangles is important when we are considering corresponding elements. Any mistake in the ordering affects the correspondence between the triangles. If two triangles are congruent then we can write this congruence in six different ways. For instance, if $\triangle ABC$ is congruent to $\triangle DEF$, the following statements are all true:

$$\begin{array}{l} \triangle ABC \cong \triangle DEF \\ \triangle ACB \cong \triangle DFE \\ \triangle BAC \cong \triangle EDF \\ \triangle BCA \cong \triangle EFD \\ \triangle CAB \cong \triangle FDE \\ \triangle CBA \cong \triangle FED. \end{array}$$



If two triangles are congruent then we can write this congruence in six different ways. For instance, if $\triangle ABC$ is congruent to $\triangle DEF$, the following statements are all true:

$$\begin{array}{l} \triangle ABC \cong \triangle DEF \\ \triangle ACB \cong \triangle DFE \\ \triangle BAC \cong \triangle EDF \\ \triangle BCA \cong \triangle EFD \\ \triangle CAB \cong \triangle FDE \\ \triangle CBA \cong \triangle FED. \end{array}$$

EXAMPLE

2

Complete each statement, given that $\triangle PRS \cong \triangle KLM$.

a. $PR \cong$ _____

b. _____ $\cong \angle K$

c. _____ $\cong SP$

d. $\angle S \cong$ _____

e. $ML \cong$ _____

f. $\angle L \cong$ _____

Solution

a. $PR \cong KL$

b. $\angle P \cong \angle K$

c. $MK \cong SP$

d. $\angle S \cong \angle M$

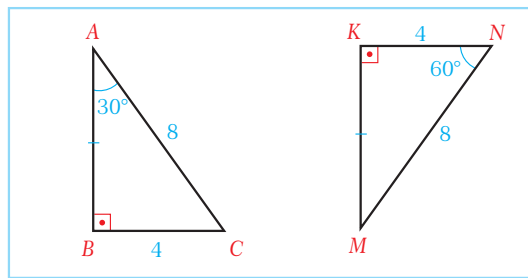
e. $ML \cong SR$

f. $\angle L \cong \angle R$

EXAMPLE

3

Decide whether or not the two triangles in the figure are congruent and give a reason for your answer.



Solution

Let us calculate the missing angles:

$$m(\angle C) = 60^\circ \quad (\text{Triangle Angle-Sum Theorem in } \triangle ABC)$$

$$m(\angle M) = 30^\circ \quad (\text{Triangle Angle-Sum Theorem in } \triangle KMN)$$

Now we can write the congruence of corresponding parts:

$$AB \cong KM \quad (\text{Given})$$

$$BC \cong KN \quad (BC = KN = 4)$$

$$AC \cong MN \quad (AC = MN = 8)$$

$$\angle A \cong \angle M \quad (m(\angle A) = m(\angle M) = 30^\circ)$$

$$\angle B \cong \angle K \quad (m(\angle B) = m(\angle K) = 90^\circ)$$

$$\angle C \cong \angle N \quad (m(\angle C) = m(\angle N) = 60^\circ)$$

Therefore, $\triangle ABC \cong \triangle KMN$ by the definition of congruent triangles.



EXAMPLE

4

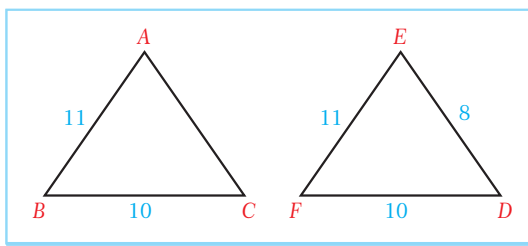
$\triangle ABC \cong \triangle EFD$ is given with $AB = 11$ cm, $BC = 10$ cm and $EF + ED = 19$ cm. Find the perimeter of $\triangle EFD$.

Solution

Since $\triangle ABC \cong \triangle EFD$, $AB = EF$, $BC = FD$ and $AC = ED$ by the definition of congruence. So by substituting the given values we get $11 = EF$, $10 = FD$ and $AC = ED$.

Since we are given that $EF + ED = 19$ cm, we have $11 + ED = 19$ cm; $ED = 8$ cm.

$$\text{So } P(\triangle EFD) = EF + ED + FD = 11 + 8 + 10 = 29 \text{ cm.}$$





What would happen if the blades of this ship's propeller or these wheels were not congruent?

Check Yourself

1. $\triangle KLM \cong \triangle XYZ$ is given. State the corresponding congruent angles and sides of the triangles.
2. State the congruence $\triangle JKM \cong \triangle SLX$ in six different ways.
3. Triangles KLM and DEF are congruent. $P(\triangle KLM) = 46$ cm, the shortest side of $\triangle KLM$ measures 14 cm, and the longest side of the $\triangle DEF$ measures 17 cm. Find the lengths of all the sides of one of the triangles.
4. Triangles DEF and KLM are congruent. If $DE = 12.5$ cm, $EF = 14.4$ cm and the perimeter of the triangle KLM is 34.6 cm, find the length of the side DF .
5. Two line segments KL and AB bisect each other at a point T . If $AL = 7$ and the lengths of the segments KL and AB are 22 and 18 respectively, find the perimeter of $\triangle KTB$.

Answers

1. $KL \cong XY$ $\angle K \cong \angle X$ 2. $\triangle PKM \cong \triangle SLN$, $\triangle KMP \cong \triangle LNS$, $\triangle MPK \cong \triangle NSL$,
 $LM \cong YZ$ $\angle L \cong \angle Y$ $\triangle PMK \cong \triangle SNL$, $\triangle KPM \cong \triangle LSN$, $\triangle MKP \cong \triangle NLS$
 $KM \cong XZ$ $\angle M \cong \angle Z$
3. 14 cm, 15 cm, 17 cm 4. 7.7 cm 5. 27

a. The Side-Angle-Side (SAS) Congruence Postulate

Postulate

Side-Angle-Side (SAS) Congruence Postulate

If two sides and their included angle in one triangle are respectively congruent to two sides and their included angle in another triangle, then the two triangles are congruent. This postulate is called the **Side-Angle-Side** (often shortened to **SAS**) **Congruence Postulate**.

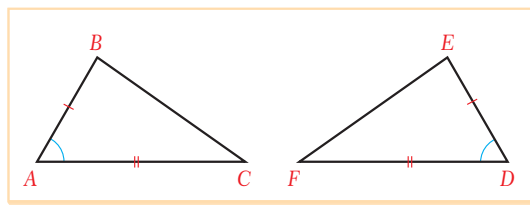
In the figure opposite,

$$AB \cong DE,$$

$$\angle A \cong \angle D \text{ and}$$

$$AC \cong DF.$$

So by the SAS Congruence Postulate,
 $\triangle ABC \cong \triangle DEF$.



Congruent triangles in construction

b. The Angle-Side-Angle (ASA) Congruence Theorem

Theorem

Angle-Side-Angle (ASA) Congruence Theorem

If two angles and their included side in a triangle are respectively congruent to two angles and their included side in another triangle, then the triangles are congruent. This theorem is called the **Angle-Side-Angle** (or **ASA**) **Congruence Theorem**.

c. The Side-Side-Side (SSS) Congruence Theorem

Theorem

Side-Side-Side (SSS) Congruence Theorem

If three sides of one triangle are congruent to three sides of another triangle, then the triangles are congruent. This theorem is called the **Side-Side-Side** (or **SSS**) **Congruence Theorem**.

d. The Hypotenuse-Leg (HL) Congruence Theorem

As we have already seen, hypotenuse-leg congruence is a particular case of SSA congruence.

Theorem

Hypotenuse-Leg (HL) Congruence Theorem

If the hypotenuse and leg of one right triangle are congruent to the hypotenuse and leg of another right triangle, then the triangles are congruent. This theorem is called the **Hypotenuse-Leg** (or **HL**) **Congruence Theorem**.

e. The Angle-Angle-Side (AAS) Congruence Theorem

Theorem

Angle-Angle-Side Congruence Theorem

If two angles and a side not shared by these angles in a triangle are congruent to two angles and a corresponding side in another triangle, then the triangles are congruent. This theorem is called the **Angle-Angle-Side** (or **AAS**) **Congruence Theorem**.

f. Theorems that Can Be Proved with Congruence

We can use congruence to prove some important results. In this section we will look at some examples of this.

Theorem

If two triangles are congruent then

1. their corresponding medians are congruent,
2. their corresponding angle bisectors are congruent, and
3. their corresponding altitudes are congruent.

Theorem

If a line parallel to one side of a triangle bisects another side of the triangle, it also bisects the third side.

EXAMPLE

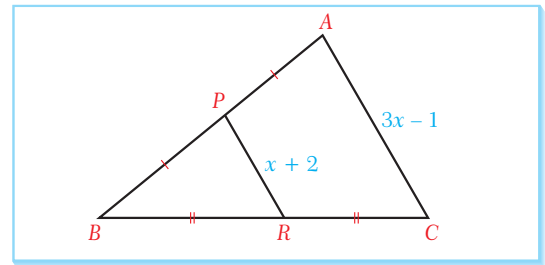
5

In a triangle ABC , P and R are the midpoints of AB and BC , respectively. $AC = 3x - 1$ and $PR = x + 2$ are given. Find PR .

Solution

- ◆ $PR = \frac{1}{2}AC$ (Triangle Midsegment Theorem)
- ◆ $x + 2 = \frac{1}{2} \cdot (3x - 1)$ (Substitute)
- ◆ $x = 5$ (Simplify)

So $PR = 5 + 2 = 7$.



Theorem

Angle Bisector Theorem

The distances from a point lying on the bisector of an angle to each side of the angle are congruent.

EXAMPLE

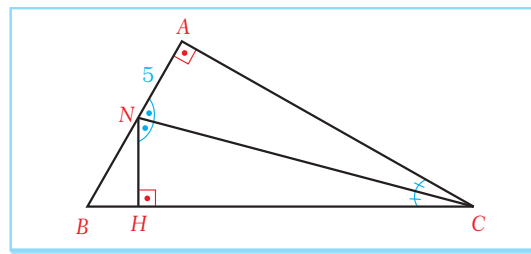


In a triangle ABC , $m(\angle BAC) = 90^\circ$ and CN is the angle bisector at vertex C . Given that $AN = 5$ cm, find the length of the altitude to BC in the triangle NBC .

Solution Let us draw NH such that $NH \perp BC$.

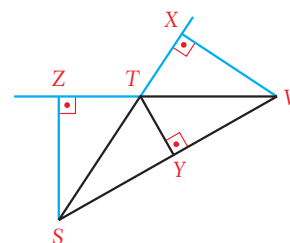
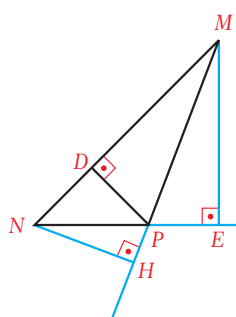
By the Angle Bisector Theorem, $NH \cong AN$.

Since $AN = 5$ cm, $NH = 5$ cm. This is the required length.

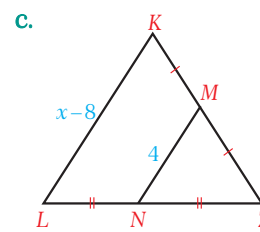
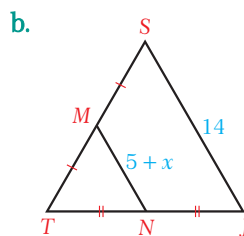
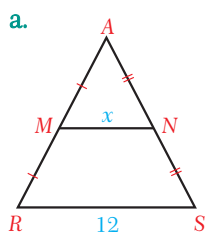


Check Yourself

- In the figure opposite, $\triangle MNP \cong \triangle SVT$. State the congruent altitudes of the two triangles.



- In each triangle below, M and N are the midpoints of the sides on which they lie. Find the value of x using the information given.



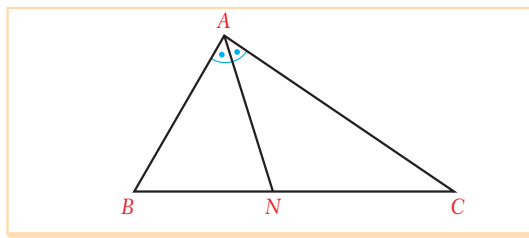
B. THE TRIANGLE ANGLE BISECTOR THEOREM

Theorem

Triangle Angle Bisector Theorem

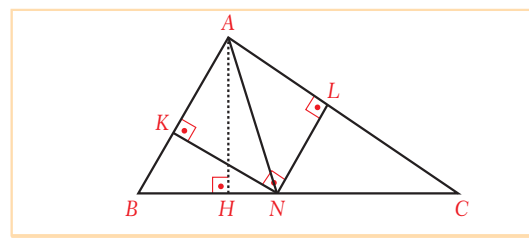
- The bisector of an interior angle of a triangle divides the opposite side in the same ratio as the sides adjacent to the angle. In other words, for a triangle ABC and angle bisector AN ,

$$\frac{AB}{AC} = \frac{BN}{CN}.$$



Proof of 1

We begin by drawing two perpendiculars NK and NL from N to the sides AB and AC respectively, then we draw the altitude $AH \perp BC$.



$$\diamond \frac{A(\triangle ABN)}{A(\triangle ANC)} = \frac{\cancel{AH} \cdot BN}{\cancel{AH} \cdot NC} = \frac{BN}{CN} \quad (1) \text{ (Definition of the area of a triangle and simplify)}$$

Now let us find the same ratio by using the sides AB and AC and the altitudes NK and NL . Since N is the point on the angle bisector, by the Angle Bisector Theorem we have $NK = NL$.

$$\diamond \frac{A(\triangle ABN)}{A(\triangle ANC)} = \frac{\cancel{NK} \cdot AB}{\cancel{NL} \cdot AC} = \frac{AB}{AC} \quad (2) \text{ (Definition of the area of a triangle and simplify)}$$

$$\diamond \frac{AB}{AC} = \frac{BN}{CN} \quad (\text{By (1), (2) and the transitive property of equality})$$

EXAMPLE

7

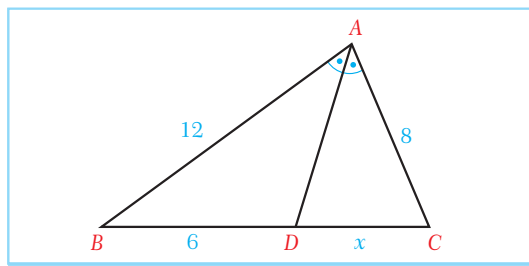
Find the length x in the figure.

Solution

$$\frac{12}{8} = \frac{6}{x} \quad (\text{Triangle Angle Bisector Theorem})$$

$$\frac{3}{2} = \frac{6}{x}$$

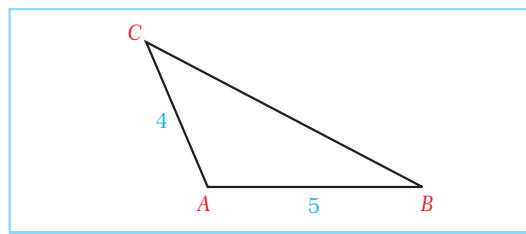
$$x = 4$$



EXAMPLE

8

In the figure, $m(\angle CAB) = 2 \cdot m(\angle ABC)$.
Given that $AC = 4$ cm and $AB = 5$ cm, find the length of BC .



Solution

Let AD be the bisector of angle A .

Then $m(\angle B) = m(\angle DAB) = m(\angle DAC)$, since $m(\angle CAB) = 2 \cdot m(\angle ABC)$.

So $\triangle DAB$ is an isosceles triangle. Let $AD = DB = x$. If $BC = a$ then $CD = a - x$.

By the Triangle Angle Bisector Theorem in $\triangle BAC$,

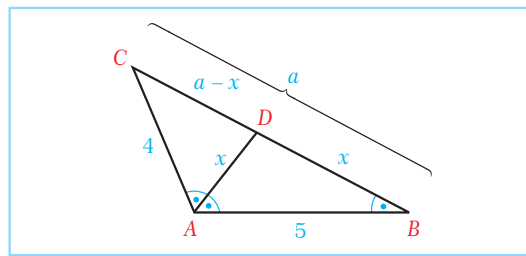
$$\begin{aligned}\frac{5}{4} &= \frac{x}{a-x} \\ 5(a-x) &= 4x \\ x &= \frac{5a}{9}. \quad (1)\end{aligned}$$

Now we can use Property 8.1:

$$\begin{aligned}x^2 &= 5 \cdot 4 - x(a-x) \\ x^2 &= 20 - ax + x^2 \\ ax &= 20. \quad (2)\end{aligned}$$

Substituting (1) into (2) gives

$$a \cdot \frac{5a}{9} = 20 ; a^2 = 36 ; a = 6 \text{ cm.}$$



INTRODUCTION TO SIMILARITY

1. Similar Figures



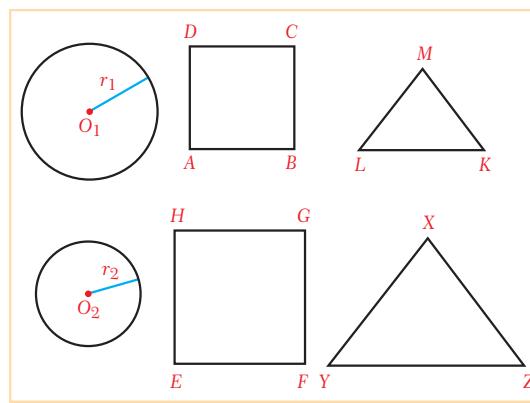
Similar figures are like the same photograph in two different sizes. The two pictures below are similar figures.



As we saw at the beginning of the previous chapter, sometimes objects are the same shape but different size. We see many things like this in nature and in daily life.

Figures which are the same shape but not necessarily the same size are called **similar figures**. For example, two circles, two squares or two equilateral triangles are always similar figures.

We can define the concept of similar polygons more formally using the language of geometry.



2. Similar Triangles

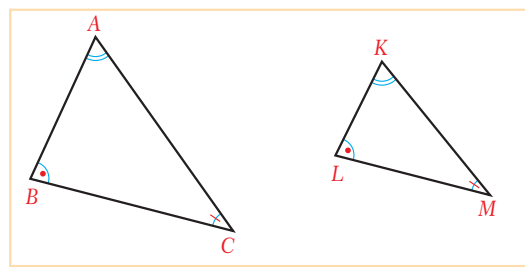
Definition

similar triangles

Two triangles are called **similar triangles** if their corresponding angles are congruent and their corresponding sides are proportional. In other words, if

$$\begin{aligned}\angle A &\cong \angle K \\ \angle B &\cong \angle L \text{ and } \frac{AB}{KL} = \frac{BC}{LM} = \frac{AC}{KM} \\ \angle C &\cong \angle M\end{aligned}$$

then $\triangle ABC \sim \triangle KLM$.

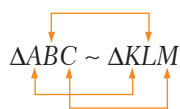


We have just seen that we need to check two conditions to identify similar polygons: their angles must be congruent and their sides must be proportional. However, to show that two triangles are similar we only need to check one condition. This is because each condition implies the existence of the other. We will prove this later in this chapter.

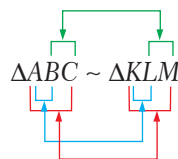
Note

We must be careful to list the vertices of similar triangles in corresponding order, so that it is clear which angles are congruent and which sides are proportional. For example, if $\triangle ABC \sim \triangle KLM$ then

$$\begin{aligned}\angle A &\cong \angle K \\ \angle B &\cong \angle L \\ \angle C &\cong \angle M\end{aligned}$$



$$\text{and } \frac{AB}{KL} = \frac{BC}{LM} = \frac{AC}{KM}.$$



Conversely, if

$$\angle A \cong \angle E$$

$$\angle B \cong \angle F \text{ and } \frac{AB}{EF} = \frac{BC}{FG} = \frac{AC}{EG} \text{ then we can conclude } \Delta ABC \sim \Delta EFG, \text{ in that order.}$$

$$\angle C \cong \angle G$$

Notice that we can also write $\Delta ABC \sim \Delta EFG$ as

$$\Delta ACB \sim \Delta EGF$$

$$\Delta BAC \sim \Delta FEG$$

$$\Delta BCA \sim \Delta FGE$$

$$\Delta CBA \sim \Delta GFE$$

$$\Delta CAB \sim \Delta GEF.$$

All of these similarities are the same. However, we cannot write $\Delta ABC \sim \Delta FEG$ or $\Delta ABC \sim \Delta EGF$ in this situation, because in these examples the vertices are not written in corresponding order.

EXAMPLE



The two triangles in the figure are similar to each other. Write this similarity using geometric notation.

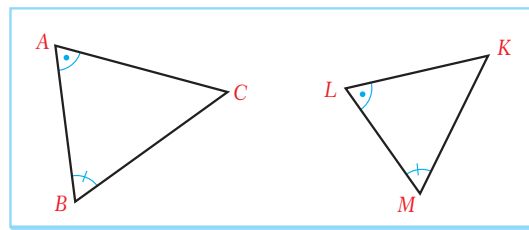
Solution

First we write the congruent angles:

$$\angle A \cong \angle L$$

$$\angle B \cong \angle M$$

$$\angle C \cong \angle K. \text{ Now we can write the similarity: } \Delta ABC \sim \Delta LMK.$$



Definition

scale factor of similarity

The ratio of the lengths of corresponding sides of similar triangles is called the **scale factor** of the similarity, and denoted by ***k***.

For example, if $\Delta ABC \sim \Delta KLM$ then $\frac{AB}{KL} = \frac{BC}{LM} = \frac{AC}{KM} = k$. We can say that ***k*** is the **scale factor of ΔABC to ΔKLM** .

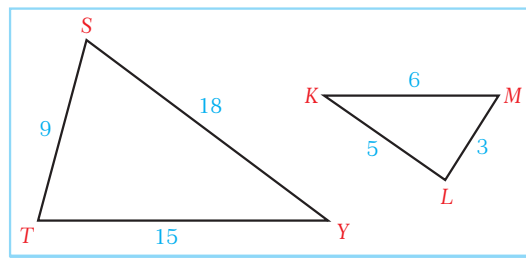


The scale factor measures the difference in size of two similar triangles.

EXAMPLE 10 In the figure, $\Delta KLM \sim \Delta YTS$. Find the scale factor of ΔKLM to ΔYTS .

Solution If $\Delta KLM \sim \Delta YTS$ then
 $\frac{KL}{YT} = \frac{5}{15}$, $\frac{LM}{TS} = \frac{3}{9}$ and $\frac{KM}{YS} = \frac{6}{18}$.
 So $\frac{KL}{YT} = \frac{LM}{TS} = \frac{KM}{YS} = k = \frac{1}{3}$.

So the scale factor of ΔKLM to ΔYTS is $\frac{1}{3}$.



EXAMPLE 11 The two triangles in the figure are similar. Write the similarity and find its scale factor, then find x and y .

Solution First let us find the corresponding angles.

Since $\angle A \cong \angle E$

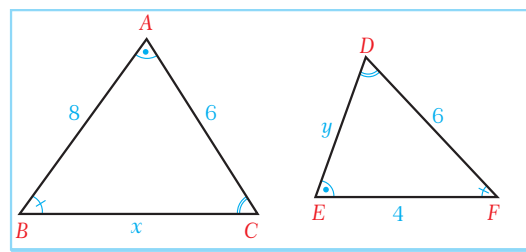
$\angle B \cong \angle F$, we can write $\Delta ABC \sim \Delta EFD$.

$\angle C \cong \angle D$

Now we find the scale factor k : if $\Delta ABC \sim \Delta EFD$ then $\frac{AB}{EF} = \frac{BC}{FD} = \frac{AC}{ED} = k$.

This means $\frac{8}{4} = \frac{x}{6} = \frac{6}{y} = k$, so the scale factor is $k = 2$.

Finally, $2 = \frac{x}{6}$ so $x = 12$; $2 = \frac{6}{y}$ so $y = 3$.

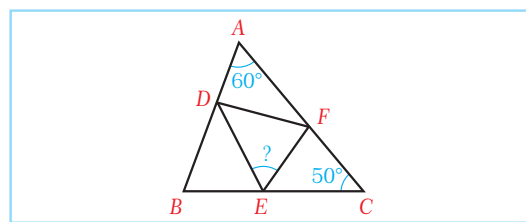


EXAMPLE 12 In the figure, $\Delta ABC \sim \Delta DEF$. Find $m(\angle DEF)$.

Solution Since $\Delta ABC \sim \Delta DEF$,
 $\angle A \cong \angle D$, $\angle B \cong \angle E$ and $\angle C \cong \angle F$.

So $m(\angle ABC) = m(\angle DEF)$. By the Interior Angle-Sum Theorem in ΔABC ,

$m(\angle B) = 180^\circ - (60^\circ + 50^\circ) = 70^\circ$. So $m(\angle E) = m(\angle B) = 70^\circ$.



Remark

If $\triangle ABC \sim \triangle DEF$ and the scale factor of similarity is 1 then $\triangle ABC \cong \triangle DEF$, i.e. the triangles are congruent.

EXAMPLE

13 In the figure, $\triangle ABC \sim \triangle CED$. Find the values of x and y .

Solution

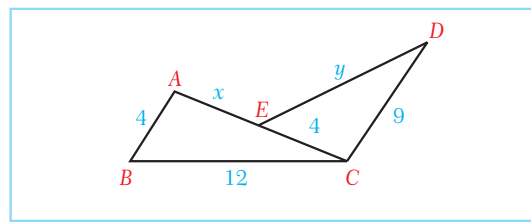
Since $\triangle ABC \sim \triangle CED$, by the definition of similarity we can write

$$\frac{AB}{CE} = \frac{BC}{ED} = \frac{AC}{CD} = k.$$

Since $\frac{AB}{CE} = \frac{4}{4}$, $k = 1$.

So the scale factor is 1 and therefore $\triangle ABC \cong \triangle CED$. Therefore, by the definition of congruence, corresponding sides are equal: $AB = CE$, $BC = ED$ and $AC = CD$.

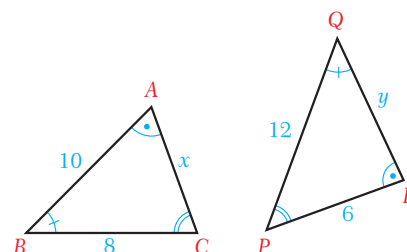
So $4 = 4$, $12 = y$, and $x + 4 = 9$, i.e. $y = 12$ and $x = 5$.



Congruent and similar triangles in construction

Check Yourself

- $\triangle MNK \sim \triangle CAB$ is given. Write the congruent angles and the ratios of the corresponding sides.
- The two triangles in the figure are similar. Write the similarity and find x and y .



- Which of the following triangles are always similar to each other?

A) isosceles triangles	B) obtuse isosceles triangles
C) right triangles	D) equilateral triangles

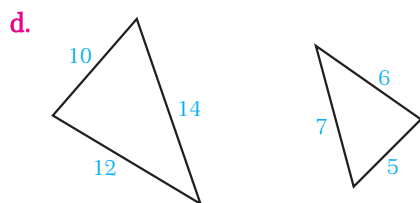
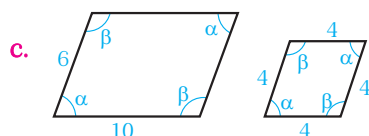
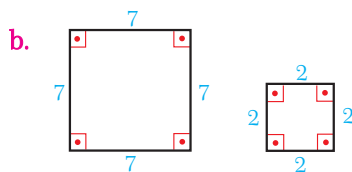
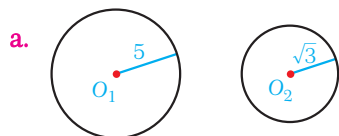
Answers

- $\triangle ABC \sim \triangle RQP$, $x = 4$, $y = 15$

EXERCISES 2.1

A. Introduction to Similarity

1. State whether the figures in each pair are similar.

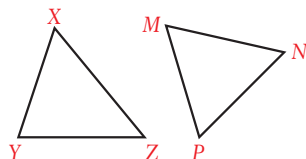


2. In the figure, $\triangle XYZ \sim \triangle PMN$.

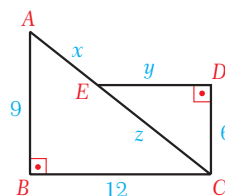
- a. List the corresponding congruent angles.

- b. Write the ratios of the corresponding sides.

- c. Find PM in terms of XY , YZ and MN .



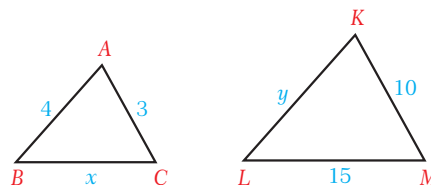
3. Find x , y and z in the figure if $\triangle ABC \sim \triangle CDE$.



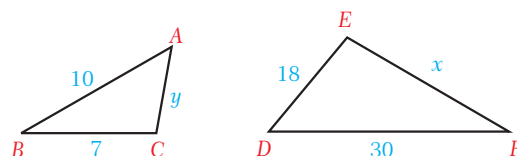
4. The sides of a triangle have lengths 4, 6 and 8. Find the length of the shortest side of a similar triangle S if the longest side of S has length 20.

5. In each question, the two triangles are similar.

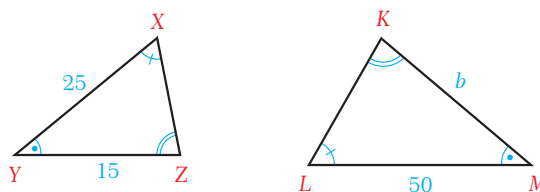
- a. If $\triangle ABC \sim \triangle KLM$, find $x + y$.



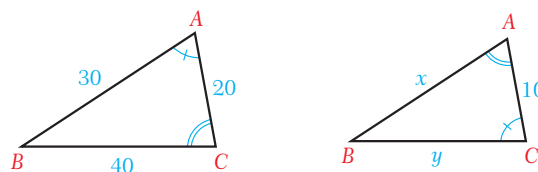
- b. If $\triangle ABC \sim \triangle DFE$, find x and y .



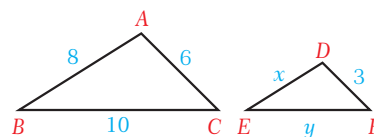
- c. Find b .



- d. Find x and y .

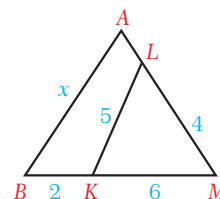


- e. Find the value of $x + y$ if $\triangle ABC \sim \triangle DEF$.



- f. Find x if

$$\triangle KLM \sim \triangle ABM.$$



THE ANGLE – ANGLE SIMILARITY POSTULATE

THE ANGLE-ANGLE (AA) SIMILARITY POSTULATE

Postulate



This easel is a physical model of AA similarity. Can you find the similar triangles?

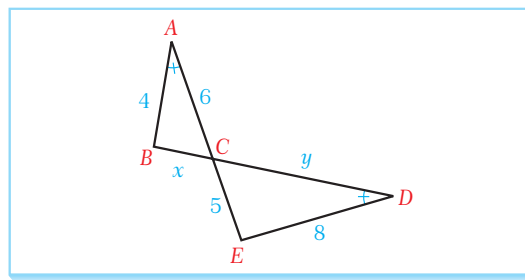
Angle-Angle (AA) Similarity Postulate

If two angles in one triangle are congruent to two angles in another triangle, then the triangles are similar. This postulate is known as the **Angle-Angle (AA) Similarity Postulate**.

In the previous section we said that three congruent corresponding angles are needed for triangles to be similar. However, the AA Similarity Postulate tells us that just two congruent angles are enough. Can you see why?

EXAMPLE

14 In the figure, $\angle A \cong \angle D$. Prove that the two triangles are similar and find x and y .



Solution

Statements	Reasons
1. $\angle A \cong \angle D$	1. Given
2. $\angle ACB \cong \angle DCE$	2. Vertical Angles Theorem
3. $\triangle ABC \sim \triangle DEC$	3. AA Similarity Postulate with 1 and 2

Now let us find x and y . Since $\triangle ABC \sim \triangle DEC$,

$$\frac{AB}{DE} = \frac{CB}{CE} = \frac{AC}{DC} = k. \text{ This means } \frac{4}{8} = \frac{x}{5} = \frac{6}{y} = k, \text{ so } k = \frac{1}{2}.$$

$$\text{So } \frac{x}{5} = \frac{1}{2} \text{ and } \frac{6}{y} = \frac{1}{2}, \text{ i.e.}$$

$$x = \frac{5}{2} = 2.5 \text{ and } y = 12.$$

EXAMPLE

15

In the figure, $KL \parallel ON$.

Find MO if $KL = 2$, $KM = x$, $NO = 6$ and $MO = 2x + 3$.



Solution If we find that the triangles are similar, we will be able to find x and y .

Statements	Reasons
1. $\angle K \cong \angle O$	1. Alternate Interior Angles Theorem ($KL \parallel ON$)
2. $\angle L \cong \angle N$	2. Alternate Interior Angles Theorem
3. $\triangle KLM \sim \triangle ONM$	3. AA Similarity Postulate with 1 and 2

Now, since $\triangle KLM \sim \triangle ONM$, we can write

$$\frac{KL}{ON} = \frac{KM}{OM} = \frac{LM}{NM} = k. \text{ So } \frac{2}{6} = \frac{x}{2x+3} = k, \text{ i.e. } k = \frac{1}{3}.$$

$$\text{Now } \frac{1}{3} = \frac{x}{2x+3}$$

$$2x + 3 = 3x$$

$$3 = x.$$

$$\text{So } MO = (2 \cdot 3) + 3 = 9.$$

EXAMPLE

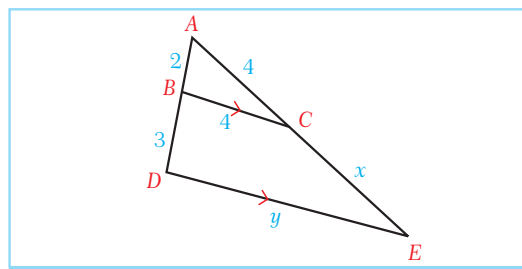
16

In the figure, $BC \parallel DE$.

Write the triangle similarity and find x and y .



If a figure contains two or more triangles, sometimes redrawing the triangles as separate figures can help you solve the problem.



Solution First prove the similarity:

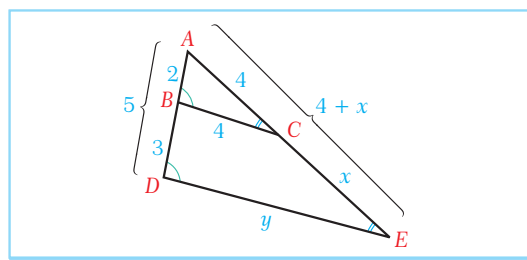
Statements	Reasons
1. $\angle B \cong \angle D$	1. Corresponding Angles Postulate ($BC \parallel DE$)
2. $\angle BCA \cong \angle E$	2. Corresponding Angles Postulate
3. $\triangle ABC \sim \triangle ADE$	3. AA Similarity Postulate

So the similarity is $\triangle ABC \sim \triangle ADE$. So

$$\frac{AB}{AD} = \frac{AC}{AE} = \frac{BC}{DE}, \text{ and } \frac{2}{5} = \frac{4}{4+x} = \frac{4}{y} = k.$$

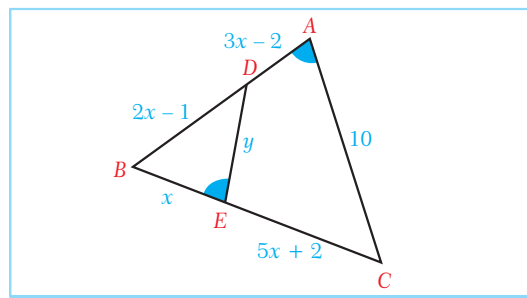


$$\begin{aligned} \text{So } \frac{2}{5} &= \frac{4}{4+x} \quad \text{and} \quad \frac{2}{5} = \frac{4}{y} \\ 2(4+x) &= 20; & 2y &= 20; \\ 8+2x &= 20; & y &= 10. \\ 2x &= 12; \\ x &= 6, \end{aligned}$$



EXAMPLE 17 In the figure, $\angle A \cong \angle E$.

Find x and y .



Solution First we find the similarity:

Statements	Reasons
1. $\angle A \cong \angle E$	1. Given
2. $\angle B \cong \angle B$	2. Common angle
3. $\triangle ABC \sim \triangle EBD$	3. AA Similarity Postulate

Now, since $\triangle ABC \sim \triangle EBD$,

$$\frac{AB}{EB} = \frac{BC}{BD} = \frac{AC}{ED}.$$

$$\text{So } \frac{5x-3}{x} = \frac{6x+2}{2x-1} = \frac{10}{y}$$

$$\frac{5x-3}{x} = \frac{6x+2}{2x-1}$$

$$\text{and } \frac{5x-3}{x} = \frac{10}{y}$$

$$(5x-3)(2x-1) = x(6x+2)$$

$$10x^2 - 6x - 5x + 3 = 6x^2 + 2x$$

$$10x^2 - 11x + 3 - 6x^2 - 2x = 0$$

$$4x^2 - 13x + 3 = 0$$

$$(4x-1)(x-3) = 0; \quad x = \frac{1}{4} \text{ or } x = 3,$$

$$\frac{12}{3} = \frac{10}{y}$$

$$12y = 30$$

$$y = \frac{5}{2}.$$



Triangles ABC and DEF are similar.

x cannot be $\frac{1}{4}$, because in this case AB would have a negative length. So $x = 3$ and $y = \frac{5}{2}$.

Notice that in Examples 7 and 8, we were able to use alternate interior angles and corresponding angles because there were two parallel lines in the given figures. In general, if there are parallel lines in a figure then it can be useful to check for similar triangles as a way of solving a problem.

EXAMPLE

18

In the figure, $ED \parallel BC$,
 $AD = 12$, $CD = 8$ and $EC = 16$. Find EF .

Solution

We can see that there are two similarities in this figure because of parallelism. One of them is $\triangle AED \sim \triangle ABC$, and the other is $\triangle EDF \sim \triangle CBF$. AD and CD are related to the first similarity, and EF is related to the second similarity. So we will use both similarities to find the answer. By the AA Similarity Postulate, $\triangle AED \sim \triangle ABC$ and $\triangle EDF \sim \triangle CBF$. So

$$\frac{AE}{AB} = \frac{ED}{CB} = \frac{AD}{AC} = k_1 \text{ and } \frac{ED}{CB} = \frac{DF}{BF} = \frac{EF}{CF} = k_2.$$

Also, $k_1 = k_2$ because $\frac{ED}{CB}$ is common in both similarity.

$$\text{So } k_1 = k_2 = k = \frac{AD}{AC} = \frac{AD}{AD + DC} = \frac{12}{12 + 8} = \frac{12}{20}.$$

$$\text{Let } EF = x, \text{ then } CF = 16 - x. \text{ Since } k = \frac{EF}{CF},$$

$$\frac{12}{20} = \frac{x}{16 - x}$$

$$3(16 - x) = 5x$$

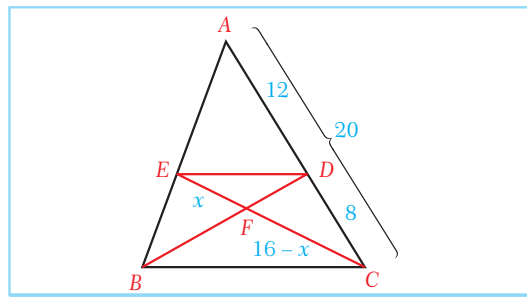
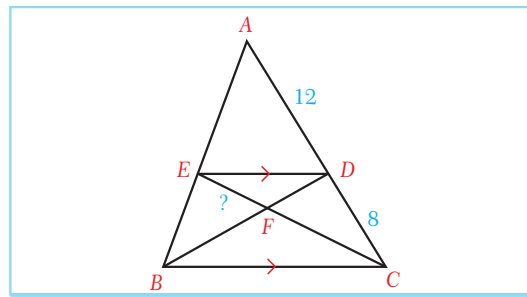
$$48 - 3x = 5x$$

$$48 = 8x$$

$$6 = x. \text{ So } EF = 6.$$



Similar triangles define the form of this ironing board.



EXAMPLE

19

In the figure, $AB \parallel EF \parallel CD$, $AB = 6$, $BF = 2$ and $FC = 1$.
 Find x and y .

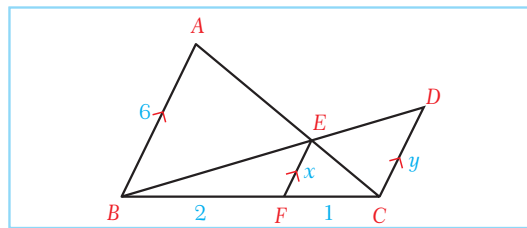
Solution

By considering the parallel lines and the AA Similarity Postulate we can find three pairs of similar triangles in the figure (can you see them all?). After this, we need to decide which similarities to use.

To find x we can use $\triangle EFC \sim \triangle ABC$, and to find y we can use $\triangle EBF \sim \triangle DBC$.

$$\text{Since } \triangle EFC \sim \triangle ABC, \frac{EF}{AB} = \frac{FC}{BC}; \frac{x}{6} = \frac{1}{3}; x = 2.$$

$$\text{Since } \triangle EBF \sim \triangle DBC, \frac{BF}{BC} = \frac{EF}{DC}; \frac{2}{3} = \frac{2}{y}; y = 3.$$



EXAMPLE

20

In the figure, $AB \parallel EF \parallel DC$. Prove that

$$\frac{1}{x} = \frac{1}{y} + \frac{1}{z}.$$

Solution $\triangle EFC \sim \triangle ABC$ (AA Similarity Postulate)

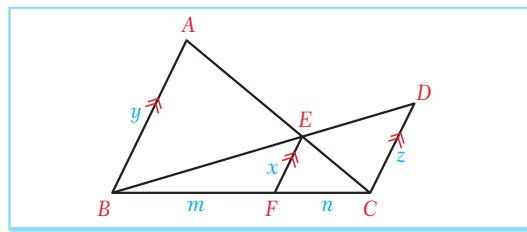
$$\text{So } \frac{x}{y} = \frac{n}{m+n}. \quad (1) \quad (\text{Definition of similarity})$$

Also, $\triangle EBF \sim \triangle DBC$. (AA Similarity Postulate)

$$\text{So } \frac{x}{z} = \frac{m}{m+n}. \quad (2) \quad (\text{Definition of similarity})$$

If we add (1) and (2) side by side, we get

$$\begin{aligned} \frac{x}{y} + \frac{x}{z} &= \frac{n}{m+n} + \frac{m}{m+n} \\ x \left(\frac{1}{y} + \frac{1}{z} \right) &= \frac{n+m}{m+n} \\ x \left(\frac{1}{y} + \frac{1}{z} \right) &= 1 \\ \frac{1}{y} + \frac{1}{z} &= \frac{1}{x}, \text{ as required.} \end{aligned}$$



EXAMPLE

21

Prove the Triangle Angle Bisector Theorem,

i.e. $\frac{c}{b} = \frac{m}{n}$ in the figure.

Solution Look at the figure.

AD is the bisector of $\angle A$. If $BA = c$, $AC = b$, $BD = m$ and $DC = n$, then we need to prove that $\frac{c}{b} = \frac{m}{n}$.

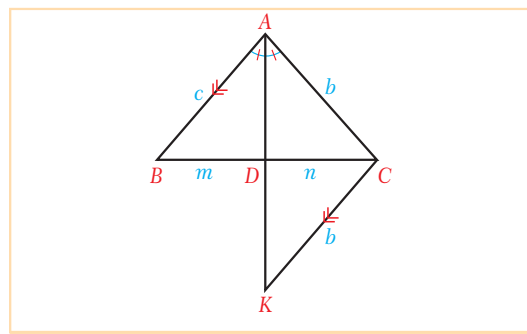
In the figure,

$$\begin{aligned} \angle BAD &\cong \angle DAC && (AD \text{ is an angle bisector}) \\ \angle BAD &\cong \angle DKC && (\text{Alternate Interior Angles Theorem}) \\ \angle DAC &\cong \angle DKC. && (\text{Transitive property of congruence}) \end{aligned}$$

So $\triangle ACK$ is an isosceles triangle and $CK = CA = b$.

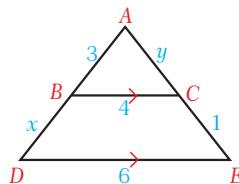
Then, since $\angle BAD \cong \angle DKC$ and $\angle BDA \cong \angle KDC$, we have $\triangle ABD \sim \triangle KCD$ by the AA Similarity Postulate.

$$\text{So } \frac{AB}{KC} = \frac{BD}{CD}, \text{ i.e. } \frac{c}{b} = \frac{m}{n} \text{ as required.}$$

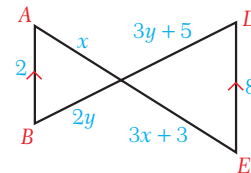


Check Yourself

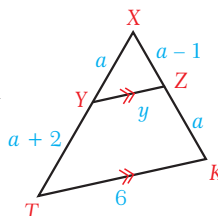
1. Find x and y in the figure if $BC \parallel DE$.



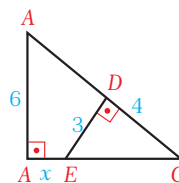
6. Find x and y in the figure if $AB \parallel DE$.



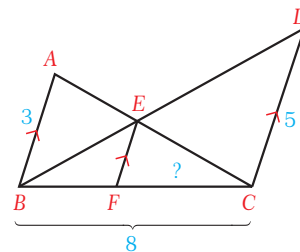
2. In the figure, $YZ \parallel TK$. Write the triangle similarity and find a and y .



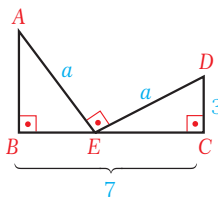
3. Find the length $AE = x$ in the figure.



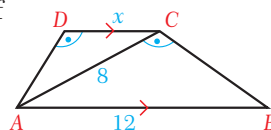
7. In the figure, $AB \parallel EF \parallel DC$, $AB = 3$, $DC = 5$ and $BC = 8$. Find the length of FC .



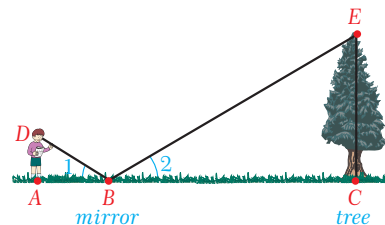
4. Find the value of a in the figure.



5. Find the value of x in the figure if $DC \parallel AB$.



8.



The boy in the picture has placed a mirror on level ground at point B so that he can see the top of the tree in it. The boy's eyes are 1.5 m from the ground, $AB = 1.2$ m and $BC = 9.6$ m. Given $\angle 1 \cong \angle 2$, find the height of the tree.

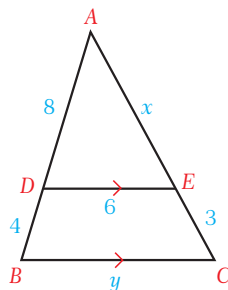
Answers

1. $x = 1.5$, $y = 2$ 2. $\triangle XYZ \sim \triangle XTK$, $a = 2$, $y = 3$ 3. 3 4. 5 5. $\frac{16}{3}$ 6. $x = 3$, $y = 1$
7. 5 8. 12 m

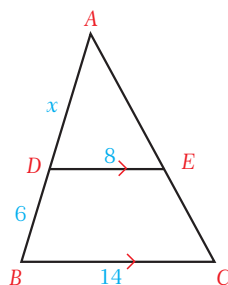
EXERCISES 2.2

A. The Angle-Angle Similarity Postulate

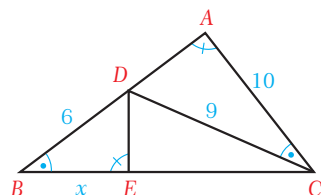
1. In the figure,
 $DE \parallel BC$.
 Find x and y .



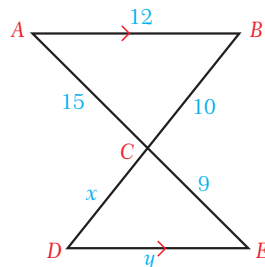
2. In the figure,
 $BC \parallel DE$.
 Find the length
 $AD = x$.



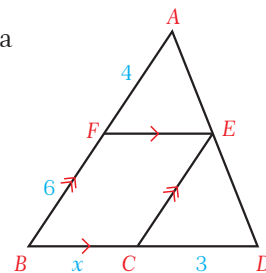
3. In the figure,
 $\angle A \cong \angle DEB$ and
 $\angle B \cong \angle ACD$.
 Find the length
 $BE = x$.



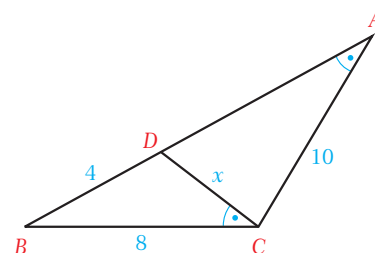
4. In the figure,
 $AB \parallel DE$. Find
 x and y .



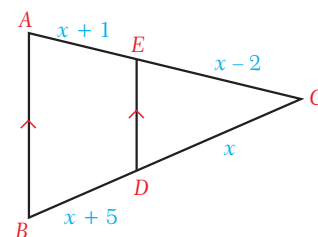
5. $BCEF$ in the figure is a
 parallelogram.
 Find x .



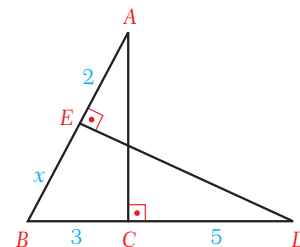
6. In the figure,
 $\angle BCD \cong \angle CAB$.
 Find x .



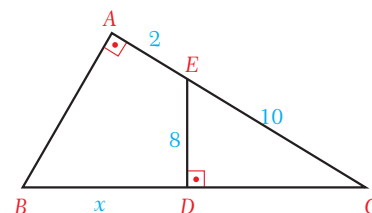
7. In the figure,
 $AB \parallel DE$.
 Find x .



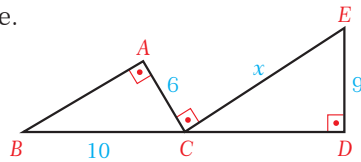
8. Find x in the
 figure.



9. Find the length
 $BD = x$ in the
 figure.

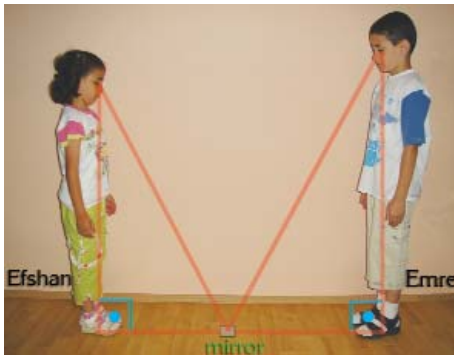


10. Find x in the figure.

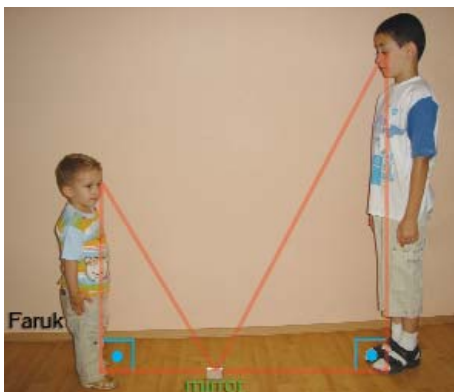


11. When you place a mirror on level ground and look at it, an angle is formed between your line of sight (from your eyes to the mirror) and the ground. This angle is the same as the angle formed between what you see in the mirror and the ground. Emre, Efshan and Faruk are brothers and sister. Emre's eyes are 140 cm from the ground. He wants to find the eye level of Efshan and Faruk by looking at their eyes in a mirror and using triangle geometry.

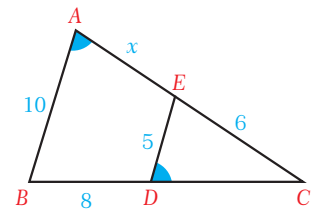
- a. Find the height of Efshan's eye level if she is 50 cm away from the mirror and Emre is 60 cm away from the mirror in the picture.



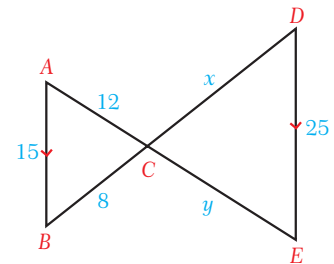
- b. Find the height of Faruk's eye level if he is 45 cm away from the mirror and Emre is 65 cm away from the mirror.



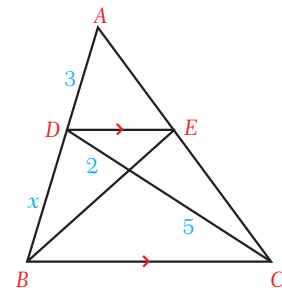
12. Find x in the figure if $\angle A \cong \angle D$.



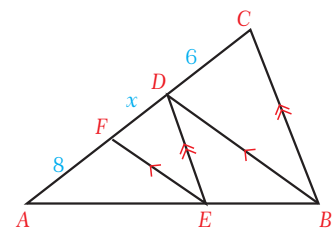
13. Find x and y in the figure.



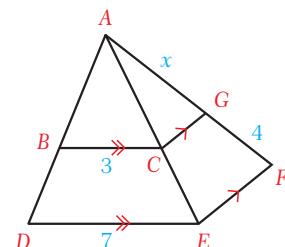
14. Find x in the figure.



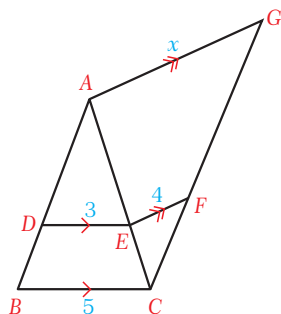
15. In the figure, $EF \parallel BD$ and $DE \parallel BC$. Find x .



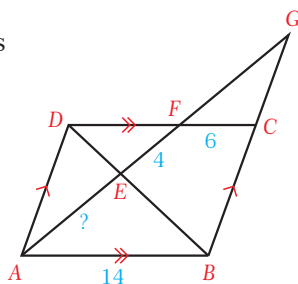
16. In the figure, $EF \parallel CG$ and $DE \parallel BC$. Find x .



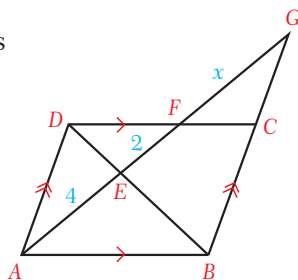
17. In the figure,
 $DE \parallel BC$ and
 $EF \parallel AG$. Find x .



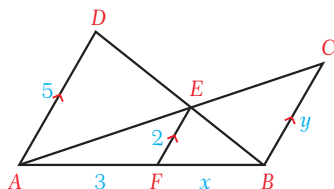
18. $ABCD$ in the figure is a parallelogram,
 $EF = 4$,
 $AB = 14$ and
 $FC = 6$.
 Find the length of AE .



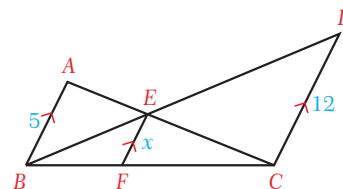
19. $ABCD$ in the figure is a parallelogram.
 Find x .



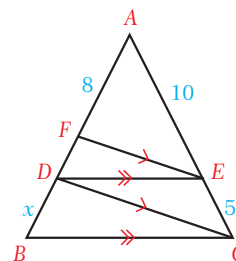
20. Find x and y in the figure.



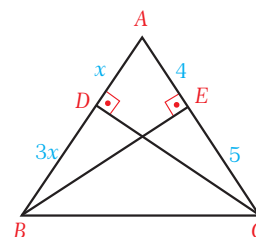
21. If $AB \parallel DC \parallel EF$
 in the figure,
 find x .



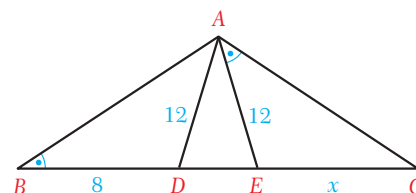
22. In the figure,
 $DE \parallel BC$ and
 $EF \parallel DC$.
 Find x .



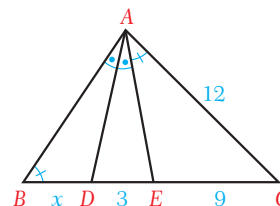
23. Find x in the figure.



24. Find x in the figure.



25. Find x in the figure.



In the previous section we looked at the Angle-Angle Similarity Postulate. We can use this postulate to prove a set of theorems which can help us decide whether two triangles are similar. In this section we will look at each theorem in turn.

A. THE SIDE-ANGLE-SIDE (SAS) SIMILARITY THEOREM

Theorem

Side-Angle-Side (SAS) Similarity Theorem

If an angle in a triangle is congruent to an angle in another triangle, and if the lengths of the sides which include these angles are in proportion, then the triangles are similar. This theorem is called the **Side-Angle-Side (SAS) Similarity Theorem**.

Proof

Let us draw an appropriate figure.

Given: $\angle A \cong \angle D$ and $\frac{AB}{DE} = \frac{AC}{DF}$.

Prove: $\triangle ABC \sim \triangle DEF$.

Let us take a point B' on DE such that $DB' = AB$.

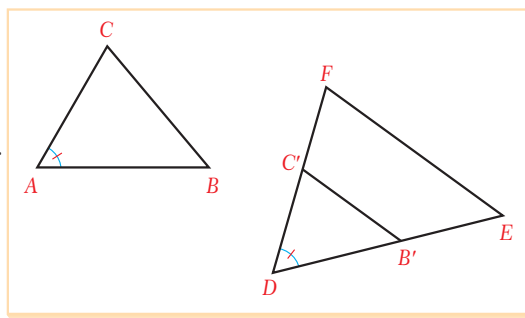
Now let us draw $B'C'$ such that $B'C'$ is parallel to EF . Then by the AA Similarity Postulate,

$\triangle DB'C' \sim \triangle DEF$. So $\frac{DB'}{DE} = \frac{DC'}{DF} = \frac{B'C'}{EF}$. (1)

Also, we know that $DB' = AB$ and $\frac{AB}{DE} = \frac{AC}{DF}$. (2)

If we replace AB with DB' then we can conclude from equations (1) and (2) that $DC' = AC$.

Thus, $\triangle ABC \cong \triangle DB'C'$ by the SAS Congruence Theorem. So $\triangle ABC \sim \triangle DB'C'$. Finally, since $\triangle ABC \sim \triangle DB'C'$ and $\triangle DB'C' \sim \triangle DEF$, we have $\triangle ABC \sim \triangle DEF$ by transitivity.



EXAMPLE

22

Find x in the figure.

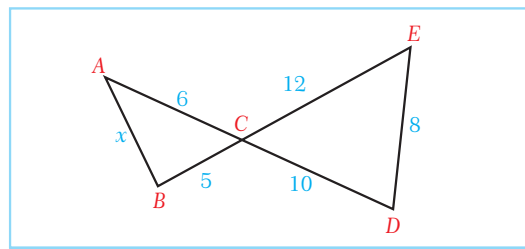
Solution Since $\frac{6}{12} = \frac{5}{10}$, we have $\frac{AC}{CE} = \frac{BC}{CD}$.

Also, $\angle ACB \cong \angle ECD$.

So by the SAS Similarity Theorem, $\triangle ACB \sim \triangle ECD$. This means

$$\frac{6}{12} = \frac{5}{10} = \frac{x}{8} = k; k = \frac{1}{2}.$$

$$\text{So } \frac{x}{8} = \frac{1}{2}, \\ x = \frac{8}{2} = 4.$$



Find the length of segment DE in the figure.

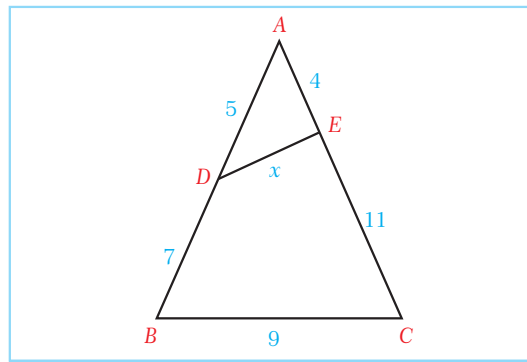
Solution There are no given angles; we only know that $\angle A$ is a common angle, so we cannot use the AA Similarity Postulate. However, if we check the ratios of the sides, we see that

$\frac{AE}{AB} = \frac{AD}{AC}$, and since $\angle A$ is a common angle, $\triangle AED \sim \triangle ABC$ by the SAS Similarity Theorem.

Thus, $\frac{4}{12} = \frac{5}{15} = \frac{x}{9} = k$. So $k = \frac{1}{3}$, which gives us

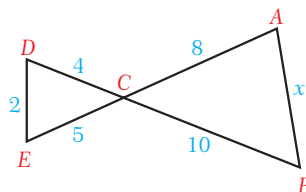
$$\frac{1}{3} = \frac{x}{9}$$

$$3 = x.$$

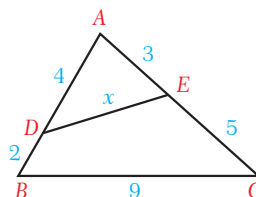


Check Yourself

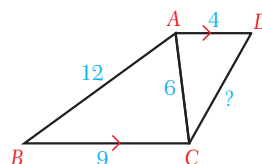
1. Find x in the figure.



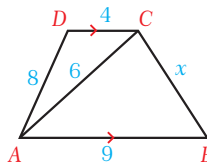
2. Find x in the figure.



3. In the figure, $AD \parallel BC$,
 $AB = 12$ cm, $BC = 9$ cm,
 $AC = 6$ cm and $AD = 4$ cm.
 Find DC .



4. Find x in the figure
 if $DC \parallel AB$.



Answers

1. $x = 4$ 2. $\frac{9}{2}$ 3. 8 cm 4. 12

B. THE SIDE-SIDE-SIDE (SSS) SIMILARITY THEOREM

Theorem

Side-Side-Side (SSS) Similarity Theorem

If the lengths of the corresponding sides of two triangles are all in proportion, then the triangles are similar.

Proof

Given: $\frac{AB}{DE} = \frac{AC}{DF} = \frac{BC}{EF}$ (1)

Prove: $\triangle ABC \sim \triangle DEF$

Let us take a point X on DE such that

$AB = DX$. (2)

Then we draw a line XY such that $XY \parallel EF$.

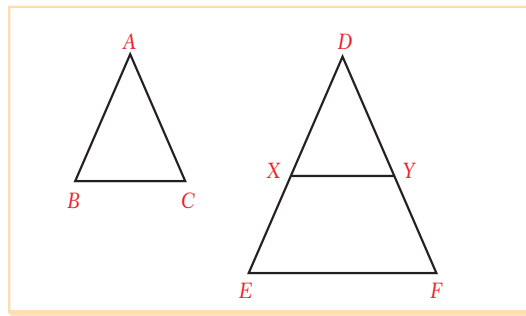
Then by the AA Similarity Postulate,
 $\triangle DXY \sim \triangle DEF$.

So $\frac{DX}{DE} = \frac{XY}{EF} = \frac{DY}{DF}$. (3)

By substituting (2) into (3) and using (1), we have $AC = DY$ and $BC = XY$.

So by the SSS Congruence Theorem, $\triangle ABC \cong \triangle DXY$.

Since $\triangle ABC \sim \triangle DXY$ and $\triangle DXY \sim \triangle DEF$ by transitivity.



EXAMPLE

24

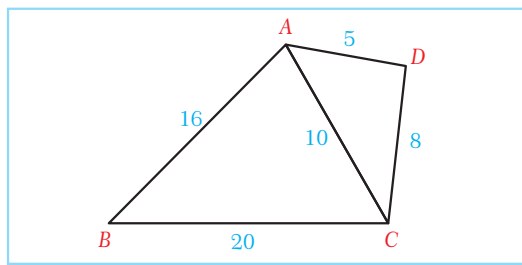
Show that $\triangle ABC \sim \triangle DCA$ in the figure.

Solution

Since $\frac{AB}{DC} = \frac{16}{8} = 2$, $\frac{AC}{AD} = \frac{10}{5} = 2$

and $\frac{BC}{AC} = \frac{20}{10} = 2$, then

by the SSS Similarity Theorem,
 $\triangle ABC \sim \triangle DCA$.



EXAMPLE

25

Find x and y in the figure.

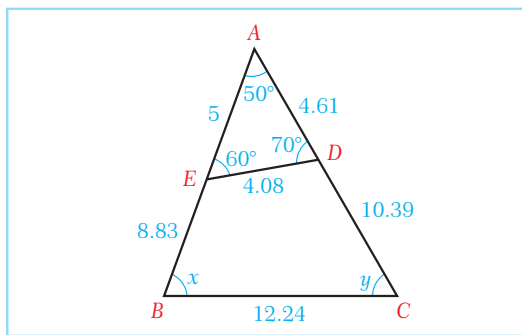
Solution

$\frac{AD}{AB} = \frac{AE}{AC} = \frac{ED}{BC}$ because $\frac{4.61}{13.83} = \frac{5}{15} = \frac{4.08}{12.24}$.

So by the SSS Similarity Theorem,
 $\triangle AED \sim \triangle ACB$.

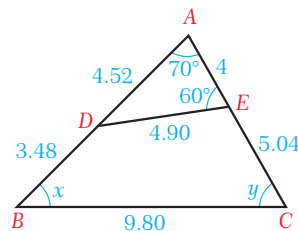
Thus $m(\angle B) = m(\angle D)$ so $x = 70^\circ$, and

$m(\angle E) = m(\angle C)$ so $y = 60^\circ$.



Check Yourself

- Find x and y in the figure.



Answers

- $x = 60^\circ$, $y = 50^\circ$

Similar triangles have many properties which are useful for solving problems. Let us look at the most useful ones.

Properties 9

- The scale factor of similarity of two triangles is equal to
 - the ratio of the lengths of corresponding altitudes,
 - the ratio of the lengths of corresponding angle bisectors, and
 - the ratio of the lengths of corresponding medians.

In other words, if $\triangle ABC \sim \triangle DEF$ then

$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF} = k = \frac{h_a}{h_d} = \frac{h_b}{h_e} = \frac{h_c}{h_f} = \frac{n_A}{n_D} = \frac{n_B}{n_E} = \frac{n_C}{n_F} = \frac{V_a}{V_d} = \frac{V_b}{V_e} = \frac{V_c}{V_f}.$$

- The ratio of the perimeters of two similar triangles is equal to the scale factor of similarity, i.e.

$$\text{if } \triangle ABC \sim \triangle DEF \text{ then } \frac{P(\triangle ABC)}{P(\triangle DEF)} = k.$$

- The ratio of the areas of two similar triangles is equal to the square of the scale factor of similarity, i.e.

$$\text{if } \triangle ABC \sim \triangle DEF \text{ then } \frac{A(\triangle ABC)}{A(\triangle DEF)} = k^2.$$

EXAMPLE

26

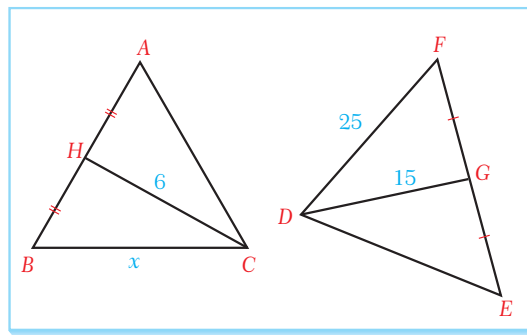
In the figure, $\triangle ABC \sim \triangle FED$. Find x .

Solution Since $\triangle ABC \sim \triangle FED$, we can use the definition of similarity and Property 9.1 and write

$$\frac{AB}{FE} = \frac{BC}{ED} = \frac{AC}{FD} = \frac{CH}{DG}.$$

$$\text{So } \frac{6}{15} = \frac{x}{25}, \text{ i.e.}$$

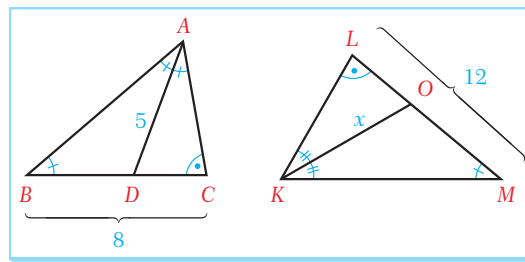
$$x = 10.$$



EXAMPLE

27

Find the length $x = OK$ in the figure at the right.



Solution

We are given $m(\angle B) = m(\angle M)$ and $m(\angle C) = m(\angle L)$.

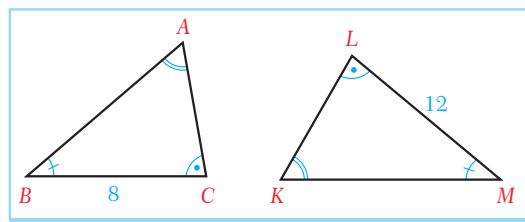
So $\triangle ABC \sim \triangle KML$. (AA Similarity Postulate)

Therefore, $m(\angle A) = m(\angle K)$.

Since AD and KO are angle bisectors, by Property 9.1 we have

$$\frac{AD}{KO} = \frac{AB}{KM} = \frac{BC}{ML} = \frac{AC}{KL}.$$

$$\text{So } \frac{5}{x} = \frac{8}{12} \\ x = 7.5.$$



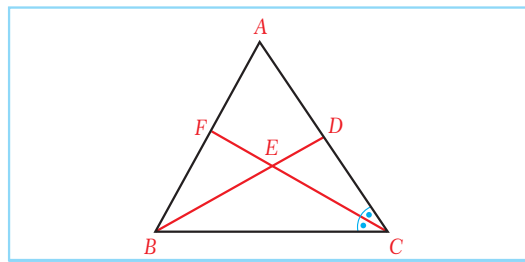
EXAMPLE

28

In the figure, $AB = AC = 9$ and $BD = BC = 6$.

Given that CF bisects $\angle C$,

find $\frac{CE}{CF}$.



Solution

$$\angle ABC \cong \angle ACB$$

(Isosceles Triangle Theorem with $AB = AC$)

$$\angle BDC \cong \angle BCD$$

(Isosceles Triangle Theorem with $BD = BC$)

So $\angle ABC \cong \angle BCD$ and $\angle ACB \cong \angle BDC$. (Identical angles)

So by the AA Similarity Postulate, $\triangle ABC \sim \triangle BDC$ and so CE and CF are angle bisectors in $\triangle BDC$ and $\triangle ABC$ respectively.

$$\text{So } \frac{CE}{CF} = \frac{BD}{AB} = \frac{DC}{BC} = \frac{BC}{AC},$$

(Definition of similarity and Property 9.1)

$$\text{i.e. } \frac{CE}{CF} = \frac{6}{9} = \frac{2}{3}.$$

EXAMPLE
29

The sum of the perimeters of two similar triangles is 49 cm and the scale factor of their similarity is $\frac{2}{5}$. Find the perimeter of the smaller triangle.

Solution

Let the perimeter of the smaller triangle be P_s , and the perimeter of the larger one be P_L .

$$\text{Then } P_s + P_L = 49 \text{ cm.} \quad (1)$$

$$\text{Also, by Property 9.2, } \frac{P_s}{P_L} = \frac{2}{5}. \quad (2)$$

$$\text{Substituting (1) into (2) gives } \frac{P_s}{49 - P_s} = \frac{2}{5}.$$

$$\text{So } P_s = 14 \text{ cm.}$$

EXAMPLE
30

$\triangle ABC$ and $\triangle MNP$ are two similar triangles with scale factor of similarity $\frac{BC}{NP} = \frac{3}{2}$. Given that $A(\triangle ABC) = 45 \text{ cm}^2$, find $A(\triangle MNP)$.

Solution

$$\text{By Property 9.3, } \frac{A(\triangle ABC)}{A(\triangle MNP)} = \left(\frac{BC}{NP}\right)^2. \text{ So } \frac{45}{A(\triangle MNP)} = \left(\frac{3}{2}\right)^2; A(\triangle MNP) = \frac{45 \cdot 4}{9};$$

$$A(\triangle MNP) = 20 \text{ cm}^2.$$

EXAMPLE
31

In the figure, $AD = DF = FB$ and the area of quadrilateral $BCGF$ is 15 cm^2 . Find the area of quadrilateral $FGED$.

Solution

Let $AD = DF = FB = x$, $A(\triangle ADE) = S_1$ and $A(\triangle AFG) = S_2$.

Since $FG \parallel BC$, we have $\triangle AFG \sim \triangle ABC$ by the AA Similarity Postulate.

$$\text{The scale factor of this similarity is } \frac{AF}{AB} = \frac{2x}{3x} = \frac{2}{3}.$$

Then, by Property 9.3,

$$\frac{A(\triangle AFG)}{A(\triangle ABC)} = \left(\frac{2}{3}\right)^2; \frac{S_2}{S_2 + 15} = \frac{4}{9};$$

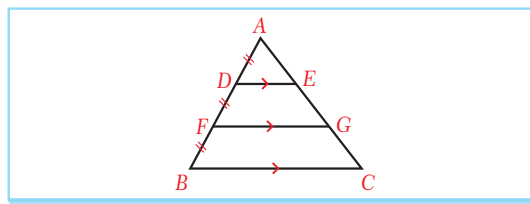
$$4S_2 + 60 = 9S_2; 60 = 5S_2; S_2 = 12 \text{ cm}^2.$$

Similarly, since $DE \parallel FG$ we have $\triangle ADE \sim \triangle AFG$ by the AA Similarity Postulate.

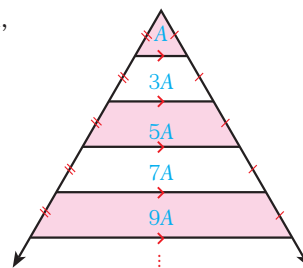
$$\text{The scale factor of this similarity is } \frac{AD}{AF} = \frac{x}{2x} = \frac{1}{2}, \text{ and so by Property 9.3,}$$

$$\frac{A(\triangle ADE)}{A(\triangle AFG)} = \left(\frac{1}{2}\right)^2; \frac{S_1}{12} = \frac{1}{4}; S_1 = 3 \text{ cm}^2.$$

$$\text{So } A(FGED) = S_2 - S_1 = 12 - 3 = 9 \text{ cm}^2.$$



Note: If the area of the top triangle in the figure opposite is A , then the areas of the other parts of the figure are as shown.



EXAMPLE

32

Prove Property 9.2: If $\triangle ABC \sim \triangle PRN$ then $\frac{P(\triangle ABC)}{P(\triangle PRN)} = k$.

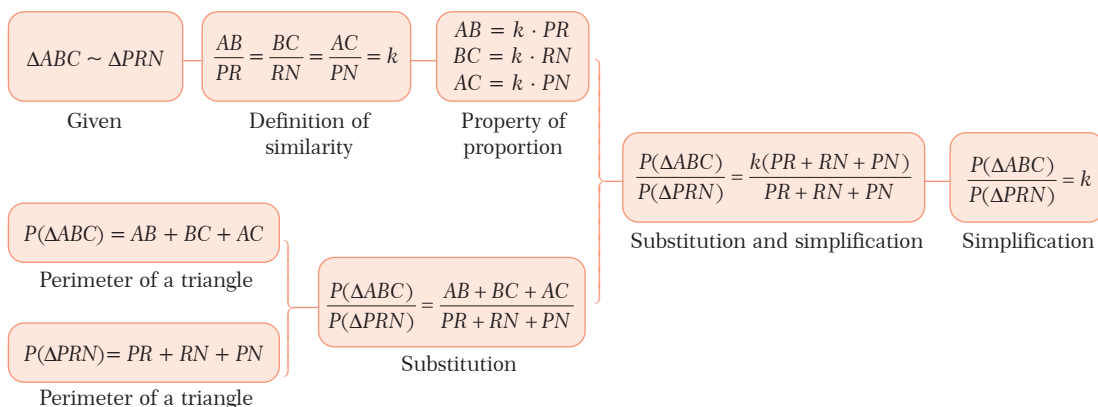
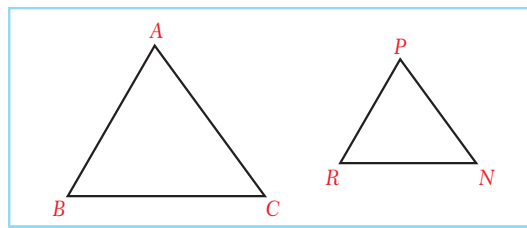
Solution Let us draw an appropriate figure.

Given: $\triangle ABC \sim \triangle PRN$

Prove: $\frac{P(\triangle ABC)}{P(\triangle PRN)} = k$

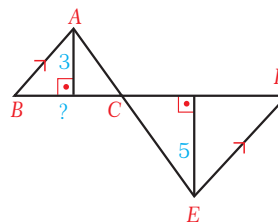
Proof:

We will write a flow-chart proof.



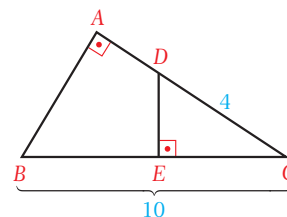
Check Yourself

- Find the length of segment BC in the figure if $BD = 24$ and $AB \parallel ED$.

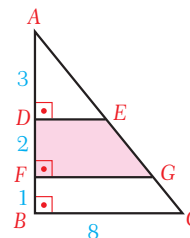


- The ratio of the areas of two similar triangles is $\frac{4}{9}$ and the perimeter of the smaller triangle is 12 cm. Find the perimeter of the bigger triangle.

3. In the figure,
 $BC = 10$ cm and $DC = 4$ cm.
 Find the value of the ratio $\frac{A(\triangle DEC)}{A(\triangle ABC)}$.



4. What is the area of quadrilateral $FGED$ in the figure?



5. Prove Property 9.3: if $\triangle ABC \sim \triangle DEF$ then $\frac{A(\triangle ABC)}{A(\triangle DEF)} = k^2$.

Answers

1. 9 2. 18 cm 3. $\frac{4}{25}$ 4. $\frac{32}{3}$

C. THE TRIANGLE PROPORTIONALITY THEOREM AND THALES' THEOREM

1. The Triangle Proportionality Theorem

Theorem

Triangle Proportionality Theorem

A line parallel to one side of a triangle which intersects the other two sides of the triangle at different points divides these two sides proportionally. In other words, in the figure below,

$$\frac{m}{n} = \frac{x}{y}.$$

Proof

Look at the figure.

Given: $DE \parallel BC$

Prove: $\frac{AD}{DB} = \frac{AE}{EC}$

$\triangle ABC \sim \triangle ADE$ (AA Similarity Postulate)

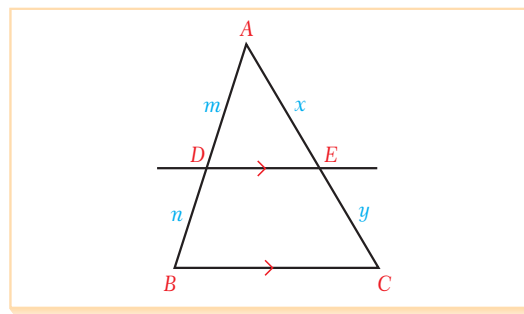
So $\frac{AB}{AD} = \frac{AC}{AE}$.

Let $AD = m$, $DB = n$, $AE = x$ and $EC = y$.

Then $\frac{m+n}{m} = \frac{x+y}{x}$

$$1 + \frac{n}{m} = 1 + \frac{y}{x}$$

$$\frac{n}{m} = \frac{y}{x}. \text{ So } \frac{DB}{AD} = \frac{EC}{AE}, \text{ and so } \frac{AD}{DB} = \frac{AE}{EC}, \text{ as required.}$$



Conclusion

Using the properties of ratio in the previous figure, we can conclude that if DE is parallel to BC then $\frac{AD}{DB} = \frac{AE}{EC}$, $\frac{AB}{DB} = \frac{AC}{EC}$ and $\frac{AB}{AD} = \frac{AC}{AE}$.

Theorem

Converse of the Triangle Proportionality Theorem

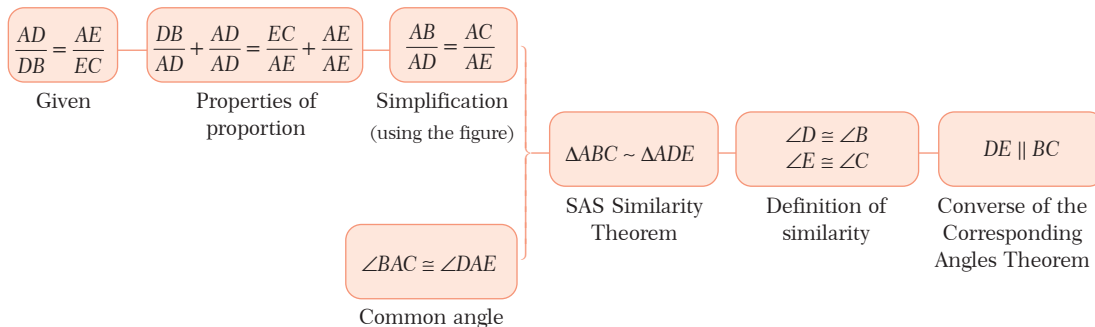
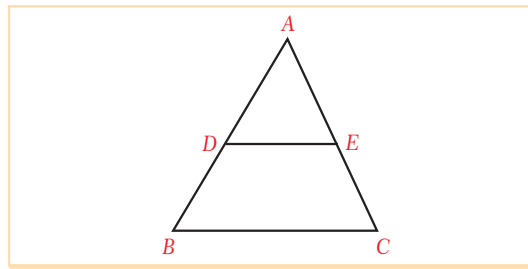
If a line divides two sides of a triangle proportionally then it is parallel to the third side of the triangle.

Proof

Look at the figure.

Given: $\frac{AD}{DB} = \frac{AE}{EC}$

Prove: $DE \parallel BC$



EXAMPLE

33

In the figure, $TS \parallel AC$, $BT = 6$ cm, $BS = 9$ cm, $AB = 2x + 4$ and $BC = 5x$. Find SC .

Solution Since $TS \parallel AC$, by the Triangle Proportionality Theorem we can write

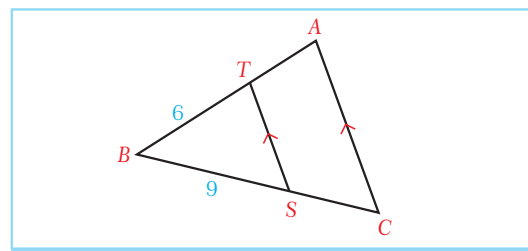
$$\frac{BT}{BA} = \frac{BS}{BC}$$

$$\frac{6}{2x+4} = \frac{9}{5x}$$

$$10x = 6x + 12; 4x = 12; x = 3 \text{ cm.}$$

So $BC = 5x = 5 \cdot 3 = 15$ cm and

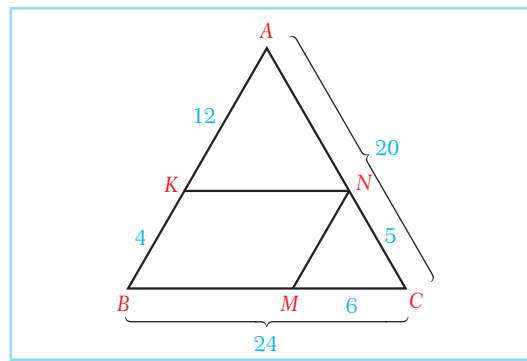
$$SC = BC - BS = 15 - 9 = 6 \text{ cm.}$$



EXAMPLE

34

In the figure, $AK = 12$ cm, $KB = 4$ cm, $AC = 20$ cm, $NC = 5$ cm, $BC = 24$ cm and $MC = 6$ cm. Show that $KN \parallel BC$ and $MN \parallel AB$.



Solution To show that the lines are parallel, it is sufficient by the Converse of the Triangle Proportionality Theorem to show that

$$\frac{AK}{KB} = \frac{AN}{NC} \text{ and } \frac{BM}{MC} = \frac{AN}{NC}.$$

Since $AN = AC - NC$,

$$AN = 20 - 5 = 15 \text{ cm.}$$

Similarly, $BM = 18$ cm.

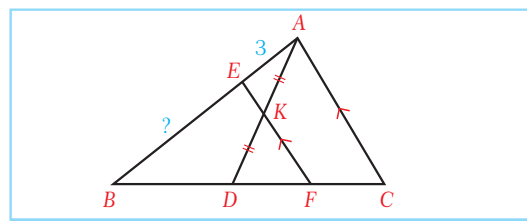
So $\frac{AK}{KB} = \frac{12}{4} = 3$ and $\frac{AN}{NC} = \frac{15}{5} = 3$. So $\frac{AK}{KB} = \frac{AN}{NC}$, and so by the Converse of the Triangle Proportionality Theorem, $KN \parallel BC$.

Also, $\frac{BM}{MC} = \frac{18}{6} = 3$ and $\frac{AN}{NC} = \frac{15}{5} = 3$, so $\frac{BM}{MC} = \frac{AN}{NC}$ and so by the same theorem, $MN \parallel AB$.

EXAMPLE

35

In the triangle ABC at the right, $EF \parallel AC$, $AK = KD$, $BD = 2DC$ and $AE = 3$ cm. Find the length of segment EB .



Solution In $\triangle ADC$, $\frac{DK}{AK} = \frac{DF}{FC} = 1$. (Triangle Proportionality Theorem and $AK = KD$)

So $DF = FC$.

Let $DF = FC = y$, then $BD = 2CD = 4y$.

So in $\triangle ABC$ we have $\frac{AE}{EB} = \frac{FC}{BF}$ (Triangle Proportionality Theorem)

$$\frac{3}{EB} = \frac{y}{5y}$$

$$EB = 15 \text{ cm.}$$

EXAMPLE

36

In $\triangle ABC$ at the right, $AF = FE$, $DB = 5$ cm, $BE = 4$ cm and $EC = 6$ cm. Find the length of AD .

Solution First we find a point K on AB such that $DC \parallel KE$. Then in $\triangle DBC$,

$$\frac{BE}{EC} = \frac{BK}{KD} \quad (\text{Triangle Proportionality Theorem})$$

$$\text{So } \frac{4}{6} = \frac{BK}{5 - BK}$$

$$6 \cdot BK = 20 - 4 \cdot BK$$

$$10 \cdot BK = 20$$

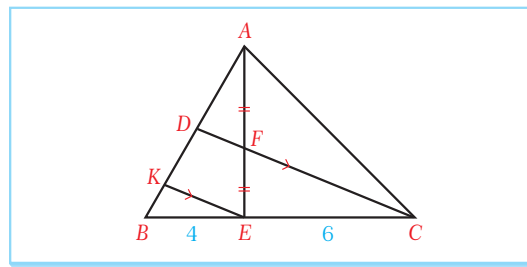
$$BK = 2 \text{ cm.}$$

Hence $KD = BD - BK = 5 - 2 = 3$ cm.

On the other hand, in $\triangle AKE$ we have

$$\frac{AF}{FE} = \frac{AD}{DK} = 1. \quad (\text{Triangle Proportionality Theorem and } AF = FE)$$

So $AD = 3$ cm.



2. Thales' Theorem of Parallel Lines

Theorem

Thales' Theorem

If two transversals intersect three or more parallel lines then the parallel lines divide the transversals proportionally. This theorem is known as **Thales' Theorem**.

Proof

Look at the figure.

Given: $AD \parallel BE \parallel CF$

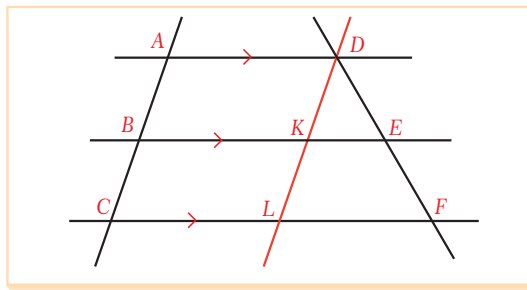
$$\text{Prove: } \frac{AB}{BC} = \frac{DE}{EF}$$

First we draw a line which is parallel to AC and passes through D . Let us label the intersection points K and L of this new line with BE and CF .

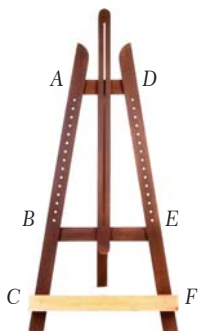
Then $BKDA$ and $CLKB$ are parallelograms, since if the opposite sides of a quadrilateral are parallel then the quadrilateral is a parallelogram. So $DK = AB$ and $KL = BC$. (1)

Since $KE \parallel LF$, by the Triangle Proportionality Theorem in $\triangle DLF$ we have $\frac{DK}{KL} = \frac{DE}{EF}$. (2)

Substituting (1) into (2) gives us $\frac{AB}{BC} = \frac{DE}{EF}$, as required.



Can you see the proportional lengths?



In the figure, $AS \parallel BR \parallel CM \parallel DN$.

Find the lengths m , n , x and y using the information in the figure.

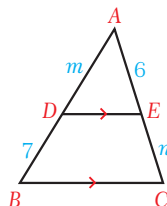
Solution Since $AS \parallel BR \parallel CM \parallel DN$ and AD , AN and NS are transversals, we can apply Thales' Theorem:

$$\frac{AK}{KP} = \frac{SR}{RM}; \quad \frac{m}{3} = \frac{3}{4}; \quad m = \frac{9}{4} \quad \text{and} \quad \frac{KP}{PN} = \frac{RM}{MN}; \quad \frac{3}{n} = \frac{4}{5}; \quad n = \frac{15}{4}.$$

$$\frac{AB}{BC} = \frac{SR}{RM}; \quad \frac{4}{x} = \frac{3}{4}; \quad x = \frac{16}{3} \quad \text{and} \quad \frac{AB}{CD} = \frac{SR}{MN}; \quad \frac{4}{y} = \frac{3}{5}; \quad y = \frac{20}{3}.$$

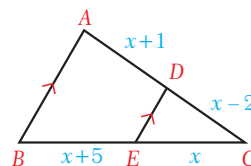
Check Yourself

1. Find the value of $m \cdot n$ in the figure.

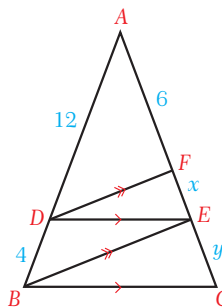


2. In the figure, $DE \parallel AB$.

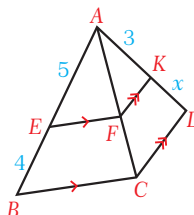
Find the value of x .



3. In the figure, $DE \parallel BC$, $DF \parallel BE$,
 $AD = 12$, $DB = 4$ and
 $AF = 6$. Find the lengths x and y .

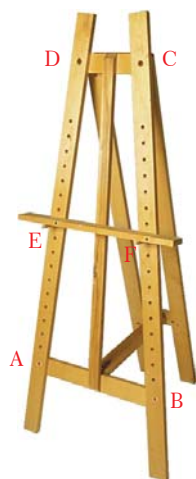


4. Find the value of x in the figure.





5. In the figure, $DC \parallel EF \parallel AB$.
 $DE = 50$ cm, $EA = 70$ cm, $CF = x$ and
 and $FB = x + 20$ cm are given. Find
 the value of x .



Answers

1. 42 2. 5 3. $x = 2$, $y = \frac{8}{3}$ 4. $\frac{12}{5}$ 5. 50 cm

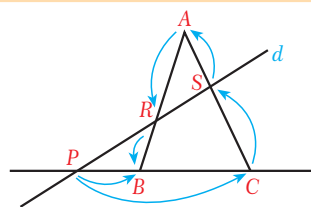
D. FURTHER APPLICATIONS

1. Menelaus' Theorem

Theorem

Menelaus' Theorem

Let ABC be a triangle. If a line d intersects the two sides AB and AC and the extension of the third side BC of $\triangle ABC$ at points R , S and P respectively, then $\frac{PB}{PC} \cdot \frac{CS}{SA} \cdot \frac{AR}{RB} = 1$.



Proof

Let us draw the line k through point B and parallel to side AC (Parallel Postulate), and let T be the intersection point of lines k and d .

Then $\triangle PBT \sim \triangle PCS$ by the AA Similarity Postulate.

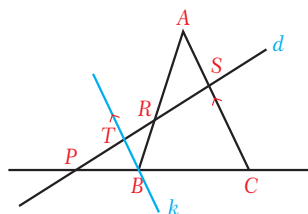
$$\text{So } \frac{PB}{PC} = \frac{BT}{CS}. \quad (1)$$

Moreover, $\triangle BRT \sim \triangle ARS$ by the AA Similarity Postulate.

$$\text{So } \frac{BR}{AR} = \frac{BT}{AS}. \quad (2)$$

Dividing (1) by (2) side by side gives $\frac{PB}{BR} = \frac{CS}{AS}$; $\frac{PB}{PC} \cdot \frac{AR}{BR} = \frac{AS}{CS}$; $\frac{PB}{PC} \cdot \frac{AR}{BR} \cdot \frac{CS}{AS} = 1$.

$$\text{So } \frac{PB}{PC} \cdot \frac{CS}{SA} \cdot \frac{AR}{RB} = 1, \text{ as required.}$$



Menelaus of Alexandria (c. 40-140 AD) was a Greek mathematician and astronomer. He was the first mathematician to describe a spherical triangle, and proved the theorem described here in his book *Sphaerica*, which is the only book he wrote that has survived.

EXAMPLE

38

In the figure, $BD = 8$ cm, $DC = 6$ cm, $AC = 22$ cm and $AF = FD$. Find the length of AE .

Solution Let $AE = x$. Then $EC = 22 - x$.

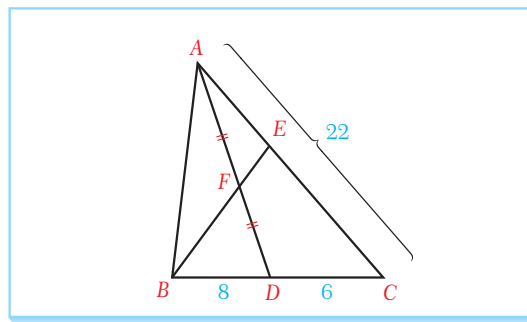
By Menelaus' Theorem in $\triangle ADC$,

$$\frac{DB}{BC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FD} = 1; \quad \frac{8}{14} \cdot \frac{22-x}{x} \cdot \frac{AF}{FD} = 1.$$

Since $AF = FD$, we have $\frac{8}{14} \cdot \frac{22-x}{x} = 1$

$$8(22-x) = 14x$$

$$x = AE = 8.$$



Note

It can also be shown that the converse of Menelaus' Theorem is true: in the figure on the previous page, if $\frac{PB}{PC} \cdot \frac{CS}{SA} \cdot \frac{AR}{RB} = 1$ then d intersects AB , AC and the extension of side BC .

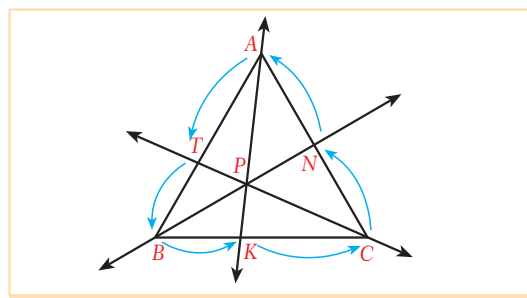
2. Ceva's Theorem

Theorem

Ceva's Theorem

Let P be a point in the interior of a triangle ABC . If the lines joining P and the vertices of the triangle intersect the sides BC , AC and AB at points K , N and T respectively, then

$$\frac{BK}{KC} \cdot \frac{CN}{NA} \cdot \frac{AT}{TB} = 1.$$



Proof

Considering $\triangle AKC$ and line BN with Menelaus' Theorem gives us

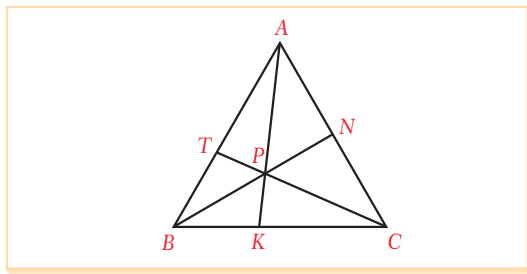
$$\frac{KB}{BC} \cdot \frac{CN}{NA} \cdot \frac{AP}{PK} = 1. \quad (1)$$

Similarly, applying Menelaus' Theorem to $\triangle ABK$ and line CT gives us

$$\frac{KC}{CB} \cdot \frac{BT}{TA} \cdot \frac{AP}{PK} = 1. \quad (2)$$

Dividing (1) and (2) side by side gives

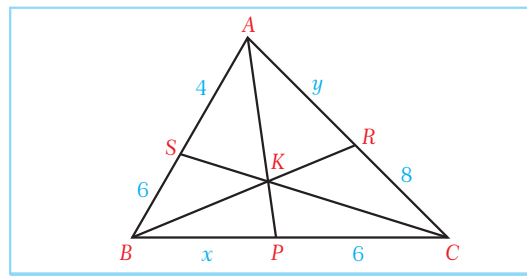
$$\frac{\frac{KB}{BC} \cdot \frac{CN}{NA} \cdot \frac{AP}{PK}}{\frac{KC}{CB} \cdot \frac{BT}{TA} \cdot \frac{AP}{PK}} = \frac{1}{1} = \frac{BK}{KC} \cdot \frac{CN}{NA} \cdot \frac{AT}{TB} = 1.$$



EXAMPLE

39

In the figure at the right, K is the intersection point of AP , BR and CS . $BP = x$, $PC = 6$ cm, $CR = 8$ cm, $AR = y$, $AS = 4$ cm and $SB = 6$ cm are given. Find the ratio $\frac{x}{y}$.



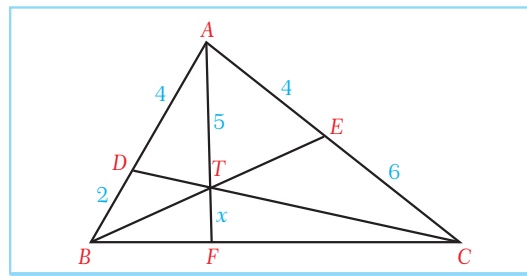
Solution Ceva's Theorem applied to $\triangle ABC$ gives us

$$\frac{BP}{PC} \cdot \frac{CR}{RA} \cdot \frac{AS}{SB} = 1. \text{ So } \frac{x}{6} \cdot \frac{8}{y} \cdot \frac{4}{6} = 1, \text{ and so } \frac{x}{y} = \frac{9}{8}.$$

EXAMPLE

40

Find the length x in the figure, using the information provided.



Solution Ceva's Theorem in $\triangle ABC$ gives us

$$\frac{BD}{DA} \cdot \frac{AE}{EC} \cdot \frac{CF}{FB} = 1. \text{ So } \frac{2}{4} \cdot \frac{4}{6} \cdot \frac{CF}{FB} = 1; \frac{CF}{FB} = 3.$$

Similarly, Menelaus' Theorem in $\triangle ABF$ with respect to line CD gives

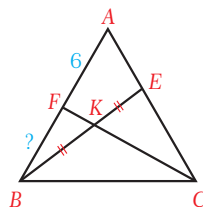
$$\frac{FC}{CB} \cdot \frac{BD}{DA} \cdot \frac{AT}{TF} = 1. \text{ Since } \frac{CF}{FB} = 3, \text{ we get } \frac{FC}{CB} = \frac{3}{4}. \text{ So } \frac{3}{4} \cdot \frac{2}{4} \cdot \frac{5}{x} = 1, \text{ i.e. } x = \frac{15}{8}.$$

Note

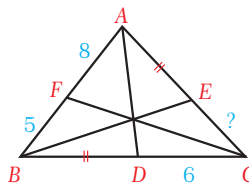
It can also be shown that the converse of Ceva's Theorem is true: in the figure on the previous page, if $\frac{BK}{KC} \cdot \frac{CN}{NA} \cdot \frac{AT}{TB} = 1$ then the lines BN , CT and AK intersect at a common point inside the triangle.

Check Yourself 8

- In the figure, $EC = 2 \cdot AE$, $BK = KE$ and $AF = 6$ cm.
Find the length of FB .



- In the figure, $AE = BD$,
 $AF = 8$,
 $FB = 5$ and
 $DC = 6$.
What is the length of EC ?



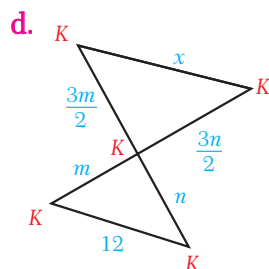
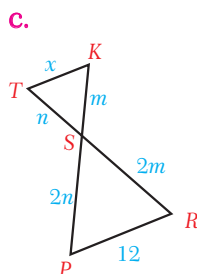
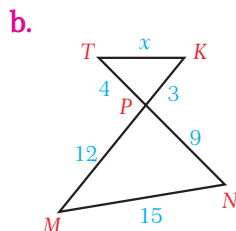
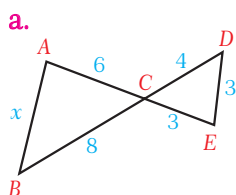
Answers

- 4 cm
- $\frac{15}{4}$

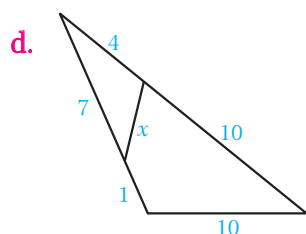
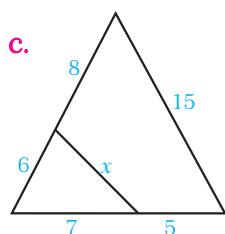
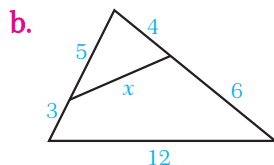
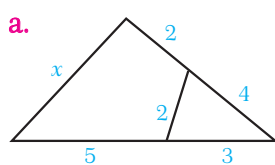
EXERCISES 2.2

A. The Side-Angle-Side (SAS) Similarity Theorem

1. Find the value of x in each figure, using the information given.



2. Find the value of x in each figure, using the information given.



3. $\triangle ABC$ and $\triangle PRS$ are triangles with $a = 12$ cm, $b = 8$ cm, $m(\angle C) = 70^\circ$, $p = 8$, $r = 6$ and $m(\angle S) = 70^\circ$. Are these triangles similar?

4. In each case, determine whether the two triangles described are similar.

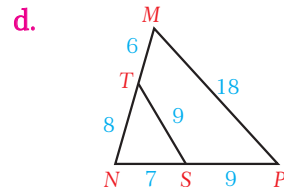
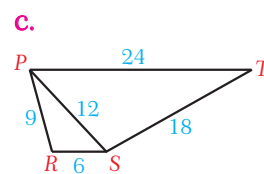
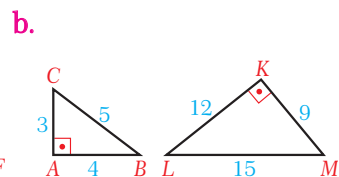
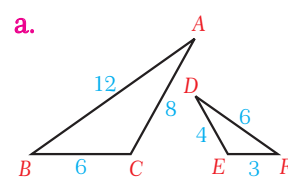
a. triangles ABC and KPS with $a = 10$ cm, $b = 8$ cm, $m(\angle C) = 65^\circ$, $k = 5$ cm, $p = 4$ cm and $m(\angle S) = 65^\circ$

b. triangles PRS and KMN with $m(\angle R) = m(\angle M) = 75^\circ$, $p = 12$ cm, $s = 10$ cm, $k = 18$ cm and $n = 15$ cm

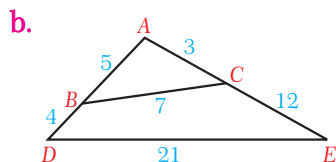
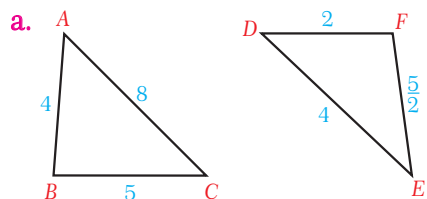
c. triangles XYZ and STK with $\angle X \cong \angle S$, $y = 3$ cm, $z = 4$ cm, $t = 9$ cm and $k = 2$ cm

B. The Side-Side-Side (SSS) Similarity Theorem

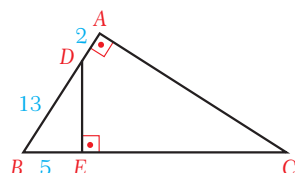
5. State whether the triangles in each pair are similar or not by considering the given lengths.



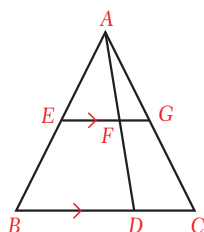
6. Find the congruent angles in each figure.



7. Find the area of $\triangle ABC$ in the figure.

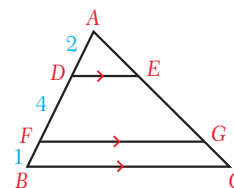


8. In the figure,
 $BC \parallel EG$.
 If $A(\triangle AFG) = 12 \text{ cm}^2$,
 $A(BDFE) = 20 \text{ cm}^2$ and
 $A(DCGF) = 15 \text{ cm}^2$, what
 is $A(\triangle AEF)$?

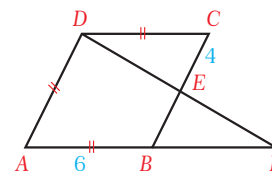


9. Find the value of the
 ratio

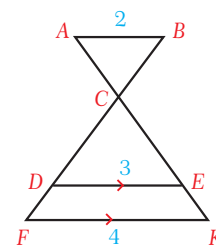
$$\frac{A(\triangle ADE) + A(BCGF)}{A(FGED)}$$
 in the figure.



10. In the figure,
 $ABCD$ is a rhombus.
 If $A(\triangle EBF) = 16 \text{ cm}^2$,
 what is $A(ABCD)$?



11. Find the value of the
 ratio $\frac{A(\triangle ABC)}{A(FKED)}$ in the
 figure.



12. Write a two-column proof to show that the ratio of
 the perimeters of two similar triangles equals the
 scale factor of their similarity.

C. The Triangle Proportionality Theorem and Thales' Theorem

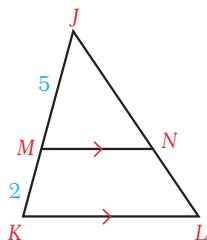
13. In the figure, $MN \parallel KL$, $JM = 5$ and $MK = 2$. Find each ratio.

a. $\frac{JN}{NL}$

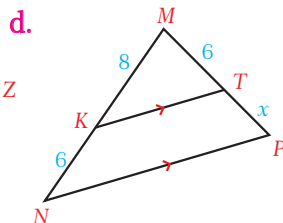
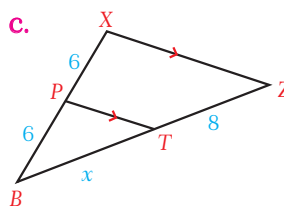
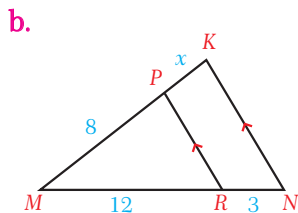
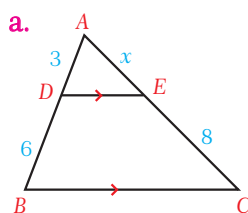
b. $\frac{JN}{JL}$

c. $\frac{NL}{JN}$

d. $\frac{NL}{JL}$



14. Find the value of x in each figure by using the information given.



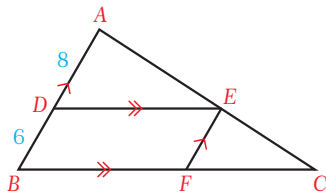
15. In the figure,
 $AD = 8$ cm,
 $DB = 6$ cm,
 $EF \parallel AB$ and $DE \parallel BC$.
 Find each ratio.

a. $\frac{AC}{AE}$

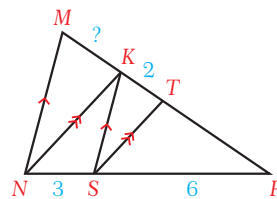
b. $\frac{AC}{EC}$

c. $\frac{BF}{FC}$

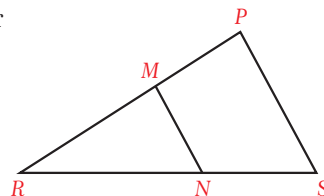
d. $\frac{FC}{BC}$



16. In the figure, MNP is a triangle and
 $MN \parallel KS$,
 $KN \parallel TS$,
 $NS = 3$ cm,
 $SP = 6$ cm and
 $KT = 2$ cm.
 What is the length of MK ?



17. Determine whether or not $MN \parallel PS$ in the figure, given each set of extra information.



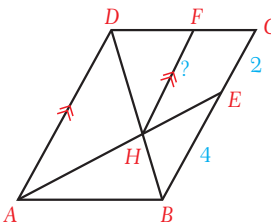
a. $PR = 18$ $MR = 6$
 $SR = 24$ $NR = 8$

b. $PR = 12$ $MP = 8$
 $SR = 16$ $NR = 12$

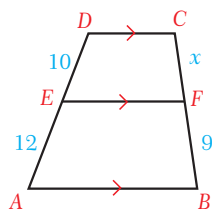
c. $MR = 5$ $MP = 4$
 $RN = 6$ $NS = \frac{24}{5}$

d. $PR = 15$ $MR = 12$
 $RN = 16$ $NS = 4$

18. $ABCD$ in the figure is a parallelogram with
 $CE = 2$ cm,
 $EB = 4$ cm and
 $FH \parallel AD$.
 What is the length of FH ?



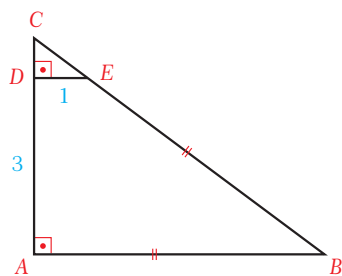
19. In the figure, $DC \parallel EF \parallel AB$.
Find the value of x .



20. Write a proof of the Converse of the Triangle Proportionality Theorem in two-column form.

21. A point on the hypotenuse of a right triangle divides the hypotenuse into two segments of lengths 12 and 16. Given that the point is equidistant to the legs of the triangle, find the lengths of the legs of the triangle.

22. In the triangle ABC at the right,
 $m(\angle A) = 90^\circ$,
 $CD \perp DE$,
 $DE = 1$,
 $AD = 3$ and
 $AB = BE$.



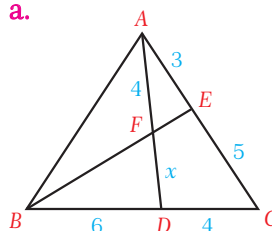
Find the length of CE .

(Hint: Draw the perpendicular $EH \perp AB$.)

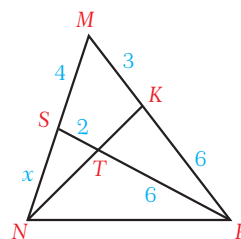
D. Further Applications

23. Find the length x in each figure by using Menelaus' Theorem.

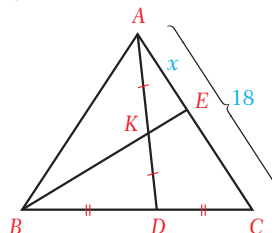
a.



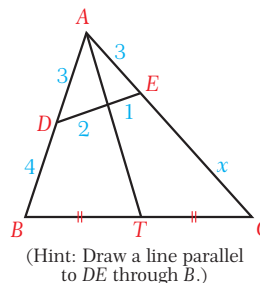
b.



c.

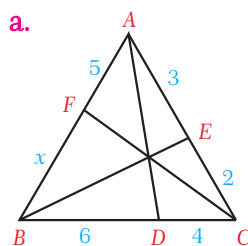


d.

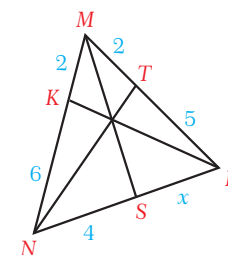


24. Find the length x in each figure by using Ceva's Theorem.

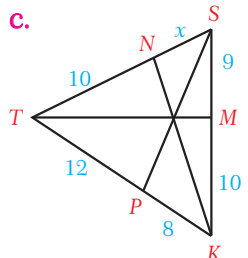
a.



b.



c.



A. EUCLIDEAN RELATIONS

Theorem

The altitude to the hypotenuse of a right triangle divides the right triangle into two smaller right triangles which are similar to the original triangle, and therefore also similar to each other.

Proof

Look at the first figure.

Given: $\triangle ABC$ is a right triangle and AH is the altitude to the hypotenuse.

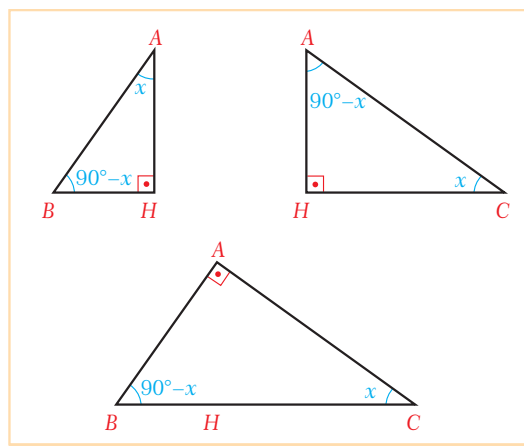
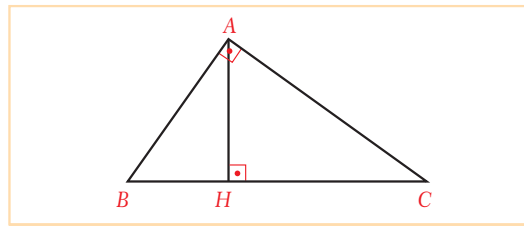
Prove: $\triangle ABC \sim \triangle HBA \sim \triangle HAC$

We will give the proof in paragraph form.

Let $m(\angle BCA) = x$.

Then, $m(\angle ABC) = 90^\circ - x$, $m(\angle HAB) = x$ and $m(\angle HAC) = 90^\circ - x$.

So each smaller triangle is similar to the larger triangle by the AA Similarity Theorem, and therefore the two smaller triangles are also similar to each other.



Remember!

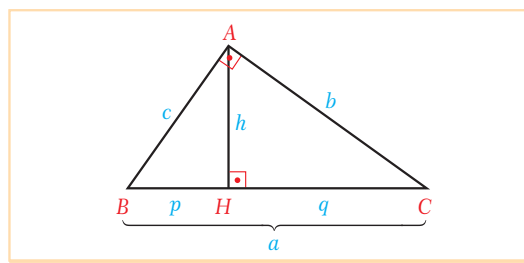
The geometric mean of two numbers a and b is a positive number x such that $\frac{a}{x} = \frac{x}{b}$,
i.e. $x = \sqrt{a \cdot b}$.

Theorem

Euclidean theorems

In any right triangle, when the altitude to the hypotenuse is drawn, the following two statements are true:

1. The length of the altitude is the geometric mean of the lengths of the two segments of the hypotenuse formed by the altitude ($AH^2 = BH \cdot CH$ in the figure).
2. The length of each leg is the geometric mean of the length of its adjacent hypotenuse segment and the length of the hypotenuse. ($CA^2 = CH \cdot CB$ in the figure).



Proof

Let us draw an appropriate figure (shown at the right).

Given: $\triangle ABC$ is a right triangle and AH is the altitude to the hypotenuse.

Prove: $AH^2 = BH \cdot CH$ (1) and

$BA^2 = BH \cdot BC$ and $CA^2 = CH \cdot CB$ (2)

We will write the proof of (1) in paragraph form.

By the theorem at the beginning of this section, $\triangle AHB \sim \triangle CHA$.

By the definition of similarity, corresponding sides are proportional:

$$\frac{BH}{AH} = \frac{AH}{CH}, \text{ i.e. } AH^2 = BH \cdot CH, \text{ as required.}$$

Now let us prove (2). By the same theorem, $\triangle HBA \sim \triangle ABC$. So by the definition of similarity, corresponding sides are proportional:

$$\frac{BH}{BA} = \frac{BA}{BC}, \text{ i.e. } BA^2 = BH \cdot BC.$$

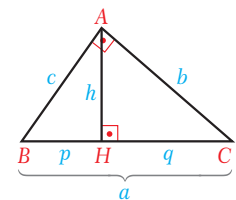
By a similar argument, $\triangle HAC \sim \triangle ABC$. So $\frac{CH}{CA} = \frac{CA}{CB}$, i.e. $CA^2 = CH \cdot CB$.



For any right triangle ABC , the relations

$h^2 = p \cdot q$, $c^2 = p \cdot a$ and $b^2 = q \cdot a$ are also called

Euclidean relations.



EXAMPLE

41

Find the lengths a , c and x in the figure.

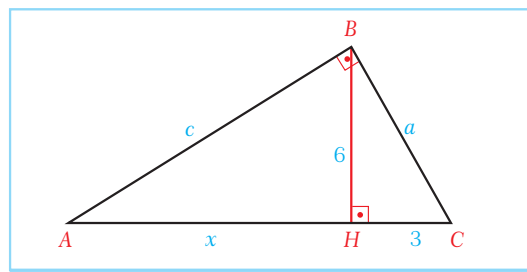
Solution

Since $\triangle ABC$ is a right triangle and BH is an altitude, we can use the Euclidean relations:

$$h^2 = p \cdot q; 6^2 = x \cdot 3; x = 12,$$

$$a^2 = 3 \cdot (3 + 12) = 3 \cdot 36; a = 6\sqrt{3},$$

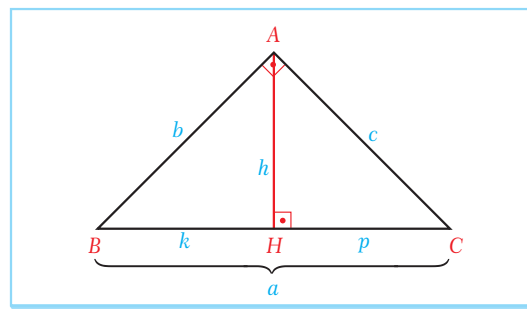
$$c^2 = 12 \cdot (12 + 3) = 180; c = 6\sqrt{5}.$$



EXAMPLE

42

Prove that $\frac{1}{h^2} = \frac{1}{b^2} + \frac{1}{c^2}$ in the figure.



Solution By the Euclidean relations,

$$b^2 = k \cdot a \text{ and } c^2 = p \cdot a.$$

If we multiply these side by side, we get

$$b^2 \cdot c^2 = \underbrace{k \cdot p}_{h^2} \cdot a^2. \text{ So } b^2 \cdot c^2 = h^2 \cdot a^2 \text{ and so}$$

$$\frac{1}{h^2} = \frac{a^2}{b^2 c^2} \quad (\text{Divide both sides by } b^2 \cdot c^2 \cdot h^2)$$

$$\frac{1}{h^2} = \frac{b^2 + c^2}{b^2 c^2} \quad (\text{Pythagorean Theorem in } \triangle ABC)$$

$$\frac{1}{h^2} = \frac{\cancel{b^2}}{\cancel{b^2} c^2} + \frac{\cancel{c^2}}{b^2 \cancel{c^2}} \quad (\text{Properties of rational numbers})$$

$$\frac{1}{h^2} = \frac{1}{c^2} + \frac{1}{b^2}.$$

EXAMPLE

43

In $\triangle ABC$ at the right, $CA \perp AB$,
 $AH \perp BC$, $AC = 3AB$, $BH = x$ and
 $HC = y$. Find the ratio $\frac{x}{y}$.

Solution

Let $AB = m$, then $AC = 3m$.

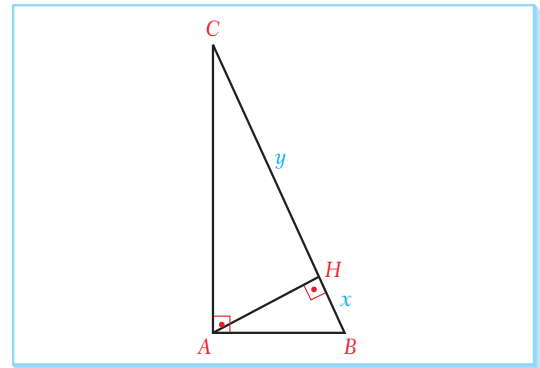
By the Euclidean relations in $\triangle ABC$,
 $AB^2 = BH \cdot BC$ and $AC^2 = CH \cdot CB$. So

$$m^2 = x(x + y) \text{ and}$$

$$9m^2 = y(x + y).$$

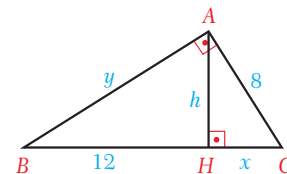
If we divide the first equality by the second, we get

$$\frac{\cancel{m^2}}{9\cancel{m^2}} = \frac{x(\cancel{x+y})}{y(\cancel{x+y})}. \text{ So } \frac{x}{y} = \frac{1}{9}.$$

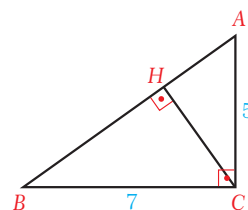


Check Yourself

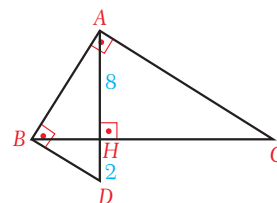
- Find x , y and h using the information in the figure.



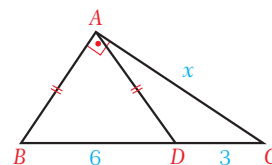
2. According to the figure, what is the value of the ratio $\frac{AH}{BH}$?



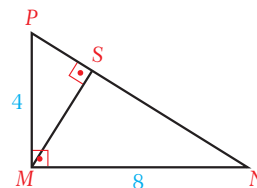
3. Find the length of BC in the figure.



4. Find the value of x in the figure.



5. Find the lengths of MS and PN using the information in the figure.



Answers

1. $x = 4$, $y = 8\sqrt{3}$, $h = 4\sqrt{3}$, 2. $\frac{25}{49}$ 3. 20 4. $3\sqrt{6}$ 5. $MS = \frac{8\sqrt{5}}{5}$, $PN = 4\sqrt{5}$

B. MEDIAN RELATIONS

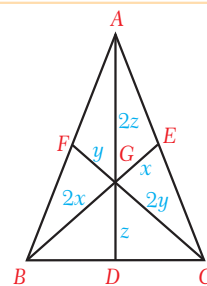
Like the angle bisectors and altitudes of a triangle, the medians in a triangle have some very useful properties.

Properties 10

Let ABC be any triangle. Then the following statements about $\triangle ABC$ are true.

1. If AD , BE and FC are medians then their common point of intersection G divides each median in the ratio $1 : 2$. For example, in the figure,

$$GD = \frac{1}{3}AD \text{ and } AG = \frac{2}{3}AD, \text{ so } \frac{GD}{AG} = \frac{1}{2}.$$



2. If AD , BE and CF are medians and points K and G are as shown in the figure, then

$$FE \parallel BC, FE = \frac{BC}{2} \text{ and } AK = KD.$$

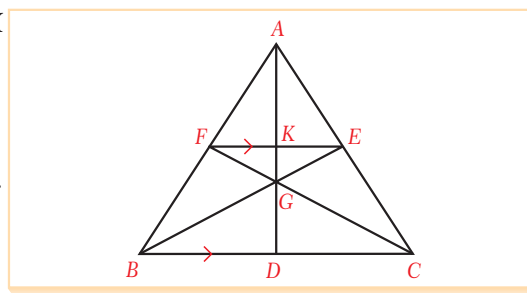
We can conclude that $KG = KD - GD$, i.e.

$$KG = \frac{AD}{2} - \frac{AD}{3} = \frac{AD}{6} \text{ and } KG = \frac{GD}{2}.$$

3. If AD , BE and FC are medians then

$$b^2 + c^2 = 2V_a^2 + \frac{a^2}{2}, \quad a^2 + b^2 = 2V_c^2 + \frac{c^2}{2} \text{ and } a^2 + c^2 = 2V_b^2 + \frac{b^2}{2}.$$

We can conclude that $3(a^2 + b^2 + c^2) = 4(V_a^2 + V_b^2 + V_c^2)$.



EXAMPLE

44

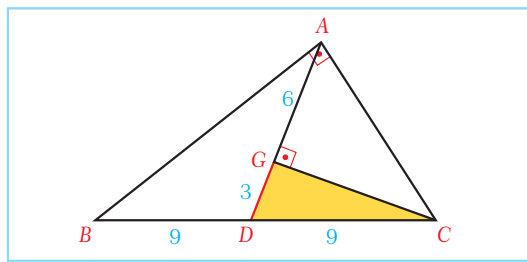
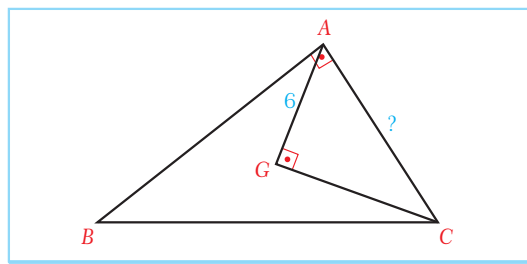
In triangle ABC in the figure, $m(\angle BAC) = 90^\circ$, $AG \perp GC$, $AG = 6$ cm and point G is the centroid of $\triangle ABC$. Find the length of AC .

Solution Let us extend AG so that it cuts BC at point D . Since G is the centroid of $\triangle ABC$, by Property 10.1 we have $GD = \frac{AG}{2} = \frac{6}{2} = 3$ cm. Also, since AD is a median to the hypotenuse (because it passes through the centroid of $\triangle ABC$), we have $AD = BD = DC = 9$ cm. The Pythagorean Theorem in $\triangle GDC$ gives us

$$\begin{aligned} GC^2 + GD^2 &= DC^2 \\ GC^2 &= 81 - 9 \\ &= 72 \\ GC &= 6\sqrt{2} \text{ cm.} \end{aligned}$$

And so by the Pythagorean Theorem in $\triangle AGC$,

$$\begin{aligned} AC^2 &= AG^2 + GC^2 \\ &= 6^2 + (6\sqrt{2})^2 \\ &= 36 + 72 \\ &= 108 \\ AC &= 6\sqrt{3} \text{ cm.} \end{aligned}$$



EXAMPLE

45

AD , BF and CE in the figure are the medians of triangle ABC . Find the lengths x and y .

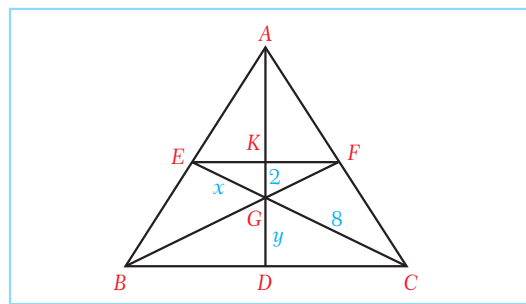
Solution $EG = \frac{GC}{2}$ (Property 10.1)

$$x = \frac{8}{2}$$

$$x = 4.$$

$$KG = \frac{GD}{2}$$
 (Property 10.2)

$$2 = \frac{y}{2}; y = 4$$



EXAMPLE

46

In the figure, G is the centroid of $\triangle ABC$, $AD = DC = 8$ cm, $GD = 4$ cm and $GC = 10$ cm. Find the length of AB .

Solution Let us draw a line from vertex A to the centroid G . Then GD is a median of $\triangle AGC$.

Let $AG = x$, then in $\triangle AGC$,

$$x^2 + 10^2 = 2 \cdot 4^2 + \frac{16^2}{2}$$
 (Property 10.3)

$$x^2 + 100 = 32 + 128$$

$$x^2 = 60$$

$$x = \sqrt{60} \text{ cm.}$$

Also, CE is a median of $\triangle ABC$. So

$$EG = \frac{GC}{2}$$
 (Property 10.1)

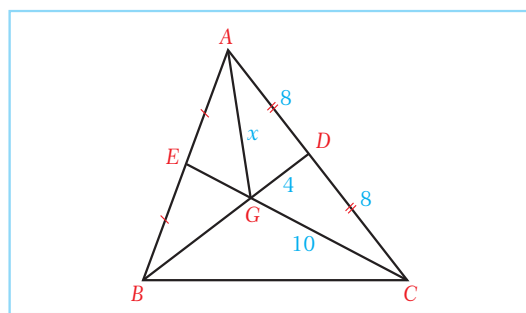
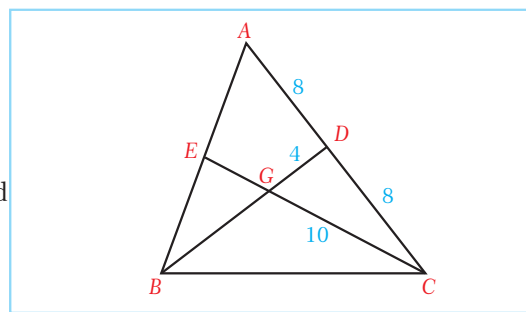
$$= \frac{10}{2}$$

$$= 5 \text{ cm.}$$

BD is also a median of $\triangle ABC$. So

$$BG = 2 \cdot GD$$
 (Property 10.1)

$$= 2 \cdot 4 = 8 \text{ cm.}$$



Finally, GE is a median of $\triangle ABG$. So

$$8^2 + (\sqrt{60})^2 = 2 \cdot 5^2 + \frac{c^2}{2} \quad (\text{Property 10.3})$$

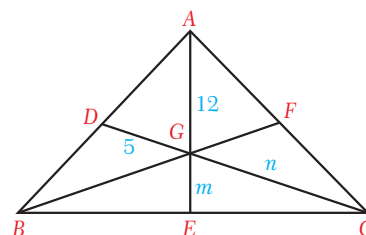
$$64 + 60 = 50 + \frac{c^2}{2}$$

$$74 \cdot 2 = c^2$$

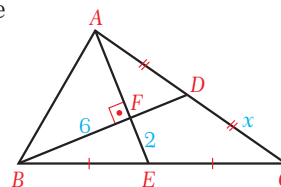
$$c = 2\sqrt{37} \text{ cm.}$$

Check Yourself

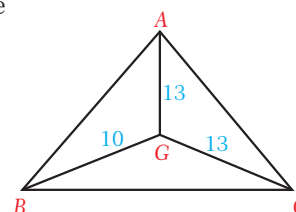
1. In the figure, G is the centroid of $\triangle ABC$. Find the value of $m + n$.



2. Find the value of x in the figure, using the information given.



3. In the figure, G is the centroid of $\triangle ABC$. Find the length of AC .



4. In a triangle ABC , $D \in BC$ and AD is a median. Given $AB = 8$ cm, $AC = 10$ cm and $AD = 6$ cm, find the length of BD .
5. V_a , V_b and V_c are the medians to the sides a , b and c of a triangle ABC . $a = 14$ cm, $V_b = 12$ cm and $V_c = 15$ cm are given. Find V_a .

Answers

1. 16
2. 5
3. 24
4. $\sqrt{46}$ cm
5. $3\sqrt{33}$ cm

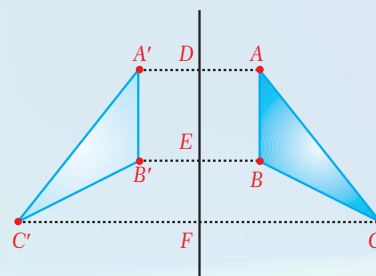
TRANSFORMATIONAL GEOMETRY

In this book we are studying congruent and similar triangles. To determine whether two triangles are congruent or similar, we have considered their vertices and side lengths. **Transformations** give us a different way of determining whether two figures are congruent or similar. A transformation means rotating, reflecting, resizing or sliding a figure in the plane.

The study of transformations in geometry is called **transformational geometry**. Transformational geometry is useful for creating repeating patterns in a plane, and has applications in genetic research, molecular biology, studies of crystals, and the decorative arts. The two most basic transformations are **reflection** and **resizing**.

Reflection

A **reflection** (also called a **flip**) is a transformation that reflects an object onto its mirror image. Mirrors and lakes create examples of reflections in the real world.

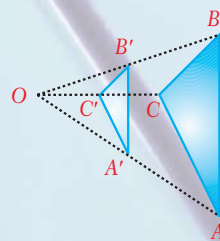


Triangle $A'B'C'$ is the reflection of triangle ABC across the mirror line DF . Notice that $AD = A'D$, $BE = B'E$, and $CF = C'F$.

A geometric reflection is defined by a central line called the **mirror line**. Every point in a geometric reflection is the same distance from this mirror line, and the reflection is the same size as the original image. The mirror line can be anywhere in the plane and the reflection will always have these properties.

Resizing

Resizing is the second basic transformation. Other words for resizing include **dilation**, **contraction**, **compression**, **enlargement** and **expansion**. In each case the transformation changes the size of a figure but not its shape. Every resizing transformation uses a fixed point called the **center of the resizing**.



O is the center of resizing. $\triangle A'B'C'$ is a resizing of $\triangle ABC$.

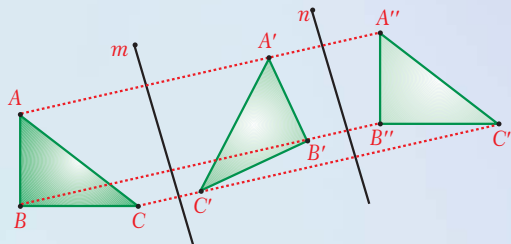
A resizing transformation makes a figure bigger or smaller but the new figure is still similar to the old one: all the angles stay the same and corresponding sides of the figure are all in proportion.

Two other transformations are **translation** and **rotation**.

Translation

A **translation** (also called a **slide**) is a transformation in which every point in a figure moves the same distance in a specified direction. A translation is like sliding a figure across a distance, without turning it or reflecting it. We can also think of a translation as two successive reflections.

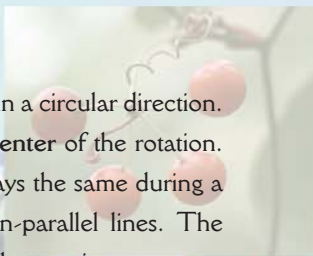
In the figure, triangle $A''B''C''$ is a translation of triangle ABC : we can slide ABC up and right across the page to match triangle $A''B''C''$ completely. Alternatively we can reflect $\triangle ABC$ across two parallel lines to get the same transformation: $\triangle A'B'C'$ is a reflection of $\triangle ABC$, and $\triangle A''B''C''$ is a reflection of $\triangle A'B'C'$. Notice that $AA'' = BB'' = CC''$ and lines m and n are parallel.



Rotation

A **rotation** (also called a **turn**) is the movement of an object in a circular direction. A rotation takes place around a point which is called the **center** of the rotation. The distance from any point on the shape to this center stays the same during a rotation. A rotation is also like two reflections across non-parallel lines. The intersection of these lines is concurrent with the center of the rotation.

In the figure, $\triangle A''B''C''$ is a rotation of $\triangle ABC$ about center O . $\triangle A''B''C''$ is also the result of two reflections of $\triangle ABC$, first across line f and then across line e . Notice that angle AOA'' measures 90° , and lines e and f are non-parallel.



Congruent or Similar?

If we can match one shape to another completely using one or more turns, flips or slides then the two shapes are congruent. To determine whether two shapes are similar we can draw lines connecting corresponding vertices. If these lines all meet at the same point then we can say that the shapes are similar.

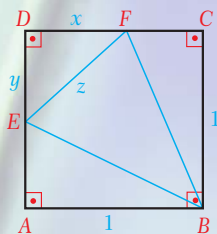
Exploring Transformational Geometry

We can use transformations to solve problems in plane geometry. Try to solve the following two problems using transformations. Can you do it?

1. If $x + y + z = 2$ in the figure, find the measure of $\angle EBF$.

(Hint: Rotate $\triangle CBF$ 90° about point B .)

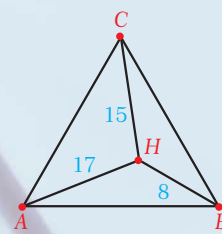
Answer: 45°



2. $\triangle ABC$ in the figure is an equilateral triangle. Find $m(\angle CHB)$.

(Hint: Consider the reflection of $\triangle CHB$ about line CB .)

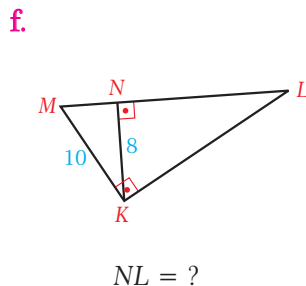
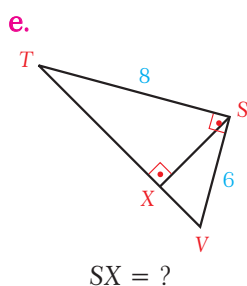
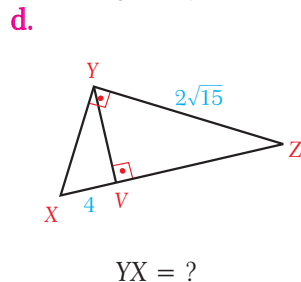
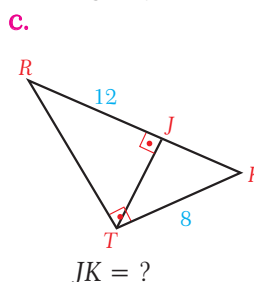
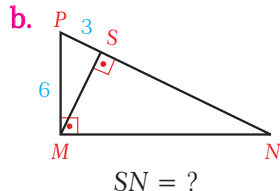
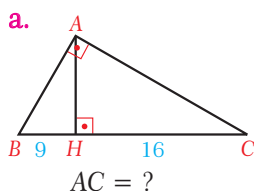
Answer: 150°



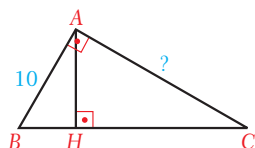
EXERCISES 2.4

A. Euclidean Relations

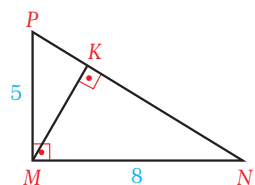
1. In each case, find the indicated length using the information in the figure.



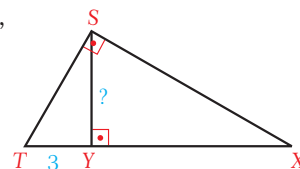
2. In the figure,
 $m(\angle BAC) = 90^\circ$,
 $AH \perp BC$,
 $BC = 4 \cdot BH$ and
 $AB = 10$.
 What is the length of segment AC ?



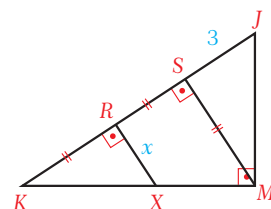
3. In the figure at the right,
 $PM \perp MN$,
 $MK \perp PN$,
 $PM = 5$ cm and
 $MN = 8$ cm.
 What is the ratio $\frac{PK}{KN}$?



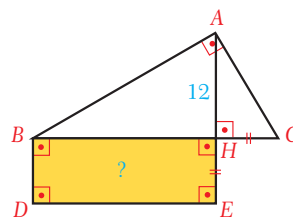
4. In the figure at the right,
 $m(\angle TSX) = 90^\circ$,
 $SY \cdot YX = 9$ and
 $TY = 3$. What is the
 length of SY ?



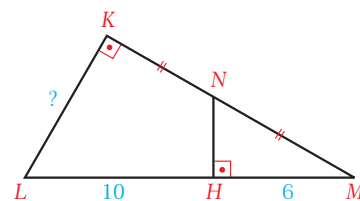
5. In the figure,
 $JM \perp KM$
 $MS \perp JK$,
 $XR \perp JK$,
 $KR = RS = SM$,
 $SJ = 3$ and $RX = x$.
 What is x ?



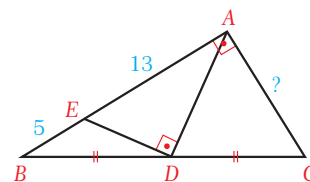
6. In the figure,
 polygon $BDEH$ is a
 rectangle.
 $m(\angle BAC) = 90^\circ$,
 $AH \perp BC$,
 $HE = HC$ and
 $AH = 12$ are given.
 What is the area of rectangle $BDEH$?



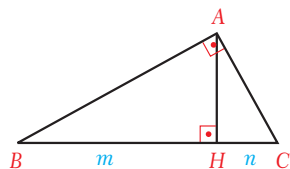
7. In the figure,
 $m(\angle K) = 90^\circ$,
 $KN = NM$,
 $LH = 10$ and
 $HM = 6$.
 What is the length
 of KL ?



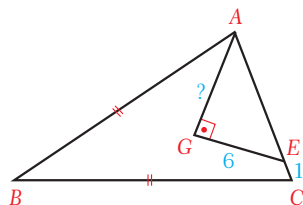
8. In the figure,
 $m(\angle BAC) = 90^\circ$,
 $ED \perp AD$,
 $BD = DC$,
 $AE = 13$ and
 $EB = 5$. Find the length of AC .



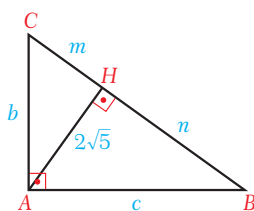
9. In the figure,
 $AH \perp BC$ and
 $m(\angle BAC) = 90^\circ$.
 $2 \cdot AB = 5 \cdot AC$,
 $BH = m$ and
 $HC = n$ are given. What is $\frac{m}{n}$?



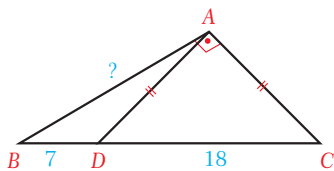
10. In the figure,
 G is the centroid
of $\triangle ABC$,
 $AG \perp GE$ and
 $AB = BC$.
If $GE = 6$ cm and
 $EC = 1$ cm, what is the length of AG ?



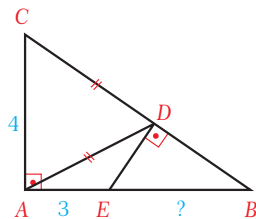
11. In the figure,
 $AH = 2\sqrt{5}$ and
 $m^2 + n^2 = 41$.
Find $b \cdot c$.



12. In the figure,
 $m(\angle BAC) = 90^\circ$,
 $AD = AC$,
 $BD = 7$ and
 $DC = 18$.
Find the length of AB .

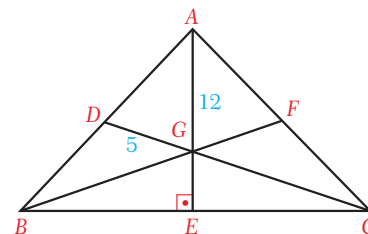


13. In the figure,
 $AC \perp AB$,
 $ED \perp BC$ and
 $AD = CD$.
Given $AC = 4$ cm
and $AE = 3$ cm,
find the length of EB .



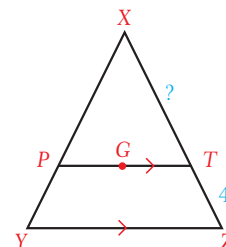
B. Median Relations

14. In the figure,
 G is the centroid
of $\triangle ABC$.
If $AE \perp BC$,
 $AG = 12$ and
 $DG = 5$,
find the length of BC .



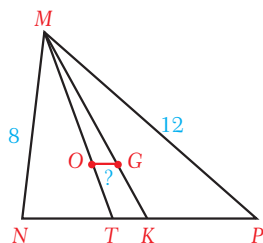
15. The sum of the lengths of the medians of a triangle ABC is 27 cm.
- Find the sum of the distances from the centroid of $\triangle ABC$ to each of its vertices.
 - Find the sum of the distances from the centroid to the midpoints of the sides of $\triangle ABC$.

16. In the figure, point G is
the centroid of $\triangle XYZ$ and
 $PT \parallel YZ$.
If $TZ = 4$, what is the
length of XT ?



17. In a triangle ABC , $D \in AB$ and CD is a median. If
 $AB = 6$, $AC = 8$ and $CD = 7$, find the length of
 CB .
18. G is the centroid of a triangle ABC with $G \in DE$,
 $DE \parallel BC$, $D \in AB$, $E \in AC$ and $BC = 12$. What is
the length of DE ?

19. In the figure, G is the centroid of $\triangle MNP$ and point O is the intersection point of its three angle bisectors.

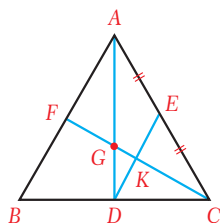


$$MN = 8,$$

$$MP = 12 \text{ and}$$

$$NP = 10 \text{ are given. Find the length of } OG.$$

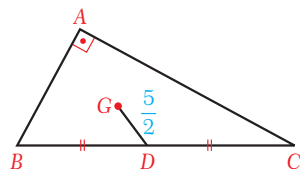
20. In $\triangle ABC$, D and E are the midpoints of sides AB and BC respectively, and G is the intersection point of AE and CD . If $AE + CD = 24$, what is $DG + EG$?



21. In the figure, G is the centroid of $\triangle ABC$, and $AE = EC$. If $FC = 18$, what is the length of GK ?

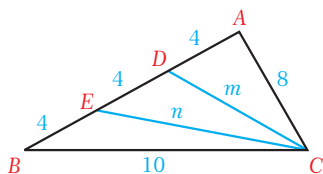
22. In a triangle ABC , D , E and F are the midpoints of the sides AB , BC and AC , respectively. Given that the perimeter of $\triangle DEF$ is 24, find the perimeter of $\triangle ABC$.

23. In the figure, G is the centroid of $\triangle ABC$, $BD = DC$ and $GD = \frac{5}{2}$.



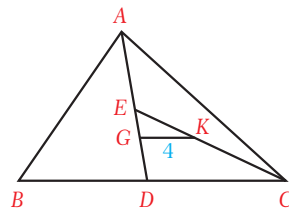
Find the length of BC .

24. Find the value of $m^2 + n^2$ by using the information in the figure.

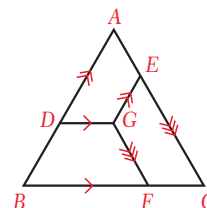


25. A triangle ABC has side lengths a , b and c such that $2a^2 = 3b^2 + c^2$ and $b = 6$. Calculate the length of the median to side c .

26. In the figure, G and K are the centroids of $\triangle ABC$ and $\triangle ADC$ respectively. Given $GK = 4$, find the length of BC .

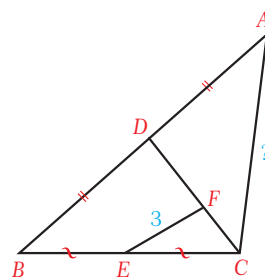


27. In the figure, G is the centroid of $\triangle ABC$, $GD \parallel BC$, $GE \parallel AB$ and $GF \parallel AC$.

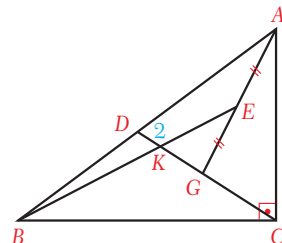


If the perimeter of $\triangle ABC$ is 48, what is $GD + GE + GF$?

28. In the figure, $AD = DB$, $BE = EC$, $DF = 2 \cdot FC$ and $V_a \perp V_c$. If $EF = 3$, what is the length of AC ?



29. In the figure, G is the centroid of $\triangle ABC$, $AE = EG$ and $DK = 2$. Find the length of AB .



CHAPTER SUMMARY

1. Introduction to Similarity

- Figures which are the same shape but not necessarily the same size are called **similar figures**.
- We use the symbol $\sim$ to show similarity: $A \sim B$ means polygon A is similar to polygon B.
- Two polygons are similar if
 - corresponding angles are congruent, and
 - corresponding sides are proportional.
- Two triangles are called **similar triangles** if their corresponding angles are congruent and their corresponding sides are proportional.
- The ratio of the lengths of corresponding sides of similar triangles is called the **scale factor** of the similarity, and denoted by k .
- If $\triangle ABC \sim \triangle DEF$ and the scale factor of similarity is 1 then $\triangle ABC \cong \triangle DEF$.

2. The Angle-Angle (AA) Similarity Postulate

- If two angles in one triangle are congruent to two angles in another triangle, then the triangles are similar.

3. The Side-Angle-Side (SAS) Similarity Theorem

- If an angle in a triangle is congruent to an angle in another triangle, and if the lengths of the sides which include these angles are in proportion, then the triangles are similar.

4. The Side-Side-Side (SSS) Similarity Theorem

- If the lengths of the corresponding sides of two triangles are all in proportion, then the triangles are similar.
- Properties of Similarity**
 - The scale factor of similarity of two triangles is equal to
 - the ratio of the lengths of corresponding altitudes,
 - the ratio of the lengths of corresponding angle bisectors,
 - the ratio of the lengths of corresponding medians.

In other words, if $\triangle ABC \sim \triangle DEF$ then

$$\frac{h_a}{h_d} = \frac{h_b}{h_e} = \frac{h_c}{h_f} = \frac{n_A}{n_D} = \frac{n_B}{n_E} = \frac{n_C}{n_F} = \frac{V_a}{V_d} = \frac{V_b}{V_e} = \frac{V_c}{V_f} = k.$$

- The ratio of the perimeters of two similar triangles is equal to the scale factor of similarity, i.e.

$$\text{if } \triangle ABC \sim \triangle DEF \text{ then } \frac{P(\triangle ABC)}{P(\triangle DEF)} = k.$$

- The ratio of the areas of two similar triangles is equal to the square of the scale factor of similarity, i.e.

$$\text{if } \triangle ABC \sim \triangle DEF \text{ then } \frac{P(\triangle ABC)}{P(\triangle DEF)} = k^2.$$

5. The Triangle Proportionality Theorem and Thales' Theorem

- (Triangle Proportionality Theorem) A line parallel to one side of a triangle which intersects the other two sides of the triangle at different points divides these two sides proportionally.
- (Converse of the Triangle Proportionality Theorem) If a line divides two sides of a triangle proportionally then it is parallel to the third side of the triangle.
- (Thales' Theorem) If two transversals intersect three or more parallel lines then the parallel lines divide the transversals proportionally.
- (Menelaus' Theorem) Let ABC be a triangle. If a line d intersects the two sides AB and AC and the extension of the third side BC of $\triangle ABC$ at points R , S and P respectively, then $\frac{BP}{PC} \cdot \frac{CS}{SA} \cdot \frac{AR}{RB} = 1$.
- (Ceva's Theorem) Let P be a point in the interior of a triangle ABC . If the lines joining P and the vertices of the triangle intersect the sides BC , AC and AB at points K , N and T respectively, then $\frac{BK}{KC} \cdot \frac{CN}{NA} \cdot \frac{AT}{TB} = 1$.

6. Euclidean Relations

- The altitude to the hypotenuse of a right triangle forms two triangles which are similar to the original triangle, and therefore also similar to each other.

- (Euclidean Theorems) In any right triangle, when the altitude to the hypotenuse is drawn, the following two statements are true:

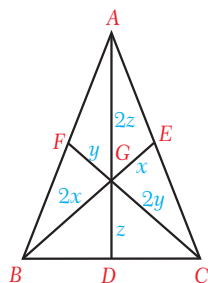
1. The length of the altitude is the geometric mean of the lengths of the two segments of the hypotenuse formed by the altitude.
2. The length of each leg is the geometric mean of the length of its adjacent hypotenuse segment and the length of the hypotenuse.

7. Median Relations

- Let ABC be a triangle with medians AD , BE and FC which intersect at a point G . Then the following statements are true:

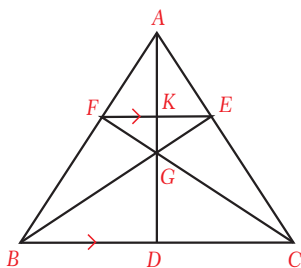
1. G divides each median in the ratio $1 : 2$, i.e. in the figure,

$$GD = \frac{AD}{3} \text{ and } AG = \frac{2}{3}AD \text{ (so } GD = \frac{AG}{2}\text{)}.$$



2. If points K and G are as shown below, then

$$FE \parallel BC, FE = \frac{BC}{2}, AK = KD \text{ and } KG = \frac{GD}{2}.$$



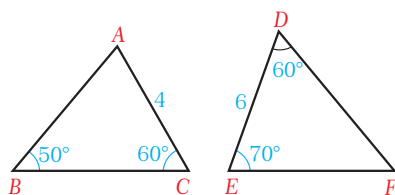
3. $b^2 + c^2 = 2V_a^2 + \frac{a^2}{2}$, $b^2 + a^2 = 2V_c^2 + \frac{c^2}{2}$ and $a^2 + c^2 = 2V_b^2 + \frac{b^2}{2}$.

Concept Check

1. State five different ways of proving that triangles are similar.
2. How many pairs of similar triangles are formed when an altitude is drawn to the hypotenuse of a right triangle?
3. What do we know about ratio of the altitudes of similar triangles?
4. Which special kind of triangle are always similar to each other?

CHAPTER REVIEW TEST 2A

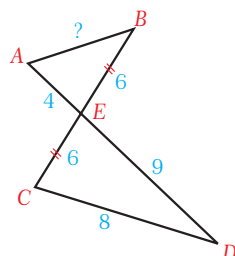
1.



$\triangle ABC \sim \triangle DEF$ is given with $m(\angle B) = 50^\circ$, $m(\angle C) = m(\angle D) = 60^\circ$, $m(\angle E) = 70^\circ$, $AC = 4$ cm and $DE = 6$ cm. What is the scale factor of similarity of these triangles?

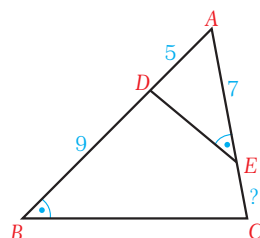
- A) $\frac{1}{2}$ B) $\frac{2}{3}$ C) $\frac{3}{4}$ D) $\frac{2}{5}$ E) $\frac{3}{5}$

2. In the figure, E is the intersection point of AD and BC . $AE = 4$, $EB = EC = 6$, $ED = 9$ and $DC = 8$ are given. Find the length of AB .



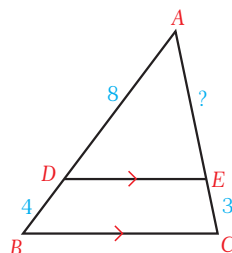
- A) 3 B) 4 C) 5 D) $\frac{16}{3}$ E) $\frac{18}{5}$

3. In the figure, $m(\angle B) = m(\angle DEA)$, $AD = 5$, $DB = 9$ and $AE = 7$. What is the length of EC ?



- A) 1 B) 2 C) 3 D) 4 E) 5

4. In the figure, $DE \parallel BC$. Find the length of AE .



- A) 9 B) 8 C) 7 D) 6 E) 4

5. In the figure,

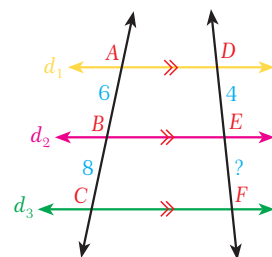
$$d_1 \parallel d_2 \parallel d_3.$$

$$\text{If } AB = 6,$$

$$BC = 8 \text{ and}$$

$$DE = 4,$$

what is the length of EF ?



- A) $\frac{8}{3}$ B) 4 C) $\frac{9}{2}$ D) 5 E) $\frac{16}{3}$

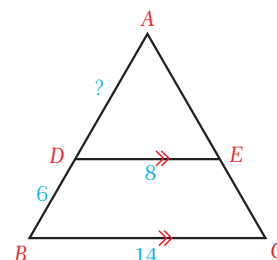
6. In the figure,

$$DE \parallel BC. \text{ If}$$

$$DB = 6,$$

$$DE = 8 \text{ and}$$

$BC = 14$, what is the length of AD ?



- A) 6 B) 7 C) 8 D) 9 E) 12

7. In the figure,

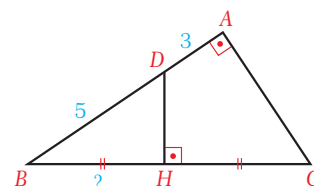
$$DH \perp BC,$$

$$AB \perp AC \text{ and}$$

$$HB = HC. \text{ If}$$

$$AD = 3 \text{ and}$$

$BD = 5$, what is the length of BH ?



- A) 3 B) 4 C) $2\sqrt{5}$ D) 5 E) $4\sqrt{2}$

8. In the figure,

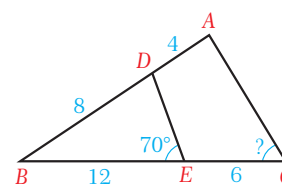
$$AD = 4,$$

$$DB = 8,$$

$$BE = 12,$$

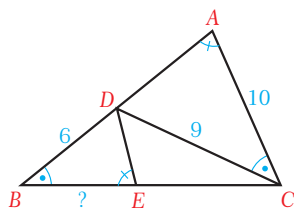
$$EC = 6 \text{ and}$$

$m(\angle DEB) = 70^\circ$. What is $m(\angle ACB)$?



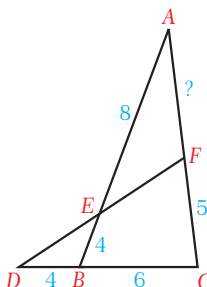
- A) 50° B) 55° C) 60° D) 65° E) 70°

9. In the figure,
 $m(\angle B) = m(\angle DCA)$,
 $m(\angle A) = m(\angle BED)$,
 $BD = 6$,
 $DC = 9$ and
 $AC = 10$. Find the
length of BE .



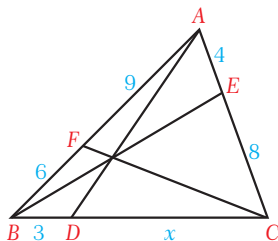
- A) $\frac{20}{3}$ B) 7 C) $\frac{24}{5}$ D) $\frac{25}{3}$ E) 8

10. In the figure,
 D, B, C and D, E, F
are two sets of
collinear points.
If $AE = 8$,
 $EB = DB = 4$,
 $BC = 6$ and
 $FC = 5$, what is the
length of AF ?



- A) 2 B) 3 C) 4 D) 5 E) 6

11. In the figure,
 $AF = 9$,
 $FB = 6$,
 $AE = 4$,
 $EC = 8$ and
 $BD = 3$.
What is the length
 $DC = x$?

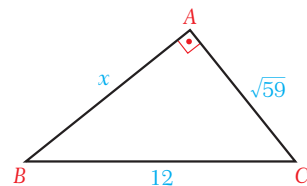


- A) 6 B) 7 C) 8 D) 9 E) 12

12. A triangle T has sides with lengths 8, 10 and 12.
The longest side of another triangle which is
similar to T has length 30. Find the perimeter of
the second triangle.

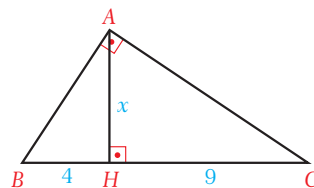
- A) 55 B) 60 C) 65 D) 70 E) 75

13. Find the length x in
the figure.



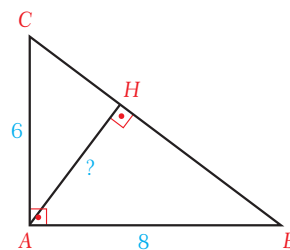
- A) $\sqrt{85}$ B) $4\sqrt{5}$ C) $6\sqrt{2}$ D) 8 E) $5\sqrt{2}$

14. Find the value of x
in the figure.



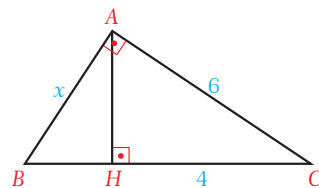
- A) 5 B) 6 C) $2\sqrt{10}$ D) $3\sqrt{5}$ E) 8

15. In the figure,
 $AC \perp AB$ and
 $AH \perp BC$. If
 $AC = 6$ cm and
 $AB = 8$ cm, what is
the length of AH ?



- A) 3 cm B) 3.6 cm C) 4 cm
D) 4.8 cm E) 5.4 cm

16. Find the length x in
the figure.

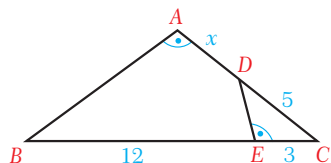


- A) 8 B) 7 C) $3\sqrt{5}$ D) 6 E) $3\sqrt{3}$

CHAPTER REVIEW TEST 2B

1. In the figure,
 $m(\angle A) = m(\angle DEC)$.

If $BE = 12$,
 $EC = 3$ and
 $DC = 5$,
 what is the value of x ?



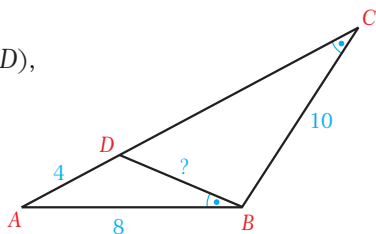
- A) 2 B) 3 C) 4 D) 5 E) 6

2. A line d which is parallel to side EF of a triangle DEF cuts DE and DF at the points M and N respectively. If $DM = 8$, $P(\triangle DMN) = 18$ and $P(\triangle DEF) = 45$, what is the length of DE ?

- A) 16 B) 20 C) 22 D) 24 E) 26

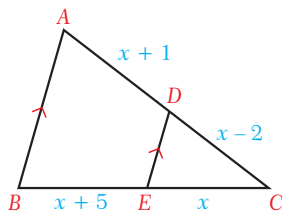
3. In the figure,
 $m(\angle C) = m(\angle ABD)$,

$AB = 8$,
 $BC = 10$ and
 $AD = 4$.
 What is the
 length of BD ?



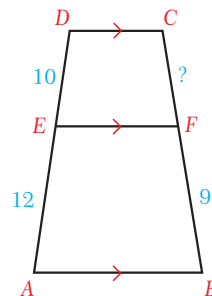
- A) 6 B) 5 C) 4 D) 3 E) 2

4. In the figure,
 $DE \parallel AB$.
 Find the value of x .



- A) 1 B) 2 C) 3 D) 4 E) 5

5. In the figure,
 $DC \parallel EF \parallel AB$,
 $DE = 10$,
 $EA = 12$ and
 $FB = 9$.
 What is the length
 of CF ?

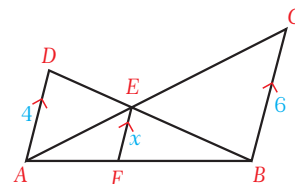


- A) 6 B) 7 C) $\frac{15}{2}$ D) 8 E) $\frac{17}{2}$

6. In a triangle KLM , points R and P lie on the sides KL and LM respectively such that MR and KP are altitudes of $\triangle KLM$. If T is the intersection point of KP and MR , $KT : MT = 2 : 3$ and $MP = 6$, what is the length of KR ?

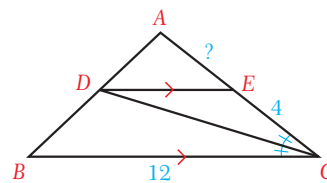
- A) 2 B) 3 C) 4 D) 5 E) 6

7. In the figure,
 $AD \parallel EF \parallel BC$,
 $AD = 4$ and
 $BC = 6$.
 Find the length
 $EF = x$.



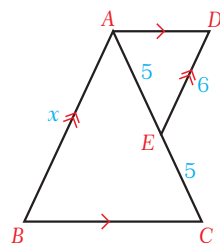
- A) 2 B) 2.4 C) 3 D) 3.6 E) 4

8. In the figure,
 CD is the bisector of
 $\angle C$ and $DE \parallel BC$.
 If $BC = 12$ and
 $EC = 4$, what is the
 length of AE ?



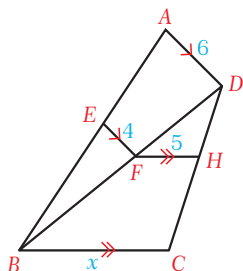
- A) 8 B) 6 C) 4 D) 3 E) 2

9. In the figure,
 $AD \parallel BC$ and
 $AB \parallel DE$. If
 $DE = 6$ cm and
 $AE = EC = 5$ cm,
 what is the length x ?



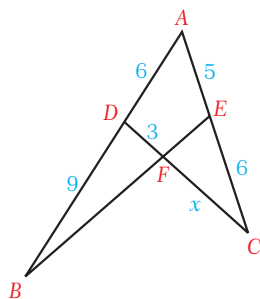
- A) 12 cm B) 13 cm C) 14 cm
 D) 16 cm E) 18 cm

10. In the figure,
 $AD \parallel EF$ and
 $FH \parallel BC$.
 Find the value of x .



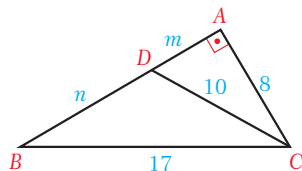
- A) 15 B) 12 C) 10 D) 9 E) 8

11. In the figure,
 $AE = 5$,
 $AD = EC = 6$,
 $DB = 9$ and
 $DF = 3$.
 What is the length
 $FC = x$?



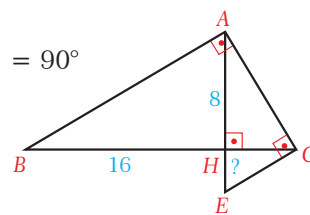
- A) 4 B) 5 C) 6 D) 8 E) 9

12. What is $\frac{m}{n}$ in the figure?



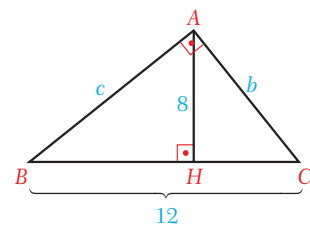
- A) $\frac{1}{2}$ B) $\frac{2}{3}$ C) $\frac{3}{2}$ D) $\frac{3}{4}$ E) $\frac{3}{5}$

13. In the figure,
 $m(\angle BAC) = m(\angle ACE) = 90^\circ$
 and $AE \perp BC$.
 If $BH = 16$ and
 $AH = 8$, what is the
 length of EH ?



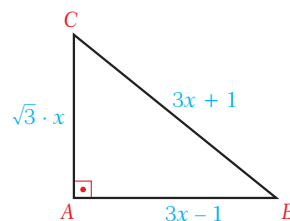
- A) 1 B) 2 C) 3 D) 4 E) 6

14. In the figure,
 $m(\angle BAC) = 90^\circ$ and
 $AH \perp BC$. If
 $BC = 12$ and
 $AH = 8$, what is the
 value of $b \cdot c$?



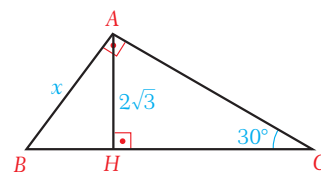
- A) 60 B) 104 C) 48 D) 96 E) 80

15. In the figure,
 $AC \perp AB$,
 $AB = 3x - 1$,
 $AC = \sqrt{3}x$ and
 $BC = 3x + 1$.
 Find the value of x .



- A) 1 B) 2 C) 3 D) 4 E) 5

16. In the figure,
 $AC \perp AB$,
 $AH \perp BC$,
 $AH = 2\sqrt{3}$ and
 $m(\angle BCA) = 30^\circ$.
 Find the value of x .



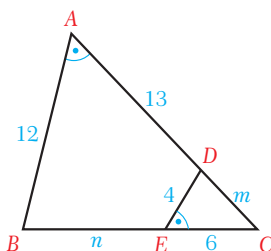
- A) 8 B) $6\sqrt{3}$ C) 6 D) $4\sqrt{3}$ E) 4

CHAPTER REVIEW TEST 2C

1. In the figure,

$$m(\angle A) = m(\angle DEC).$$

Find the value of $\frac{m}{n}$.



- A) $\frac{5}{9}$ B) $\frac{5}{8}$ C) $\frac{5}{6}$ D) $\frac{4}{5}$ E) $\frac{3}{5}$

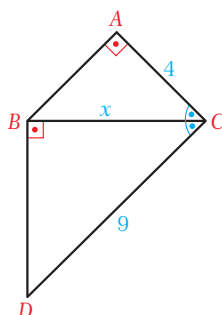
2. In the figure,

$$AB \perp AC,$$

$$BC \perp BD \text{ and } BC \text{ bisects } \angle ACD.$$

$$\text{If } AC = 4 \text{ and } DC = 9,$$

what is the value of x ?



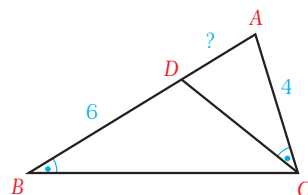
- A) 5 B) 6 C) 7 D) 8 E) $\frac{25}{8}$

3. In the figure,

$$m(\angle ACD) = m(\angle B),$$

$$BD = 6 \text{ and } AC = 4.$$

Find the length of AD.



- A) 1 B) $\frac{3}{2}$ C) 2 D) 3 E) 4

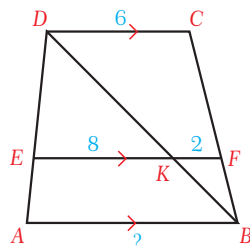
4. In the figure,

$$DC \parallel EF \parallel AB.$$

$$\text{If } DC = 6,$$

$$EK = 8 \text{ and } KF = 2,$$

what is the length of AB?



- A) 12 B) 15 C) 16 D) 18 E) 24

5. The base KM of an isosceles triangle KLM measures 24 cm and one leg measures 27 cm. Points A and B are situated on the sides KL and LM respectively, such that $LA = LB$. If $P(\triangle LAB) = 26$ cm, find AK.

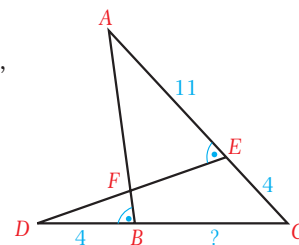
- A) 9 cm B) 12 cm C) 15 cm
D) 18 cm E) 27 cm

6. In the figure,

$$m(\angle DBF) = m(\angle AEF),$$

$$AE = 11 \text{ and } EC = DB = 4.$$

Find the length of BC.



- A) 5 B) 6 C) 7 D) 8 E) 10

7. In the figure,

$$AD \parallel EF \parallel BC.$$

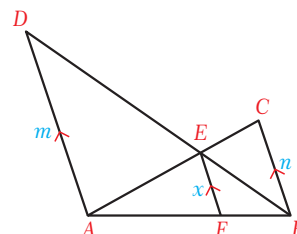
$$AD = m,$$

$$BC = n,$$

$$EF = x,$$

$$m + n = 15 \text{ and } m \cdot n = 54$$

are given. Find the value of x .



- A) 3 B) 3.6 C) 4 D) 4.2 E) 5.4

8. In the figure,

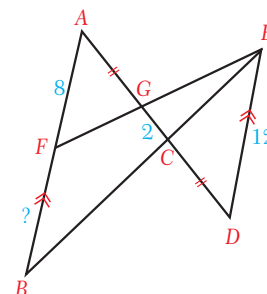
$$AB \parallel ED$$

$$AG = CD,$$

$$AF = 8,$$

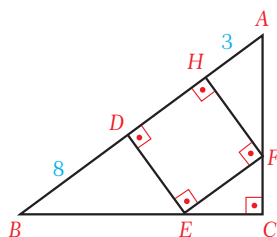
$$GC = 2 \text{ and } DE = 12.$$

What is the length of FB?



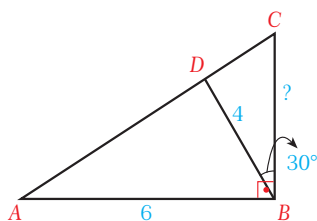
- A) 9 B) 10 C) 12 D) 15 E) 16

9. In the figure, $\triangle ABC$ is a right triangle and $DEFH$ is a square. If $AH = 3$ and $BD = 8$, what is the length of one side of $DEFH$?



- A) $2\sqrt{6}$ B) 5 C) $4\sqrt{2}$ D) 6 E) $3\sqrt{5}$

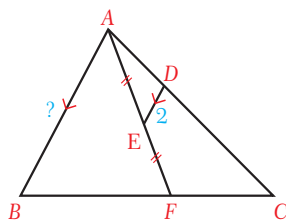
10. In the figure, $\triangle ABC$ is a right triangle and $m(\angle CBD) = 30^\circ$. If $AB = 6$ and $DB = 4$, what is the length of BC ?



(Hint: Draw an altitude from D to AB .)

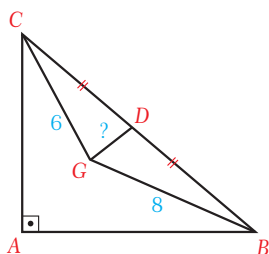
- A) $4\sqrt{3}$ B) 6 C) $3\sqrt{3}$ D) $2\sqrt{6}$ E) $3\sqrt{2}$

11. In the figure, $DC = 2 \cdot AD$ and $AE = EF$. If $DE \parallel AB$ and $DE = 2$, what is the length of AB ?



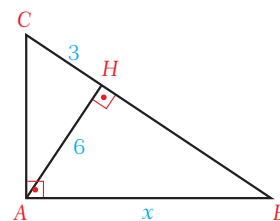
- A) 8 B) 9 C) 12 D) 15 E) 16

12. In the figure, G is the centroid of $\triangle ABC$. If $CD = DB$, $GC = 6$ and $GB = 8$, what is the length of GD ?



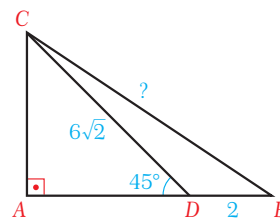
- A) 2 B) $\sqrt{5}$ C) 3 D) $2\sqrt{3}$ E) $2\sqrt{5}$

13. Find the length x in the figure.



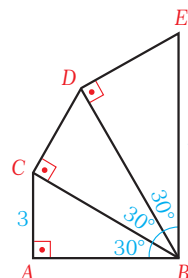
- A) 10 B) 12 C) 13 D) $4\sqrt{10}$ E) $6\sqrt{5}$

14. In the figure, $AB \perp AC$. If $m(\angle ADC) = 45^\circ$, $DB = 2$ and $CD = 6\sqrt{2}$, what is the length of BC ?



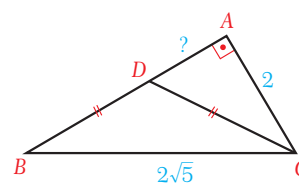
- A) 9 B) $3\sqrt{10}$ C) 10 D) $3\sqrt{13}$ E) 13

15. In the figure, $AC = 3$. Find the length of BE .



- A) 12 B) 10 C) 8 D) 6 E) 4

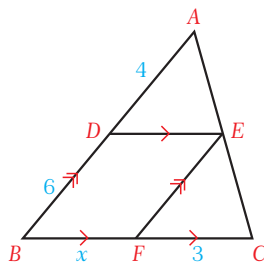
16. In the figure, $m(\angle A) = 90^\circ$, $BD = DC$, $AC = 2$ and $BC = 2\sqrt{5}$. What is the length of AD ?



- A) 1 B) $\frac{3}{2}$ C) 2 D) $\frac{5}{2}$ E) 3

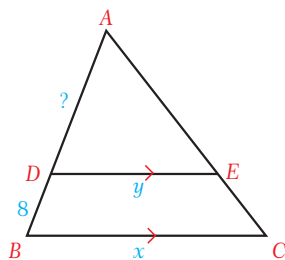
CHAPTER REVIEW TEST 2C

1. In the figure,
 $DE \parallel BC$ and
 $AB \parallel EF$.
 If $AD = 4$,
 $DB = 6$ and
 $FC = 3$, what is the
 value of x ?



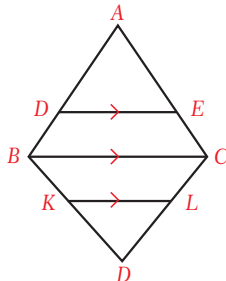
- A) 1 B) $\frac{3}{2}$ C) 2 D) $\frac{5}{2}$ E) 3

2. In the figure, $DE \parallel BC$,
 $DE = y$,
 $BC = x$,
 $BD = 8$ cm and
 $\frac{x-y}{x+y} = \frac{2}{7}$.
 Find the length of AD .



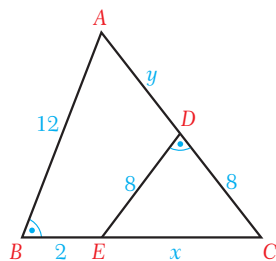
- A) 16 B) 15 C) 12 D) 10 E) 9

3. In the figure,
 $BC \parallel DE \parallel KL$.
 If $AD = 2 \cdot BD$ and
 $BK = KD$, what is the
 value of $\frac{DE}{KL}$?



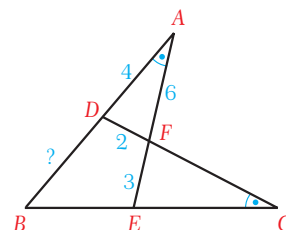
- A) 1 B) $\frac{1}{2}$ C) $\frac{2}{3}$ D) $\frac{3}{2}$ E) $\frac{4}{3}$

4. In the figure,
 $m(\angle B) = m(\angle EDC)$,
 $AB = 12$,
 $BE = 2$,
 $DE = DC = 8$,
 $AD = y$ and $EC = x$.
 What is the value of
 $x + y$?



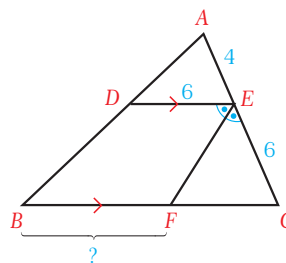
- A) 12 B) 13 C) 15 D) 17 E) 18

5. In the figure,
 $m(\angle A) = m(\angle C)$,
 $AD = 4$,
 $AF = 6$,
 $DF = 2$ and
 $EF = 3$.
 Find the length of BD .



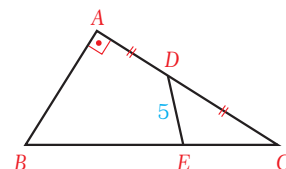
- A) 6 B) 4 C) $\frac{9}{2}$ D) $\frac{11}{4}$ E) 2

6. In the figure,
 $DE \parallel BC$ and EF is
 the angle bisector of
 $\angle DEC$.
 If $AE = 4$ and
 $DE = EC = 6$, find
 the length of BF .



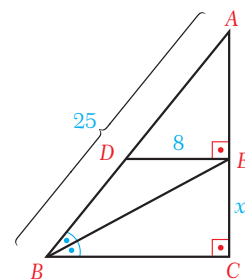
- A) 8 B) 9 C) 10 D) 12 E) 15

7. In the figure,
 $AB \perp AC$,
 $AD = DC$ and
 $BC = 4 \cdot EC$.
 Given $DE = 5$, find
 the length of BC .



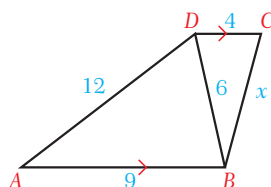
- A) 10 B) 15 C) 20 D) 25 E) 30

8. Find the length
 $EC = x$ in the figure.



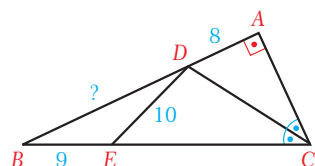
- A) $\frac{120}{17}$ B) $\frac{90}{17}$ C) $\frac{75}{17}$ D) $\frac{75}{13}$ E) $\frac{60}{13}$

9. In the figure,
 $DC \parallel AB$,
 $AB = 9$,
 $AD = 12$,
 $BD = 6$ and
 $DC = 4$. What is the value of x ?



- A) 6 B) 7 C) 8 D) 9 E) 10

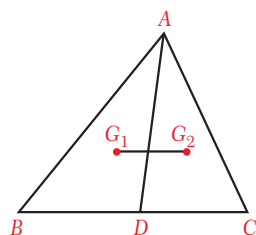
10. In the right triangle ABC in the figure, CD is the angle bisector of $\angle C$.



Find the length of BD if $AD = 8$, $DE = 10$ and $BE = 9$.

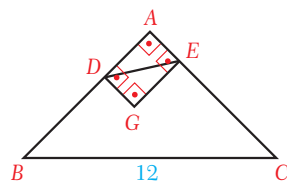
- A) 11 B) 13 C) 15 D) 16 E) 17

11. In the figure,
 G_1 is the centroid of $\triangle ABD$ and G_2 is the centroid of $\triangle ADC$.
If $BC = 24$ cm,
what is the length of G_1G_2 ?



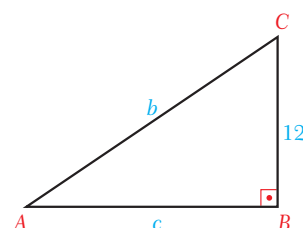
- A) 8 cm B) 10 cm C) 12 cm
D) 14 cm E) 16 cm

12. In the figure, $ADGE$ is a rectangle and G is the centroid of the triangle ABC .
If $BC = 12$,
find the length of DE .



- A) 2 B) 3 C) 4 D) 5 E) 6

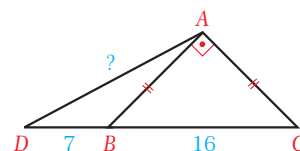
13. In the figure,
 $AB \perp BC$,
 $BC = 12$ cm,
 $AB = c$ and
 $AC = b$.



If $b - c = 4$ cm, what is the perimeter of $\triangle ABC$?

- A) 60 cm B) 54 cm C) 48 cm
D) 40 cm E) 28 cm

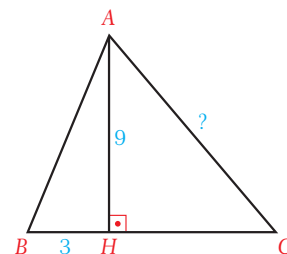
14. In the figure,
 $AB = AC$,
 $m(\angle BAC) = 90^\circ$,
 $DB = 7$ and
 $BC = 16$.



What is the length of AD ?

- A) 15 B) 16 C) 17 D) 18 E) 20

15. In the figure,
 $AC = BC$.
Find the length of AC .



- A) 10 B) 12 C) 13 D) 14 E) 15

16. A right triangle KLM has $m(\angle L) = 90^\circ$. A line is drawn through K parallel to LM , and another line is drawn through M perpendicular to KM . E is the intersection of these two lines. If $KE = 25$ and $KM = 15$, what is the length of LM ?

- A) 3 B) 6 C) 9 D) 12 E) 15

BASIC CONCEPTS

Arc Length and the Circumference of a Circle

In the previous section we considered the degree measure of an arc. We can also measure an arc in terms of its length.

Definition

circumference

The length of the complete arc of any circle is called the **circumference** of the circle.

Property

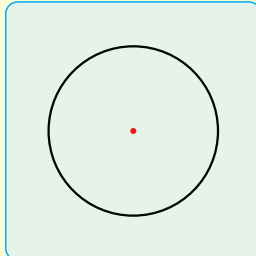
The ratio of the circumference of any circle to the length of its diameter is always the same number, called **pi** (π) and pronounced like the English word 'pie'.

Corollary

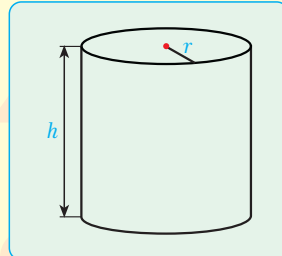
The circumference of a circle with radius r is $2\pi r$.

THE HISTORY OF PI (π)

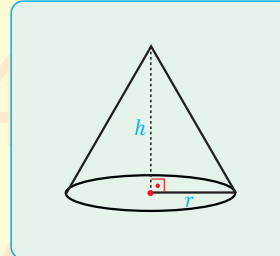
Whenever we see a circular object, the constant number π is always there. For example, the circumference of any circular disk with radius r is $2\pi r$ and its area is πr^2 . The volume and surface area of a cylinder and a circular cone are also related to π . We can use this mysterious number to calculate things such as the orbits of planets and the shortest airline route between two cities.



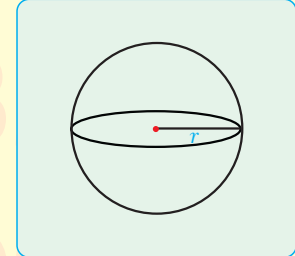
$$\begin{aligned}\text{Area} &= \pi r^2 \\ \text{Circumference} &= 2\pi r\end{aligned}$$



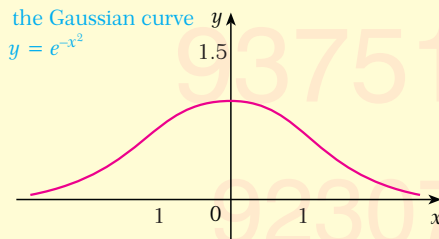
$$\begin{aligned}\text{Surface Area} &= 2\pi r^2 + 2\pi rh \\ \text{Volume} &= \pi r^2 h\end{aligned}$$



$$\text{Volume} = \frac{\pi r^2 h}{3}$$



$$\begin{aligned}\text{Surface Area} &= 4\pi r^2 \\ \text{Volume} &= \frac{4\pi r^3}{3}\end{aligned}$$



However, it is even more remarkable that we find π in many other applications that have nothing to do with circles. The area under a Gaussian curve is $\sqrt{\pi}$, and electrical engineers also use π in calculations of alternate currents and radiation. In calculus, mathematicians have found different relations between π and integer numbers. Here are some examples:

$$\begin{aligned}\frac{\pi}{4} &= 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \frac{1}{11} + \frac{1}{13} - \dots \\ \frac{\pi^2}{6} &= 1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \frac{1}{25} + \frac{1}{36} + \frac{1}{49} + \dots \\ \frac{\pi}{2} &= \frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5} \cdot \frac{6}{5} \cdot \frac{6}{7} \cdot \frac{8}{7} \cdot \frac{8}{9} \cdot \frac{10}{9} \cdot \frac{10}{11} \cdot \frac{12}{11} \cdot \dots\end{aligned}$$

Abraham Sharp:

$$\pi = \sum_{k=0}^{\infty} \frac{2(-1)^k 3^{\frac{1}{2}-k}}{2k+1}$$

Newton:

$$\pi = \frac{3}{4}\sqrt{3} + 24 \int_0^{1/4} \sqrt{x-x^2} dx$$

$$\frac{2}{\pi} = \sqrt{\frac{1}{2}} \sqrt{\frac{1}{2} + \frac{1}{2}} \sqrt{\frac{1}{2}} \sqrt{\frac{1}{2} + \frac{1}{2}} \sqrt{\frac{1}{2}} \sqrt{\frac{1}{2} + \frac{1}{2}} \dots$$

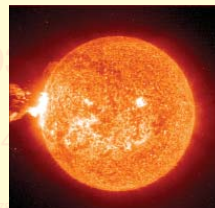
$$\pi = \lim_{n \rightarrow \infty} 2^{n+1} \sqrt{2 - \underbrace{\sqrt{2 + \sqrt{2 + \sqrt{2 + \dots + \sqrt{2}}}}}_n}$$

$$\pi = \prod_{n=1}^{\infty} \left(1 + \frac{1}{4n^2 - 1} \right)$$

$$\pi = \sum_{n=1}^{\infty} \frac{1}{4n^2 - 1}$$

So what is the history of this magic number?

Before the invention of the wheel, early peoples identified the circle as a powerful symbol of nature. This was because they could see circular objects everywhere in nature, such as in the sun, the moon, and some flowers.

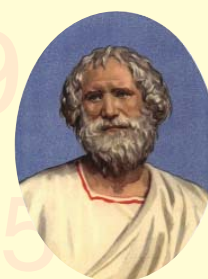
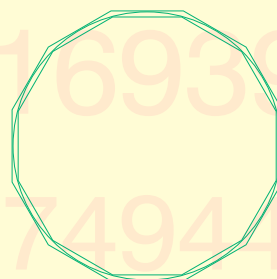
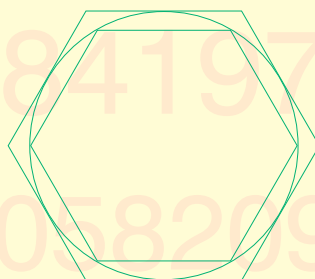


In order to build circular buildings or temples, people needed to estimate the total distance around a circle in terms of the total distance across it. Early civilizations realized that the ratio of the circumference to the diameter is the same in all circles. After careful calculation, they found that this ratio was slightly bigger than 3. The Babylonians used $\frac{25}{8}$ for this constant, and the Egyptians used $\frac{256}{81}$.



Since π was so important, many people tried to find its approximate numerical value. The first serious effort was made by Archimedes (287-212 BC). He obtained an approximate value by constructing regular polygons inside and around a circle. When he compared the perimeter of a hexagon inside a unit circle and the circumference of the circle, he concluded that π is greater than 3. When he compared the perimeter of the hexagon around the circle and the circumference, he found that π is less than $2\sqrt{3}$. Then he doubled the number of sides of the polygon to 12 and made similar comparisons. He continued this process until the number of sides of the polygon reached 96. He concluded that π is between $\frac{223}{71}$ and $\frac{22}{7}$. The upper bound $\frac{22}{7}$ is still used today as a fairly good approximation for π .

After this, Egypt, India and China used Archimedes' method and increased the number of sides to obtain more accurate results. An amazing Chinese calculation using polygons with more than 3000 sides gave five correct decimals of π (3.14159). The Chinese also found the fraction $\frac{355}{113}$, which approximates π to six correct decimal places.



With this number, the Chinese held the world record calculation of π for more than a thousand years, until Arabic numerals provided a more efficient way to use arithmetic.

After the discovery of trigonometric functions and infinite series, formulas were found which made it possible to approximate π without using geometry. By the end of the 19th century, mathematicians had calculated the value of π to hundreds of decimal places with these formulas. In the 20th century they began to use new computers and other mathematical methods. In 1989 more than one billion decimal places were known. In 1999 more than 206 billion decimal places were determined, and further calculations are expected.

For a long time, people wondered whether π was an exact fraction such as $\frac{22}{7}$. In other words, is π a rational number or an irrational number?

Recall that if a decimal number is also a rational number, its decimal part always has a repeating pattern. For example, $\frac{22}{7} = 3.\underline{142857}\underline{142857}\underline{142857}\underline{142857}\dots$



With this in mind, people searched for a repeating pattern in the decimal expansion of π . However they couldn't find any pattern. In the 18th century the German mathematician J. Lambert proved that no repeating pattern can be found. He used continued fractions to prove that there are no integers whose ratio is equal to π . In other words, although we can show π easily with a circle, we will never be able to write it exactly in numbers.

EXAMPLE

- 1** What is the circumference C of a circle with radius 3 cm?

Solution

$$\begin{aligned} C &= 2\pi r \\ &= 2 \cdot \pi \cdot (3) \\ &= 6\pi \text{ cm.} \end{aligned}$$

**EXAMPLE**

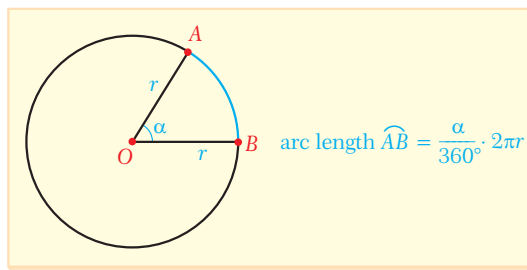
- 2** What is the radius of a circle with circumference 2π ?

Solution

$$\begin{aligned} C &= 2\pi r \\ 2\pi &= 2\pi r \\ r &= 1. \end{aligned}$$

Theorem

A circle with radius r is given. If the measure of a central angle is α , the length of its intercepted arc is $\frac{\alpha}{360^\circ} \cdot 2\pi r$.

**Proof**

In order to find the length of the intercepted arc, first we need to know what fractional part of circle is intercepted. Since 360° represents a whole circle, $\frac{\alpha}{360^\circ}$ is the fractional part. Therefore the length of the intercepted arc is $\frac{\alpha}{360^\circ} \cdot 2\pi r$.

EXAMPLE

- 3** What is the length of the arc intercepted by a 40° central angle of a circle with radius 6 cm?

Solution

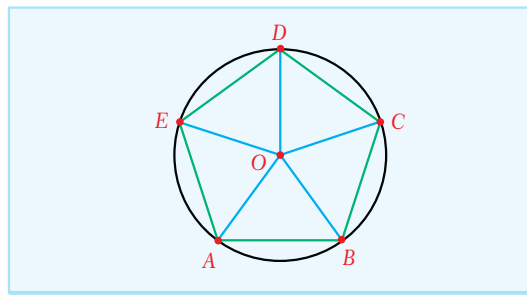
Let the length of the intercepted arc be l .

$$\begin{aligned} \text{Then by the theorem above, } l &= \frac{\alpha}{360^\circ} \cdot 2\pi r \\ &= \frac{40^\circ}{360^\circ} \cdot 2\pi \cdot 6 \\ &= \frac{4}{3}\pi \text{ cm.} \end{aligned}$$

EXAMPLE

- 4** The figure shows a regular pentagon inscribed in a circle with radius 4 cm. Find the following.

- a. $m(\angle EOD)$ b. $m(\widehat{DE})$
c. $m(\widehat{ABC})$ d. the length of $\widehat{EDC}$



Solution a. All the triangles inside the circle are congruent, so $m(\angle EOD) = m(\angle EOA) = m(\angle AOB) = m(\angle BOC) = m(\angle COD) = \alpha$.
So $5\alpha = 360^\circ$, which means $\alpha = m(\angle EOD) = 72^\circ$.

b. Since by definition $m(\angle EOD) = m(\widehat{DE})$, $m(\widehat{DE}) = 72^\circ$.

c. $m(\widehat{ABC}) = m(\angle AOC) = 2\alpha$, so $m(\widehat{ABC}) = 2 \cdot 72^\circ = 144^\circ$. (Can you also calculate this using part b?)

d. $\widehat{EDC} = \frac{144}{360} \cdot 2\pi r = \frac{2}{5} \cdot 2\pi r \cdot 4 = \frac{16\pi}{5}$ cm.



In this book, we write $\widehat{AB}$ to represent the arc AB and its length. We write $m(\widehat{AB})$ to mean the angle measure of $\widehat{AB}$.

EXAMPLE

- 5** The figure shows a section of a stone floor. O is the center of the sector, $m(\angle AOB) = 90^\circ$ and $OC = CA = AE = 2$ m.

Given that the straight lines divide the arcs into equal lengths, find the following arc lengths.

- a. $\widehat{AB}$ b. $\widehat{CD}$
c. $\widehat{EF}$ d. $\widehat{FG}$



Solution a. $\widehat{AB} = \frac{90^\circ}{360^\circ} \cdot 2\pi \cdot 4 = 2\pi$ m.

b. $\widehat{CD} = \frac{67.5^\circ}{360^\circ} \cdot 2\pi \cdot 2 = \frac{3\pi}{4}$ m.

c. $\widehat{EF} = \frac{45^\circ}{360^\circ} \cdot 2\pi \cdot 6 = \frac{3\pi}{2}$ m.

d. $\widehat{FG} = \frac{22.5^\circ}{360^\circ} \cdot 2\pi \cdot 6 = \frac{3\pi}{4}$ m.

EXAMPLE



The lanes around the circular road junction in the figure form circles with radii 24 m, 27 m and 30 m. $m(\angle AOB) = 120^\circ$ and $m(\angle COD) = 20^\circ$ are given.

What are the circular distances along the lanes between the cars at

- A and B?
- C and D?

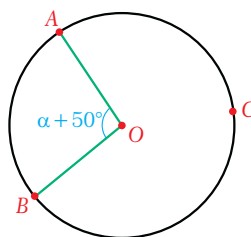
Solution

- $\frac{120^\circ}{360^\circ} \cdot 2\pi \cdot 24 = 16\pi$ m.
- $\frac{20^\circ}{360^\circ} \cdot 2\pi \cdot 27 = 3\pi$ m.



Check Yourself

- In the figure,
 $m(\angle AOB) = \alpha + 50^\circ$ and
 $m(\widehat{ACB}) = 3\alpha + 10^\circ$.
 Find the actual degree measure $m(\angle AOB)$.



- A central angle of a circle with radius 27 cm creates an intercepted arc of length 18π cm. Find the measure of the central angle.
- What is the length of the arc which is intercepted by a 30° central angle of a circle with radius 12 cm?

Answers

- 125°
- 120°
- 2π cm

ANGLES AND CIRCLES

A. INSCRIBED ANGLES AND ARCS

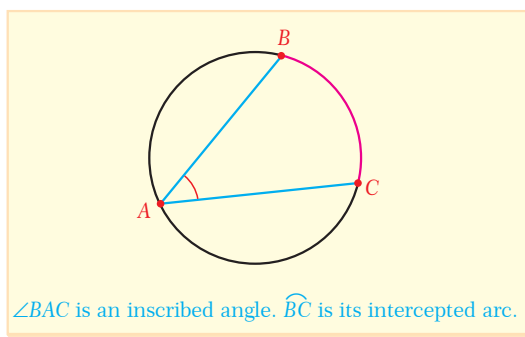
Recall that a central angle is an angle whose vertex lies at the center of a circle. In this section we will look at another type of angle in a circle.

Definition

inscribed angle, intercepted arc

An angle whose vertex lies on a circle and whose sides contain chords of the circle is called an **inscribed angle**. The arc of the circle between the sides of the inscribed angle is called the **intercepted arc** of the inscribed angle.

An arc can also be intercepted by a central angle or an angle formed by a chord and a tangent. We will examine these two cases later.



Theorem

The measure of an inscribed angle is equal to half the angle measure of its intercepted arc.

Proof

Look at the figure. We can see that $\triangle AOC$ and $\triangle AOB$ are isosceles triangles.

Let us name the angles

$$m(\angle OAC) = m(\angle OCA) = \alpha \text{ and}$$

$$m(\angle OAB) = m(\angle OBA) = \beta.$$

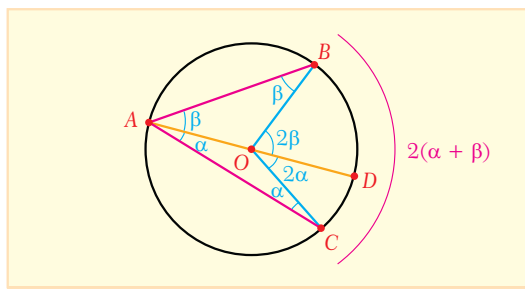
Then $m(\angle COD) = 2\alpha$ and $m(\angle BOD) = 2\beta$ by the triangle exterior angle theorem.

Since $\angle COD$ and $\angle BOD$ are central angles,

$$m(\widehat{CD}) = 2\alpha \text{ and } m(\widehat{BD}) = 2\beta.$$

Therefore $m(\widehat{BC}) = 2\alpha + 2\beta$, which we can write as $m(\widehat{BC}) = 2(\alpha + \beta)$.

So $m(\angle BAC) = \frac{m(\widehat{BC})}{2}$, as required.



EXAMPLE

7

In the figure, $m(\widehat{AB}) = 124^\circ$ and $m(\angle B) = 65^\circ$. Find the angle measures.

- a. $m(\angle C)$ b. $m(\widehat{AC})$
c. $m(\angle A)$ d. $m(\widehat{BC})$

Solution a. $m(\angle C) = \frac{m(\widehat{AB})}{2}$ by the theorem we have just seen.

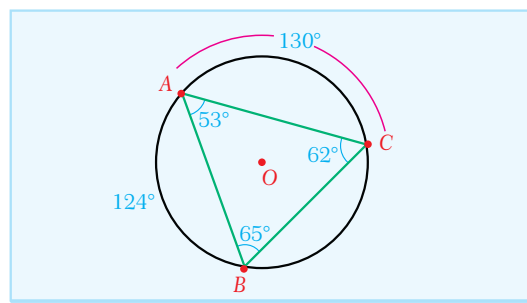
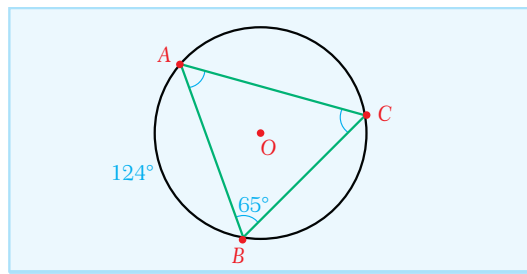
$$\text{So } m(\angle C) = \frac{124^\circ}{2} = 62^\circ.$$

$$\begin{aligned} \text{b. } \frac{m(\widehat{AC})}{2} &= m(\angle B) \Rightarrow \frac{m(\widehat{AC})}{2} = 65^\circ \\ &\Rightarrow m(\widehat{AC}) = 130^\circ \end{aligned}$$

c., d.

$$\begin{aligned} \text{In } \triangle ABC, m(\angle A) + m(\angle B) + m(\angle C) &= 180^\circ \\ m(\angle A) + 65^\circ + 62^\circ &= 180^\circ \\ m(\angle A) &= 53^\circ. \end{aligned}$$

$$\text{So } \frac{m(\widehat{BC})}{2} = 53^\circ, \text{ which means } m(\widehat{BC}) = 106^\circ.$$

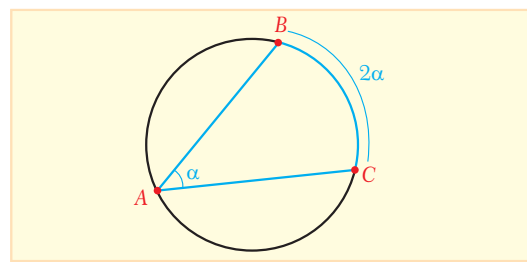


Corollary

The measure of the arc intercepted by an inscribed angle is twice the measure of the inscribed angle.

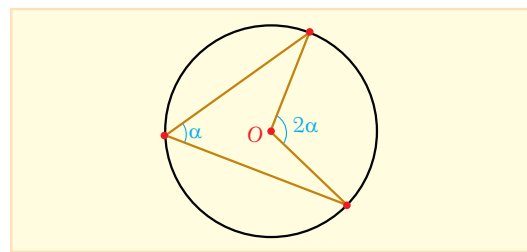
For example, in the figure,

$$m(\widehat{BC}) = 2 \cdot m(\angle BAC).$$



Corollary

The measure of an inscribed angle is half the measure of the central angle which intercepts the same arc.



EXAMPLE

8

Find the value of α in the figure.

Solution Since $\angle BAC$ is an inscribed angle,

$$m(\widehat{BC}) = 2 \cdot m(\angle BAC)$$

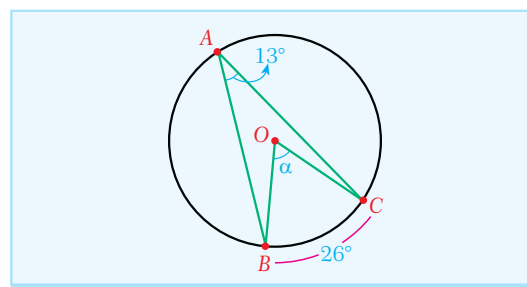
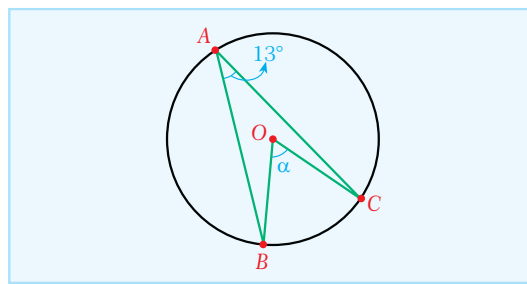
$$m(\widehat{BC}) = 2 \cdot 13^\circ$$

$$m(\widehat{BC}) = 26^\circ.$$

Also, $\angle BOC$ is a central angle,

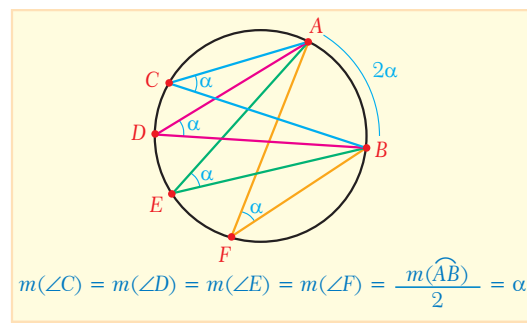
so $m(\angle BOC) = m(\widehat{BC})$ and so

$$m(\widehat{BC}) = \alpha = 26^\circ.$$



Corollary

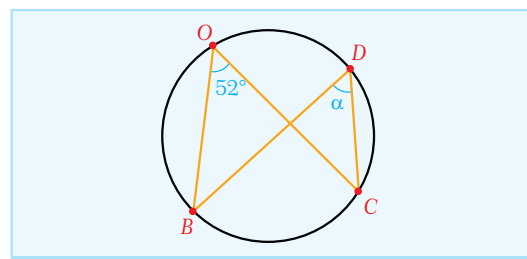
The measures of all the inscribed angles which intercept the same arc of a circle are equal.



EXAMPLE

9

Find the value of α in the figure.



Solution $\angle BOC$ and $\angle BDC$ are inscribed angles intercepting $\widehat{BC}$,

so $m(\angle BOC) = m(\angle BDC) = \alpha = 52^\circ$ by the corollary we have just seen.

ANGLE OF DANGER

Sailors use navigation charts that contain information about the horizontal angle of danger for different places. An angle of danger helps ships to avoid rocks and other dangers in the water. Here is one example of how the angle of danger works.

The map below shows two lighthouses on the coast and some dangerous rocks in the water. On a chart, navigators draw the smallest circle that passes through the lighthouses and contains all of the rocks. Joining the two lighthouses to any point on the circle creates an angle, which is the angle of danger for the rocks. The angle of danger is the same for any point on the circle (can you see why?).

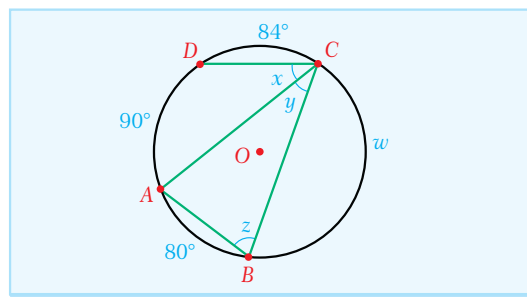
When the ship is sailing, the navigator checks the angle created by his ship and the two lighthouses. If the angle is smaller than the angle of danger then the ship is safe. If the angle is greater than the angle of danger then the ship is in a dangerous area, too close to the rocks.



EXAMPLE

10

The figure shows a circle $\odot O$ with $m(\widehat{AB}) = 80^\circ$, $m(\widehat{BC}) = w$, $m(\widehat{CD}) = 84^\circ$ and $m(\widehat{DA}) = 90^\circ$. Find the values of x , y , z and w .



Solution

$$\begin{aligned} x &= \frac{m(\widehat{AD})}{2} & y &= \frac{m(\widehat{AB})}{2} & z &= \frac{m(\widehat{ADC})}{2} \\ &= \frac{90^\circ}{2} & &= \frac{80^\circ}{2} & &= \frac{90^\circ + 84^\circ}{2} \\ &= 45^\circ. & &= 40^\circ. & &= 87^\circ. \end{aligned}$$

Also, $m(\widehat{BA}) + m(\widehat{AD}) + m(\widehat{DC}) + m(\widehat{CB}) = 360^\circ$. So

$$80^\circ + 90^\circ + 84^\circ + w = 360^\circ$$

$$w = 106^\circ.$$

So $x = 45^\circ$, $y = 40^\circ$, $z = 87^\circ$ and $w = 106^\circ$.

EXAMPLE

11

A , B , C and D are four points located counter-clockwise on a circle and E is the intersection point of AC and BD . P is a point outside the circle such that P , A , B and P , D , C are collinear. Given that $m(\angle P) = 45^\circ$ and $m(\angle BEC) = 85^\circ$, find $m(\angle C)$.

Solution

Let us draw the figure. Since $\angle C$ and $\angle B$ intercept the same arc, we can write $m(\angle C) = m(\angle B) = \alpha$.

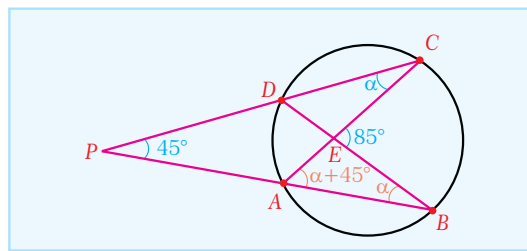
In $\triangle APC$,

$$\begin{aligned} m(\angle BAC) &= m(\angle P) + m(\angle C) \\ &= 45^\circ + \alpha. \end{aligned}$$

Similarly, in $\triangle AEB$,

$$\begin{aligned} m(\angle BEC) &= m(\angle BAE) + m(\angle B) \\ 85^\circ &= \alpha + 45^\circ + \alpha \\ 40^\circ &= 2\alpha \\ \alpha &= 20^\circ. \end{aligned}$$

So $m(\angle C) = \alpha = 20^\circ$.



EXAMPLE

12

A, B and C are three points on a circle centered at O such that $m(\angle BAC) = 80^\circ$. What is $m(\angle OBC)$?

Solution Look at the figure. Let $m(\angle OBC) = \alpha$.

We have $m(\angle BAC) = \frac{m(\widehat{BC})}{2}$, so

$$80^\circ = \frac{m(\widehat{BC})}{2}, \text{ which gives us } m(\widehat{BC}) = 160^\circ.$$

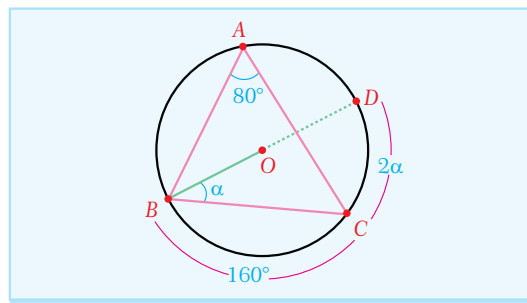
Now let us extend the radius BO to D.

Then BD is a diameter.

Since $\angle CBD$ is an inscribed angle, $m(\widehat{CD}) = 2\alpha$. Therefore,

$$160 + 2\alpha = 180^\circ.$$

$$\alpha = 10^\circ.$$

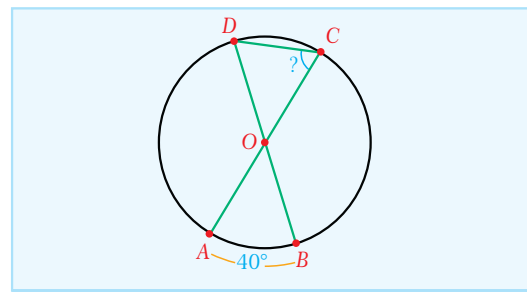


EXAMPLE

13

In the figure, O is the center of the circle and $m(\widehat{AB}) = 40^\circ$.

Find $m(\angle OCD)$.

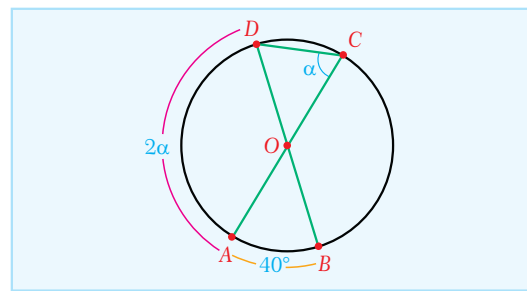


Solution Let $m(\angle OCD) = \alpha$.

So $m(\widehat{AD}) = 2\alpha$.

Since $\widehat{BAD}$ is a semicircle, $2\alpha + 40^\circ = 180^\circ$.

So $\alpha = 70^\circ = m(\angle OCD)$.

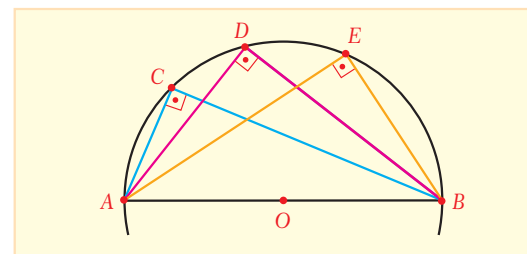


Property

Any inscribed angle α which intercepts a semicircle is a right angle

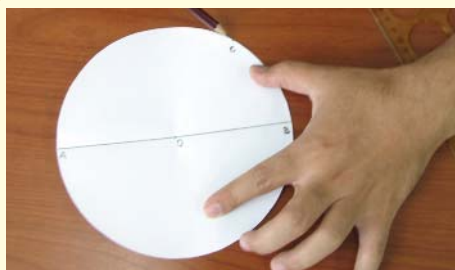
(since $\alpha = 180^\circ \div 2 = 90^\circ$).

Conversely, if an inscribed angle is a right angle then it intercepts a semicircle.



Activity

Folding a Right Angle Inscribed in a Semicircle



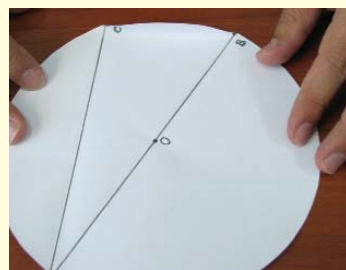
Fold and draw the diameter AB .



Fold a chord AC .



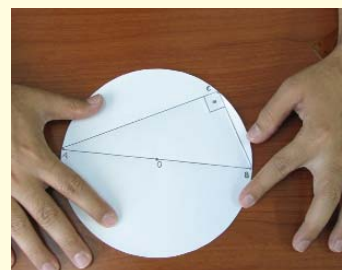
Draw the chord AC .



Fold the chord BC .



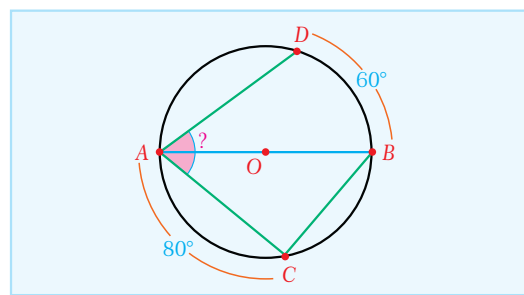
Draw the chord BC . Check that $\angle ACB$ is a right angle.



EXAMPLE

14

In the figure, O is the center of the circle,
 $m(\widehat{BD}) = 60^\circ$ and
 $m(\widehat{AC}) = 80^\circ$.
 Find $m(\angle CAD)$.



Solution By the properties of inscribed angles,

$$m(\angle BAD) = \frac{60^\circ}{2} = 30^\circ \text{ and}$$

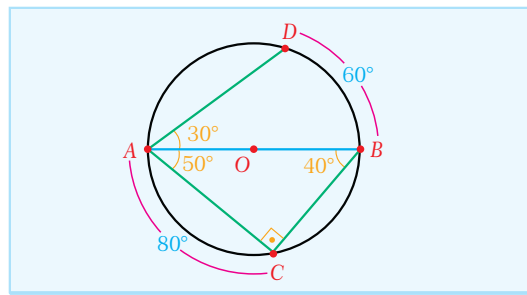
$$m(\angle ABC) = \frac{80^\circ}{2} = 40^\circ.$$

In $\triangle ABC$, $m(\angle C) = 90^\circ$ since it intercepts a semicircle.

$$\text{So } m(\angle BAC) + 90^\circ + 40^\circ = 180^\circ$$

$$m(\angle BAC) = 50^\circ.$$

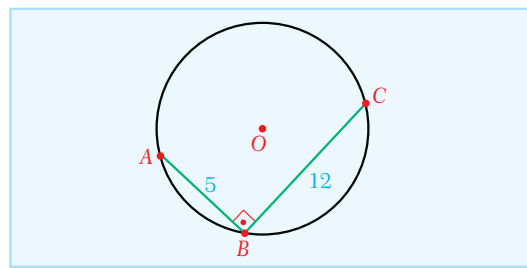
$$\text{Finally, } m(\angle CAD) = 50^\circ + 30^\circ = 80^\circ.$$



EXAMPLE

15

What is the radius of $\odot O$ in the figure?



Solution Since $\angle ABC$ is an inscribed right angle, it intercepts a semicircle.

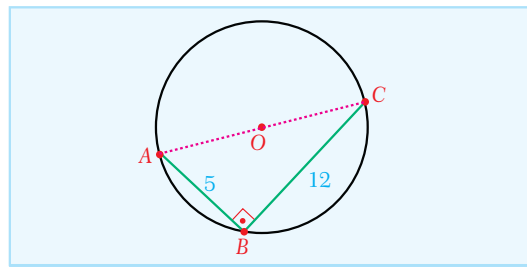
So AC is a diameter. In $\triangle ABC$,

$$AC^2 = AB^2 + BC^2$$

$$AC^2 = 5^2 + 12^2$$

$$AC = 13.$$

Therefore the radius of the circle is $\frac{13}{2}$.



EXAMPLE

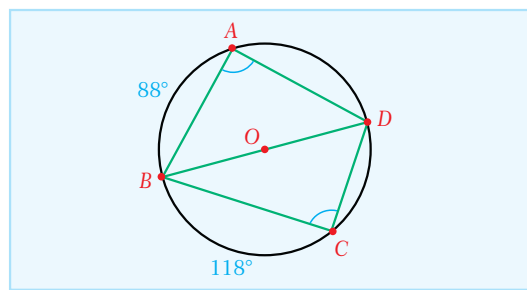
16

In the figure, O is the center of the circle, $m(\widehat{BC}) = 118^\circ$ and $m(\widehat{AB}) = 88^\circ$.

Find

a. $m(\angle A)$. b. $m(\angle C)$.

c. $m(\angle CDA)$. d. $m(\angle ABC)$.



Solution a., b.

Since $\angle A$ and $\angle C$ intercept semicircles, they are both right angles.

$$\begin{aligned} \text{c. } m(\angle CDA) &= \frac{m(\widehat{ABC})}{2} \\ &= \frac{88^\circ + 118^\circ}{2} \\ &= 103^\circ. \end{aligned}$$

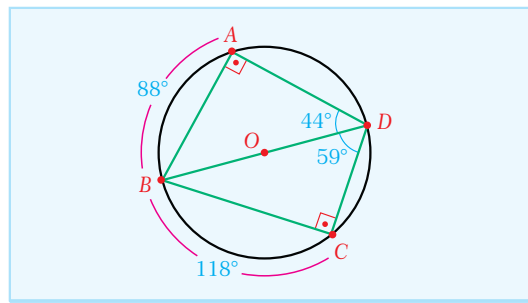
$$\text{d. } m(\angle BDA) = \frac{m(\widehat{AB})}{2} = 44^\circ,$$

$$\text{so } m(\angle ABD) = 180^\circ - 90^\circ - 44^\circ = 46^\circ.$$

$$\text{Similarly, } m(\angle BDC) = \frac{m(\widehat{BC})}{2} = 59^\circ,$$

$$\text{so } m(\angle CBD) = 180^\circ - 90^\circ - 59^\circ = 31^\circ.$$

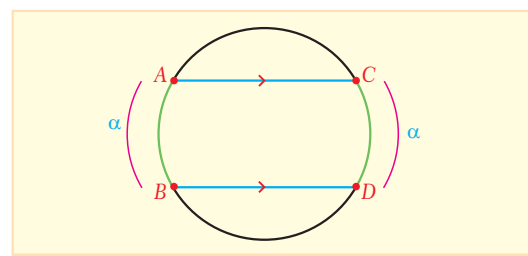
$$\text{So } m(\angle ABC) = 46^\circ + 31^\circ = 77^\circ.$$



Property

The arcs between two parallel chords are congruent, i.e. in the figure,

$$m(\widehat{AB}) = m(\widehat{CD}).$$



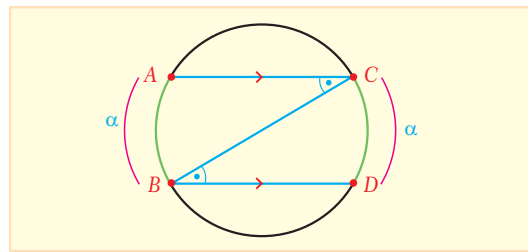
Proof

Let us draw BC .

Since $AC \parallel BD$, $m(\angle ACB) = m(\angle CBD)$.

Since the inscribed angles $\angle ACB$ and $\angle CBD$ are congruent, the arcs intercepted by these angles are congruent.

So $m(\widehat{AB}) = m(\widehat{CD})$.

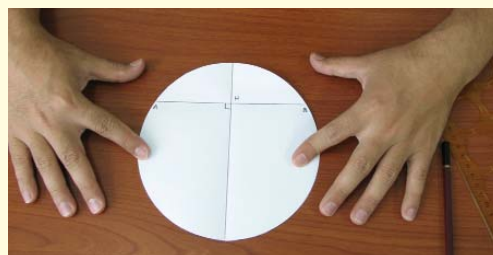
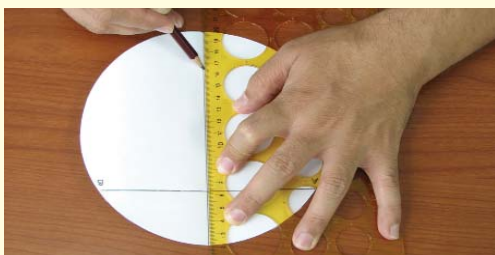


Activity

Folding Parallel Chords



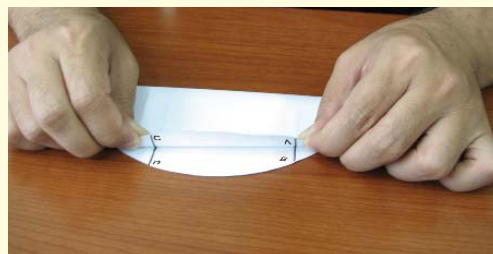
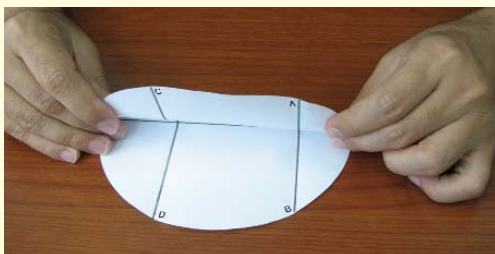
Fold and draw a chord AB.



Fold and draw the perpendicular bisector of AB, which is a diameter of the circle.



Fold and draw another chord that is perpendicular to the diameter.

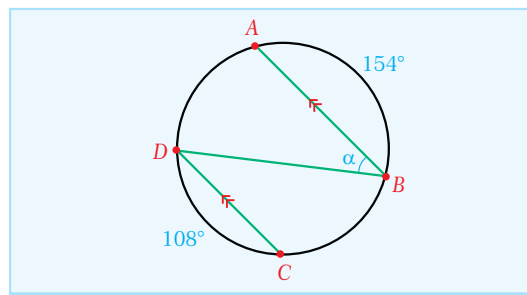


Compare AB and CD by folding. Are they parallel?

EXAMPLE

17

In the figure, $AB \parallel CD$.
Find $m(\angle DBA) = \alpha$.



Solution

Since $AB \parallel CD$,

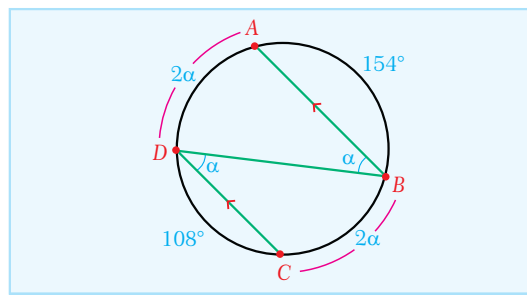
$m(\angle ABD) = m(\angle BDC) = \alpha$ by the property
we have just seen.

Therefore $m(\widehat{AD}) = m(\widehat{BC}) = 2\alpha$.

So $154^\circ + 108^\circ + 4\alpha = 360^\circ$

$$4\alpha = 98^\circ$$

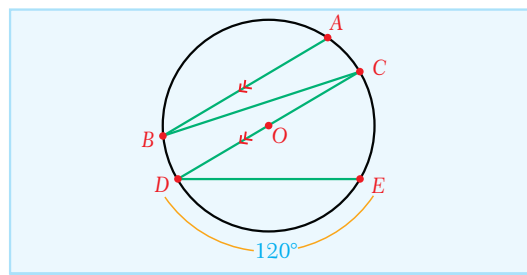
$$\alpha = 24.5^\circ.$$



EXAMPLE

18

In the figure, $AB \parallel CD$,
 $m(\widehat{DE}) = 120^\circ$ and
 $m(\angle CDE) = 3 \cdot m(\angle ABC)$.
Find $m(\widehat{AB})$.



Solution

Let $m(\angle ABC) = \alpha$. Then $m(\angle CDE) = 3\alpha$.

Therefore $m(\widehat{AC}) = 2\alpha$ and $m(\widehat{CE}) = 6\alpha$.

Since $\widehat{CED}$ is a semicircle,

$$6\alpha + 120^\circ = 180^\circ$$

$$6\alpha = 60^\circ$$

$$\alpha = 10^\circ.$$

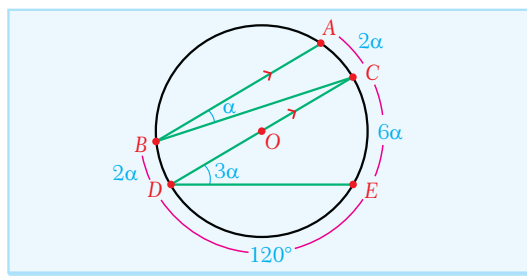
Since $AB \parallel CD$, $m(\widehat{AC}) = m(\widehat{BD}) = 2\alpha$.

Hence

$$m(\widehat{BD}) + m(\widehat{AB}) + m(\widehat{AC}) = 180^\circ$$

$$4\alpha + m(\widehat{AB}) = 180^\circ$$

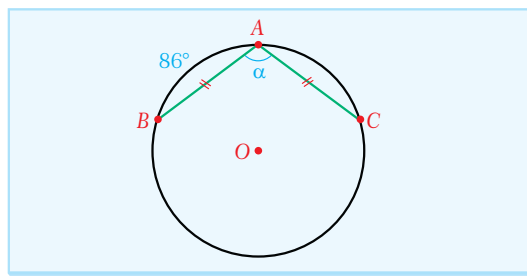
$$m(\widehat{AB}) = 140^\circ.$$



EXAMPLE

19

In the figure, $AC = AB$ and $m(\widehat{AB}) = 86^\circ$.
Find $m(\angle BAC) = \alpha$.



Solution

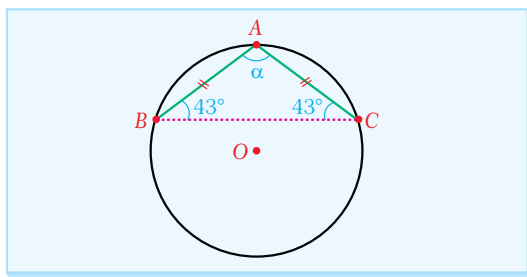
Since $m(\widehat{AB}) = 86^\circ$, $m(\angle C) = 43^\circ$.

Since $AB = AC$,

$$m(\angle ACB) = m(\angle ABC) = 43^\circ.$$

In $\triangle ABC$, $\alpha + 43^\circ + 43^\circ = 180^\circ$.

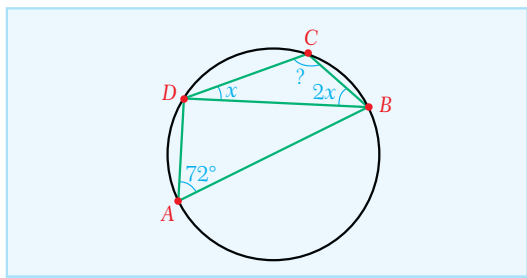
So $\alpha = 94^\circ$.



EXAMPLE

20

Find $m(\angle C)$ in the figure.



Solution

By the properties of inscribed angles,

$$\frac{m(\widehat{BC})}{2} = x \text{ and } \frac{m(\widehat{DC})}{2} = 2x.$$

So $m(\widehat{BC}) = 2x$ and $m(\widehat{DC}) = 4x$.

However, since $m(\angle DAB) = 72^\circ$ we have

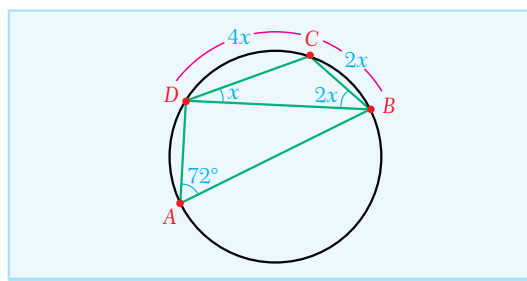
$$\frac{m(\widehat{DCB})}{2} = 72^\circ, \text{ which means}$$

$$\frac{6x}{2} = 3x = 72^\circ \Rightarrow x = 24^\circ.$$

Finally, $m(\angle C) + 3x = 180^\circ$

$$m(\angle C) + 72^\circ = 180^\circ$$

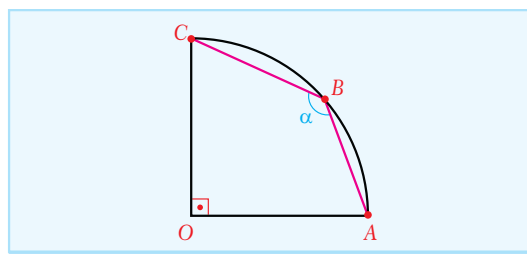
$$m(\angle C) = 108^\circ.$$



EXAMPLE

21

AC in the figure is an arc of $\odot O$.
 $CO \perp AO$ is given.
 Find the angle measure α .



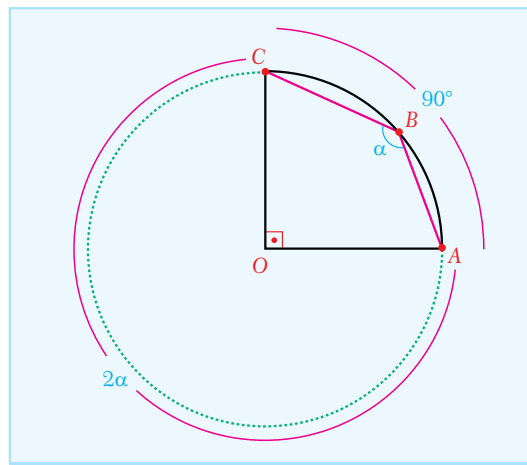
Solution

Let us complete the circle.

Since $m(\angle O) = 90^\circ$, $m(\widehat{ABC}) = 90^\circ$.

Since the major arc $\widehat{AC}$ is an intercepted arc,
 $m(\widehat{AC}) = 2\alpha$. So $90^\circ + 2\alpha = 360^\circ$

$$\alpha = 135^\circ.$$



EXAMPLE

22

Prove that the interior angles of a triangle add up to 180° by using inscribed angles in a circle.

Solution

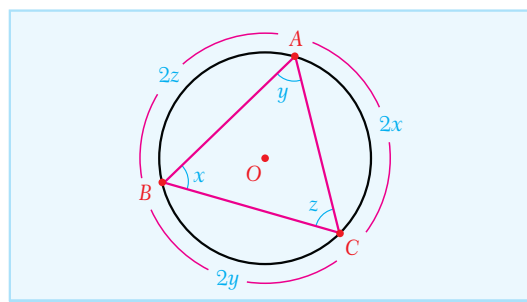
Let us draw a triangle ABC and its circumscribed circle, and then name the interior angles as shown in the figure.

By the property of inscribed angles,
 $m(\widehat{AC}) = 2x$, $m(\widehat{BC}) = 2y$ and $m(\widehat{AB}) = 2z$.

We know that $2x + 2y + 2z = 360^\circ$, so

$$x + y + z = 180^\circ.$$

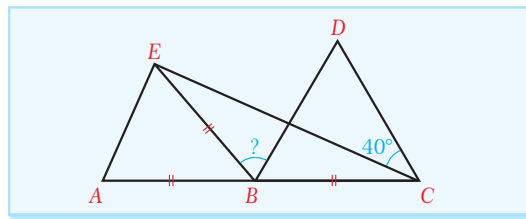
Therefore the sum of the interior angles of a triangle is 180° .



EXAMPLE

23

In the figure, $AB = BC = BE = BD$.
Find $m(\angle EBD)$.

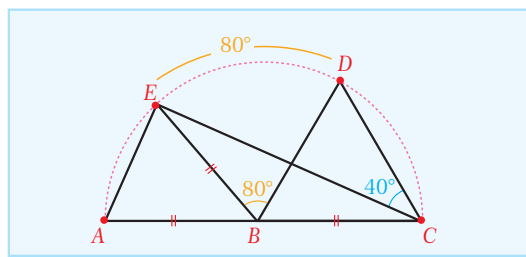


Solution

We can draw a semicircle with diameter AC and center B . The semicircle passes through D and E (can you see why?).

Since $\widehat{ED}$ is an intercepted arc, $m(\widehat{ED}) = 80^\circ$.

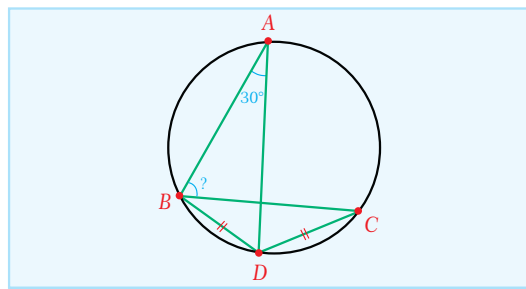
Since $\angle EBD$ is the central angle of this arc, $m(\angle EBD) = 80^\circ$.



EXAMPLE

24

In the figure,
 $AB = BC$,
 $BD = CD$
and $m(\angle A) = 30^\circ$.
Find $m(\angle ABC)$.



Solution

Let us draw the segment AC . Since $\angle BAD$ and $\angle BCD$ intercept the same arc $\widehat{BD}$,

$$m(\angle BAD) = m(\angle BCD) = 30^\circ.$$

$BD = DC$ implies $m(\angle CBD) = 30^\circ$.

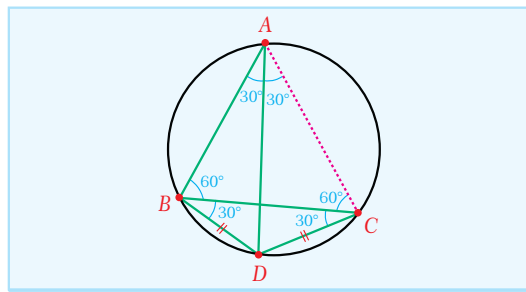
Since $\angle CAD$ and $\angle CBD$ both intercept the arc CD , $m(\angle CAD) = 30^\circ$.

So $m(\angle BAC) = 60^\circ$.

Since $AB = BC$ and $m(\angle BAC) = 60^\circ$,

$\triangle ABC$ is an equilateral triangle.

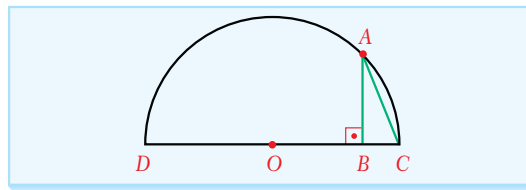
Therefore $m(\angle ABC) = 60^\circ$.



EXAMPLE

25

O is the center of the semicircle in the figure. Given that $AC = 8$ cm and $BD = 12$ cm, find $m(\widehat{AC})$.



Solution

Let us connect A and D, as in the figure. Since $\angle DAC$ intercepts a semicircle, it is a right angle. By the Euclidean Theorem in $\triangle DAC$,

$$8^2 = x \cdot (x + 12)$$

$$x^2 + 12x - 64 = 0$$

$$x = 4 \text{ cm.}$$

Applying the Euclidean Theorem once more gives us

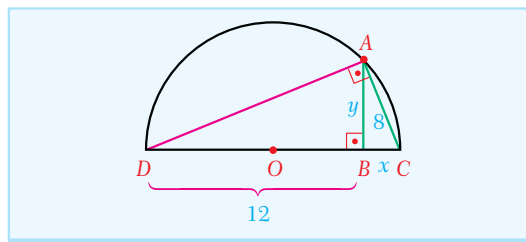
$$y^2 = 12 \cdot x$$

$$= 12 \cdot 4$$

$$y = 4\sqrt{3} \text{ cm.}$$

Therefore the sides of $\triangle ABC$ are $AB = 4\sqrt{3}$ cm, $BC = 4$ cm and $AC = 8$ cm. These are side lengths of a 30° - 60° - 90° triangle, so $m(\angle ACB) = 60^\circ$ and $m(\angle ADC) = 30^\circ$.

It follows that $m(\widehat{AC}) = 60^\circ$.



EXAMPLE

26

A line segment AB is the diameter of a semicircle centered at O. C and D are two points located counter-clockwise on the semicircle such that $m(\angle BCD) = 110^\circ$. What is $m(\angle BOD)$?

Solution

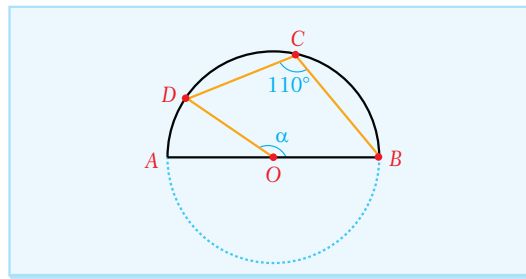
Look at the figure. Since $m(\angle BCD) = 110^\circ$, $m(\widehat{DAB}) = 220^\circ$.

$$m(\widehat{BCD}) + m(\widehat{DAB}) = 360^\circ$$

$$m(\widehat{BCD}) + 220^\circ = 360^\circ$$

$$m(\widehat{BCD}) = 140^\circ.$$

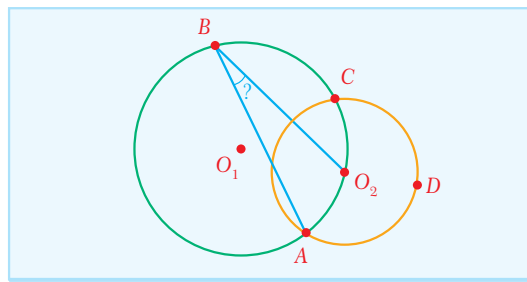
Notice that $\angle BOD$ is a central angle, and we know that the measure of a central angle is same as the angle measure of its intercepted arc. So $m(\angle BOD) = 140^\circ$.



EXAMPLE

27

In the figure, O_1 and O_2 are the centers of the circles and $m(\widehat{ADC}) = 220^\circ$. Find $m(\angle ABO_2)$.



Solution

Let us draw a diameter of $\odot O_1$ passing through O_2 and name its other endpoint E . $\angle ABO_2$ and $\angle AEO_2$ both intercept $\widehat{AO_2}$.

So $m(\angle ABO_2) = m(\angle AEO_2) = \alpha$.

In $\odot O_2$, $AO_2 = CO_2$ (radii).

Also, $AE \perp AO_2$ and $EC \perp CO_2$ because the inscribed angles intercept semicircles.

Therefore $\triangle ECO_2$ and $\triangle EAO_2$ are congruent by the AAS congruence theorem.

So $m(\angle EO_2C) = m(\angle EO_2A)$.

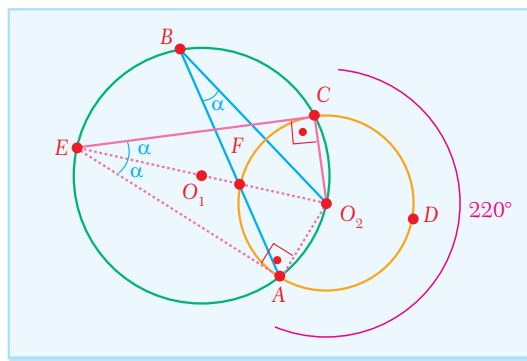
Since $m(\widehat{ADC}) = 220^\circ$, $m(\widehat{AFC}) = 360^\circ - 220^\circ = 140^\circ$.

In $\odot O_2$, $m(\angle EO_2C) = m(\angle EO_2A)$ gives us $m(\widehat{AF}) = m(\widehat{FC}) = \frac{140^\circ}{2} = 70^\circ$.

Finally, in $\triangle EAO_2$ we have $\alpha + 70^\circ + 90^\circ = 180^\circ$

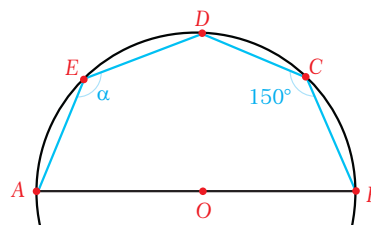
$$\alpha = 20^\circ.$$

So $m(\angle ABO_2) = 20^\circ$.

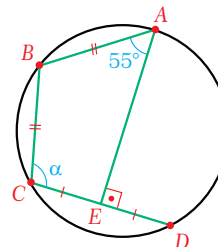


Check Yourself

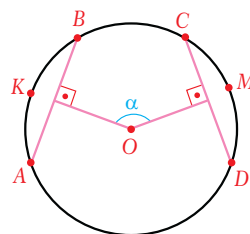
- In the figure, O is the center of the semicircle. Find the value of α .



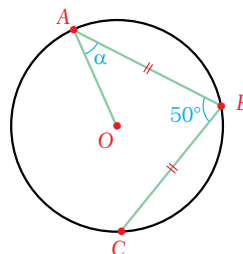
- Find the value of α in the figure.



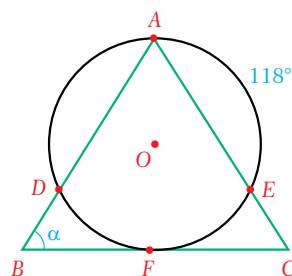
3. In the figure,
 $m(\widehat{AKB}) + m(\widehat{DMC}) = 160^\circ$ and
 $m(\widehat{BC}) = 70^\circ$.
 Find the value of α .



4. Find the value of α in the figure.



5. In the figure, $3 \cdot AD = 2 \cdot AB$
 and $AE = 2 \cdot EC$.
 Find the value of α .



Answers

1. 120° 2. 105° 3. 150° 4. 25° 5. 59°

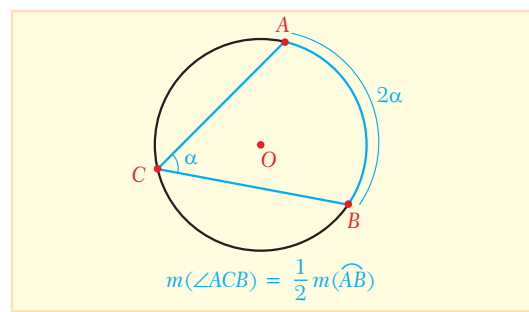
B. ANGLES FORMED BY SECANTS, TANGENTS AND CHORDS

A point in the plane can have three possible positions relative to a circle: it can be inside the circle, on the circle or outside the circle. This gives us three possible positions for the vertex of an angle associated with a circle.

1. Angles on a Circle

a. The angle between two chords

We have already seen that the measure of an inscribed angle formed by two chords is half the measure of its intercepted arc.



Theorem

b. The angle between a secant (or chord) and a tangent

The measure of the acute angle formed by a secant and a tangent is half the measure of the intercepted arc of the corresponding central angle.

Proof

Look at the figure. Let $m(\angle APB) = \alpha$.

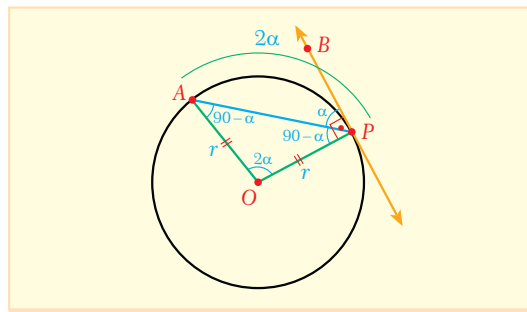
Since $OP \perp BP$, $m(\angle OPA) = 90^\circ - \alpha$.

Also, $\triangle AOP$ is an isosceles triangle, which gives us $m(\angle OAP) = 90^\circ - \alpha$.

So $m(\angle AOP) = 180^\circ - (90^\circ - \alpha + 90^\circ - \alpha) = 2\alpha$.

Since $\angle AOP$ is a central angle, $m(\widehat{AP}) = 2\alpha$.

Therefore $m(\angle APB) = \frac{m(\widehat{AP})}{2}$.



Remember!

A secant line is a line that passes through two distinct points on a circle.

EXAMPLE

28

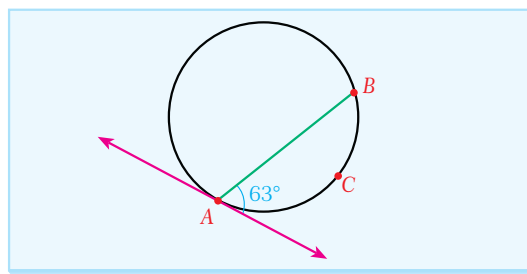
In the figure, A is the point of tangency of the line to the circle. If $m(\angle A) = 63^\circ$, find $m(\widehat{ACB})$.

Solution

Since $m(\angle A) = \frac{m(\widehat{ACB})}{2}$,

$$63^\circ = \frac{m(\widehat{ACB})}{2}.$$

So $m(\widehat{ACB}) = 126^\circ$.



EXAMPLE

29

In the figure, the line is tangent to the circle $\odot O$ at B. Find $m(\angle ABC)$.

Solution

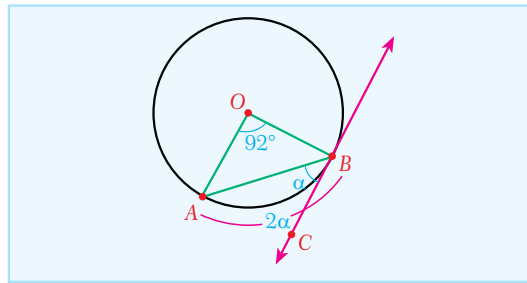
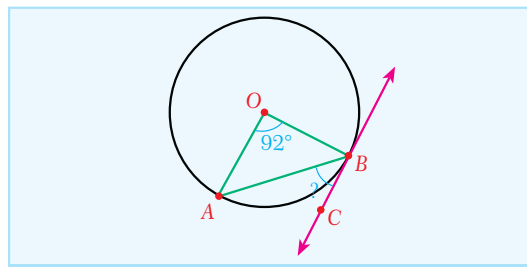
Let $m(\angle ABC) = \alpha$, then

$m(\widehat{AB}) = 2\alpha$.

Since $\angle AOB$ is a central angle,

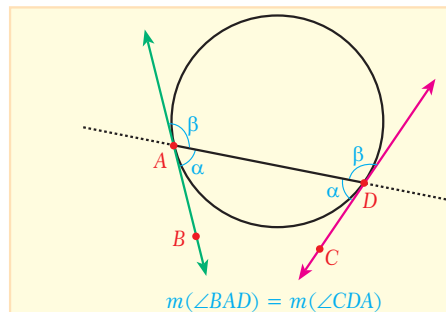
$m(\angle AOB) = m(\widehat{AB}) = 2\alpha$.

So $92^\circ = 2\alpha$, and so $m(\angle ABC) = \alpha = 46^\circ$.



Property

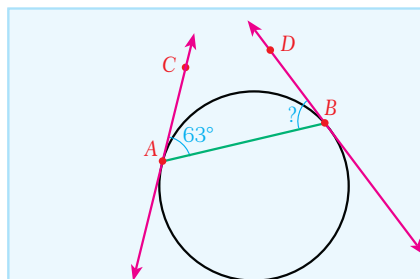
The two corresponding angles formed by a chord (or a secant) and the two tangents which intercept the same arc are congruent.



EXAMPLE 30 In the figure, AC and DB are tangent lines. Find $m(\angle ABD)$.

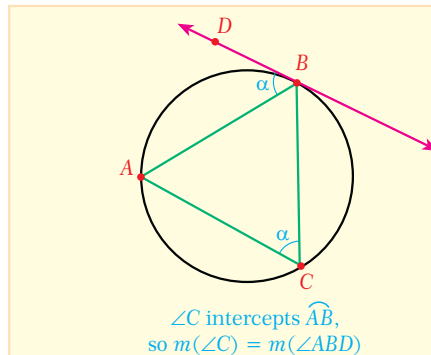
Solution Since $\angle CAB$ and $\angle ABD$ intercept the same arc, by the property we have just seen they are congruent.

So $m(\angle ABD) = 63^\circ$.

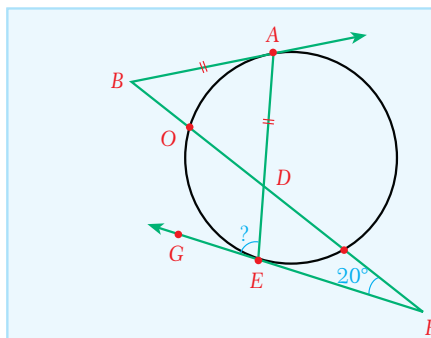


Property

The acute angle between a chord and a tangent is congruent to the inscribed angle which intercepts the same arc.



EXAMPLE 31 In the figure, EF and BA are tangent to the circle, $AB = AD$ and $m(\angle F) = 20^\circ$. Find $m(\angle AEG)$.



Solution Let $m(\angle B) = \alpha$, then $m(\angle ADB) = \alpha$ since $\triangle ADB$ is isosceles.

Then $m(\angle EDF) = \alpha$.

By the triangle exterior angle theorem in $\triangle DEF$, $m(\angle DEG) = 20^\circ + \alpha$.

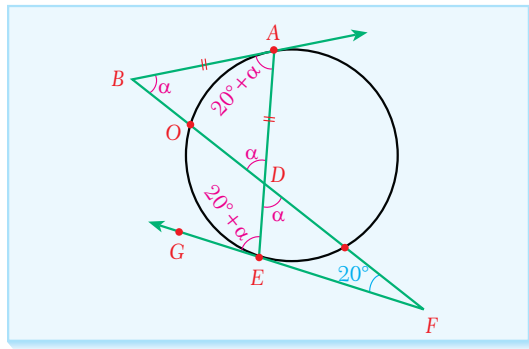
Since $\angle BAD$ and $\angle DEG$ both intercept $\widehat{AOE}$,
 $m(\angle BAD) = 20^\circ + \alpha$.

Finally, in $\triangle ADB$,

$$20^\circ + \alpha + \alpha + \alpha = 180^\circ$$

$$20^\circ + 3\alpha = 180^\circ$$

$$\alpha = \frac{160^\circ}{3}.$$

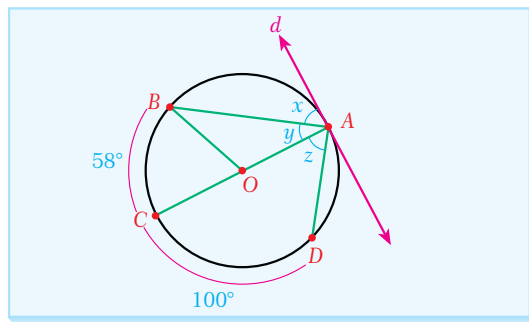


EXAMPLE

32

In the figure, d is tangent to the circle at A and O is the center of the circle.

Given $m(\widehat{BC}) = 58^\circ$ and $m(\widehat{CD}) = 100^\circ$, find the angle measures x , y and z .



Solution $y = \frac{m(\widehat{BC})}{2} = \frac{58^\circ}{2} = 29^\circ$

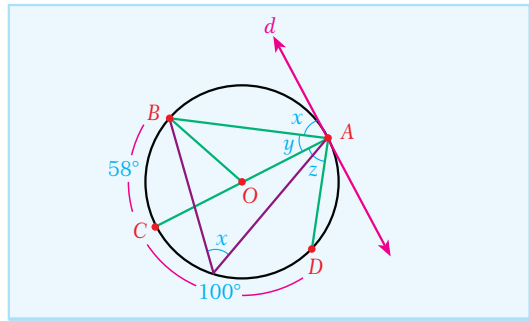
$$z = \frac{m(\widehat{CD})}{2} = \frac{100}{2} = 50^\circ$$

Since ABC is a semicircle,

$$m(\widehat{BC}) + m(\widehat{AB}) = 180^\circ$$

$$58^\circ + m(\widehat{AB}) = 180^\circ$$

$$m(\widehat{AB}) = 122^\circ.$$



By the property of secant and tangent lines, $x = \frac{m(\widehat{AB})}{2} = \frac{122^\circ}{2} = 61^\circ$.

EXAMPLE

33

Two lines l_1 and l_2 are tangent to a circle at points A and C respectively. B is a point on the circle such that $AB = BC$ and $m(\angle A) = 130^\circ$. What is $m(\angle ABC)$?

Solution Look at the figure. Since $m(\angle EAB) = 130^\circ$,
 $m(\angle DAB) = 50^\circ$.

Therefore $m(\widehat{AB}) = 100^\circ$. Since the two
 chords are congruent,

$$m(\widehat{AB}) = m(\widehat{BC}) = 100^\circ.$$

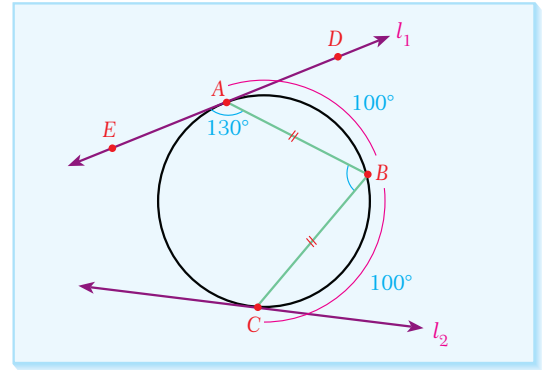
We know

$$m(\widehat{AC}) + m(\widehat{AB}) + m(\widehat{BC}) = 360^\circ. \text{ So}$$

$$m(\widehat{AC}) + 100^\circ + 100^\circ = 360^\circ$$

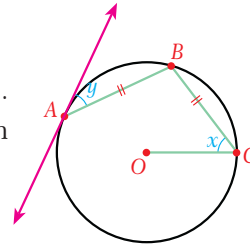
$$m(\widehat{AC}) = 160^\circ.$$

$$\text{Hence } m(\angle ABC) = \frac{160^\circ}{2} = 80^\circ.$$

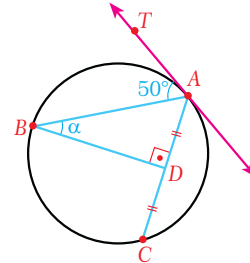


Check Yourself

1. O is the center of the circle in the figure.
 What is the value of the angle measure x in
 terms of y ?



2. In the figure, $m(\angle BAT) = 50^\circ$,
 $AD = DC$ and $BD \perp AC$.
 Find the value of the angle measure α .



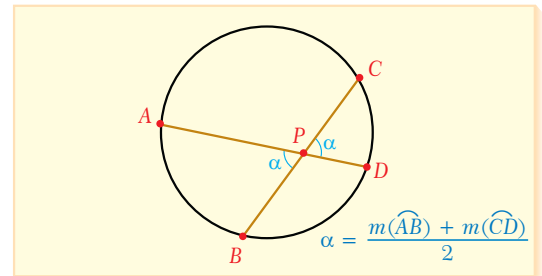
Answers

1. $x = 90^\circ - y$
2. 40°

2. Angles Inside a Circle

Theorem

The measure of the angle formed by two
 secants (or two chords) that intersect in the
 interior of a circle is half the sum of the
 measures of the arcs intercepted by the
 angle and its vertical angle.



Proof 1 Let $m(\angle ADB) = y$ and $m(\angle CBD) = x$.

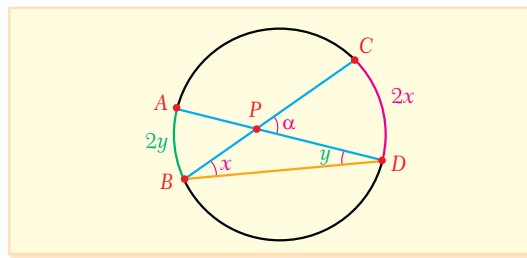
In $\triangle BPD$, $\alpha = x + y$.

$$\begin{aligned} m(\widehat{AB}) + m(\widehat{CD}) &= 2x + 2y \\ &= 2(x + y) \quad (1) \end{aligned}$$

$$m(\angle CPD) = \alpha = x + y = \frac{2 \cdot (x + y)}{2} \quad (2)$$

By (1) and (2),

$$m(\angle CPD) = \frac{m(\widehat{AB}) + m(\widehat{CD})}{2}, \text{ as required.}$$



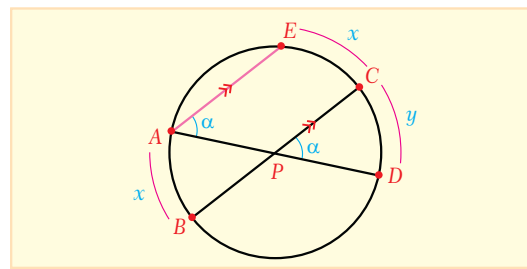
Proof 2 Let $m(\widehat{AB}) = x$ and $m(\widehat{CD}) = y$.

If we construct AE such that $AE \parallel BC$, then $m(\widehat{EC}) = x$ by the property of parallel chords.

Also, $m(\angle EAP) = m(\angle CPD) = \alpha$ because these are corresponding angles.

$$\begin{aligned} \text{Therefore } m(\angle EAD) &= \frac{m(\widehat{ECD})}{2} \\ &= \frac{(x + y)}{2}. \end{aligned}$$

So $\alpha = \frac{x + y}{2}$, as required.

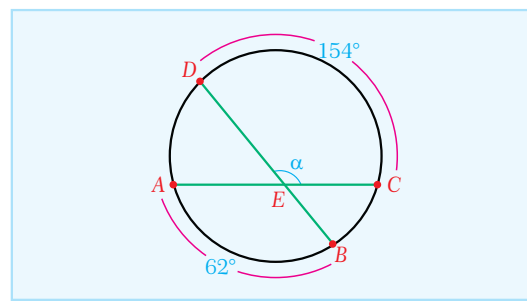


EXAMPLE 34 In the figure, $m(\widehat{AB}) = 62^\circ$ and $m(\widehat{CD}) = 154^\circ$.

Find $m(\angle DEC) = \alpha$.

Solution By the theorem we have just seen,

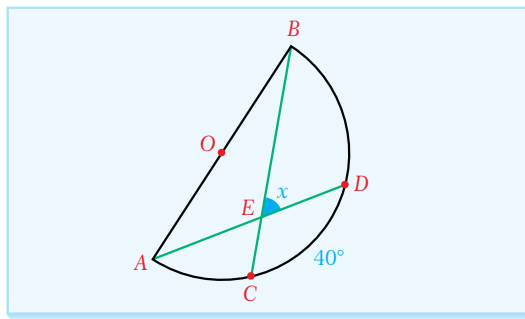
$$\begin{aligned} \alpha &= \frac{m(\widehat{AB}) + m(\widehat{CD})}{2} \\ &= \frac{62^\circ + 154^\circ}{2} \\ &= 108^\circ. \end{aligned}$$



EXAMPLE

35

The figure shows a semicircle with center O . Find $m(\angle BED) = x$.

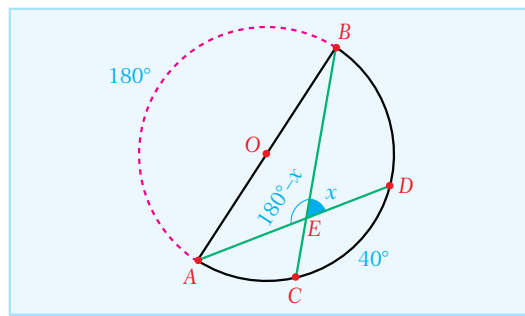


Solution If we complete the circle we can write

$$180^\circ - x = \frac{180^\circ + 40^\circ}{2}$$

$$180^\circ - x = 110^\circ$$

$$x = 70^\circ.$$



EXAMPLE

36

A, B, C and D are four points located counter-clockwise on a circle such that $m(\widehat{AB}) = 60^\circ$ and $m(\widehat{CD}) = 80^\circ$. If K is the intersection point of the two chords AC and BD , find $m(\angle AKD)$.

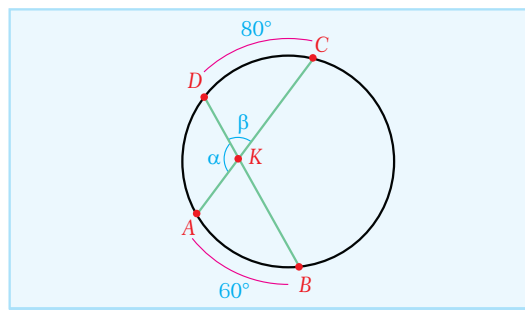
Solution Look at the figure. Let $m(\angle CKD) = \beta$.

$$\text{Then } \beta = \frac{80^\circ + 60^\circ}{2} = 70^\circ.$$

$$\text{Since } \alpha + \beta = 180^\circ,$$

$$\alpha + 70^\circ = 180^\circ$$

$$\alpha = 110^\circ = m(\angle AKD).$$

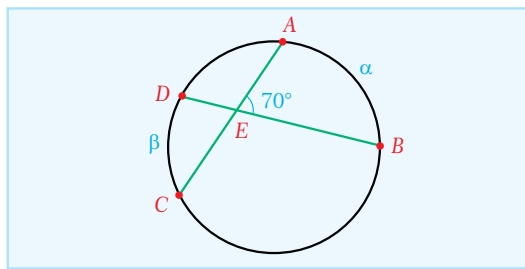


EXAMPLE

37

In the figure, $\alpha - \beta = 70^\circ$ and $m(\angle AEB) = 70^\circ$.

Find the values of α and β .



Solution $70^\circ = \frac{\alpha + \beta}{2}$ so $\alpha + \beta = 140^\circ$.

We are also told that $\alpha - \beta = 70^\circ$. Adding these two results gives us

$$\begin{array}{r} \alpha + \beta = 140^\circ \\ + \alpha - \beta = 70^\circ \\ \hline 2\alpha = 210^\circ \\ \alpha = 105^\circ. \end{array}$$

Finally,

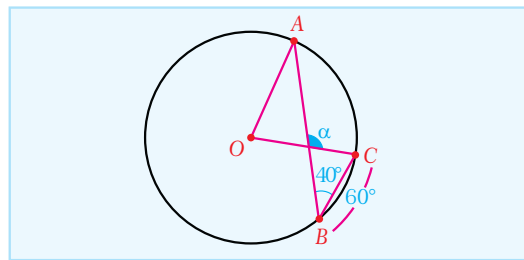
$$\begin{array}{r} \alpha + \beta = 140^\circ \\ 105^\circ + \beta = 140^\circ \\ \beta = 35^\circ. \end{array}$$



EXAMPLE

38

In the figure, O is the center of the circle, $m(\angle ABC) = 40^\circ$ and $m(\widehat{BC}) = 60^\circ$. Find the angle measure α .



Solution

Since $m(\angle ABC) = 40^\circ$,

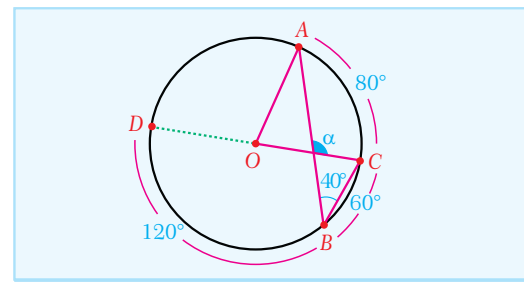
$$m(\widehat{AC}) = 2 \cdot 40^\circ = 80^\circ.$$

Now let us extend CO to the diameter CD .

$$\begin{aligned} \text{Then } m(\widehat{BD}) &= 180^\circ - 60^\circ \\ &= 120^\circ. \end{aligned}$$

Finally,

$$\begin{aligned} \alpha &= \frac{m(\widehat{AC}) + m(\widehat{BD})}{2} \\ &= \frac{80^\circ + 120^\circ}{2} \\ &= 100^\circ. \end{aligned}$$



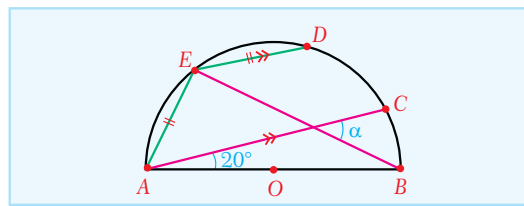
EXAMPLE

39

The figure shows a semicircle with center O .

$ED \parallel AC$ and $AE = ED$ are given.

Find the angle measure α .



Solution Since $m(\angle CAB) = 20^\circ$, $m(\widehat{BC}) = 40^\circ$.

Since chords AE and DE are congruent, we can write $m(\widehat{AE}) = m(\widehat{ED}) = x$.

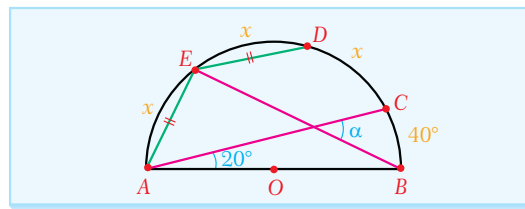
Since $ED \parallel AC$, $m(\widehat{AE}) = m(\widehat{DC})$. So $m(\widehat{DC}) = x$.

In the semicircle, $40^\circ + x + x + x = 180^\circ$

$$3x = 140^\circ$$

$$x = \frac{140^\circ}{3}.$$

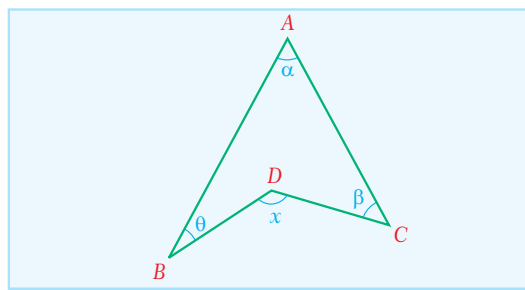
$$\begin{aligned} \text{So } \alpha &= \frac{m(\widehat{BC}) + m(\widehat{AE})}{2} = \frac{40^\circ + \frac{140^\circ}{3}}{2} \\ &= \frac{260^\circ}{6} \\ &= \frac{130^\circ}{3}. \end{aligned}$$



EXAMPLE

40

Show that $x = \alpha + \beta + \theta$ in the figure at the right.



Solution Let us draw a circle passing through A , B and C and then extend CD and BD as shown.

By inscribed angles,

$$m(\widehat{AE}) = 2\theta,$$

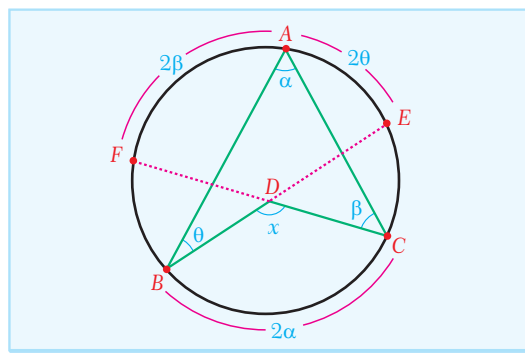
$$m(\widehat{AF}) = 2\beta \text{ and}$$

$$m(\widehat{BC}) = 2\alpha.$$

By intersecting chords,

$$\begin{aligned} x &= \frac{m(\widehat{BC}) + m(\widehat{EF})}{2} \\ &= \frac{2\alpha + 2\beta + 2\theta}{2}. \end{aligned}$$

So $x = \alpha + \beta + \theta$, as required.



EXAMPLE

41

Four points are placed around a circle and the midpoints of the arcs between each adjacent pair of points are marked. Finally, the opposite pairs of midpoints are joined by two chords. Prove that these chords intersect at right angles.

Solution Let A, B, C and D be the four points and let X, Y, Z and W be the midpoints. Then let

$$m(\widehat{AX}) = m(\widehat{XB}) = a$$

$$m(\widehat{BY}) = m(\widehat{YC}) = b$$

$$m(\widehat{CZ}) = m(\widehat{ZD}) = c$$

$$m(\widehat{DW}) = m(\widehat{WA}) = d, \text{ so}$$

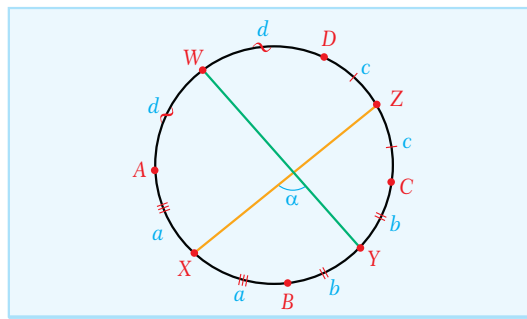
$$2a + 2b + 2c + 2d = 360^\circ, \text{ i.e.}$$

$$a + b + c + d = 180^\circ. \text{ Then}$$

$$\alpha = \frac{m(\widehat{XY}) + m(\widehat{ZW})}{2} = \frac{a + b + c + d}{2}$$

$$= \frac{180^\circ}{2}$$

$$\alpha = 90^\circ, \text{ as required.}$$



EXAMPLE

42

Show that in a triangle, the measure of the angle formed by the altitude and the angle bisector at a given vertex is equal to the half of the absolute difference of the other two angles in the triangle.

Solution Look at the figure.

$$\text{We want to show that } x = \frac{|y - z|}{2}.$$

Without loss of generality, let us assume that $y > z$.

$$\text{So we want to show that } 2x = y - z.$$

$$m(\angle ABC) = y \Rightarrow m(\widehat{AC}) = 2y.$$

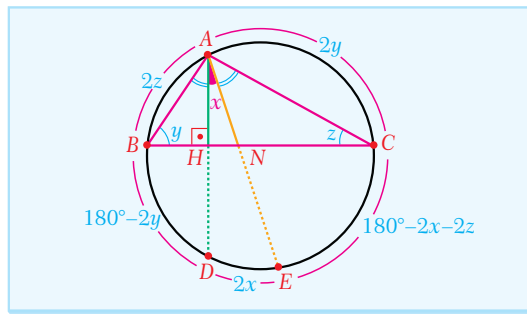
$$m(\angle ACB) = z \Rightarrow m(\widehat{AB}) = 2z.$$

Now extend AH and AN , so $m(\widehat{DE}) = 2x$. We know

$$m(\angle AHB) = 90^\circ = \frac{m(\widehat{AC}) + m(\widehat{BD})}{2} \Rightarrow 90^\circ = \frac{2y + m(\widehat{BD})}{2} \Rightarrow m(\widehat{BD}) = 180^\circ - 2y.$$

$$\text{Also, } m(\widehat{BA}) + m(\widehat{AC}) + m(\widehat{CE}) + m(\widehat{ED}) + m(\widehat{DB}) = 360^\circ.$$

$$2z + 2y + m(\widehat{CE}) + 2x + 180^\circ - 2y = 360^\circ.$$



So $m(\widehat{CE}) = 180^\circ - 2x - 2z$.

Since AN is an angle bisector, $m(\angle BAN) = m(\angle NAC)$. So

$$m(\widehat{BE}) = m(\widehat{CE})$$

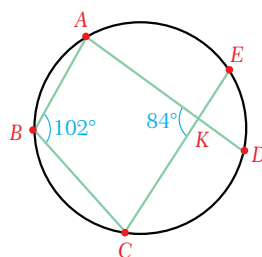
$$180^\circ - 2y + 2x = 180^\circ - 2x - 2z$$

$$4x = 2y - 2z$$

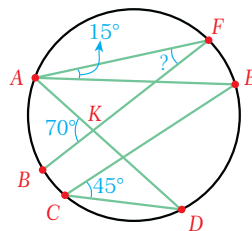
$$2x = y - z, \text{ as required.}$$

Check Yourself

1. Find $m(\widehat{DE})$ in the figure.



2. Find $m(\angle AFB)$ in the figure, using the information given.



3. A , B , C , D and E are five points located counter-clockwise on a circle such that $m(\widehat{CD}) = 50^\circ$ and $m(\angle BFE) = 130^\circ$. F is the intersection point of EC and BD . What is $m(\angle BAE)$?

Answers

1. 12° 2. 10° 3. 75°

3. Angles Outside a Circle

a The angle between two tangents

Theorem

The measure of an angle formed by two tangents to a circle is half the difference of the intercepted arcs.

Proof

Look at the figure.

Let $m(\angle PAB) = m(\angle PBA) = \alpha$ because they intercept the same arc, $\widehat{AB}$. So $AP = PB$.

Then $m(\angle APB) = 180^\circ - 2\alpha$.

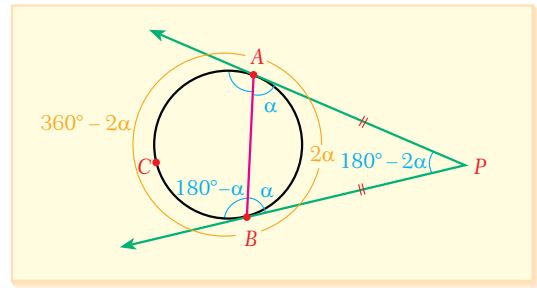
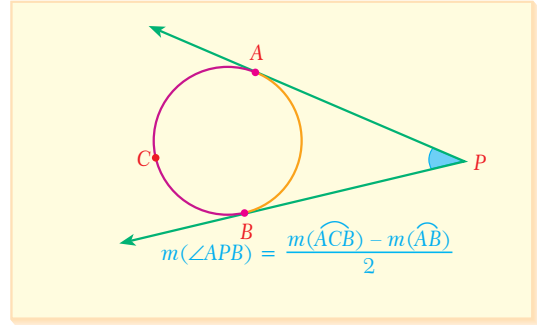
Also, $m(\widehat{AB}) = 2\alpha$ because $m(\angle PAB) = \alpha$.

So $m(\widehat{ACB}) = 360^\circ - 2\alpha$.

Half the difference of the intercepted arcs is

$$\begin{aligned} \frac{m(\widehat{ACB}) - m(\widehat{AB})}{2} &= \frac{360^\circ - 2\alpha - (2\alpha)}{2} \\ &= \frac{360^\circ - 4\alpha}{2} \\ &= 180^\circ - 2\alpha = m(\angle APB). \end{aligned}$$

So $m(\angle APB) = \frac{m(\widehat{ACB}) - m(\widehat{AB})}{2}$, as required.



EXAMPLE

43

In the figure, PA and PB are tangent to the circle at A and B . Given $m(\widehat{ACB}) = 250^\circ$, find $m(\angle P) = \alpha$.

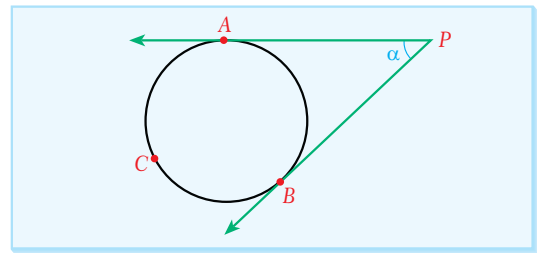
Solution

Since $m(\widehat{ACB}) = 250^\circ$,

$$m(\widehat{AB}) = 360^\circ - 250^\circ = 110^\circ.$$

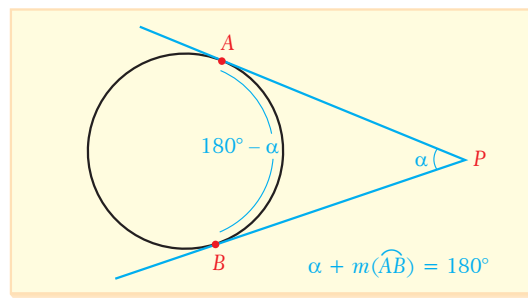
By the theorem we have just seen,

$$\alpha = \frac{m(\widehat{ACB}) - m(\widehat{AB})}{2} = \frac{250^\circ - 110^\circ}{2} = 70^\circ.$$



Property

The angle formed by two tangents and the angle of the minor arc intercepted by these tangents are supplementary.



EXAMPLE

44

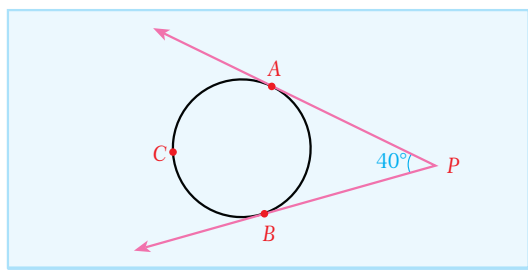
In the figure, PA and PB are tangent to the circle at A and B . If $m(\angle P) = 40^\circ$, find $m(\widehat{ACB})$.

Solution

$m(\widehat{AB}) + 40^\circ = 180^\circ$ by the property just given. So

$$\begin{aligned} m(\widehat{AB}) &= 180^\circ - 40^\circ \\ &= 140^\circ. \end{aligned}$$

$$\begin{aligned} \text{So } m(\widehat{ACB}) &= 360^\circ - 140^\circ \\ &= 220^\circ. \end{aligned}$$



EXAMPLE

45

Lines PA and PB are tangent to a circle at points A and B , and C is a point on the minor arc AB . Find $m(\angle ACB)$ if $m(\angle APB) = 50^\circ$.

Solution

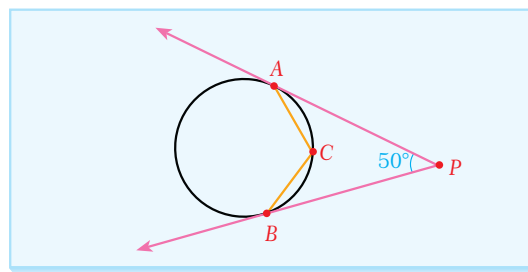
Look at the figure.

$$\begin{aligned} m(\widehat{ACB}) + 50^\circ &= 180^\circ \\ m(\widehat{ACB}) &= 130^\circ. \end{aligned}$$

$$\begin{aligned} \text{Also, } m(\widehat{AB}) + m(\widehat{ACB}) &= 360^\circ \\ m(\widehat{AB}) + 130^\circ &= 360^\circ \\ m(\widehat{AB}) &= 230^\circ. \end{aligned}$$

Since $m(\angle ACB)$ is an inscribed angle,

$$m(\angle ACB) = \frac{m(\widehat{AB})}{2} = \frac{230^\circ}{2} = 115^\circ.$$



EXAMPLE

46

In the figure, A and B are points of tangency on the larger circle, and D and E are points of tangency on the smaller circle. Given $m(\angle APB) = 80^\circ$, find $m(\angle F)$.

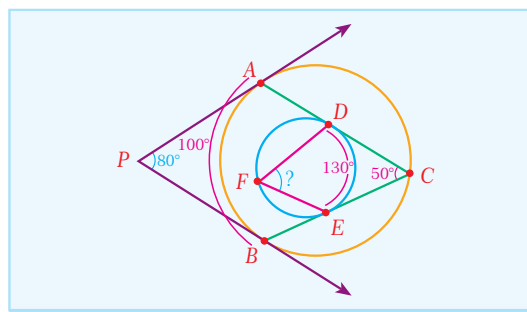
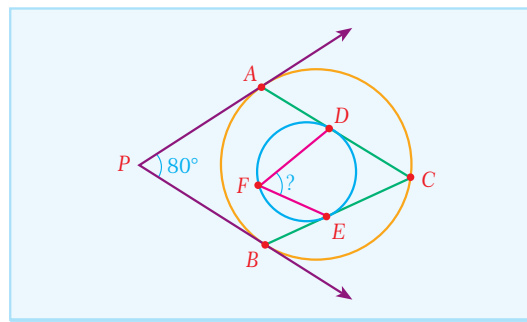
Solution

$$\begin{aligned} m(\widehat{AB}) &= 180^\circ - m(\angle P) \\ &= 180^\circ - 80^\circ \\ &= 100^\circ. \end{aligned}$$

$$m(\angle ACB) = \frac{m(\widehat{AB})}{2} = \frac{100^\circ}{2} = 50^\circ.$$

$$\begin{aligned} m(\widehat{DE}) &= 180^\circ - m(\angle ACB) \\ &= 180^\circ - 50^\circ \\ &= 130^\circ. \end{aligned}$$

$$\text{Finally, } m(\angle DFE) = \frac{m(\widehat{DE})}{2} = \frac{130^\circ}{2} = 65^\circ.$$



EXAMPLE

47

In the figure, AB is tangent to both circles and $m(\angle ABC) = 40^\circ$. Find $m(\angle ADC) = \alpha$.

Solution

Let us draw the common tangent d , as shown.

$$\frac{m(\widehat{BC})}{2} = m(\angle ABC) \Rightarrow m(\widehat{BC}) = 80^\circ.$$

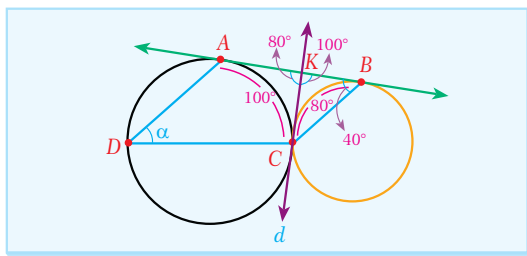
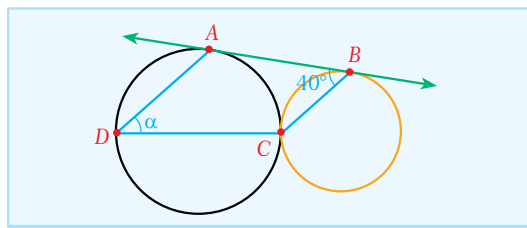
By complementarity,

$$m(\angle BKC) = 180^\circ - 80^\circ = 100^\circ.$$

So $m(\angle AKC) = 80^\circ$.

$$\begin{aligned} m(\widehat{AC}) &= 180^\circ - m(\angle AKC) \\ &= 180^\circ - 80^\circ \\ &= 100^\circ. \end{aligned}$$

$$\text{So } \alpha = \frac{m(\widehat{AC})}{2} = \frac{100^\circ}{2} = 50^\circ.$$



b. The angle between two secants

Theorem

The measure of the angle formed by two secants of a circle is half the difference of the intercepted arcs.

Proof



As we have seen, this theorem is also true for tangents: the measure of the angle formed by two tangents to a circle is half the difference of the intercepted arcs.

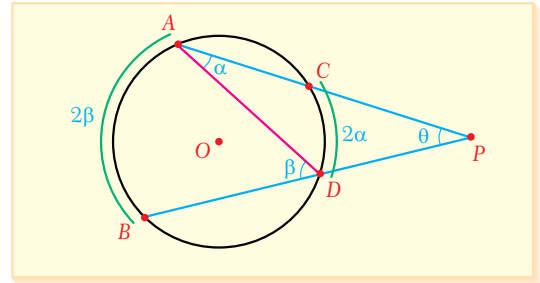
Look at the figure. In $\triangle ADP$, $\beta = \alpha + \theta$.

So $\theta = \beta - \alpha$.

$$m(\widehat{AB}) - m(\widehat{CD}) = 2\beta - 2\alpha = 2(\underbrace{\beta - \alpha}_{\theta})$$

Therefore $m(\widehat{AB}) - m(\widehat{CD}) = 2\theta$, and so

$$m(\angle P) = \theta = \frac{m(\widehat{AB}) - m(\widehat{CD})}{2}, \text{ as required.}$$



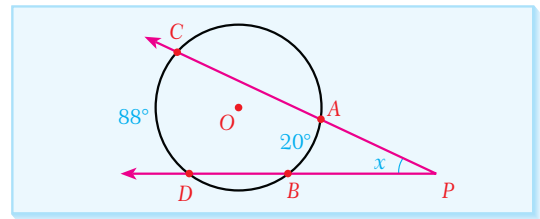
EXAMPLE

48

Find $m(\angle CPD) = x$ in the figure.

Solution

$$x = \frac{m(\widehat{CD}) - m(\widehat{AB})}{2} = \frac{88^\circ - 20^\circ}{2} = 34^\circ$$



EXAMPLE

49

Find $m(\widehat{AB})$ in the figure, using the information given.

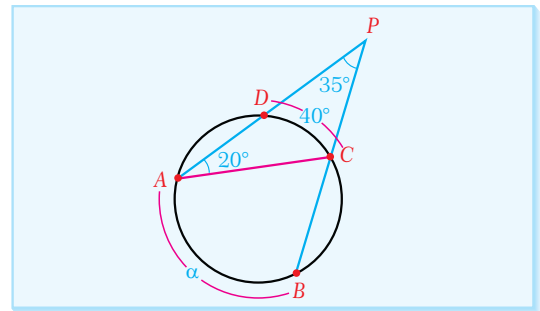
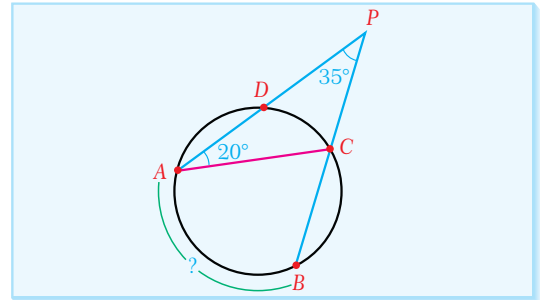
Solution 1 $m(\angle DAC) = 20^\circ \Rightarrow m(\widehat{DC}) = 40^\circ$

Let $m(\widehat{AB}) = \alpha$. Then

$$35^\circ = \frac{\alpha - 40^\circ}{2} \quad (\text{by the theorem above})$$

$$\alpha - 40^\circ = 70^\circ$$

$$\alpha = 110^\circ.$$



Solution 2 By the triangle exterior angle theorem in $\triangle PAC$, $m(\angle ACB) = 20^\circ + 35^\circ = 55^\circ$.

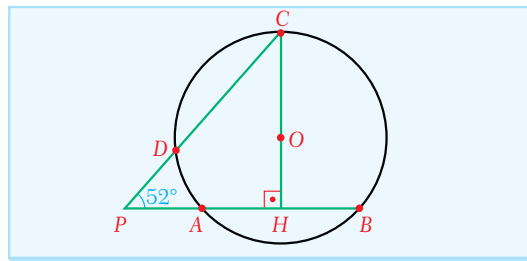
So $m(\widehat{AB}) = 2 \cdot 55^\circ = 110^\circ$.

EXAMPLE

50

In the figure, $AB = CD$, O is the center of the circle and $AB \perp CH$.

Find $m(\widehat{BC})$.



Solution

Let us extend CH to make it a diameter.

Since OE bisects AB , let us write

$$m(\widehat{AE}) = m(\widehat{EB}) = \alpha.$$

Since $AB = CD$, $m(\widehat{AB}) = m(\widehat{CD}) = 2\alpha$.

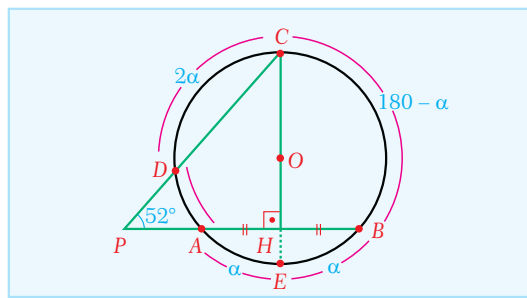
$$m(\widehat{EB}) = \alpha \Rightarrow m(\widehat{BC}) = 180^\circ - \alpha.$$

Now, from the theorem we have just seen,

$$m(\angle P) = \frac{m(\widehat{BC}) - m(\widehat{AD})}{2}.$$

$$52^\circ = \frac{180^\circ - \alpha - (180^\circ - 3\alpha)}{2}. \text{ So } \alpha = 52^\circ.$$

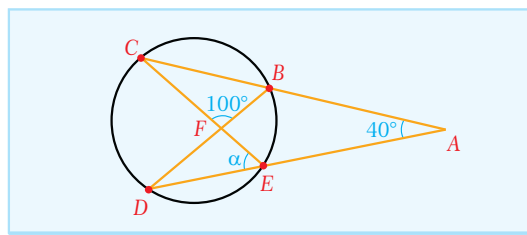
Therefore, $m(\widehat{BC}) = 180^\circ - \alpha = 180^\circ - 52^\circ = 128^\circ$.



EXAMPLE

51

Find the angle measure α in the figure.



Solution

$$m(\angle CED) = \alpha \Rightarrow m(\widehat{CD}) = 2\alpha.$$

In $\triangle DEF$, $m(\angle EDF) = 80^\circ - \alpha$.

$$m(\widehat{BE}) = 2 \cdot m(\angle EDF) = 160^\circ - 2\alpha.$$

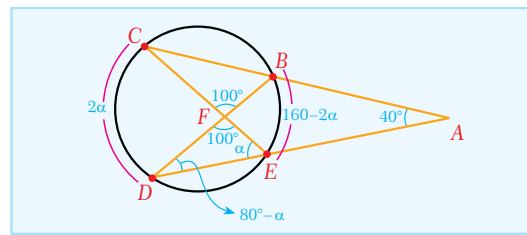
$$\text{Now, } m(\angle A) = \frac{m(\widehat{CD}) - m(\widehat{BE})}{2}$$

$$40^\circ = \frac{2\alpha - (160^\circ - 2\alpha)}{2}$$

$$40^\circ = 2\alpha - 80^\circ$$

$$120^\circ = 2\alpha$$

$$60^\circ = \alpha.$$



EXAMPLE

52

In the figure, $BD = DC$,
 $AB \perp CF$ and $AC \perp BE$.
 Find $m(\angle FDE) = \alpha$.

Solution

Since ED and FD are medians to the hypotenuse in the right triangles $\triangle BEC$ and $\triangle BFC$, $BD = DC = ED = FD$.

This means that we can draw a circle centered at D in which BD , DC , ED and FD are all radii.

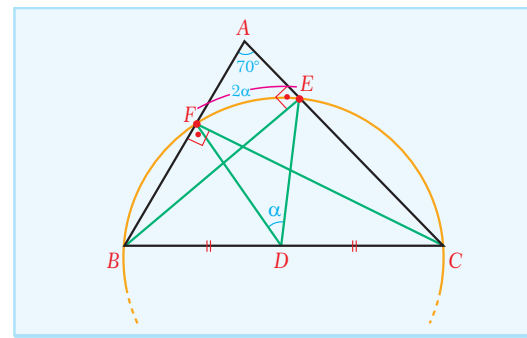
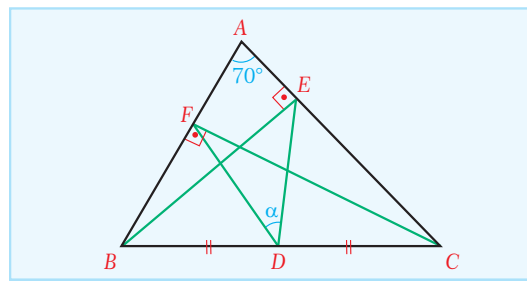
So $m(\widehat{EF}) = 2\alpha$.

Finally, $m(\angle A) = \frac{180^\circ - 2\alpha}{2}$

$$70^\circ = \frac{180^\circ - 2\alpha}{2}$$

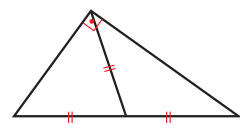
$$70^\circ = 90^\circ - \alpha$$

$$\alpha = 20^\circ.$$



Remember!

The length of the median to the hypotenuse of a right triangle is equal to half the length of the hypotenuse.



c. The angle between a secant and a tangent

Theorem

The measure of the angle formed by a secant line and a tangent is half the difference of the intercepted arcs.

Proof

Look at the figure. We need to prove

$$m(\angle P) = \frac{m(\widehat{AB}) - m(\widehat{AC})}{2}.$$

In $\triangle ABP$, $\alpha = \beta + \theta$. So $\theta = \alpha - \beta$.

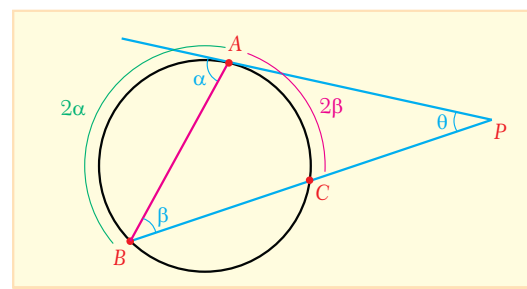
$$m(\widehat{AB}) - m(\widehat{AC}) = 2\alpha - 2\beta = 2(\alpha - \beta)$$

Therefore,

$$\frac{m(\widehat{AB}) - m(\widehat{AC})}{2} = \frac{2 \cdot (\alpha - \beta)}{2} = \alpha - \beta = \theta.$$

$$\text{So } \frac{m(\widehat{AB}) - m(\widehat{AC})}{2} = m(\angle P) \text{ as required.}$$

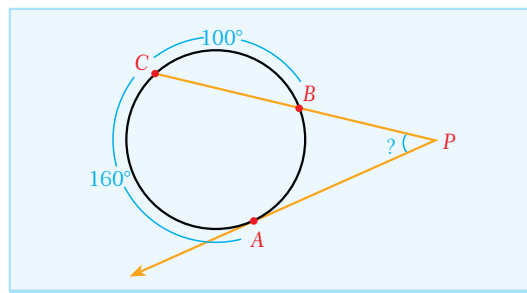
As you can see, this proof is very similar to the proof that the angle formed by two secants is half the difference of the intercepted arcs.



EXAMPLE

53

In the figure, PA is tangent to the circle at A ,
 $m(\widehat{AC}) = 160^\circ$ and $m(\widehat{BC}) = 100^\circ$.
 Find $m(\angle P)$.



Solution

$$m(\widehat{AC}) + m(\widehat{BC}) + m(\widehat{AB}) = 360^\circ$$

$$160^\circ + 100^\circ + m(\widehat{AB}) = 360^\circ$$

$$m(\widehat{AB}) = 100^\circ$$

$$\text{So } m(\angle P) = \frac{m(\widehat{AC}) - m(\widehat{AB})}{2}$$

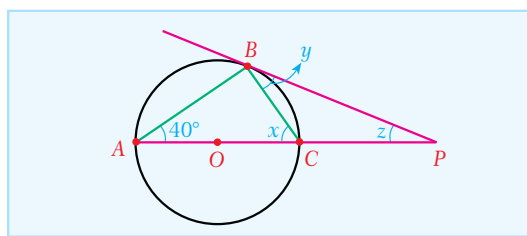
$$= \frac{160^\circ - 100^\circ}{2}$$

$$= 30^\circ.$$

EXAMPLE

54

In the figure, O is the center of the circle and
 B is a point of tangency. Find the values of x ,
 y and z .



Solution

Since $m(\angle BAC)$ and $m(\angle CBP)$ intercept the
 same arc, they are congruent.

$$\text{So } y = 40^\circ.$$

Also, $m(\angle ABC) = 90^\circ$ because it intercepts a
 semicircle.

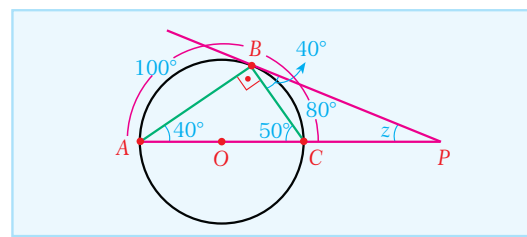
$$\text{So } x = 50^\circ, \text{ which gives us } m(\widehat{AB}) = 100^\circ.$$

$$\text{Finally, } z = \frac{m(\widehat{AB}) - m(\widehat{BC})}{2}$$

$$= \frac{100^\circ - 80^\circ}{2}$$

$$= 10^\circ.$$

In summary, $x = 50^\circ$, $y = 40^\circ$ and $z = 10^\circ$.



EXAMPLE

55

In the figure, PA is tangent to the circle at A . Find the values of x and y .

Solution $40^\circ = \frac{140^\circ - x}{2}$

$$80^\circ = 140^\circ - x$$

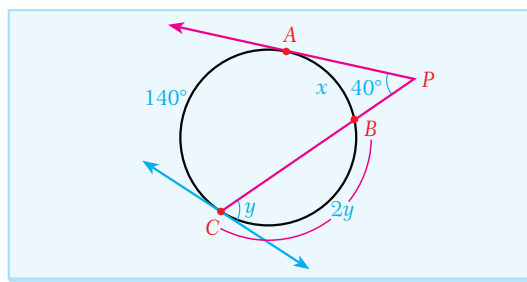
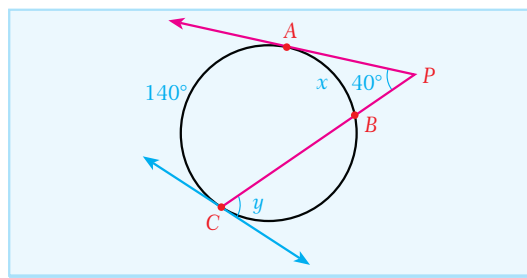
$$x = 60^\circ.$$

$$140^\circ + x + 2y = 360^\circ$$

$$140^\circ + 60^\circ + 2y = 360^\circ$$

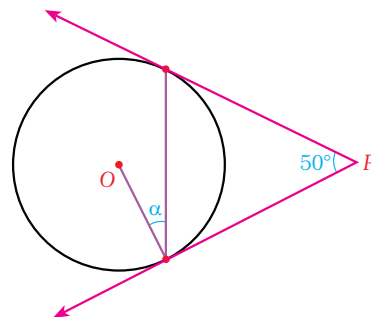
$$2y = 160^\circ$$

$$y = 80^\circ.$$



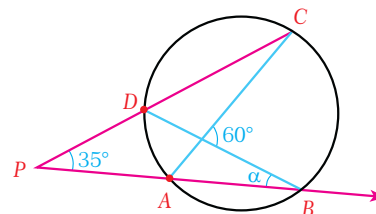
Check Yourself

1. A, B, C and D are four points located counter-clockwise on a circle. PA and PC are tangent to the circle, and $2 \cdot m(\angle ABC) = 3 \cdot m(\angle ADC)$. What is $m(\angle APC)$?
2. In the figure, the two lines are tangent to $\odot O$. Find the angle measure α .



3. AB is the diameter of a semicircle centered at O . C and D are two points on the semicircle, and P is a point outside the circle such that P, B, A and P, C, D are collinear. Given $m(\angle DAP) = 60^\circ$ and $m(\angle P) = 22^\circ$, find $m(\angle DCO)$.

4. Find the angle measure α in the figure.



Answers

1. 36°
2. 25°
3. 38°
4. 12.5°

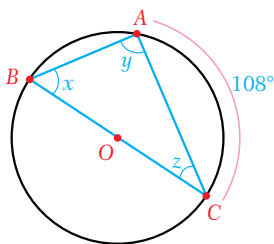
EXERCISES 3.1

A. Inscribed Angles and Arcs

1. In the figure,
 $m(\widehat{AC}) = 108^\circ$.

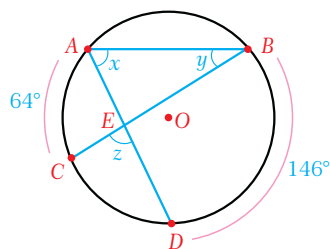
Find the angle measures

x , y and z .



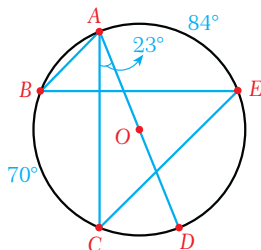
2. In the figure,
 $m(\widehat{AC}) = 64^\circ$ and
 $m(\widehat{BD}) = 146^\circ$.

Find the angle measures x ,
 y and z .

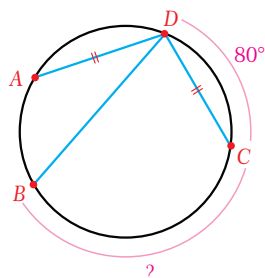


3. Find the angle measures using the information in the figure.

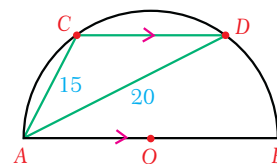
- $m(\angle C)$
- $m(\widehat{CD})$
- $m(\angle BAC)$
- $m(\angle E)$
- $m(\angle BAD)$



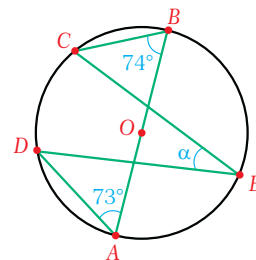
4. In the figure, $AD = CD$,
 $m(\widehat{CD}) = 80^\circ$ and
 $3 \cdot m(\angle ADB) = m(\angle BDC)$.
 Find $m(\widehat{BC})$.



5. O is the center of the semicircle shown at the right. $CD \parallel AB$,
 $AD = 20$ cm and
 $AC = 15$ cm are given. Find CD .

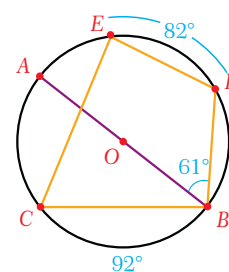


6. Find the value of α in the figure.

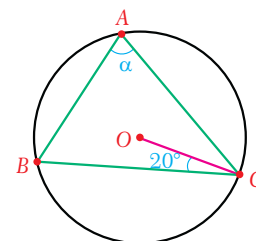


7. Find the indicated measures by using the information given.

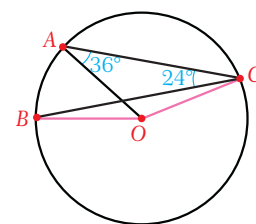
- $m(\angle E)$
- $m(\widehat{AE})$
- $m(\widehat{BD})$
- $m(\angle D)$



8. Find the value of α in the figure.



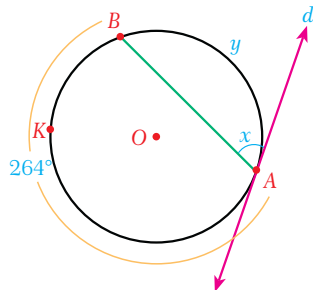
9. O is the center of the circle shown at the right. Given
 $m(\angle OAC) = 36^\circ$ and
 $m(\angle ACB) = 24^\circ$, find
 $m(\angle BOC)$.



10. A, B, C, D and E are five points located counter-clockwise on a circle. $m(\angle AED) = 145^\circ$ and $m(\angle BCD) = 100^\circ$ are given. Find $m(\angle ADB)$.

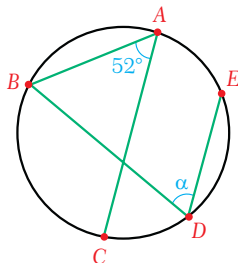
B. Angles Formed by Secants, Tangents and Chords

11. In the figure, $m(\widehat{AKB}) = 264^\circ$ and d is tangent to the circle at A . Find the degree measures x and y .

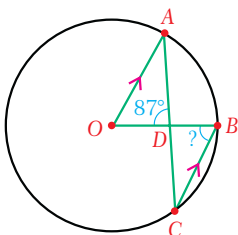


12. Two circles are externally tangent to each other at point C . A , B and C are located counter-clockwise on the smaller circle, and B , C and D are collinear such that DE is the diameter of the larger circle. Given $m(\angle BAC) = 62^\circ$, find $m(\angle CDE)$.

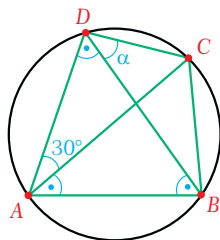
13. In the figure, $m(\angle BAC) = 52^\circ$ and $m(\widehat{CE}) = 110^\circ$. Find $m(\angle BDE) = \alpha$.



14. O is the center of the circle in the figure and $OA \parallel BC$. Find $m(\angle CBO)$.

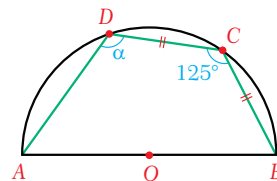


15. In the figure, $m(\angle ADB) = m(\angle BAC) = m(\angle DBA)$ and $m(\angle DAC) = 30^\circ$. Find the value of α .

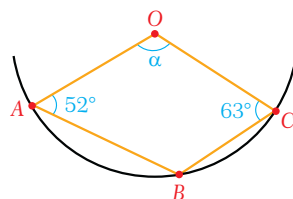


16. A and B are two points on a circle centered at O . A line PA is tangent to the circle, and the points B , O and P are collinear. Find $m(\angle ABP)$ if $m(\angle APB) = 44^\circ$.

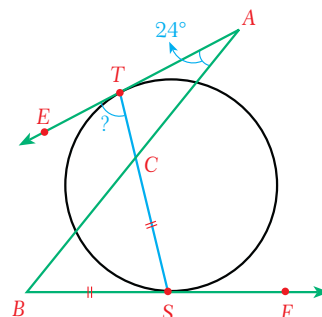
17. AB is the diameter of the semicircle shown at the right. If $m(\angle BCD) = 125^\circ$ and $BC = CD$, find the value of α .



18. Given that O is the center of the arc in the figure, find $m(\angle AOC) = \alpha$.

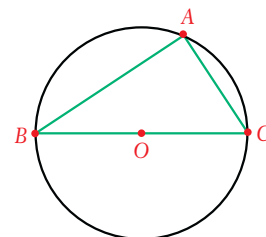


19. In the figure, T and S are points of tangency, $BS = CS$ and $m(\angle A) = 24^\circ$. Find $m(\angle ETS)$.

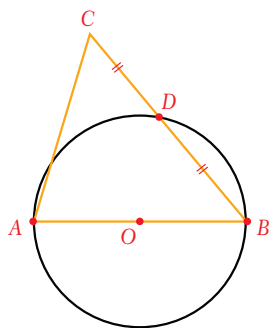


20. O is the center of the circle shown at the right.

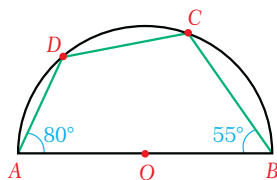
$AB = x + 5$ cm,
 $AC = x - 2$ cm and
 $BC = x + 6$ cm are given. Find the radius of the circle.



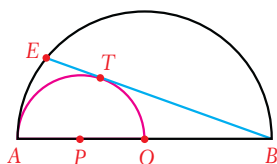
21. In the figure, AB is the diameter of the circle and $CD = BD$. Find the radius of the circle if $AC = 10$ cm.



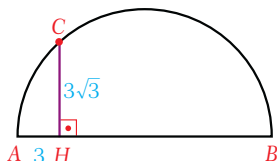
22. P is a point outside a circle $\odot O$ and PA is tangent to the circle at A . A, B, C are located clockwise around $\odot O$, and PC is a secant line where P, C and B are collinear. If $PA = AC$ and $m(\angle P) = 80^\circ$, find $m(\angle OCB)$.



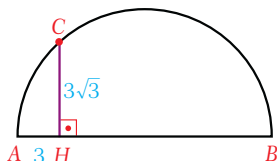
23. AB is the diameter of the semicircle at the right. Given $AB = 12$ cm, $m(\angle DAB) = 80^\circ$ and $m(\angle ABC) = 55^\circ$, find DC .



24. P and O are the centers of the two semicircles shown opposite. Given $TB = 6\sqrt{2}$ cm, find OB .

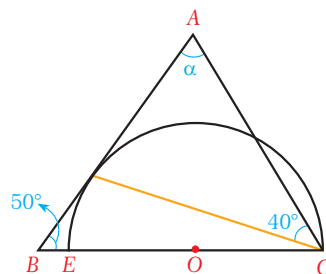


25. In the figure, AB is a diameter, $CH = 3\sqrt{3}$ cm and $AH = 3$ cm. Find the radius of the circle.

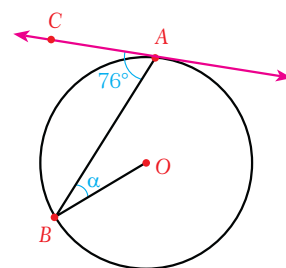


26. A, B, C and D are four points located clockwise on a circle, and P is a point outside the circle such that P, D and C are collinear. PA is tangent to the circle, $AD = AB$, $m(\angle P) = 40^\circ$ and $m(\widehat{ABC}) = 140^\circ$. Find $m(\angle BAC)$.

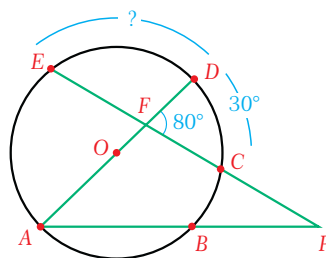
27. AB is tangent to the semicircle shown at the right. Find the angle measure α by using the information in the figure.



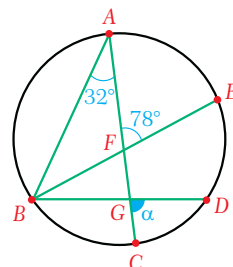
28. In the figure, O is the center of the circle, A is a point of tangency and $m(\angle BAC) = 76^\circ$. Find the angle measure α .



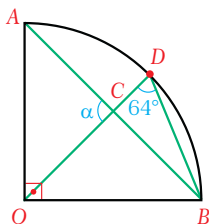
29. In the figure, O is the center of the circle, $m(\angle DFC) = 80^\circ$ and $m(\widehat{CD}) = 30^\circ$. Find $m(\widehat{DE})$.



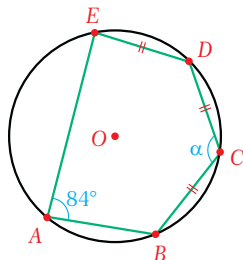
30. In the figure, $m(\widehat{AE}) = 2 \cdot m(\widehat{ED})$. Find the value of α .



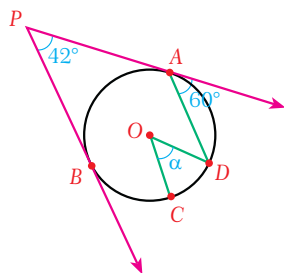
31. In the figure, O is the center of the quarter circle and $m(\angle ODB) = 64^\circ$. Find $m(\angle ACO) = \alpha$.



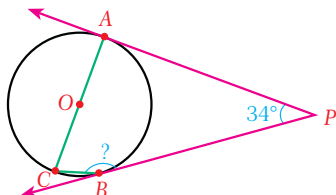
32. In the figure, $BC = CD = DE$ and $m(\angle BAE) = 84^\circ$. Find the value of α .



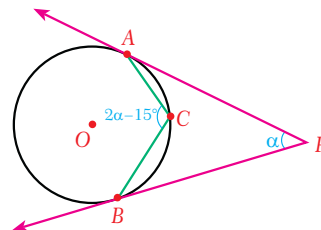
33. A and B in the figure are points of tangency. Given $m(\widehat{BC}) = m(\widehat{CD})$ and the angle measures shown, find the value of α .



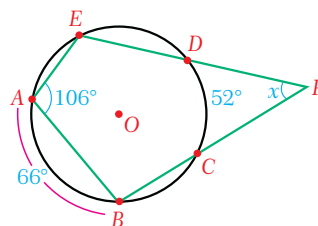
34. A and B in the figure are points of tangency on $\odot O$. Find $m(\angle PBC)$ if $m(\angle APB) = 34^\circ$.



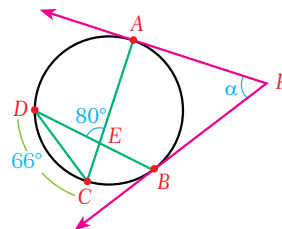
35. The figure shows two tangents to $\odot O$. Find the value of α .



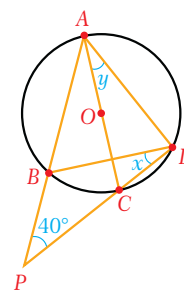
36. Find the angle measure x using the information in the figure.



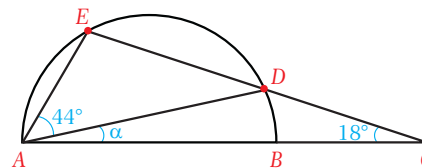
37. In the figure, PA and PB are tangents, $m(\angle AED) = 80^\circ$ and $m(\widehat{CD}) = 66^\circ$. Find the value of α .



38. Find the value of y in terms of x using the information given in the figure.

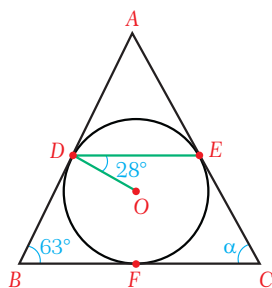


- 39.

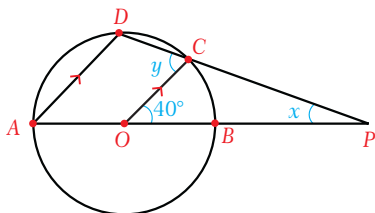


AB is the diameter of the semicircle shown. Find $m(\angle DAB) = \alpha$.

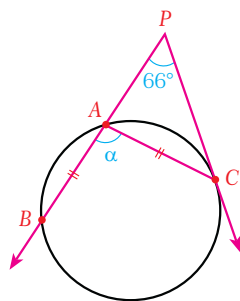
40. The points D , E and F are points of tangency on the circle at the right, and O is the center of the circle. Find the value of α .



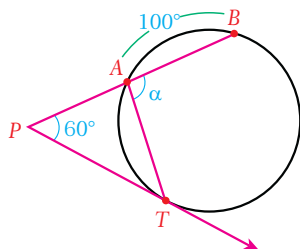
41. The figure shows $\odot O$ with $AD \parallel OC$ and $m(\angle COB) = 40^\circ$. Find the value of $x + y$.



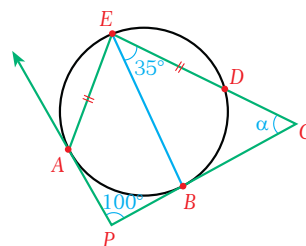
42. In the figure, C is a point of tangency and $AB = AC$. Find the value of α .



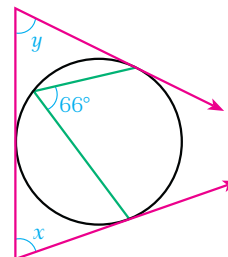
43. In the figure, T is a point of tangency, $m(\angle BPT) = 60^\circ$ and $m(\widehat{AB}) = 100^\circ$. Find the value of α .



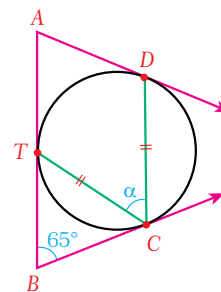
44. In the figure, PA and PC are tangents, $m(\angle APB) = 100^\circ$, $m(\angle BEC) = 35^\circ$ and $AE = DE$. Find the value of α .



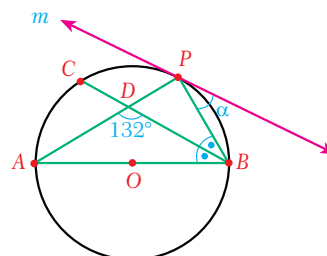
45. Given that the three lines are tangents, find the value of $x + y$ in the figure.



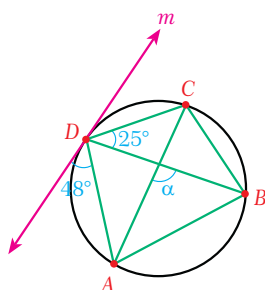
46. Find the value of α in the figure.



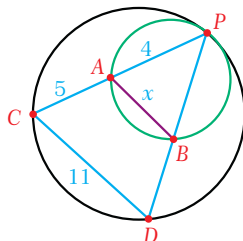
47. In the figure, P is a point of tangency on $\odot O$. Find the value of α using the information given.



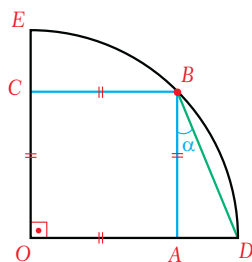
48. m is tangent to the circle in the figure at D . Find the value of α .



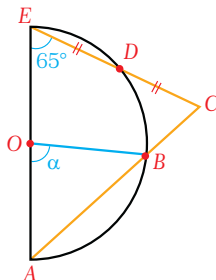
49. The circles in the figure are internally tangent at P . Find the length $AB = x$.



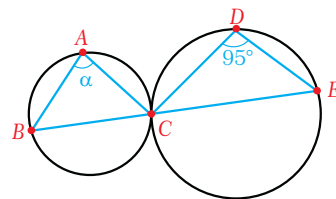
50. In the figure, DE is an arc of $\odot O$ and $OABC$ is a square. Find $m(\angle DBA) = \alpha$.



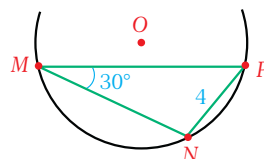
51. In the figure, O is the center of the semicircle, $ED = DC$ and $m(\angle AED) = 65^\circ$. Find the angle measure α .



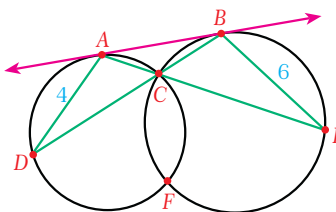
52. The two circles in the figure are tangent to each other at C , and $m(\angle CDE) = 95^\circ$. Find the value of α .



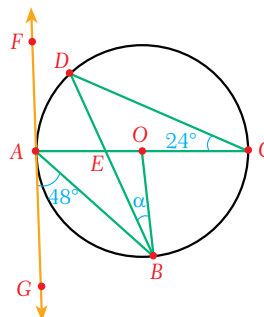
53. Find the diameter of $\odot O$ in the figure.



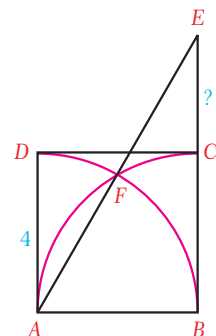
54. In the figure, AB is a common tangent, $AD = 4$ cm and $BE = 6$ cm. Find the length of the segment AB .



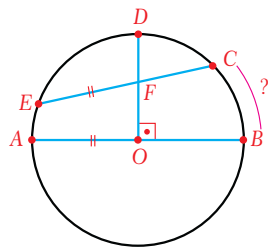
55. The figure shows $\odot O$ and a tangent line FG . Find the value of α .



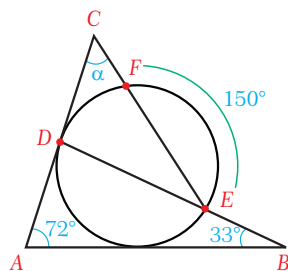
56. A and B are the centers of the quarter circles in the figure shown. Find EC .



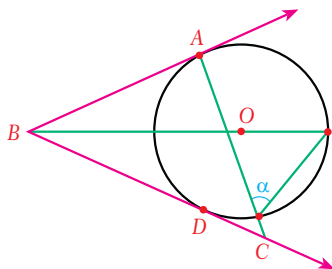
57. In the figure, O is the center of the circle and $EF = AO$. Given $m(\widehat{AE}) = 15^\circ$, find $m(\widehat{BC})$.



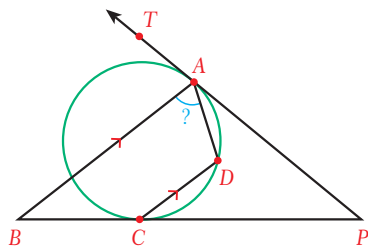
58. In the figure, AB and AC are tangents and $m(\widehat{EF}) = 150^\circ$. Find the value of α .



59. In the figure, O is the center of the circle and $BD = AC$. If $m(\angle ACB) = 56^\circ$, find the value of α .



60. In the figure, A and C are points of tangency, $AB \parallel CD$, $AB = AP$ and $AD = CD$. Find $m(\angle BAD)$.



SEGMENTS FORMED BY SECANTS AND CHORDS

1. The Power of a Point

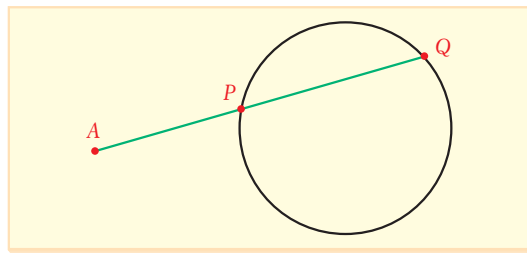
Definition

power of a point

A point A and a circle are given. A line that passes through A intersects the given circle at P and Q . The product $AP \cdot AQ$ is called the **power of the point A** with respect to the given circle.

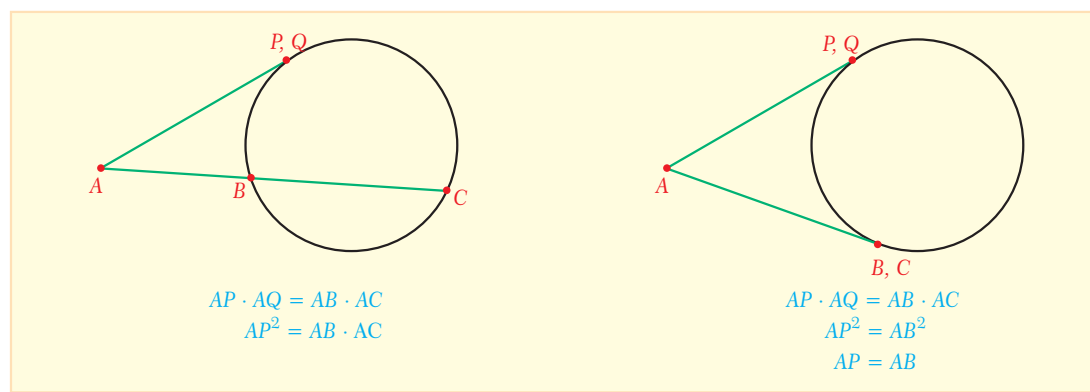
It is an amazing fact that P is independent of the choice of the line APQ . In other words, the power of a point is unique for a given point and a given circle. For example, in the figure opposite we can write the power p as

$$p = AP \cdot AQ = AB \cdot AC = AD \cdot AE.$$



The concept of the power of a point was first described by the Swiss mathematician Jakob Steiner in 1826.

The line APQ (or the line ABC , or both) may be tangent to the circle. In this case, P and Q (or B and C) become the same point. However, the property of the power of a point still holds:



The point A may also lie either inside or outside the circle. When the point is inside the circle, the lines through the point become chords and the power equality becomes

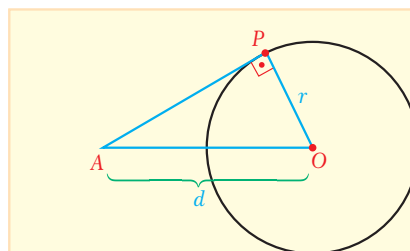
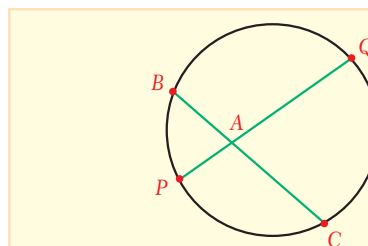
$$AP \cdot AQ = AB \cdot AC.$$

In all cases, the same equality holds. It is known as the **power of a point theorem**.

When the point is inside the circle, the theorem is also called the **intersecting chords theorem**. When the point is outside the circle, the theorem can be called the **intersecting secants theorem**. The proof of each theorem will be shown later.

Now consider a point A outside a circle centered at O with radius r . If the distance between A and O is d then $AP^2 + r^2 = d^2$ by the Pythagorean Theorem. So $AP^2 = d^2 - r^2$.

We know that the power P of A with respect to $\odot O$ is AP^2 . Therefore we can write the equality as $p = AP^2 = d^2 - r^2$ and $p = d^2 - r^2$.



2. Segments Formed by Intersecting Secants

Theorem

intersecting secants theorem

If two secants intersect outside a circle, the product of the lengths of one secant segment and its external segment is equal to the same product for the other segment.

We can summarize this rule informally as '**outer $\times$ whole = outer $\times$ whole**'.

Proof

Look at the figure. We need to show that $PC \cdot PA = PD \cdot PB$.

In the figure, $m(\angle A) = m(\angle B)$ because they intercept the same arc, namely CD .

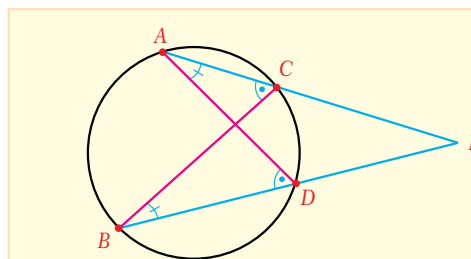
Similarly, $m(\angle ACB) = m(\angle ADB)$.

So $\triangle ADP$ and $\triangle BCP$ are similar triangles by the AAA similarity theorem.

So we can write $\frac{PD}{PC} = \frac{PA}{PB}$, which means

$PC \cdot PA = PD \cdot PB$, as required.

Remember that this result is one case of the power of a point theorem.



EXAMPLE

56

Find the length x in the figure.

Solution By the intersecting secants theorem,

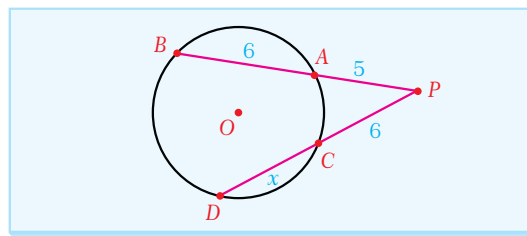
$$PA \cdot PB = PC \cdot PD$$

$$5 \cdot (5 + 6) = 6 \cdot (6 + x)$$

$$55 = 36 + 6x$$

$$19 = 6x$$

$$x = \frac{19}{6}.$$



EXAMPLE

57

$\triangle ABC$ in the figure is an equilateral triangle.

$AE = 6$ cm and $CD = 2$ cm are given.

Find the length $AB = x$.

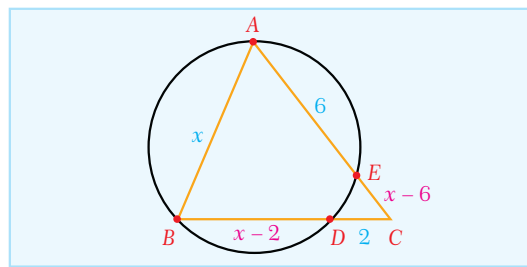
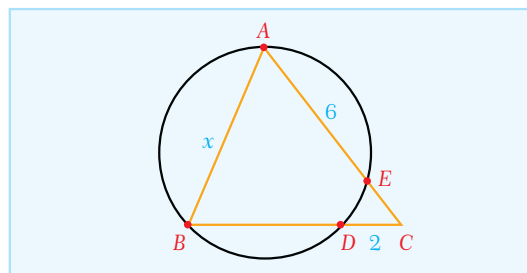
Solution Since $\triangle ABC$ is equilateral,

$$EC = x - 6 \text{ cm and } BD = x - 2 \text{ cm.}$$

$$\text{So } CD \cdot CB = CE \cdot CA$$

$$2 \cdot x = (x - 6) \cdot x$$

$$x = 8 \text{ cm.}$$



Theorem

If a secant and a tangent intersect outside a circle then the product of the secant segment and its external segment equals the square of the length of the tangent segment.

We can summarize this rule informally as '**outer $\times$ whole = tangent squared**'.

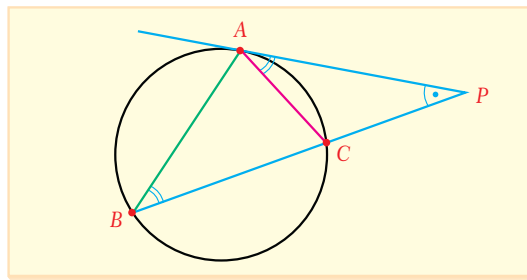
Proof Look at the figure. We need to show that

$$PA^2 = PC \cdot PB.$$

Since $\angle CAP$ and $\angle ABC$ intercept the same arc, they are congruent. As $\angle P$ is a common angle for $\triangle ACP$ and $\triangle BAP$, by the AAA similarity theorem the triangles are similar.

$$\text{So } \frac{PA}{PB} = \frac{PC}{PA}, \text{ which gives us } PA^2 = PC \cdot PB,$$

as required. Recall that this is also one case of the power of a point theorem.



EXAMPLE

58

Point A in the figure is a point of tangency. Find the length $BC = x$.

Solution By the theorem we have just seen,

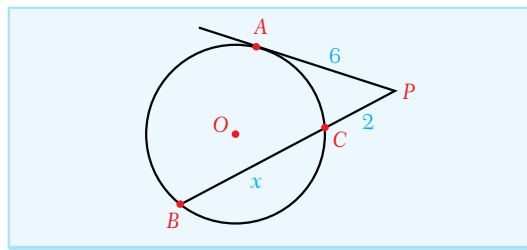
$$PA^2 = PC \cdot PB$$

$$6^2 = 2 \cdot (2 + x)$$

$$36 = 2 \cdot (2 + x)$$

$$18 = 2 + x$$

$$x = 16.$$



EXAMPLE

59

In the figure, O is the center of the semicircle and E and F are points of tangency. Given $AF = CD$, find the length $DE = x$.

Solution $AF^2 = AB \cdot AC$

$$= 4 \cdot 16$$

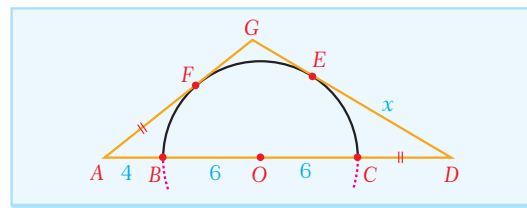
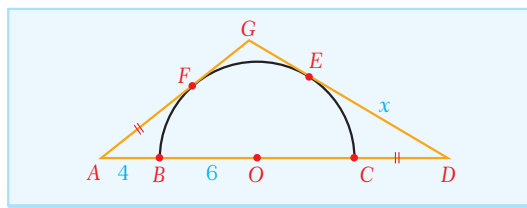
$$AF = CD = 8.$$

$$DE^2 = x^2 = CD \cdot BD$$

$$x^2 = 8 \cdot (6 + 6 + 8) \quad (CD = AF \text{ is given})$$

$$x = \sqrt{160}$$

$$= 4\sqrt{10}.$$



EXAMPLE

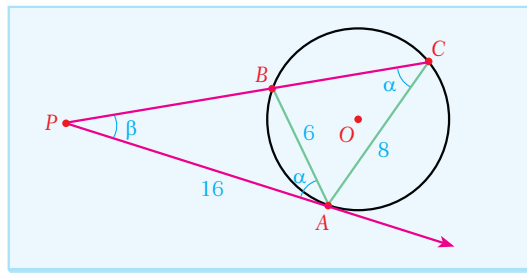
60

A, B and C are three points located clockwise on a circle. P is a point outside the circle such that PA is a tangent and P, B and C are collinear. If AB = 6 cm, AC = 8 cm and PA = 16 cm, find PB.

Solution Look at the figure. $m(\angle PAB) = m(\angle PCA)$ since they intercept the same arc. The angle P is common to both $\triangle PAB$ and $\triangle PCA$.

So $\triangle PAB \sim \triangle PCA$ by AAA similarity.

$$\text{So } \frac{6}{8} = \frac{PB}{16} \Rightarrow PB = 12 \text{ cm.}$$



EXAMPLE

61

In the figure, $AE = EB$ and $DE = DC = 12$ cm. Find DF.

Solution Since $AB = CD = 12$ and $AE = EB$,
 $AE = EB = 6$ cm.

By the power of E, $AE^2 = EF \cdot ED$

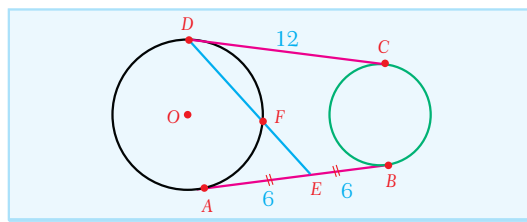
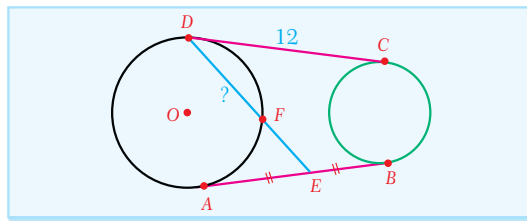
Also, $ED = DC = 12$ cm (given).

$$\text{So } 6^2 = EF \cdot 12$$

$$36 = EF \cdot 12$$

$$EF = 3 \text{ cm.}$$

$$\text{Finally, } DF = DE - EF = 12 - EF = 12 - 3 = 9 \text{ cm.}$$



EXAMPLE

62

In the figure, C is a point of tangency. Find the lengths x and y.

Solution $PC^2 = PD \cdot PE$ (1)

$$x^2 = 2 \cdot 9$$

$$x = 3\sqrt{2}.$$

Similarly, $PC^2 = PA \cdot PB$. (2)

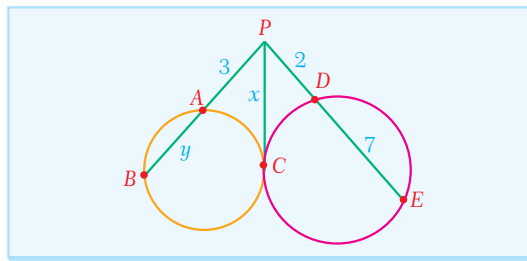
By (1) and (2) we can write

$$PD \cdot PE = PA \cdot PB$$

$$2 \cdot 9 = 3 \cdot (3 + y)$$

$$18 = 3 \cdot (3 + y)$$

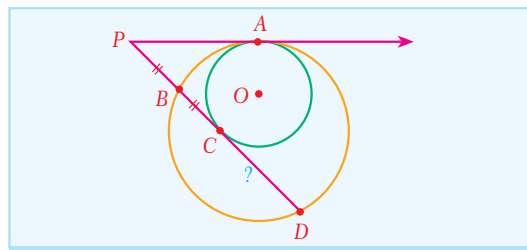
$$y = 3.$$



EXAMPLE

63

In the figure, A is a point of tangency, $PA = 8$ cm and $PB = BC$. Find CD .



Solution $PA = PC = 8$ cm, so $PB = BC = 4$ cm.

In the larger circle,

$$PA^2 = PB \cdot PD$$

$$8^2 = 4 \cdot PD$$

$$64 = 4 \cdot PD$$

$$PD = 16 \text{ cm.}$$

Finally,

$$CD = PD - PC$$

$$= 16 - 8$$

$$= 8 \text{ cm.}$$

EXAMPLE

64

AB is the diameter of a semicircle centered at O , and C and D are two points located counter-clockwise on the semicircle. PD is tangent to the semicircle at D and PO intersects the circle at C . If $PC = 3$ cm and $PD = 6$ cm, find the radius of the semicircle.

Solution The figure illustrates the problem.

Let $BO = r$ and extend PO to point K .

Then $CO = OK = r$.

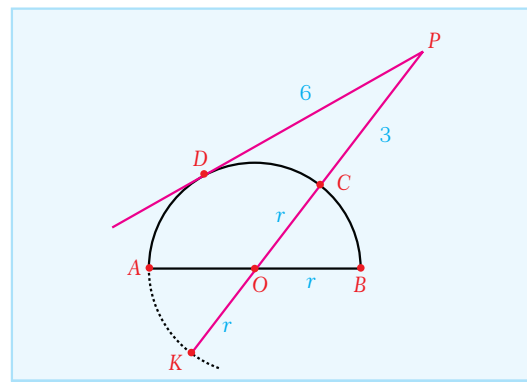
By the power of point P ,

$$6^2 = 3(3 + 2r)$$

$$36 = 3(3 + 2r)$$

$$12 = 3 + 2r$$

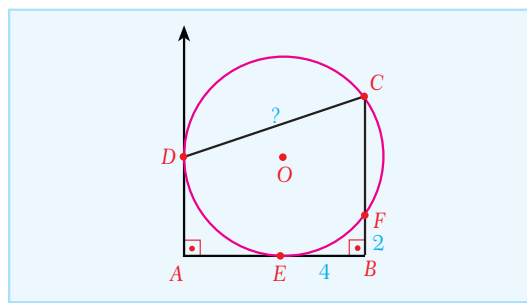
$$r = 4.5 \text{ cm.}$$



EXAMPLE

65

In the figure, O is the center of the circle, $BF = 2$ cm and $BE = 4$ cm. Find the length of chord CD .



$$BE^2 = BF \cdot BC$$

$$4^2 = 2 \cdot BC$$

$BC = 8 \text{ cm.}$

Since $FC = BC - BF$,

$$FC = 8 - 2$$

$$= 6 \text{ cm.}$$

If we draw DO and extend through BC ,
 $DH \perp BC$ and $CH = HF$ (can you see why?).

So $AEOD$ becomes a square, and $DH = AB = 9$.

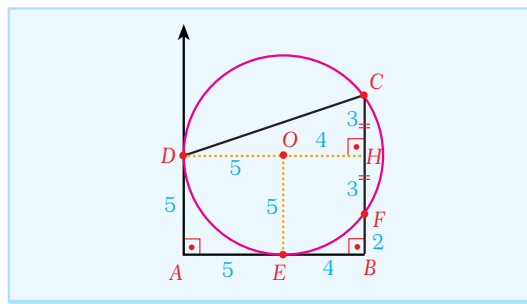
Finally, in ΔCDH , $DH^2 + CH^2 = CD^2$

$$9^2 + 3^2 = CD^2$$

$$90 = CD^2$$

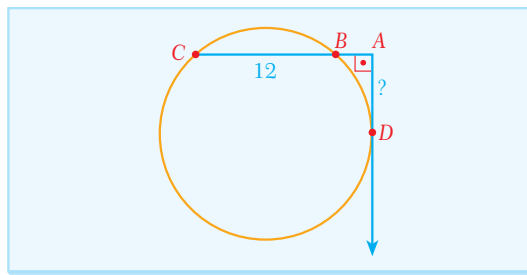
$$CD = \sqrt{90}$$

$$= 3\sqrt{10} \text{ cm.}$$



The radius of the circle in the figure is 10 cm, and $BC = 12$ cm.

Find AD .



Solution Let O be the center of the circle, and let us draw OH such that $OH \perp BC$.

So $CH = HB = 6$ cm.

Since $ADOH$ is a rectangle (can you see why?), $OD = AH = 10$ cm.

Then $AB = AH - HB = 10 - 6 = 4$ cm.

Then by the power of point A ,

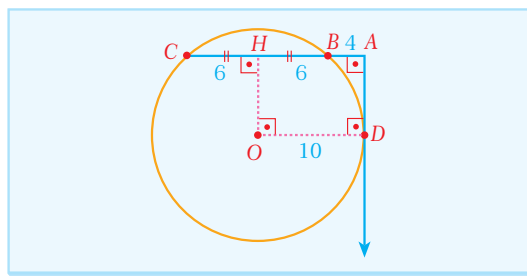
$$AD^2 = AB \cdot AC.$$

$$= 4 \cdot 16$$

$$= 64.$$

So $AD = 8$ cm.

As an exercise, try to find another solution to this problem without using the power of a point. Can you do it?



EXAMPLE

67

In the figure, $DP = DB$, PA and DB are tangent to $\odot O$. $BC = 11$ cm and $PB = 9$ cm are given. What is the radius of the circle?

Solution By the power of P ,

$$PA^2 = PB \cdot PC$$

$$PA^2 = 9 \cdot 20$$

$$PA = 6\sqrt{5} \text{ cm.}$$

Since AD and BD are tangents from a common point, they are congruent.

Notice that $OA \perp PA$. Constructing the line AB , we can also say that $m(\angle ABC) = 90^\circ$ because of the congruency of BD , AD and DP in $\triangle APB$. So the segment AC is a diameter of the circle.

Now, in $\triangle ACP$,

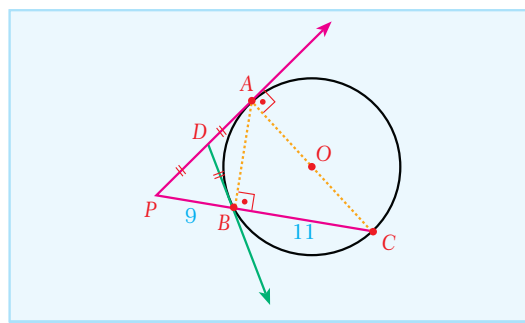
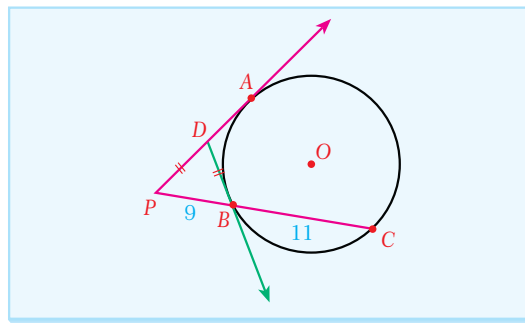
$$AP^2 + AC^2 = PC^2$$

$$(6\sqrt{5})^2 + AC^2 = 20^2$$

$$180 + AC^2 = 400$$

$$AC^2 = 220$$

$$AC = 2\sqrt{55} \text{ cm. Therefore the radius is } \sqrt{55} \text{ cm.}$$



EXAMPLE

68

In the figure, PB is a common tangent, $PA = AB$, $CD = 5$ cm and $DE = 3$ cm. Find AB .

Solution Let $AB = AP = x$ and $PC = y$.

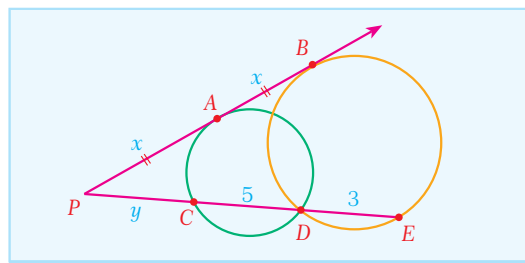
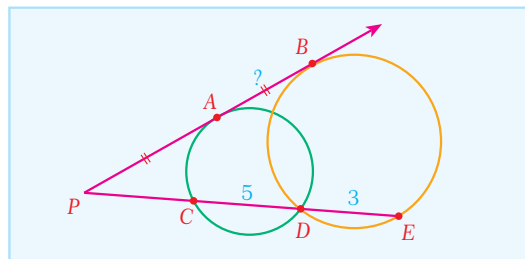
$$\text{Then } x^2 = y(y + 5). \quad (1)$$

$$\text{Also, } (2x)^2 = (y + 5)(y + 8). \quad (2)$$

$$\frac{(2)}{(1)} \Rightarrow 4 = \frac{y+8}{y} \Rightarrow 4y = y+8$$

$$3y = 8$$

$$y = \frac{8}{3} \text{ cm.}$$



$$\begin{aligned}
 \text{Finally, } x^2 &= y(y+5) \\
 &= \frac{8}{3} \left(\frac{8}{3} + 5 \right) \\
 &= \frac{8 \cdot 23}{9} \\
 x = AB &= \frac{2}{3} \sqrt{46} \text{ cm.}
 \end{aligned}$$

EXAMPLE

69

In the figure, DE and AC are the diameters of the semicircles and F and G are points of tangency. Given $AF = 3$ cm and $CG = 12$ cm, find AC .

Solution

Look at the figure.

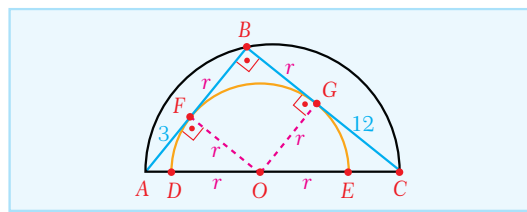
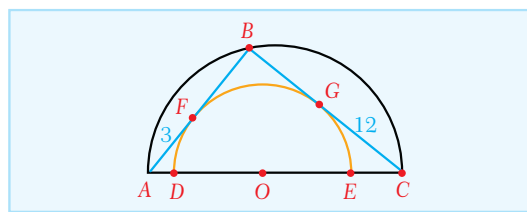
$$m(\angle B) = m(\angle OGB) = m(\angle OFA) = 90^\circ$$

(Can you see why?)

So $\triangle OGC \sim \triangle ABC$.

$$\begin{aligned}
 \text{Then } \frac{r}{r+3} &= \frac{12}{12+r} \Rightarrow 12r + 36 = 12r + r^2 \\
 & \qquad \qquad \qquad r = 6 \text{ cm.}
 \end{aligned}$$

$$\begin{aligned}
 \text{In } \triangle ABC, \quad AB^2 + BC^2 &= AC^2 \\
 9^2 + 18^2 &= AC^2 \\
 AC &= 9\sqrt{5} \text{ cm.}
 \end{aligned}$$



EXAMPLE

70

The figure shows $\odot O$ with a point of tangency at D . $AB \perp AE$, $AB = 4$ cm, $AD = 8$ cm and $OE = 15$ cm are given. Find DE .

Solution

Let us extend AB to C as shown. Then

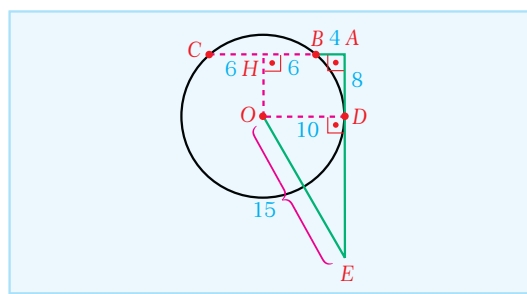
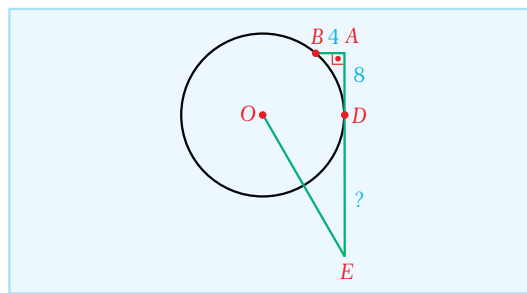
$$AD^2 = AB \cdot AC$$

$$8^2 = 4 \cdot AC \Rightarrow AC = 16 \text{ cm.}$$

$$\text{So } BC = 16 - 4 = 12 \text{ cm.}$$

Let us draw OH so that $OH \perp BC$.

Then $CH = HB$. (Can you see why?)



Since $ADOH$ is a rectangle, $AH = DO = 10$ cm.

In $\triangle DOE$, $DO^2 + DE^2 = OE^2$

$$10^2 + DE^2 = 15^2$$

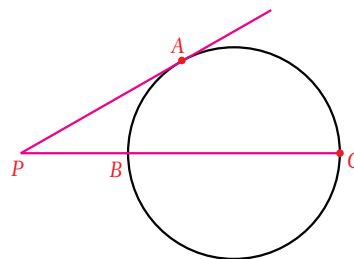
$$DE^2 = 125$$

$$DE = 5\sqrt{5} \text{ cm.}$$

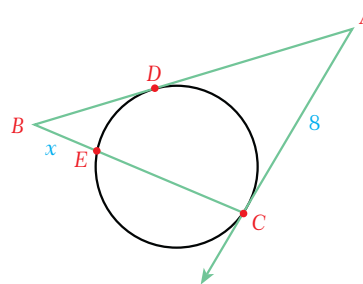
Check Yourself

- Two secant lines passing through a point P intersect a circle at points A, B, C and D such that P, D, A and P, C, B are respectively collinear. A, B, C and D are located clockwise on the circle, and $PC = 4$ cm, $BC = 8$ cm and $AD = 3 \cdot DP$ are given. Find PA .
- ABC is a triangle and AE and CD are two altitudes inside the triangle. $BE = 6$ cm, $EC = 8$ cm and $BD = 9$ cm are given. Find AD .

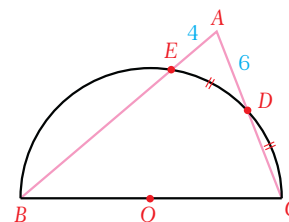
- In the figure, $PC = 4 \cdot PA$. What is $\frac{PB}{BC}$?



- In the figure, $AB = 14$ cm, $AC = 8$ cm and $BC = 9$ cm. Find the length $BE = x$.



- In the figure, $m(\widehat{DE}) = m(\widehat{DC})$, $AD = 6$ cm and $AE = 4$ cm. Find the radius of the semicircle with diameter BC .



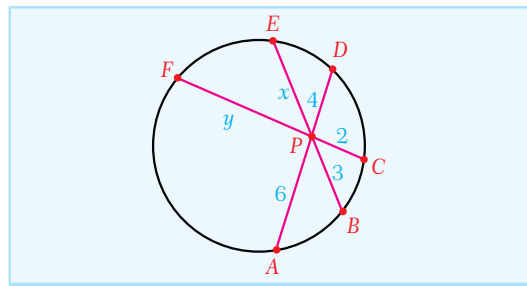
Answers

- $8\sqrt{3}$ cm
- $\frac{1}{3}$ cm
- $\frac{1}{15}$
- 4 cm
- 9 cm

EXAMPLE 71 Find the value of $x + y$ in the figure.

Solution By the corollary we have just seen,
 $PA \cdot PD = PB \cdot PE = PC \cdot PF$, i.e.
 $6 \cdot 4 = 3 \cdot x = 2 \cdot y$.

So $x = 8$ cm and $y = 12$ cm, and $x + y = 8 + 12 = 20$ cm.



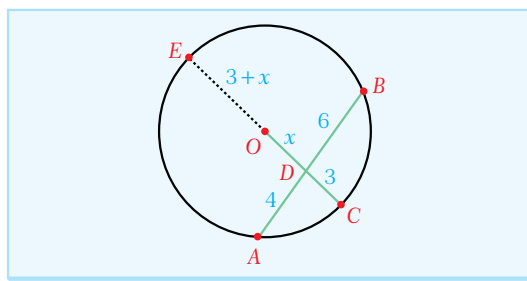
EXAMPLE 72 A, C and B are three points located counter-clockwise on a circle centered at O, and D is the intersection point of AB and OC. If $AD = 4$ cm, $CD = 3$ cm and $BD = 6$ cm, find OD.

Solution Look at the figure. Let $OD = x$ and let us extend CO to point E on the circle.

Then $OC = OE = 3 + x$.

By the power of a point theorem,

$$\begin{aligned} 6 \cdot 4 &= 3(3 + 2x) \\ 8 &= 3 + 2x \\ x &= \frac{5}{2} = 2.5 \text{ cm.} \end{aligned}$$



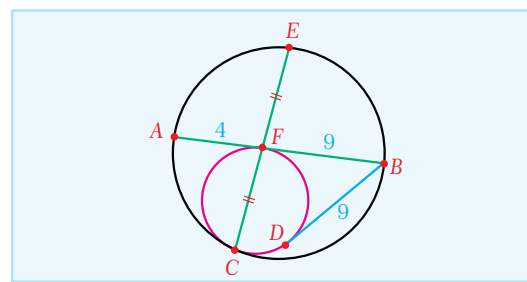
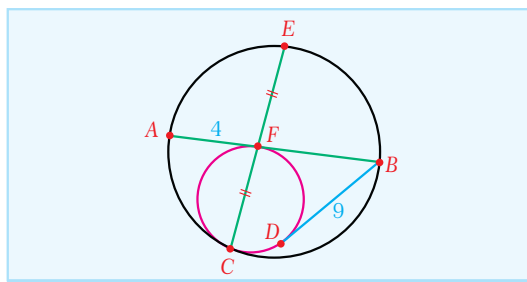
EXAMPLE 73 The two circles in the figure are internally tangent at C. $EF = FC$, $AF = 4$ cm and $BD = 9$ cm are given. Find the length EC.

Solution $BD = BF = 9$ cm (common tangent segments).

So

$$\begin{aligned} AF \cdot BF &= EF \cdot FC \\ 4 \cdot 9 &= EF^2 \\ EF &= 6 \text{ cm.} \end{aligned}$$

So $EC = 2 \cdot 6 = 12$ cm.



EXAMPLE

74

Find the values of x and y using the information in the figure, given that A is a point of tangency.

Solution

$$AP^2 = PE \cdot PD$$

$$8^2 = 4(4 + x + 3)$$

$$16 = 7 + x$$

$$x = 9.$$

$$DF \cdot FE = BF \cdot FC$$

$$3 \cdot x = 5 \cdot y$$

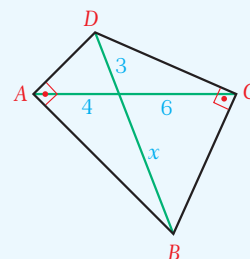
$$3 \cdot 9 = 5y$$

$$y = \frac{27}{5}.$$

EXAMPLE

75

In the figure, $m(\angle BAD) = m(\angle BCD) = 90^\circ$. Find the length x .



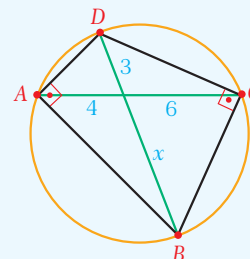
Solution

Since $m(\angle A) = m(\angle C) = 90^\circ$, they are supplementary.

So $ABCD$ is a cyclic quadrilateral and AC and DB are chords.

$$\text{Hence } 4 \cdot 6 = 3 \cdot x$$

$$x = 8.$$



EXAMPLE

76

O is the center of the semicircle in the figure and $AH \perp BC$. Find the length AB.

Solution When we reflect the semicircle along BC we get

$$BH \cdot HC = AH \cdot HA'$$

$$8 \cdot 2 = AH^2$$

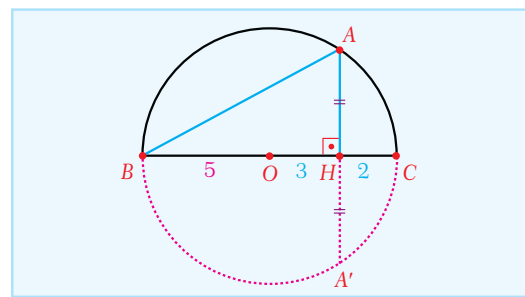
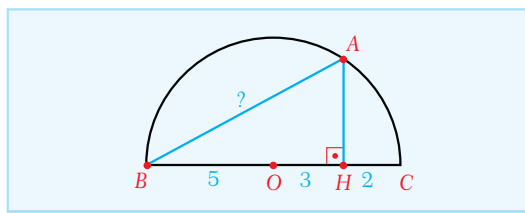
$$AH = 4.$$

In $\triangle ABH$,

$$AB^2 = AH^2 + BH^2$$

$$AB^2 = 4^2 + 8^2$$

$$AB = 4\sqrt{5}.$$



EXAMPLE

77

The figure shows $\odot O$ with $OH \perp BC$, $BP = 4$ cm, $AP = 8$ cm and $5 \cdot OH = 3 \cdot OA$. Find OA.

Solution

$$PA^2 = PB \cdot PC$$

$$8^2 = 4 \cdot PC$$

$$PC = 16 \text{ cm.}$$

$$\text{So } BC = 16 - 4 = 12 \text{ cm.}$$

$$\text{Since } OH \perp BC, BH = HC = 6 \text{ cm.}$$

Let us extend OH to make it a diameter.

$$\text{Then let } OH = 3k \text{ and } OA = 5k.$$

$$\text{So } HE = 2k \text{ and } DO = 5k.$$

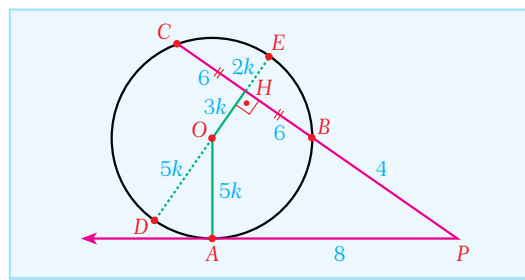
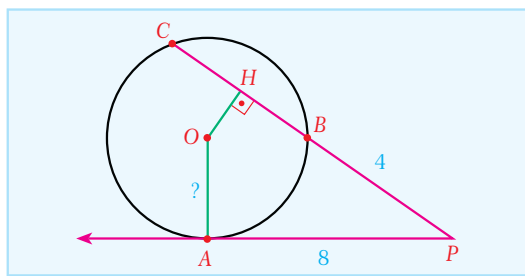
$$\text{Then } BH \cdot HC = DH \cdot HE$$

$$6 \cdot 6 = 8k \cdot 2k$$

$$k^2 = \frac{36}{16}$$

$$k = \frac{3}{2}.$$

$$\text{Finally, } OA = 5k = 5 \cdot \frac{3}{2} = \frac{15}{2} \text{ cm.}$$



EXAMPLE

78

O is the center of the semicircle in the figure. Find the length BD.

Solution Let us draw AC. Then $AC \perp BC$. (Can you see why?)

In $\triangle ABC$, since BE is an angle bisector, by the angle bisector theorem we have

$$\frac{BC}{AB} = \frac{CE}{AE} \Rightarrow \frac{12}{20} = \frac{3}{5} = \frac{CE}{AE}.$$

Now let $CE = 3k$ and $AE = 5k$. Also,

$$AB^2 = BC^2 + AC^2$$

$$20^2 = 12^2 + AC^2$$

$$AC = 16 \text{ cm.}$$

So $16 = 5k + 3k = 8k \Rightarrow k = 2 \text{ cm}$. Hence $CE = 6 \text{ cm}$ and $AE = 10 \text{ cm}$.

In $\triangle BCE$, $BC^2 + CE^2 = BE^2$

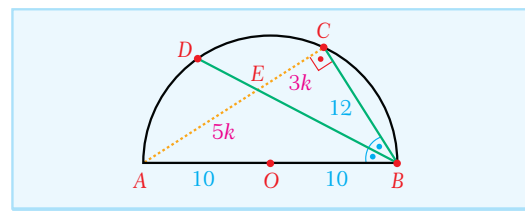
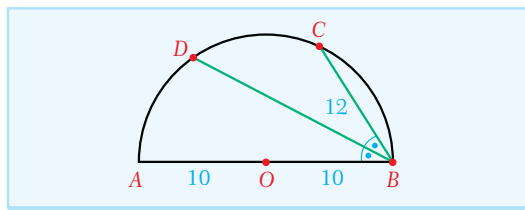
$$12^2 + 6^2 = BE^2$$

$$BE = 6\sqrt{5} \text{ cm.}$$

Then $AE \cdot CE = BE \cdot DE$

$$10 \cdot 6 = 6\sqrt{5} \cdot DE \Rightarrow DE = \frac{60}{6\sqrt{5}} = 2\sqrt{5} \text{ cm.}$$

Finally, $BD = BE + ED = 6\sqrt{5} + 2\sqrt{5} = 8\sqrt{5} \text{ cm}$.



EXAMPLE

79

What is the radius of the circle in the figure?

Solution Let us draw BE such that $BE \perp AC$.

Since $BH \perp AC$ and $\triangle ABC$ is an isosceles triangle, $AH = HC = 5$.

In $\triangle ABH$, $AH^2 + BH^2 = AB^2$

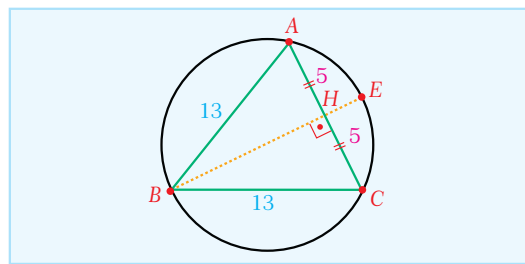
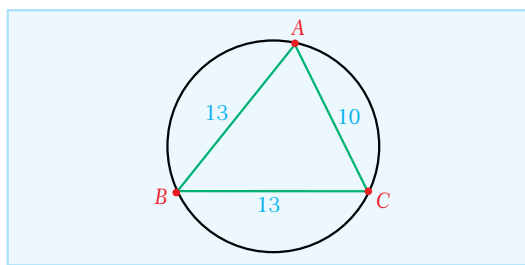
$$BH^2 + 5^2 = 13^2$$

$$BH = 12.$$

So $BH \cdot HE = AH \cdot HC$

$$12 \cdot HE = 5 \cdot 5$$

$$HE = \frac{25}{12}.$$



Then $BE = BH + HE = 12 + \frac{25}{12} = \frac{169}{12}$.

Since BE is a diameter (can you see why?), $r = \frac{169}{12} \cdot \frac{1}{2} = \frac{169}{24}$.

EXAMPLE

80

In the figure, $AH \perp BC$.
Prove the Euclidean theorem
 $h^2 = p \cdot k$.

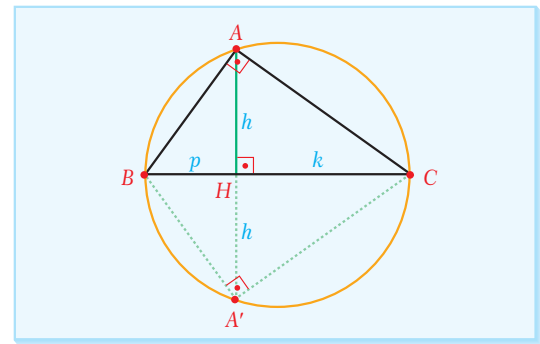
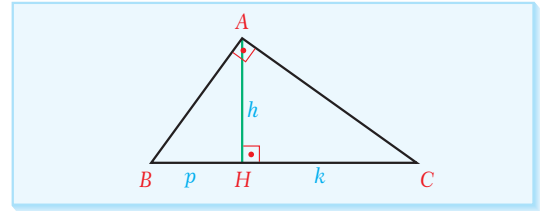
Solution Since $m(\angle A) = 90^\circ$, we can construct a circumscribed circle such that BC is a diameter. Reflecting $\triangle ABC$ along BC gives us $AH = HA' = h$.

By the power of H ,

$$AH \cdot HA' = BH \cdot HC$$

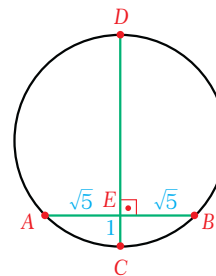
$$h \cdot h = p \cdot k$$

$$h^2 = p \cdot k, \text{ as required.}$$



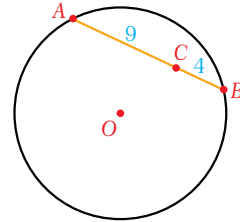
Check Yourself

- Find the radius of the circle by using the given measures.



2. AB is a diameter of a circle centered at O and CD is chord perpendicular to AB . E is the intersection point of AB and CD . If $AO = 6$ cm and $CD = 10$ cm, find the lengths EB and OE ($EB < \text{radius}$).

3. What is the length of the shortest chord which passes through C in the figure?



Answers

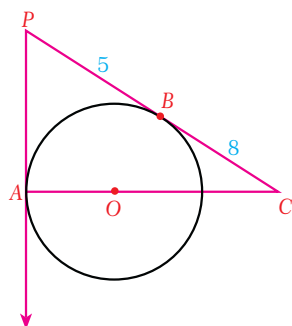
1. 3 2. $EB = (6 - \sqrt{11})$ cm and $OE = \sqrt{11}$ cm 3. 12



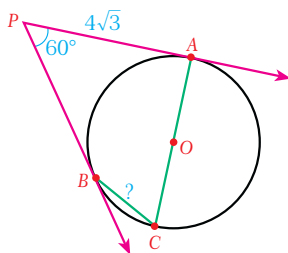
EXERCISES 3.2

A. Segments Formed by Tangents

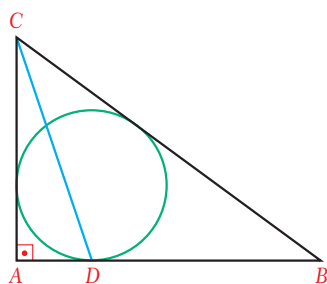
1. The figure shows $\odot O$ and two tangents. Find the radius of the circle.



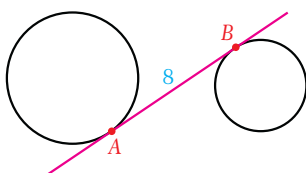
2. A and B in the figure are points of tangency on $\odot O$. Find the length BC.



3. In the figure, $AB = 8$ cm and $AC = 6$ cm. Find CD.

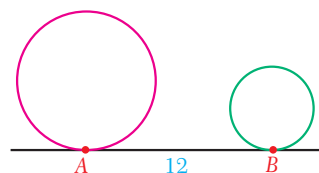


4. The radii of the two circles shown at the right are 4 cm and 2 cm.

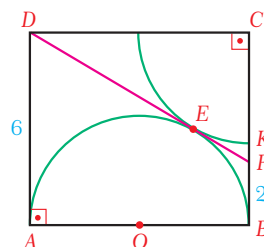


What is the shortest distance between the two circles?

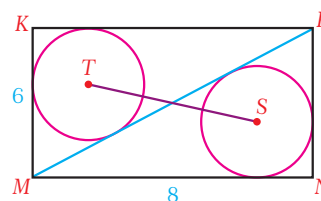
5. The radii of the circles opposite are 8 cm and 3 cm. What is the distance between the circles at their closest point?



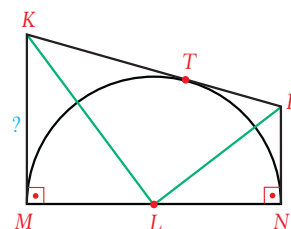
6. In the figure, $ABCD$ is a rectangle and O and C are the centers of the arcs. Given $AD = 6$ cm and $BF = 2$ cm, find FK .



7. In the figure, $\odot T$ and $\odot S$ are circles and $MNPK$ is a rectangle with $MN = 8$ cm and $MK = 6$ cm. Find TS .

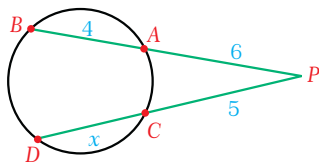


8. In the figure, $KL = 20$ cm and $PL = 15$ cm. Find KM .



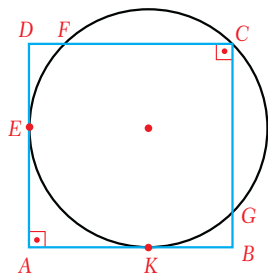
B. Segments Formed by Secants and Chords

9. Find the length $DC = x$ in the figure.

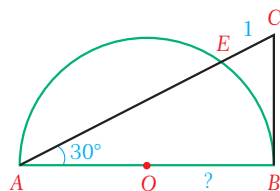


10. A , B , C and D are four points located counter-clockwise on a circle. P is point outside the circle and PA is tangent to the circle at A . P , D and B are collinear and E is the intersection point of AC and BD . If $AP = 8$ cm, $PD = 4$ cm and $DE = 5$ cm, find $AE \cdot EC$.

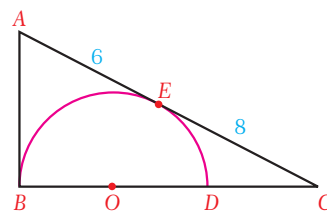
11. $ABCD$ in the figure is a rectangle and K and E are points of tangency. Given $CG = 16$ cm and $BG = 2$ cm, find DF .



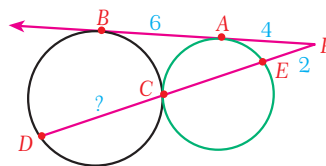
12. CB in the figure is tangent to the circle $\odot O$. Find the radius of $\odot O$.



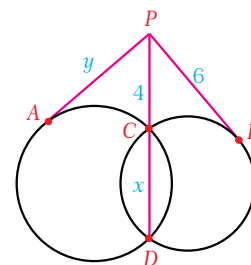
13. AB and AC at the right are tangent to $\odot O$. Find the radius of $\odot O$.



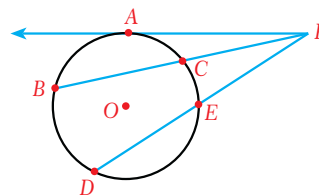
14. Find the length CD in the figure.



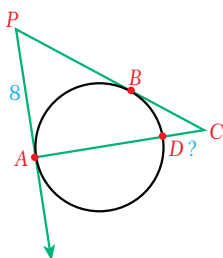
15. Find the lengths x and y in the figure.



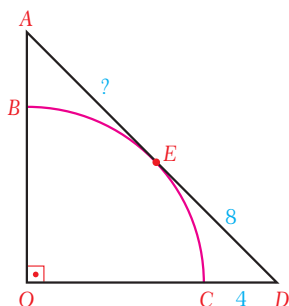
16. In the figure, $PA = 4\sqrt{3}$ cm, $BC = PC$ and $2PE = DE$. Find $PE + BC$.



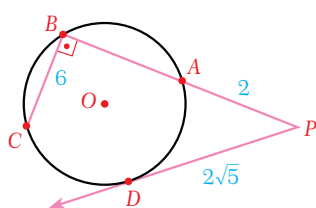
17. In the figure,
 $PA = 8$ cm,
 $PC = 14$ cm and
 $AC = 12$ cm.
 Find CD .



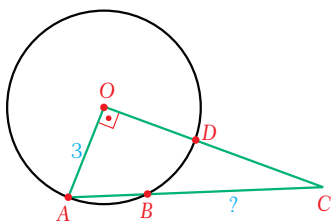
18. O is the center of the quarter circle shown at the right. Find the length of segment AE using the information given.



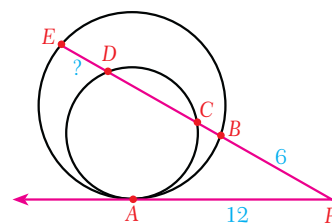
19. Find the radius of the circle $\odot O$ in the figure.



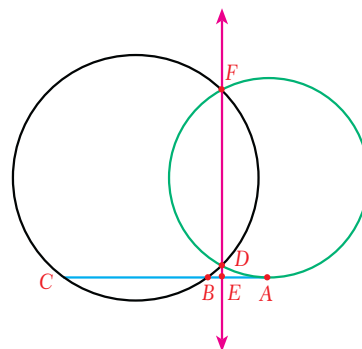
20. O is the center of the circle in the figure. Given $AC = 5$ cm and $OA = 3$ cm, find BC .



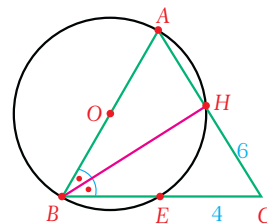
21. In the figure,
 $PA = 12$ cm,
 $PB = 6$ cm and
 $BC = 2$ cm.
 Find DE .



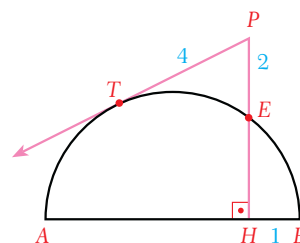
22. In the figure,
 $AE = 4$ cm and
 $EB = 2$ cm.
 Find the length of segment AC .



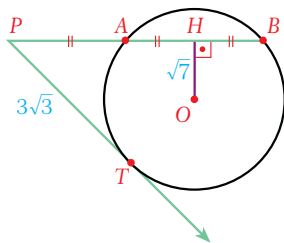
23. Find BE in the figure, using the information given.



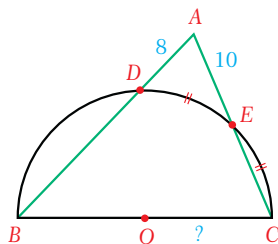
24. AB is a diameter of the semicircle at the right. Find AB .



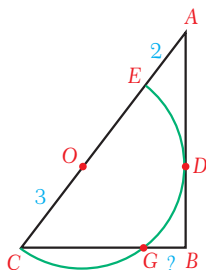
25. In the figure,
 $PA = AH = HB$,
 $PT = 3\sqrt{3}$ cm and
 $OH = \sqrt{7}$ cm.
 What is the radius
 of $\odot O$?



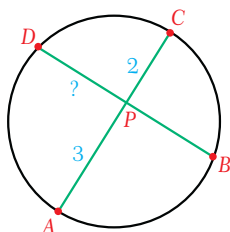
26. $\star$ O is the center of
 the semicircle
 shown at the right.
 $AE = 10$ cm,
 $AD = 8$ cm and
 $m(\widehat{DE}) = m(\widehat{EC})$
 are given.
 Find the radius of $\odot O$.



27. $\star$ In the figure, O is
 the center of the
 semicircle,
 $AE = 2$ cm and
 $OC = 3$ cm.
 Find BG.

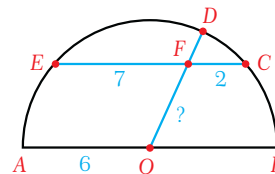


28. In the figure,
 $m(\widehat{ADC}) = m(\widehat{BCD})$,
 $CP = 2$ cm and
 $AP = 3$ cm.
 Find DP.

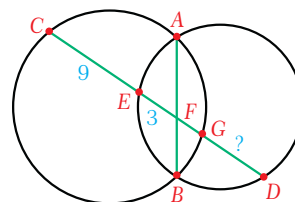


29. OA is a radius of a circle centered at O, and BC is
 a chord of this circle such that $AO \perp BC$. D is the
 intersection point of OA and BC. Given $AD = 3$ cm
 and $CD = 6$ cm, find OD.

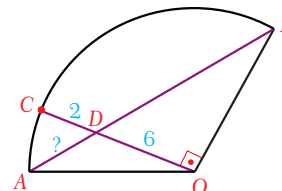
30. O is the center of
 the semicircle shown
 in the figure.
 Find OF by using the
 information given.



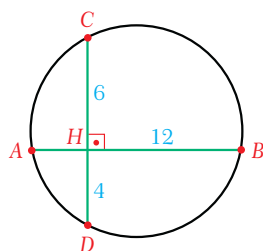
31. In the figure,
 $CE = 9$ cm,
 $EF = 3$ cm and
 $FG = 2$ cm.
 Find GD.



32. $\widehat{ACB}$ is an arc of $\odot O$
 in the figure. If
 $CD = 2$ cm and
 $DO = 6$ cm,
 find AD.

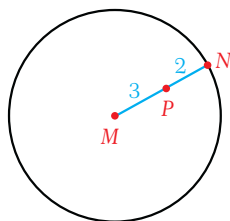


33. Find the radius of the circle in the figure, using the given information.



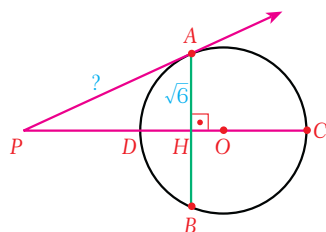
34. A, B, C and D are four points located clockwise on a circle centered at O such that BD is a diameter of the circle. AC and BD intersect at E, $m(\angle BAC) = 60^\circ$, $DE = 4$ cm and $BE = 8$ cm. Find BC.

35. In the figure, M is the center of the circle, $MP = 3$ cm and $PN = 2$ cm.

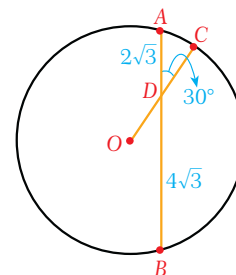


What is the length of the shortest chord which passes through P?

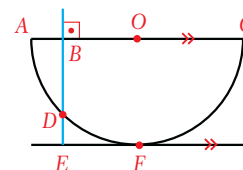
36. O is the center of the circle shown at the right. $AH = \sqrt{6}$ cm and $HC = 3$ cm are given. Find AP.



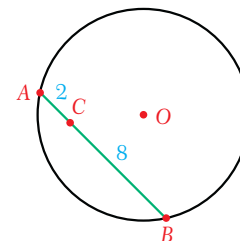
37. O is the center of the circle in the figure. $m(\angle ADC) = 30^\circ$, $AD = 2\sqrt{3}$ cm and $BD = 4\sqrt{3}$ cm are given. Find CD.



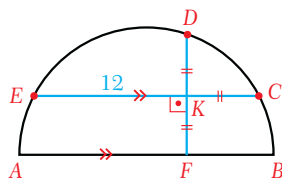
38. In the figure, $AC \parallel EF$ and EF is tangent to the semicircle at F. If $AB = 8$ cm and $BD = 12$ cm, find DE.



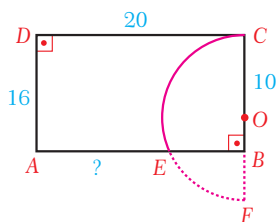
39. In the figure, $AC = 2$ cm and $BC = 8$ cm. Find the length of the shortest chord which can be drawn through C.



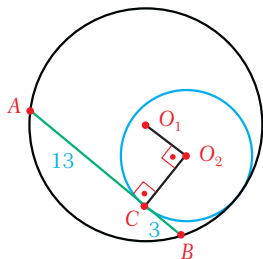
40. AB is a diameter of the semicircle at the right. $EC \parallel AB$, $DK = FK = KC$ and $EK = 12$ cm are given. Find the radius of the semicircle.



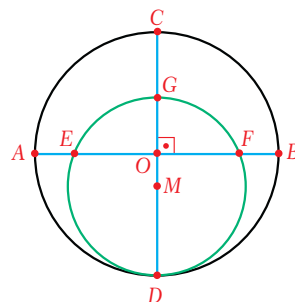
41. $ABCD$ in the figure is a rectangle and O is the center of the semicircle. If $AD = 16$ cm, $CD = 20$ cm and $OC = 10$ cm, find AE .



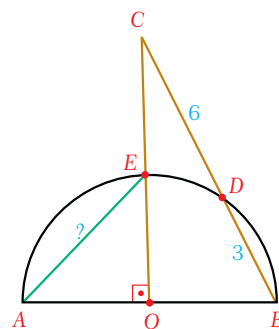
42. O_1 and O_2 are the centers of the circles shown opposite. If $BC = 3$ cm and $AC = 13$ cm, what is the sum of the radii of the circles?



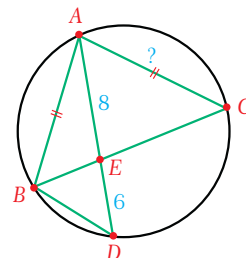
43. O and M are the centers of the two circles in the figure and D is a point of tangency. Find OF if $OG = 12$ cm and $AE = 6$ cm.



44. In the figure, O is the center of the semicircle, $BD = 3$ cm and $CD = 6$ cm. Find AE .



45. In the figure, $AB = AC$, $AE = 8$ cm and $DE = 6$ cm. Find AC .



TRIGONOMETRIC THEOREMS AND FORMULAS

A. TRIGONOMETRIC THEOREMS

We now know how to calculate one or more angles or side lengths in a right triangle from given information about the triangle. However, there are also relations between the angles and sides of any triangle (not just right triangles). In this section we will study these general relations and use them to solve triangle problems.

1. The Law of Cosines

Theorem

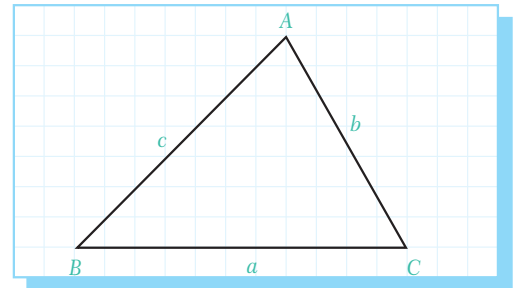
law of cosines

In a triangle ABC with side lengths a , b and c ,

$$a^2 = b^2 + c^2 - 2bc \cdot \cos A$$

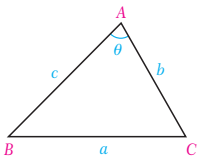
$$b^2 = a^2 + c^2 - 2ac \cdot \cos B$$

$$c^2 = a^2 + b^2 - 2ab \cdot \cos C$$



Proof

In a triangle ABC , a is the side opposite A , b is the side opposite B and c is the side opposite C .



a , b and c can also mean the lengths of sides a , b and c respectively.

We also write A to mean the angle at A , B to mean the angle at B and C to mean the angle at C : in the figure, $m(A) = \theta$.

First we draw CH as an altitude of the triangle. Point H now divides AB into two segments such that $AH = x$ and $HB = c - x$. Consequently there exist two right triangles: $\triangle CAH$ and $\triangle CHB$.

Applying the Pythagorean Theorem to each right triangle gives us

$$h^2 + (c - x)^2 = a^2 \quad (1)$$

$$h^2 + x^2 = b^2. \quad (2)$$

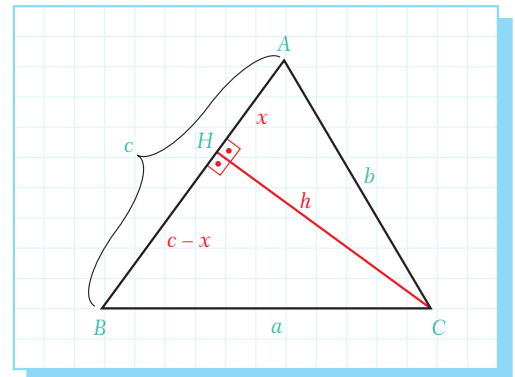
We can subtract these side by side:

$$\begin{array}{r} h^2 + (c - x)^2 = a^2 \\ - \quad h^2 + x^2 = b^2 \\ \hline \end{array}$$

$$h^2 + (c - x)^2 - h^2 - x^2 = a^2 - b^2$$

$$c^2 - 2cx + \cancel{x^2} - \cancel{x^2} = a^2 - b^2.$$

$$\text{So } a^2 = b^2 + c^2 - 2cx. \quad (3)$$



In the right triangle CAH ,

$$\cos A = \frac{x}{b}, \text{ i.e. } x = b \cdot \cos A.$$

Substituting this value of x in (3) gives us

$$a^2 = b^2 + c^2 - 2cb \cdot \cos A.$$

We can rewrite this as

$$a^2 = b^2 + c^2 - 2bc \cdot \cos A,$$

which is the required result.

The proofs of the other two identities are similar.

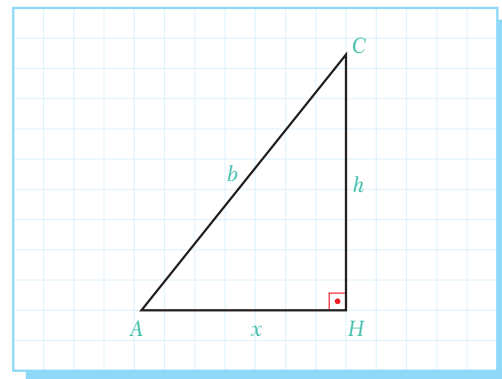
They are left as an exercise for you.

Notice that by rewriting the three parts of the law of cosines we obtain the relations

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}.$$



Note

By the law of cosines:

1. If two sides of a triangle and the angle between them are known, it is possible to calculate the length of the third side of the triangle.
2. If the three sides of a triangle are known, it is possible to calculate the trigonometric values of the angles in the triangle.

EXAMPLE

1 In the figure, ABC is a triangle with $AC = 4$, $BC = 6$ and $m(\angle ACB) = 60^\circ$. Find AB .

Solution By the law of cosines,

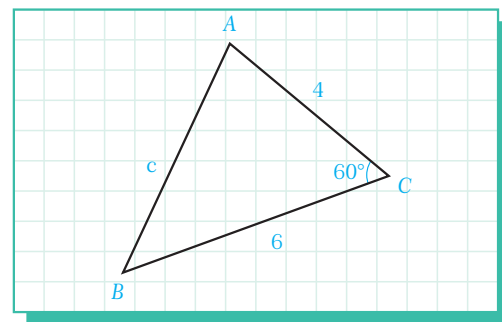
$$c^2 = a^2 + b^2 - 2ab \cdot \cos C$$

$$c^2 = 6^2 + 4^2 - 2 \cdot 6 \cdot 4 \cdot \cos 60^\circ$$

$$c^2 = 36 + 16 - \cancel{2} \cdot 6 \cdot 4 \cdot \frac{1}{\cancel{2}}$$

$$c^2 = 28$$

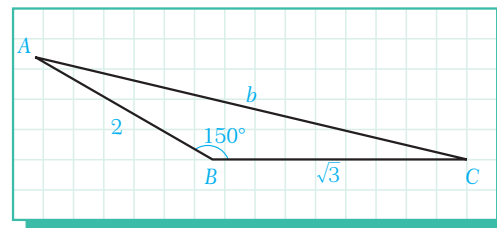
$$c = \sqrt{28} = 2\sqrt{7}.$$



EXAMPLE

2

In the figure, ABC is a triangle with $AB = 2$, $BC = \sqrt{3}$ and $m(\angle ABC) = 150^\circ$. Find b .



Solution By the law of cosines,

$$b^2 = a^2 + c^2 - 2ac \cdot \cos B$$

$$b^2 = (\sqrt{3})^2 + 2^2 - 2 \cdot \sqrt{3} \cdot 2 \cdot \left(-\frac{\sqrt{3}}{2}\right) \quad \left(\cos 150^\circ = -\frac{\sqrt{3}}{2}\right)$$

$$b^2 = 3 + 4 + 6$$

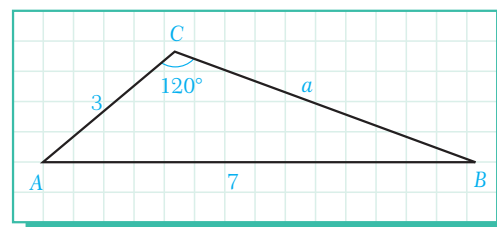
$$b^2 = 13$$

$$b = \sqrt{13}.$$

EXAMPLE

3

The figure shows a triangle ABC with $AC = 3$, $AB = 7$ and $m(\angle ACB) = 120^\circ$. Find a .



Solution By the law of cosines,

$$c^2 = a^2 + b^2 - 2ab \cdot \cos C$$

$$7^2 = a^2 + 3^2 - 2 \cdot a \cdot 3 \cdot \left(-\frac{1}{2}\right) \quad (\cos 120^\circ = -\frac{1}{2})$$

$$49 = a^2 + 9 + 3a; \quad (a-5)(a+8) = 0;$$

$$a^2 + 3a - 40 = 0; \quad a = 5, a = -8.$$

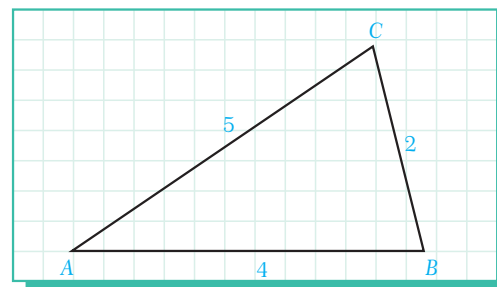
$$\begin{matrix} \nearrow & \searrow \\ (-5) & (8) \end{matrix}$$

Since the length of a side cannot be negative, $a = 5$.

EXAMPLE

4

Find $\cos A$ in the triangle in the figure.



Solution By the law of cosines,

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$= \frac{5^2 + 4^2 - 2^2}{2 \cdot 5 \cdot 4}$$

$$= \frac{25 + 16 - 4}{40}$$

$$= \frac{37}{40}.$$

EXAMPLE

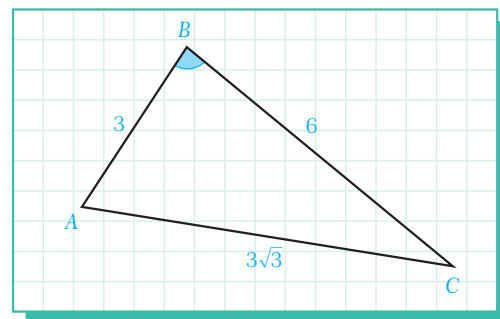
5

Find the measure of angle B in the figure.

Solution By the law of cosines,

$$\begin{aligned}\cos B &= \frac{a^2 + c^2 - b^2}{2ac} \\ &= \frac{3^2 + 6^2 - (3\sqrt{3})^2}{2 \cdot 3 \cdot 6} \\ &= \frac{9 + 36 - 27}{36} \\ &= \frac{18}{36} = \frac{1}{2}.\end{aligned}$$

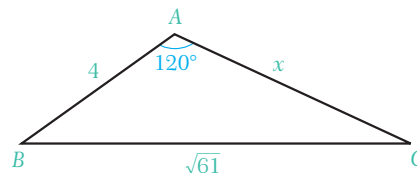
So $\cos B = \frac{1}{2}$. We know that $m(\angle B) < 90^\circ$ and $\cos 60^\circ = \frac{1}{2}$. Therefore $m(\angle B) = 60^\circ$.



Check Yourself

- The side lengths of a triangle are 4, 5 and 6 units respectively. Find the cosine of the smallest angle in the triangle.

- ABC is a triangle with $AB = 4$, $BC = \sqrt{61}$ and $m(\angle BAC) = 120^\circ$. Find AC .



Answers

- 0.75
- 5

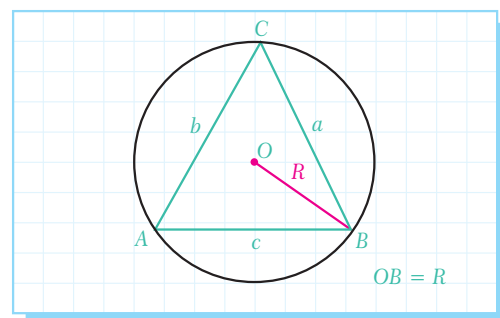
2. The Law of Sines

Theorem

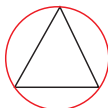
law of sines

Let ABC be a triangle with side lengths a , b and c which is inscribed in a circle with radius R .

$$\text{Then } \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R.$$



Inscribed and circumscribed:



The black line shows an inscribed triangle.
The red line shows a circumscribed circle.

Proof Look at the figure.

In the right triangle CDA ,

$$\sin A = \frac{h_c}{b}, \text{ i.e. } h_c = b \cdot \sin A. \quad (1)$$

In the right triangle CDB ,

$$\sin B = \frac{h_c}{a}, \text{ i.e. } h_c = a \cdot \sin B. \quad (2)$$

From (1) and (2)

we obtain $b \cdot \sin A = a \cdot \sin B$, i.e.

$$\frac{a}{\sin A} = \frac{b}{\sin B}. \quad (I)$$

Similarly, in the right triangle BFA we have $\sin A = \frac{h_b}{c}$, i.e. $h_b = c \cdot \sin A$. (3)

Finally, in the right triangle BFC , $\sin C = \frac{h_b}{a}$ which gives $h_b = a \cdot \sin C$. (4)

From (3) and (4) we obtain $c \cdot \sin A = a \cdot \sin C$, i.e. $\frac{a}{\sin A} = \frac{c}{\sin C}$. (II)

From (I) and (II) we obtain $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$. (Result 1)

To show the relation with the radius R of the circumscribed circle, we can use the figure opposite.

M is a point on the circle such that MC is the diameter of the circumscribed circle and $m(\angle MBC) = 90^\circ$. Since $\angle BAC$ and $\angle BMC$ are circumscribed angles having the same arc $\widehat{BC}$ on the circle, $m(\angle BAC) = m(\angle BMC)$.

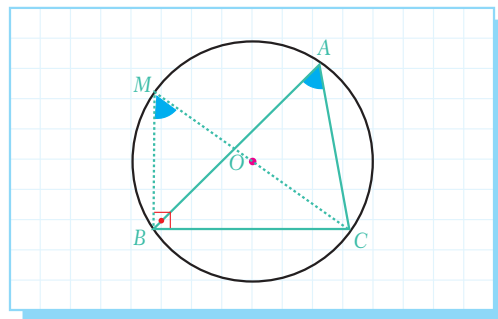
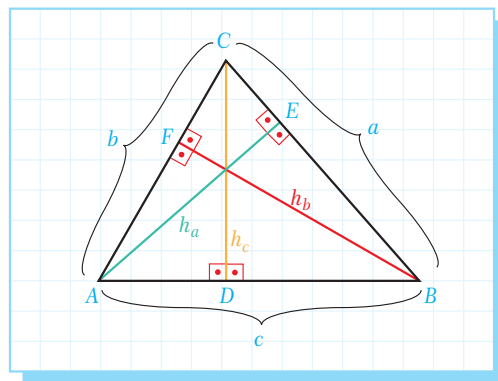
Since $m(\angle M) = m(\angle A)$ we have $\sin M = \sin A$.

In the right triangle MBC ,

$$\sin M = \frac{a}{MC} = \frac{a}{2R}, \text{ i.e. } \sin A = \frac{a}{2R}.$$

So $\frac{a}{\sin A} = 2R$. (Result 2)

Combining Result 1 and Result 2 gives us $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$.



Note

Let ABC be an inscribed triangle. By the law of sines:

1. If the measures of the three angles and the radius of circumscribed circle are known, it is possible to find the side lengths of the triangle.
2. If the radius of the circumscribed circle and the side lengths of the triangle are known, it is possible to calculate the trigonometric values of the angles of the triangle.



EXAMPLE

6

In a triangle ABC , $m(\angle A) = 30^\circ$ and the length of side a is 8 cm. Find the area of the circumscribed circle of $\triangle ABC$.

Solution

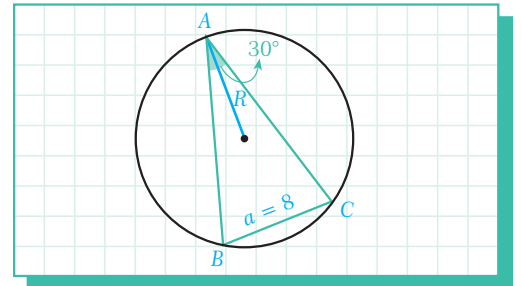
By the law of sines,

$$\frac{a}{\sin A} = 2R \quad \text{so} \quad \frac{8}{\sin 30^\circ} = 2R.$$

$$\text{So } \frac{8}{\frac{1}{2}} = 2R, \text{ i.e. } R = 8 \text{ cm.}$$

We can calculate the area A of a circle with the formula $A = \pi R^2$.

$$\text{So the area is } \pi \cdot 8^2 = 64\pi \text{ cm}^2.$$



EXAMPLE

7

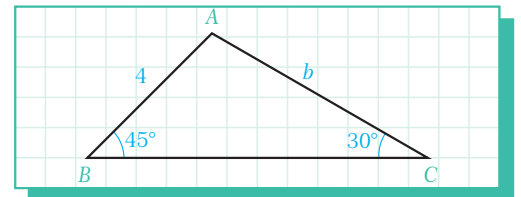
In a triangle ABC , $m(\angle B) = 45^\circ$, $m(\angle C) = 30^\circ$ and the length of side c is 4 cm. Find the length of side b .

Solution

By the law of sines,

$$\frac{c}{\sin C} = \frac{b}{\sin B}, \text{ i.e. } \frac{4}{\sin 30^\circ} = \frac{b}{\sin 45^\circ}.$$

$$\text{So } \frac{4}{\frac{1}{2}} = \frac{b}{\frac{\sqrt{2}}{2}} \quad \text{and} \quad b = 4\sqrt{2} \text{ cm.}$$



EXAMPLE**8**

In a triangle ABC , the measures of the interior angles are 30° , 60° and 90° respectively. The radius of the triangle's circumscribed circle is 6 cm. Find the perimeter of the triangle ABC .

Solution By the law of sines,

$$\frac{a}{\sin 30^\circ} = \frac{b}{\sin 60^\circ} = \frac{c}{\sin 90^\circ} = 2 \cdot 6 = 12. \text{ This gives}$$

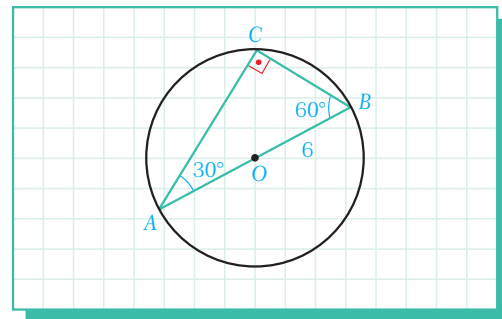
$$a = \sin 30^\circ \cdot 12 = \frac{1}{2} \cdot 12 = 6 \text{ cm,}$$

$$b = \sin 60^\circ \cdot 12 = \frac{\sqrt{3}}{2} \cdot 12 = 6\sqrt{3} \text{ cm and}$$

$$c = \sin 90^\circ \cdot 12 = 12 \text{ cm.}$$

Since the perimeter $P(\triangle ABC) = a + b + c$ we can write

$$P(\triangle ABC) = 6 + 6\sqrt{3} + 12 = 18 + 6\sqrt{3} \text{ cm.}$$

**Check Yourself**

1. In a triangle ABC , $a = 5$ cm and the radius of the circumscribed circle is 5 cm. Find $m(\angle A)$.
2. In a triangle ABC , $m(\angle A) = 30^\circ$, $m(\angle B) = 135^\circ$ and $b = 4\sqrt{2}$ cm. Find the length of side a .

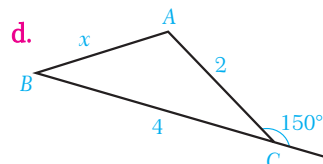
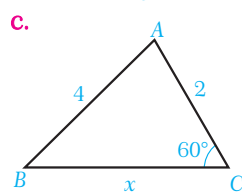
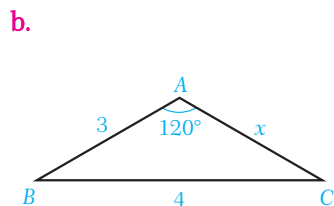
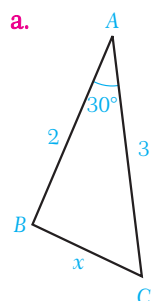
Answers

1. 30° or 150°
2. 4 cm

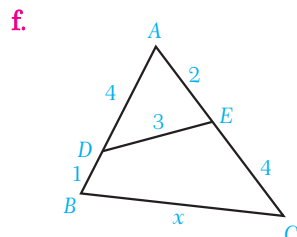
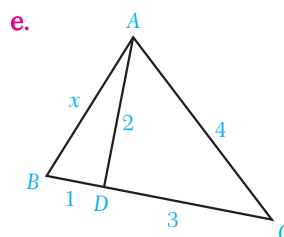
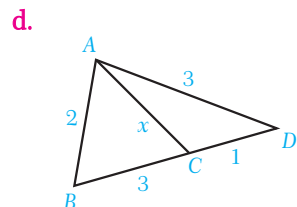
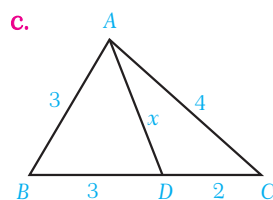
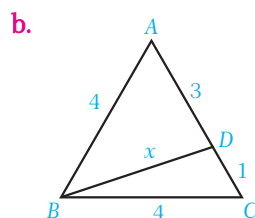
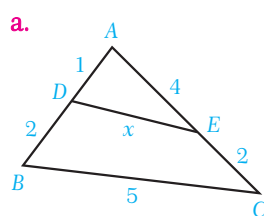
EXERCISES 4

A. Trigonometric Theorems

1. Find the length of side x in each triangle.



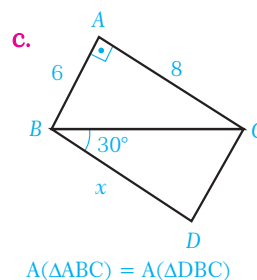
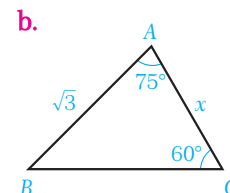
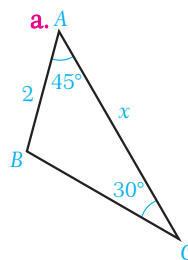
2. Find the length x in each figure.



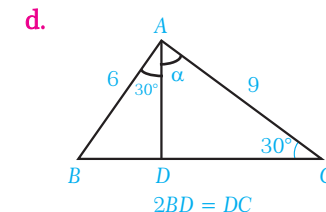
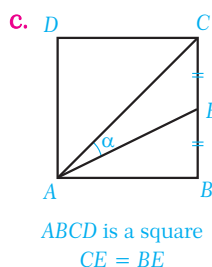
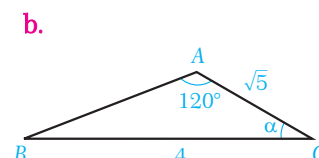
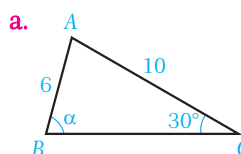
3. A triangle ABC has sides a , b and c such that $a^2 - b^2 - c^2 - (b \cdot c) = 0$. Find $m(\angle A) = \alpha$.

4. A triangle ABC has sides a , b and c such that $c^2 - b^2 = a^2 + \sqrt{2} \cdot b \cdot a$. Find $m(\angle C) = \alpha$.

5. Find the length x in each figure.



6. Calculate $\sin \alpha$ in each figure.



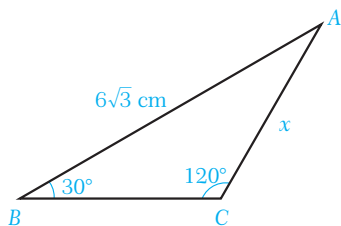
7. The side lengths of a triangle ABC are 3, 7 and 8 units respectively. What is the circumference of the circumscribed circle of the triangle?

8. In a triangle ABC , $m(\angle A) = 75^\circ$, $m(\angle B) = 60^\circ$ and $AB = c = 10$ cm.

a. Find the length of the side b .

b. Find the radius of the circumscribed circle of the triangle.

9. In a triangle ABC ,
 $m(\angle ACB) = 120^\circ$,
 $m(\angle ABC) = 30^\circ$
and $AB = 6\sqrt{3}$ cm.
Find AC .



10. A triangle ABC has side lengths $a = 3$, $b = 3\sqrt{3}$ and $c = 6$. Its circumscribed circle has radius $R = 3$. Find the measure of each interior angle of the triangle.

11. A triangle ABC has sides $a = 7$ cm, $b = 24$ cm and $c = 25$ cm. Find $A(\triangle ABC)$.

12. A triangle ABC has sides of length 8, 15 and 17 units respectively. Find the radius of its circumscribed circle.



POLYGONS



INTRODUCTION TO POLYGONS

In this section we will study geometric figures with many sides, which we call polygons.

A. BASIC CONCEPTS

1. Simple and Closed Curves



The basic undefined concepts in Euclidean geometry are **point**, **line**, **straight line** and **plane**. We use the term 'curve' to describe what Euclid meant by 'line', since a Euclidean line was not necessarily straight.

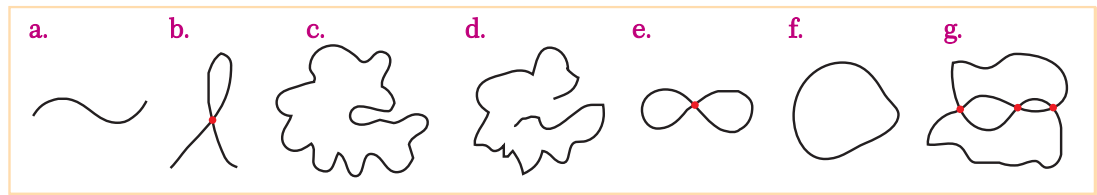
A curve is one of the basic undefined concepts in geometry. We can define a curve informally as any figure which you can draw without lifting your pencil from a piece of paper.

A curve can be called simple, closed, or neither simple nor closed. A simple curve is a curve which never crosses itself in the plane. In other words, it is a curve which you can draw without lifting your pencil from the paper and without passing through any point twice.

A closed curve begins and ends at the same point. In other words, if the final position of your pencil is the same as its starting position, and if you do not lift your pencil from the paper in between, then you have drawn a closed curve.

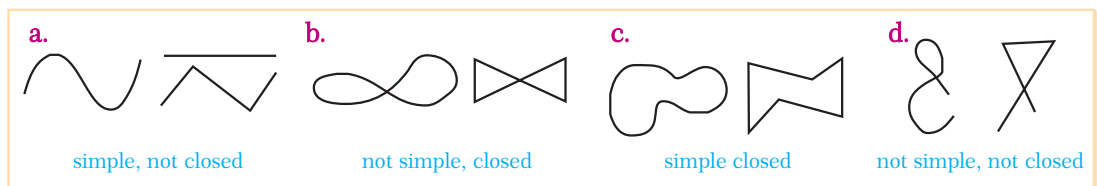
The curves a, c, d and f below are simple curves. The curves b, e and g are not simple, since they pass through the red points shown more than once.

Curves c, e, f and g are closed, and curves a, b and d are not closed.



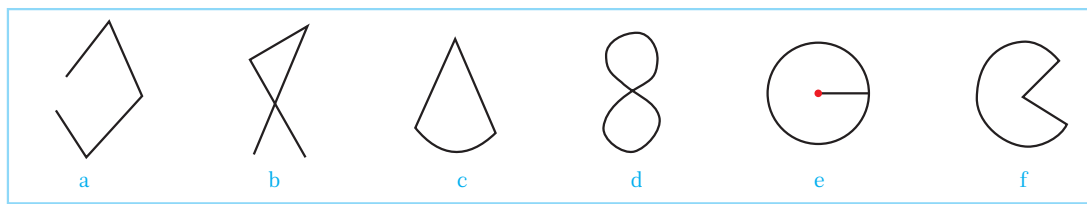
A curve which is both simple and closed is called a **simple closed curve**. Curves **c** and **f** above are simple closed curves.

A curve can also contain straight line segments. Some examples are shown below.



EXAMPLE**1**

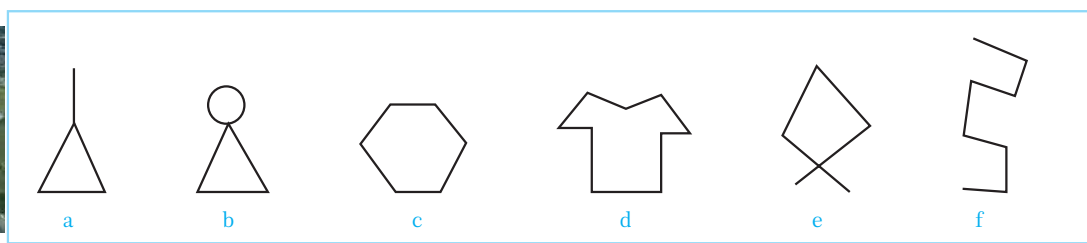
State which curves in the figure are

a. simple.**b.** closed.**c.** simple closed.

Solution **a.** simple: **a**, **c**, **e** and **f**; not simple: **b** and **d**
b. closed: **c**, **d** and **f**; not closed: **a**, **b** and **e**
c. **c** and **f** are simple closed curves

EXAMPLE**2**

Decide which curves in the figure are

a. simple.**b.** closed.**c.** simple closed.

Solution **a.** simple: **a**, **c**, **d** and **f**; not simple: **b** and **e**
b. closed: **b**, **c** and **d**; not closed: **a**, **e** and **f**
c. **c** and **d** are simple closed curves

2. Definitions

Now we are ready to define a polygon as a special type of curve.

Definition**polygon**A **polygon** is a simple closed curve that is made up entirely of straight line segments such that

1. each line segment intersects exactly two other line segments and the segments intersect only at their endpoints, and
2. no two segments with a common endpoint are collinear.



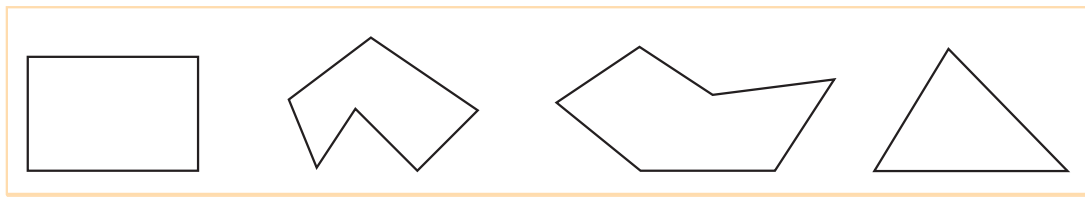
Polygons make a striking design on this soccer ball.

Polygons are the most important type of simple closed curve.

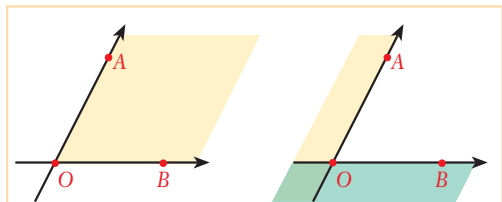
These shapes are examples of polygons:



The word 'polygon' comes from the Greek words *poly* (meaning 'many') and *gonia* (meaning 'angle'). So 'polygon' means 'many angles'.

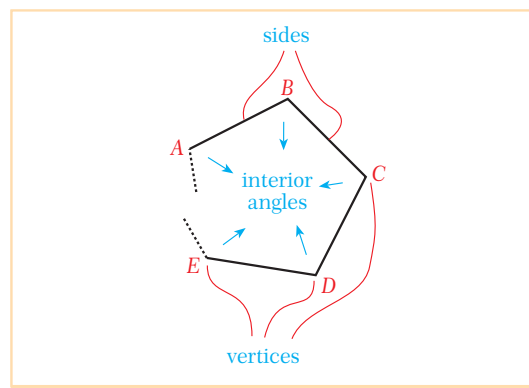


These shapes are not polygons:

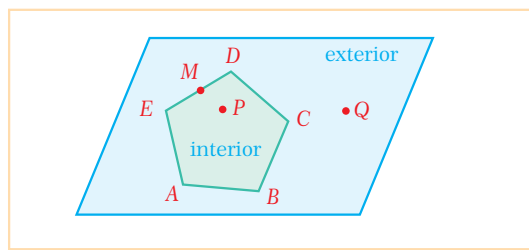


Each line segment in a polygon is called a **side** of the polygon. The intersection point of two line segments is called a **vertex** (plural **vertices**) of the polygon. In a polygon, the number of sides is equal to the number of vertices. Two vertices which share the same side of a polygon are called **consecutive vertices**, and the angles they form with the side are called **consecutive angles**. Two sides which share a common vertex are called **consecutive sides**.

In the adjacent figure, line segments AB , BC and CD are three sides of the polygon. Points A , B , C and D are some of the vertices of the polygon. The vertex pairs $\{A, B\}$, $\{B, C\}$ and $\{C, D\}$ are three examples of consecutive vertices. The pairs $\{A, C\}$ and $\{B, D\}$ are not consecutive vertices. The pairs of sides $\{AB, BC\}$, $\{BC, CD\}$ and $\{CD, DE\}$ are three examples of consecutive sides. The pairs of sides $\{AB, CD\}$ and $\{BC, DE\}$ are not consecutive. The angle pairs $\{\angle ABC, \angle BCD\}$ and $\{\angle BCD, \angle CDE\}$ are two examples of consecutive angles.



A polygon separates a plane into three regions: the **polygon itself**, the **interior region** of the polygon, and the **exterior region** of the polygon. For example, point P in the figure opposite is in the interior of the polygon, point Q is in the exterior of the polygon, and point M is on the polygon.

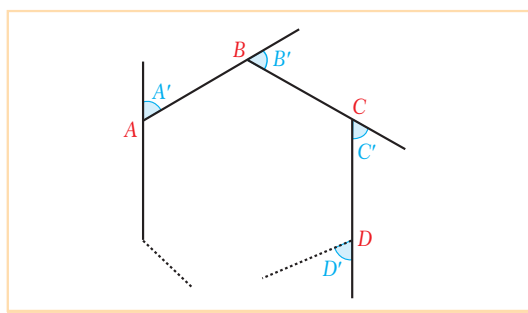




In a polygon with vertices A, B, C , etc. the angles $\angle ABC, \angle BCD, \angle CDE$, etc. are called **interior angles**. We often name these angles simply with the letter of their vertex: $\angle A, \angle B, \angle C$, etc.

We form an **exterior angle** of a polygon by extending one side of the polygon in one direction. In other words, the supplementary angle of the interior angle at a given vertex is called the exterior angle at that vertex. An exterior angle is often represented by a capital letter with a dash ($\angle A', \angle B', \angle C'$, etc.) according to its vertex.

As we can see, $m(\angle A) + m(\angle A') = 180^\circ$, $m(\angle B) + m(\angle B') = 180^\circ$, etc.

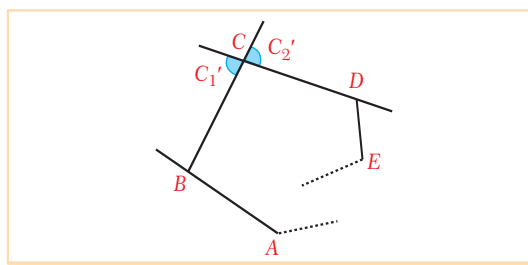


Note

Each vertex of a polygon has two exterior angles and the measures of these two angles are equal. For example, in the figure,

$$m(\angle C_1') = m(\angle C_2').$$

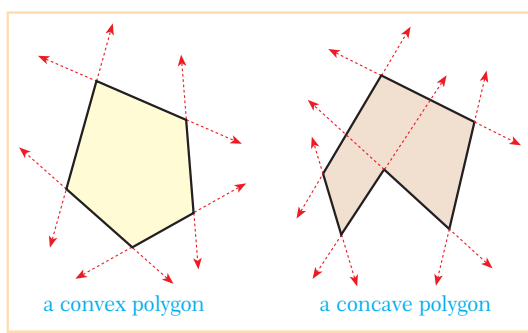
because these are vertical angles.



3. Convex and Concave Polygons

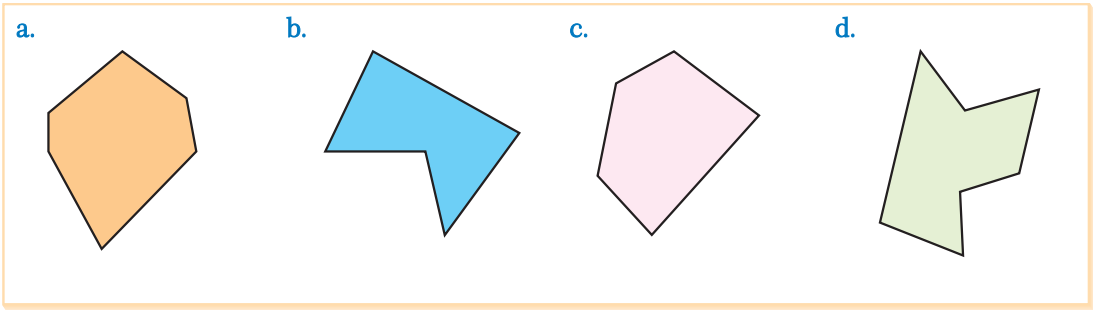
A polygon is called **convex** if the lines which contain its sides do not contain points in the interior of the polygon. It is called **concave** if any of the lines which contain the sides also contain points in the interior of the polygon. The figure at the right shows the difference between a convex polygon and a concave polygon.

We can also determine whether a polygon is convex or concave by checking its angles. If every interior angle of a polygon measures less than 180° , the polygon is convex. If one or more of the interior angles is greater than 180° , the polygon is concave.

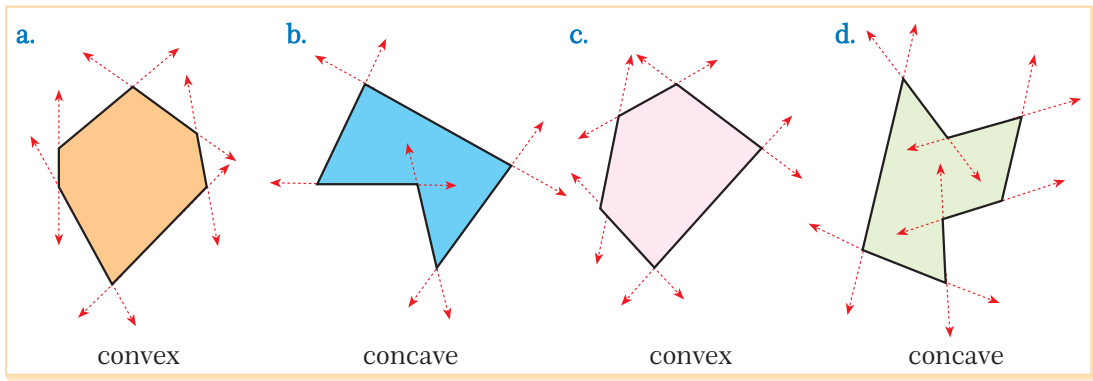


EXAMPLE

3 State whether each polygon is convex or concave.



Solution



4. Types of Polygon

We can classify polygons according to how many sides they have. A **triangle** (three sides) is the simplest type of polygon. A polygon with four sides is called a **quadrilateral**. A polygon with fifteen sides is called a **15-gon**, a polygon with twenty-three sides is called a **23-gon**, and a polygon with n sides is called an **n -gon**. The table shows some more special names.

number of sides	name
3	triangle
4	quadrilateral
5	pentagon
6	hexagon

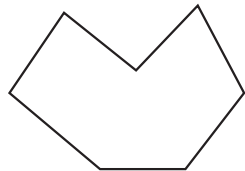
number of sides	name
7	heptagon
8	octagon
9	nonagon
10	decagon

number of sides	name
11	undecagon
12	dodecagon
n	n -gon

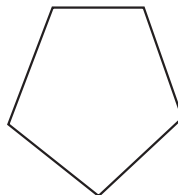
EXAMPLE

4 Classify each polygon by counting its sides.

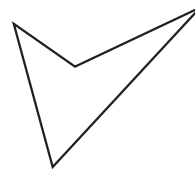
a.



b.



c.



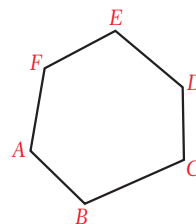
Solution a. a heptagon (seven sides)

b. a pentagon (five sides)

c. a quadrilateral (four sides)

We name a polygon by listing the capital letters of the vertices in order, beginning with any vertex and moving either clockwise or counterclockwise around the polygon.

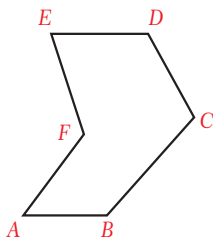
For example, we can refer to the shape opposite as hexagon $ABCDEF$, polygon $ABCDEF$ or polygon $BAFEDC$. We cannot call it polygon $ACDEFB$ or hexagon $FDEABC$ because in these cases the vertices are not written in order.



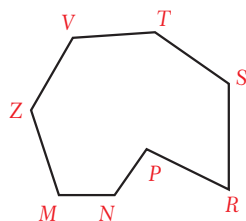
EXAMPLE

5 Name each polygon by listing its vertices.

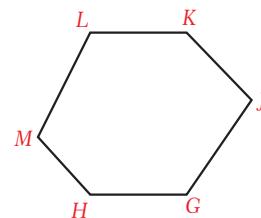
a.



b.



c.



Solution a. polygon $ABCDEF$ b. polygon $TVZMNPRS$ c. polygon $MHGJKL$

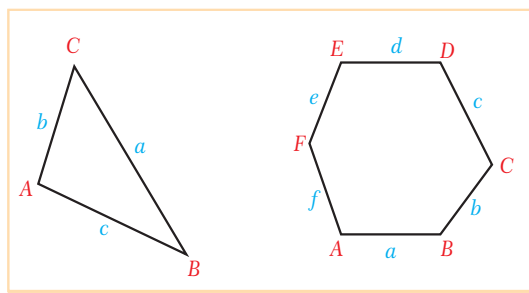
Note that these are not the only possible answers to the question. As an exercise, try finding one more name for each polygon in the example.

We name the sides of a polygon by referring to its vertices. In a triangle, each side is named by the lower-case form of the opposite vertex. In other polygons, each side is usually named by the lower-case form of the vertex at the beginning of the side, moving counterclockwise around the polygon.



The lower-case letters a, b, c , etc. also indicate the lengths of the sides of a polygon.

For example, the sides of the triangle in the figure opposite are a, b and c , which are respectively opposite the vertices A, B and C . In the second polygon, the sides a, b, c, d, e and f take their names from the vertices A, B, C, D, E and F , counted in a counterclockwise direction.

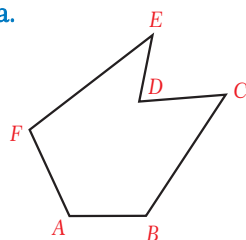


EXAMPLE

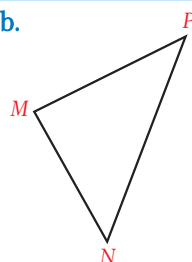


Label the sides of each polygon by referring to its vertices.

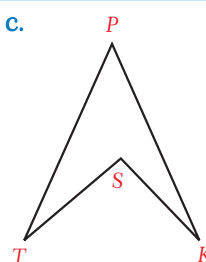
a.



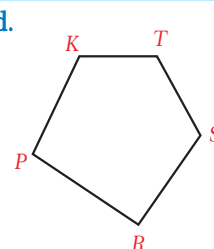
b.



c.

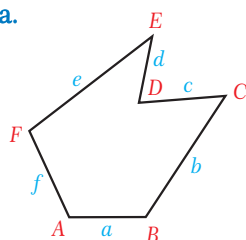


d.

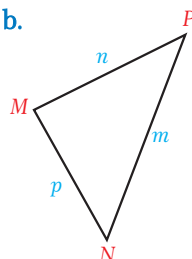


Solution

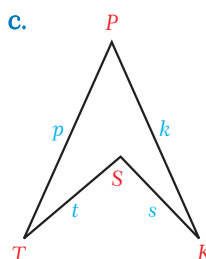
a.



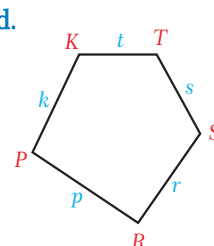
b.



c.



d.



5. Diagonals in a Polygon

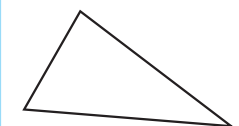
Definition

diagonal

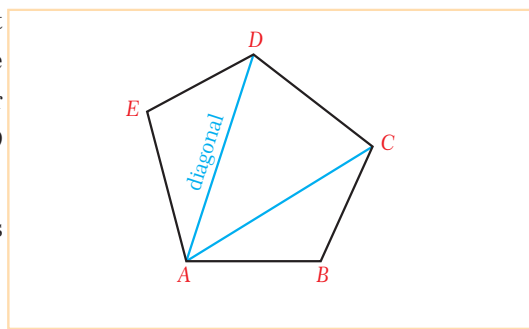
A **diagonal** of a polygon is a line segment that joins two non-consecutive vertices.

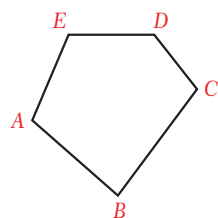
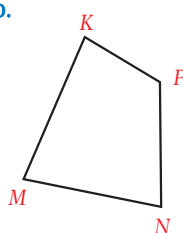
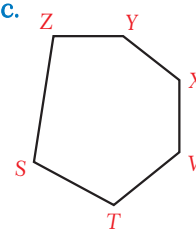
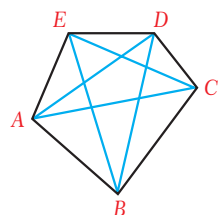
In the polygon $ABCDE$ opposite, line segment AD is a diagonal. It joins the non-consecutive vertices A and D . Line segment AC is another diagonal that joins vertices A and C . BE , BD and CE are other diagonals in the polygon.

Notice that a triangle has no diagonals because it has no non-consecutive vertices.

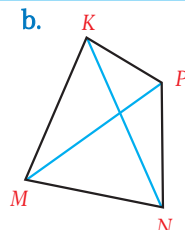


A triangle has no diagonal.

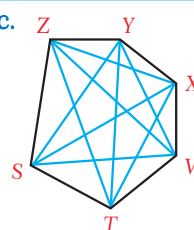


EXAMPLE**7** Draw and name the diagonals in each polygon.**a.****b.****c.****Solution****a.**

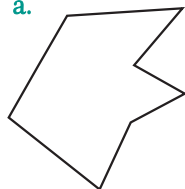
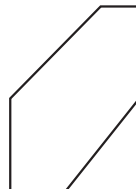
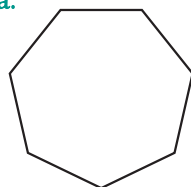
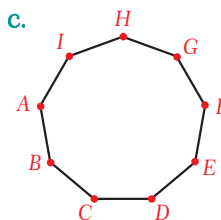
AD, AC, BE, BD, CE

b.

KN, MP

c.

SY, SX, SV, TZ, TY, TX, VZ, VY, XZ

Check Yourself**1.** Which of the following curves are polygons? If a curve is a polygon, is it convex or concave?**a.****b.****c.****2.** Classify each polygon by counting its sides.**a.****b.****c.****Answers****2. a.** a heptagon**b.** an octagon**c.** a nonagon

B. BASIC PROPERTIES OF A CONVEX POLYGON

In the rest of this book we will use the word ‘polygon’ to mean a convex polygon, unless stated otherwise.



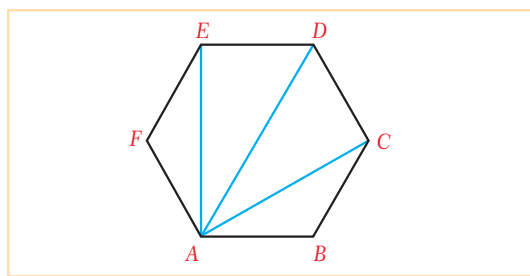
1. Number of Diagonals

What is the total number of diagonals that we can draw from every single vertex of an n -sided polygon? Let us try to answer this question.

Imagine a polygon with n sides. If we choose one vertex then there are $n - 1$ other vertices left on the polygon. By the definition of a diagonal, we cannot draw a diagonal through two adjacent vertices. So there are $(n - 1) - 2 = n - 3$ vertices left, and so the number of diagonals which we can draw from a single vertex of an n -sided polygon is $n - 3$.

For example, consider the hexagon in the figure opposite. From vertex A we can draw diagonals only to vertices C , D and E : since vertices B and F are adjacent to vertex A , we cannot draw a diagonal to these vertices.

So the number of the diagonals which we can draw from vertex A is 3. We can check this in the formula above: $n - 3 = 6 - 3 = 3$.



Theorem 1

A convex polygon with n sides has $\frac{n(n-3)}{2}$ diagonals.

Proof 1

An n -sided polygon has n vertices, and from each vertex we can draw $n - 3$ diagonals. So the total number of diagonals that we can draw is $n(n - 3)$. But this means that we are counting each diagonal twice, since a diagonal joins two vertices. So the expression $n(n - 3)$ must be divided by 2.

So the number of diagonals in an n -sided polygon is $\frac{n(n-3)}{2}$.

Proof 2

An n -sided polygon has n vertices. Consider all the lines determined by these n points. Remember that there is only one line which passes through two distinct points, and no three vertices of the polygon are collinear. So the number of lines determined by the n noncollinear vertices of the polygon is equal to $C(n, 2)$ (i.e. the number of combinations of n points taken two at a time). Since n of these lines include the sides of the polygon, the rest of the lines are the diagonals of the polygon.

So the number of diagonals is $C(n, 2) - n = \frac{n(n-1)}{2} - n = \frac{n(n-3)}{2}$.



Remember:

$C(n, r)$ means the number of possible combinations of n elements taken r at a time:

$$C(n, r) = \frac{n!}{r!(n-r)!}$$

EXAMPLE**8**

- a. Find the number of diagonals in a heptagon (a 7-sided polygon).
 b. Find the number of diagonals in an 11-sided polygon.

Solution

- a. We know that the number of diagonals in an n -sided polygon is $\frac{n(n-3)}{2}$. Substituting $n = 7$ in the formula we get $\frac{7 \cdot (7-3)}{2} = \frac{7 \cdot 4}{2} = 14$. So a heptagon has 14 diagonals.
 b. Using the same formula, the number of diagonals in an 11-sided polygon is $\frac{11 \cdot (11-3)}{2} = \frac{11 \cdot 8}{2} = 44$.

EXAMPLE**9**

- a. Find the number of sides of a polygon which has 9 diagonals.
 b. Find the number of diagonals which pass through one vertex of the polygon in part a.

Solution

- a. Let n be the number of sides. The number of diagonals of an n -sided polygon is $\frac{n(n-3)}{2}$.

$$\text{So } \frac{n(n-3)}{2} = 9$$

$$n^2 - 3n = 18$$

$$n^2 - 3n - 18 = 0, \text{ which factors as } (n + 3)(n - 6) = 0, \text{ so } n = -3 \text{ or } n = 6.$$

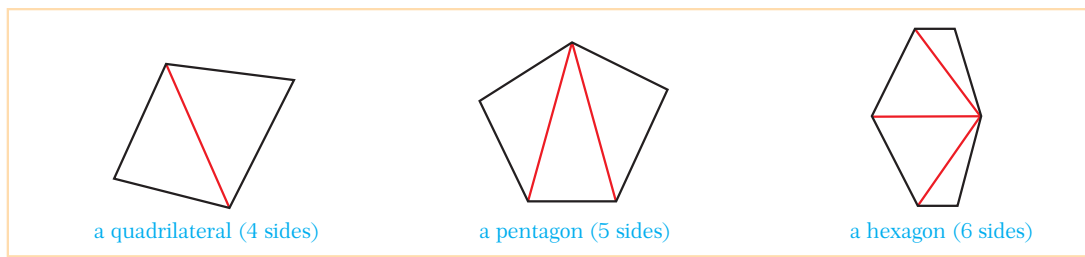
Since the number of sides cannot be negative, the polygon must have six sides.

- b. We know that there are $n - 3$ diagonals through one vertex of an n -sided polygon, and substituting 6 for n gives us $n - 3 = 6 - 3 = 3$. So the answer is three diagonals.

2. Angle Measures

We know that the sum of the interior angles of a triangle is 180° . What about the sum of the interior angles of other polygons? Before trying to find a general rule, think about this question: If you pick any vertex of a polygon and draw all the possible diagonals from this vertex, how many triangles are formed?

Look at the figures and count the triangles.



In each figure we have drawn diagonals from one vertex to each of the other vertices. The diagonals form triangles. In each case, the number of triangles formed is two less than the number of sides of the polygon. In fact, it can be shown that for an n -sided polygon the number of triangles formed is $n - 2$.

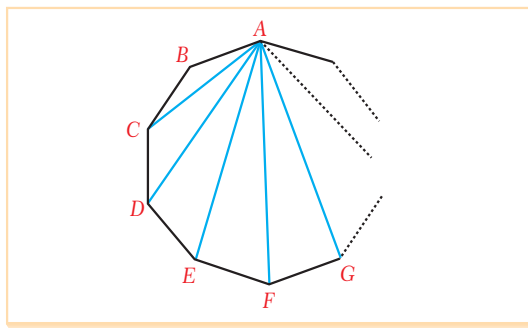
Theorem 2

The sum of the measures of the interior angles of an n -sided polygon is $(n - 2) \cdot 180^\circ$.

Proof



As we have seen, drawing all the possible diagonals of an n -sided polygon from one vertex makes $n - 2$ triangles. The sum of the measures of the interior angles in a polygon is the sum of the measures of the angles of these triangles. Since the sum of the measures of the interior angles of a triangle is 180° , the sum of the measures of the interior angles of an n -sided polygon is $(n - 2) \cdot 180^\circ$.



EXAMPLE

10

- Find the sum of the measures of the interior angles of a polygon with 9 sides.
- The sum of the interior angles of a polygon is 1620° . How many sides does the polygon have?

Solution

- Since the sum of the interior angles of a polygon with n sides is $(n - 2) \cdot 180^\circ$, by substituting $n = 9$ we obtain $(9 - 2) \cdot 180^\circ = 1260^\circ$.

So the answer is 1260° .

- Using the same formula gives us the equation

$$(n - 2) \cdot 180^\circ = 1620^\circ$$

$$n - 2 = 9$$

(dividing both sides by 180°)

$$n = 11. \text{ So the answer is 11.}$$

Theorem 3

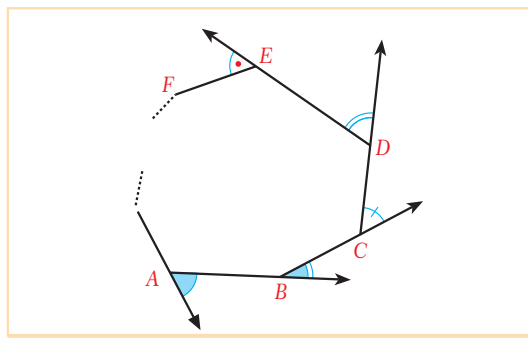
The sum of the measures of the exterior angles of any polygon is 360° .

Proof

Let S be the sum of the angle measures. A polygon with n sides has n vertices. Since the interior and exterior angles at one vertex are supplementary and there are n vertices, the sum of the measures of all the interior and exterior angles of the polygon is $n \cdot 180^\circ$. Since the sum of the measures of the interior angles is $(n - 2) \cdot 180^\circ$, we can write

$$S = n \cdot 180^\circ - ((n - 2) \cdot 180^\circ) = (n \cdot 180^\circ) - (n \cdot 180^\circ) + (2 \cdot 180^\circ) = 360^\circ, \text{ as required.}$$

The next two activities will also help us to understand why the sum of the exterior angles of any polygon is 360° .

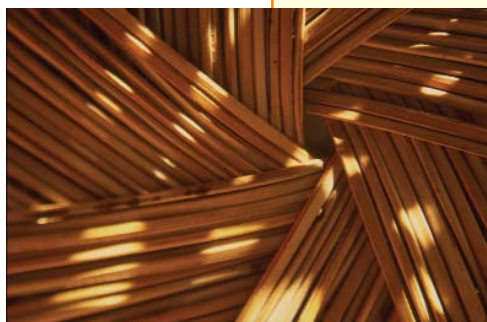
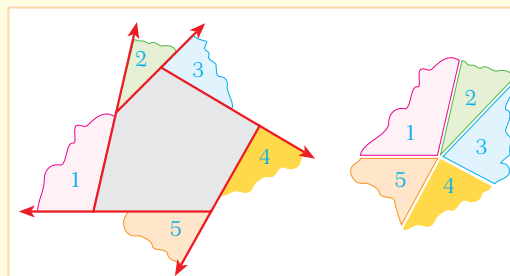


The sum of the measures of the interior angles of a polygon depends on the number of sides, but the sum of the measures of the exterior angles is always 360° .

Activity 1

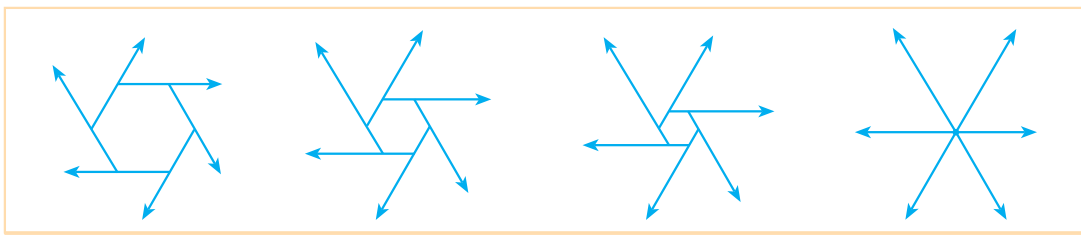
Draw a convex quadrilateral and extend the sides to form an exterior angle at each vertex. Then cut out each exterior angle and join them all together, as shown in the second figure opposite.

What can you conclude?



Activity 2

The sequence of polygons shown below shows the polygon getting smaller and smaller. What is the sum of the measures of the exterior angles of the polygon?



EXAMPLE

11

The sum of the measures of the interior angles of a polygon is 1800° .

- How many sides does this polygon have?
- How many diagonals does it have?

Solution a. The sum of the interior angles of a polygon with n sides is $(n - 2) \cdot 180^\circ$, so we need to solve the equation $(n - 2) \cdot 180^\circ = 1800^\circ$. Dividing both sides by 180° gives us

$$n - 2 = 10; n = 12.$$

So the polygon has 12 sides.

- b. The number of diagonals in an n -sided polygon is $\frac{n(n-3)}{2}$.

$$\text{For } n = 12, \text{ we get } \frac{12 \cdot (12 - 3)}{2} = \frac{12 \cdot 9}{2} = 54.$$

So the polygon has 54 diagonals.

EXAMPLE

12

The measures of the interior angles of a heptagon are in the ratio $4 : 5 : 6 : 7 : 7 : 8 : 8$. Find the measure of each interior and exterior angle of this heptagon.

Solution

Let x be the constant of proportionality, then the interior angles are $4x, 5x, 6x, 7x, 7x, 8x$ and $8x$. If we add these together we get $45x$, which is the sum of the measures of the interior angles. Using the formula $(n - 2) \cdot 180^\circ$ for $n = 7$ (i.e. a heptagon), we find that the sum of the measures of the interior angles of a heptagon is 900° .

So $45x = 900^\circ$, i.e. $x = 20^\circ$.

So the interior angles are $80^\circ, 100^\circ, 120^\circ, 140^\circ, 140^\circ, 160^\circ$ and 160° , and the supplementary angles $100^\circ, 80^\circ, 60^\circ, 40^\circ, 40^\circ, 20^\circ$ and 20° are the exterior angles.

EXAMPLE

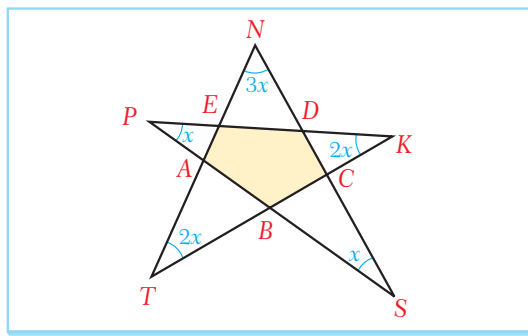
13

The figure which is constructed when we extend the sides of a convex n -gon for any $n > 4$ is called an **n -pointed star**.

The figure at the right shows a 5-pointed star.

$m(\angle T) = m(\angle K) = 2x$,

$m(\angle S) = m(\angle P) = x$ and $m(\angle N) = 3x$ are given. Find the value of x .



Solution

In $\triangle ETK$,

$$\begin{aligned} m(\angle DEN) &= m(\angle T) + m(\angle K) \\ &= 2x + 2x \\ &= 4x. \end{aligned}$$

($\angle DEN$ is an exterior angle of $\triangle ETK$)

Similarly, in $\triangle DPS$,

$$\begin{aligned} m(\angle EDN) &= m(\angle P) + m(\angle S) \\ &= x + x \\ &= 2x. \end{aligned}$$

($\angle EDN$ is an exterior angle of $\triangle DPS$)

So in $\triangle NED$,

$$\begin{aligned} 3x + 4x + 2x &= 180^\circ \\ 9x &= 180^\circ \\ x &= 20^\circ. \end{aligned}$$

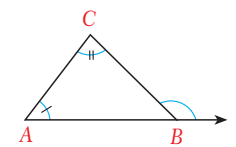
(sum of the interior angles of a triangle)

Note

The sum of the measures of the interior angles at the corners of an n -pointed star is $(n - 4) \cdot 180^\circ$.



The measure of an exterior angle in a triangle is the sum of the measures of the two other interior angles.



$$m(\angle B') = m(\angle A) + m(\angle C)$$



The arms of this starfish form a five-pointed star.

EXAMPLE**14**

The sum of the measures of the interior angles of a polygon is 540° more than the sum of the measures of the exterior angles. How many sides does the polygon have?

Solution

The sum of the measures of the exterior angles of a polygon is 360° , and the sum of the measures of the interior angles of a polygon with n sides is $(n - 2) \cdot 180^\circ$.

So $(n - 2) \cdot 180^\circ = 360^\circ + 540^\circ$

$$(n - 2) \cdot 180^\circ = 900^\circ \quad (\text{divide both sides by } 180^\circ)$$

$n - 2 = 5$; $n = 7$. So the polygon has 7 sides.

EXAMPLE**15**

In a polygon, the ratio of the sum of the measures of the exterior angles to the sum of the measures of the interior angles is $\frac{1}{3}$.

- How many sides does the polygon have?
- How many diagonals does the polygon have?
- Find the sum of the measures of the interior angles of the polygon.

Solution

a. The sum of the measures of the exterior angles of a polygon is 360° and the sum of the measures of the interior angles of an n -sided polygon is $(n - 2) \cdot 180^\circ$.

So $\frac{360^\circ}{(n - 2) \cdot 180^\circ} = \frac{1}{3}$ with $n \neq 2$. By simplifying and cross multiplying we get

$$n - 2 = 6; n = 8.$$

So the polygon has 8 sides.

b. The number of diagonals in an n -sided polygon is $\frac{n(n - 3)}{2}$.

$$\text{For } n = 8, \text{ we get } \frac{8 \cdot (8 - 3)}{2} = \frac{8 \cdot 5}{2} = 20.$$

So there are 20 diagonals.

c. Using the formula $(n - 2) \cdot 180^\circ$ for $n = 8$ gives us $(8 - 2) \cdot 180^\circ = 1080^\circ$.

So the sum of the measures of the interior angles of the polygon is 1080° .

EXAMPLE**16**

Is it possible for the smallest interior angle of a 9-sided polygon to measure 141° ?

Solution

A 9-gon has nine interior angles. If the smallest angle measures 141° , each other interior angle will have to measure more than 141° . So the sum of the measures of the interior angles will be greater than $9 \cdot 141^\circ = 1269^\circ$.

We also know that the sum of the interior angles of an n -sided polygon is $(n - 2) \cdot 180^\circ$.

For nine sides, this becomes $(9 - 2) \cdot 180^\circ = 1260^\circ$.

Since 1269° is greater than 1260° , the smallest interior angle of a 9-sided polygon cannot measure 141° .

What is the maximum number of acute interior angles that a polygon can have? Explain your answer.

Solution

A polygon can have at most three acute interior angles, because if an interior angle is acute then its exterior angle is obtuse and the sum of the measures of more than three obtuse angles is more than 360° (since $360^\circ \div 4 = 90^\circ$, which is not an obtuse angle). These obtuse angles would not form a polygon, since the sum of the measures of the exterior angles of a polygon is 360° . So the answer is three.

Check Yourself 2

- A given polygon has 20 sides.
 - Find the sum of the measures of the interior angles of the polygon.
 - Find the number of diagonals in the polygon.
- The measures of the interior angles of a hexagon are in the ratio $3 : 5 : 6 : 7 : 7 : 8$. Find the measure of each interior and exterior angle of this polygon.
- The sum of the measures of the interior angles of a polygon is 1080° more than the sum of the measures of its exterior angles.
 - How many sides does this polygon have?
 - How many diagonals does it have?

Answers

- a. 3240° b. 170
- interior angles: $60^\circ, 100^\circ, 120^\circ, 140^\circ, 140^\circ, 160^\circ$
exterior angles: $120^\circ, 80^\circ, 60^\circ, 40^\circ, 40^\circ, 20^\circ$
- a. 10 b. 35

C. REGULAR POLYGONS**1. Definition**

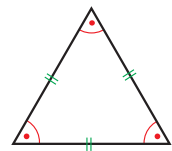
A polygon is called equilateral if all its sides are congruent. It is called equiangular if all its interior angles are congruent. A polygon may be equilateral, equiangular, both equilateral and equiangular, or neither equilateral nor equiangular.

Definition**regular polygon**

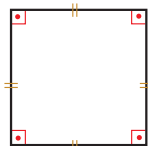
A polygon which is both equilateral and equiangular is called a **regular polygon**.

For example, the polygons in the next figure are regular because all the sides of each polygon are the same length, and all the angles of each polygon are congruent.

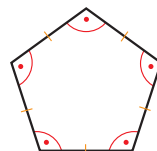




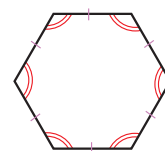
an equilateral triangle



a square

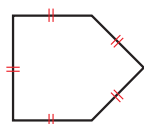


a regular pentagon



a regular hexagon

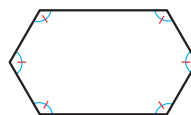
The following polygons are not regular:



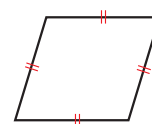
equilateral but
not equiangular



equiangular but
not equilateral



equiangular but
not equilateral



equilateral but
not equiangular

2. Basic Properties of a Regular Polygon

a. Interior angle measures

We know that the sum of the measures of the interior angles of an n -sided polygon is $(n - 2) \cdot 180^\circ$.

Since the interior angles of a regular polygon are all congruent, the measure of each interior angle is $\frac{(n - 2) \cdot 180^\circ}{n}$.

EXAMPLE

18

What is the measure of each interior angle of a regular 8-sided polygon?

Solution

Substituting $n = 8$ in the formula $\frac{(n - 2) \cdot 180^\circ}{n}$, we get the measure of each interior angle as $\frac{(8 - 2) \cdot 180^\circ}{8} = \frac{6 \cdot 180^\circ}{8} = 135^\circ$.

b. Exterior angle measures

We know that the measures of the interior angles of a regular polygon are equal, and so the measures of the exterior angles are also equal. We also know that the sum of the measures of the exterior angles of any polygon is 360° . So the measure of each exterior angle of any regular n -gon is $\frac{360^\circ}{n}$.

EXAMPLE

19

What is the measure of each exterior angle of a regular 12-sided polygon?

Solution

Substituting $n = 12$ in the formula $\frac{360^\circ}{n}$, we get the measure of each exterior angle as $\frac{360^\circ}{12} = 30^\circ$.

EXAMPLE

20

Determine the number of sides of a regular polygon if

- one interior angle measures 140° .
- one exterior angle measures 36° .

Solution a. Using the formula $\frac{(n-2) \cdot 180^\circ}{n}$ for a polygon with n sides gives us the equation

$$\frac{(n-2) \cdot 180^\circ}{n} = 140^\circ, n \neq 0$$

$$(n-2) \cdot 9 = n \cdot 7 \quad (\text{by cross multiplication and simplification})$$

$$9n - 18 = 7n; 2n = 18; n = 9.$$

So the polygon has 9 sides.

b. Using the formula $\frac{360^\circ}{n}$ where n is the number of sides gives us

$$\frac{360^\circ}{n} = 36^\circ, n \neq 0$$

$$n = 10.$$

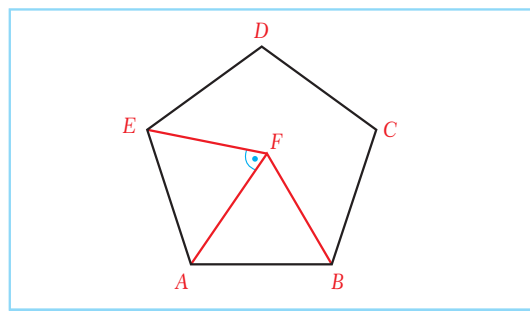
(by cross multiplication and simplification)

So the polygon has 10 sides.

EXAMPLE

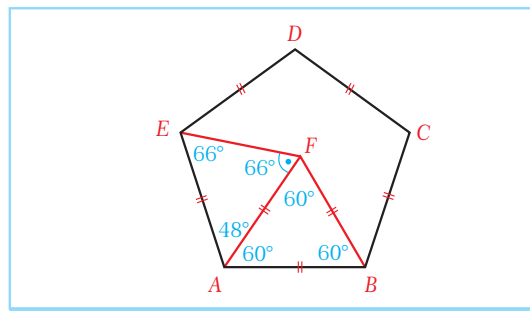
21

In the figure opposite, $ABCDE$ is a regular pentagon and $\triangle ABF$ is an equilateral triangle. What is $m(\angle AFE)$?



Solution The measure of an interior angle of a regular pentagon is $\frac{(5-2) \cdot 180^\circ}{5} = \frac{3 \cdot 180^\circ}{5} = 108^\circ$, and each interior angle of an equilateral triangle measures 60° .

So $m(\angle EAB) = 108^\circ$ and $m(\angle FAB) = 60^\circ$, and so $m(\angle EAF) = 108^\circ - 60^\circ = 48^\circ$.



The Pentagon in Washington is a remarkable architectural example of a regular polygon.



Since $ABCDE$ is a regular pentagon, the line segments AB and AE are congruent. Line segments AB and AF are also congruent because $\triangle ABF$ is equilateral. And so line segments AF and AE must be congruent. We can conclude that $\triangle AFE$ is an isosceles triangle and the base angles $\angle AFE$ and $\angle AEF$ are equal.

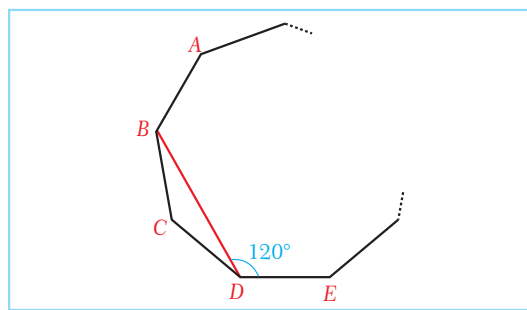
$$\text{So } m(\angle AFE) = \frac{180^\circ - 48^\circ}{2} = 66^\circ.$$

EXAMPLE

22

In the figure, $ABCDE\dots$ is a regular polygon and $m(\angle BDE) = 120^\circ$.

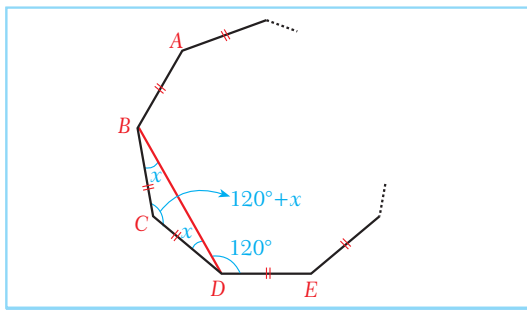
How many sides does this polygon have?



Solution

We know the formula $a = \frac{(n-2) \cdot 180^\circ}{n}$ for the measure a of an interior angle of a regular polygon with n sides. Let us try to find a using the information we are given.

Since $ABCDEF\dots$ is a regular polygon, it is equilateral and equiangular. This means that the line segments AB , BC and CD are congruent and $\angle A$, $\angle B$ and $\angle C$ have equal measures.



$\triangle BCD$ is isosceles, so we can write $m(\angle CBD) = m(\angle CDB) = x$.

So $m(\angle BCD) = m(\angle CDE) = x + 120^\circ$.

Thus in $\triangle BCD$, $x + x + (120^\circ + x) = 180^\circ$ (sum of the interior angles of a triangle)

$$x = 20^\circ.$$

So the measure of one interior angle of the regular polygon is $a = 120^\circ + 20^\circ = 140^\circ$.

Substituting this in the formula $a = \frac{(n-2) \cdot 180^\circ}{n}$ where n is the number of sides, we get the

$$\text{equation } \frac{(n-2) \cdot 180^\circ}{n} = 140^\circ, n \neq 0$$

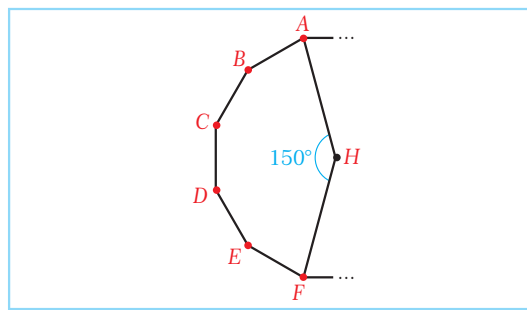
$$(n-2) \cdot 9 = n \cdot 7 \quad (\text{by cross multiplication and simplification})$$

$$9n - 18 = 7n; 2n = 18; n = 9. \text{ So the polygon has 9 sides.}$$

EXAMPLE

23

In the figure, $ABCDEF\dots$ is a regular polygon and line segments AH and FH bisect $\angle A$ and $\angle F$ respectively. If $m(\angle AHF) = 150^\circ$, find the number of sides of the polygon.



Solution Since $ABCDEF\dots$ is a regular polygon, $\angle A$, $\angle B$, $\angle C$, $\angle D$, $\angle E$ and $\angle F$ are all congruent and $m(\angle BAH) = m(\angle EFH) = \frac{m(\angle A)}{2}$ because AH and FH are angle bisectors.

In the figure,

$$x = 150^\circ \div 5 = 30^\circ.$$

All the triangles are isosceles, so

$$m(\angle BAH) = \frac{(180^\circ - 30^\circ)}{2} = 75^\circ \text{ and}$$

$$m(\angle B) = m(\angle C) = \dots = 2 \cdot 75^\circ = 150^\circ.$$

So one interior angle of the regular polygon measures 150° .

Now, using the formula $\frac{(n-2) \cdot 180^\circ}{n}$ we can write

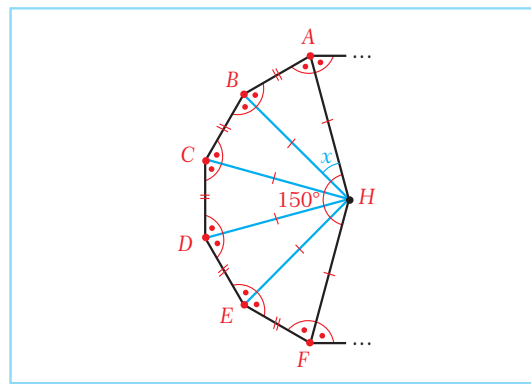
$$\frac{(n-2) \cdot 180^\circ}{n} = 150^\circ, n \neq 0$$

$$(n-2) \cdot 6 = n \cdot 5 \quad (\text{by cross multiplication and simplification})$$

$$6n - 12 = 5n$$

$$n = 12.$$

So $ABCDEF\dots$ has 12 sides.

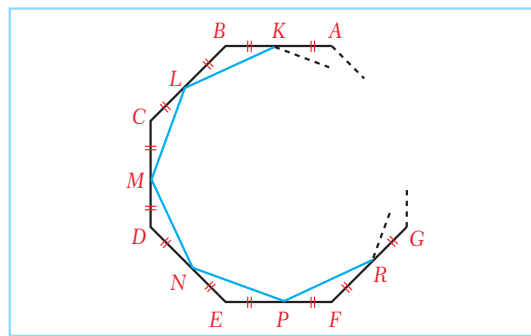


EXAMPLE

24

Prove that the midpoints of the sides of a regular polygon form the vertices of another regular polygon with the same number of sides.

Solution Let $ABCDEF\dots$ be an n -sided regular polygon and let points K, L, M, N, P , etc. be the midpoints of the sides AB, BC, CD, DE, EF , etc. respectively. First we will show that polygon $KLMPN\dots$ is a regular polygon. In other words, we will show that it is both equilateral and equiangular.



Statements	Reasons
1. $AB \cong BC \cong CD \cong \dots$	Polygon $ABCDEF\dots$ is equilateral.
2. $\angle A \cong \angle B \cong \angle C \cong \dots$	Polygon $ABCDEF\dots$ is equiangular.
3. $AK \cong KB \cong BL \cong LC \cong \dots$	Points $K, L, M, N, \dots$ are midpoints.
4. $\triangle KBL \cong \triangle LCM \cong \triangle MDN \cong \dots$	By 2 and 3
5. $KL \cong LM \cong MN \cong \dots$	Corresponding sides of congruent triangles
6. $\angle BKL \cong \angle BLK \cong \angle CLM \cong \angle CML \dots$	Corresponding angles of congruent triangles
7. $\angle AKB \cong \angle BLC \cong \angle CMD \dots$	Straight angles
8. $\angle KLM \cong \angle LMN \cong \angle MNP \dots$	By 6 and 7
9. Polygon $KLMNP\dots$ is equilateral.	By 5
10. Polygon $KLMNP\dots$ is equiangular.	By 8

So polygon $KLMNP\dots$ is a regular polygon. Also, each vertex of $KLMNP\dots$ lies on exactly one side of the polygon $ABCDE\dots$. In other words, $KLMNP\dots$ has n vertices. Since the number of vertices of a polygon is equal to the number of sides, $KLMNP\dots$ has n sides.

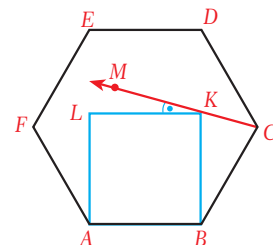
In conclusion, $KLMNP\dots$ is an n -sided regular polygon.



The pattern on this cowfish and the structure of this honeycomb are made up of hexagons. Can you think of a reason why the hexagon is such a useful shape?

Check Yourself

- Determine the number of sides of a regular polygon if
 - one interior angle measures 168° .
 - one exterior angle measures 18° .
- In the figure opposite, $ABCDEF$ is a regular hexagon, $ABKL$ is a square and points C, K and M are collinear. Find $m(\angle MKL)$.



Answers

- a. 30 b. 20 2. 15°

3. Inscribed and Circumscribed Polygons

Definition

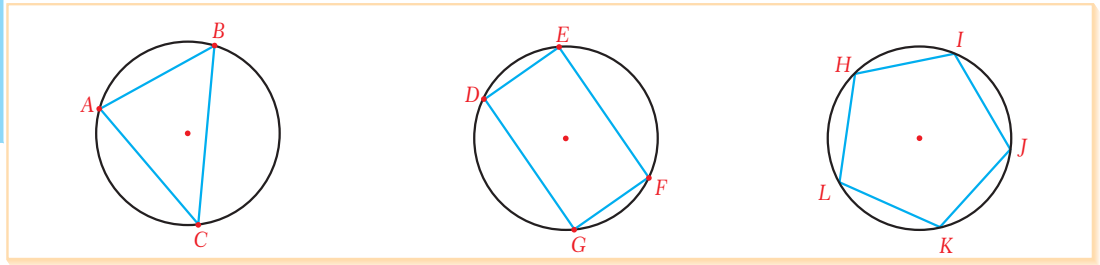
Remember:

$x \cong y$ means x is congruent to y . Congruent **angles** have the same measure. Congruent **line segments** have the same length. Congruent **triangles** have congruent angles and congruent sides.

inscribed polygon, cyclic polygon, circumscribed circle, circumcircle

A polygon is called an **inscribed polygon** (or **cyclic polygon**) if all of its vertices lie on the same circle. This circle is called the **circumscribed circle** (or **circumcircle**) of the polygon.

Look at some examples of inscribed polygons:

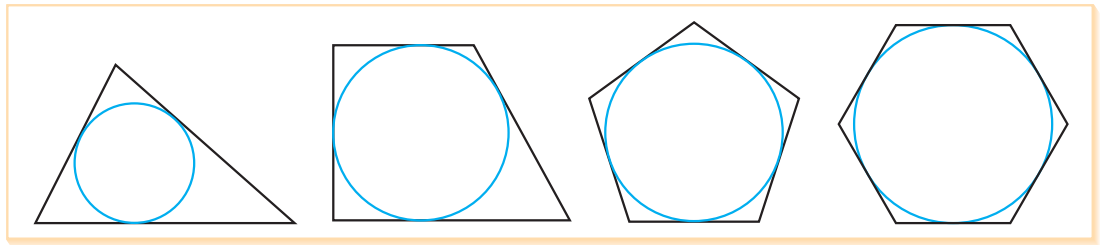


Definition

circumscribed polygon, inscribed circle

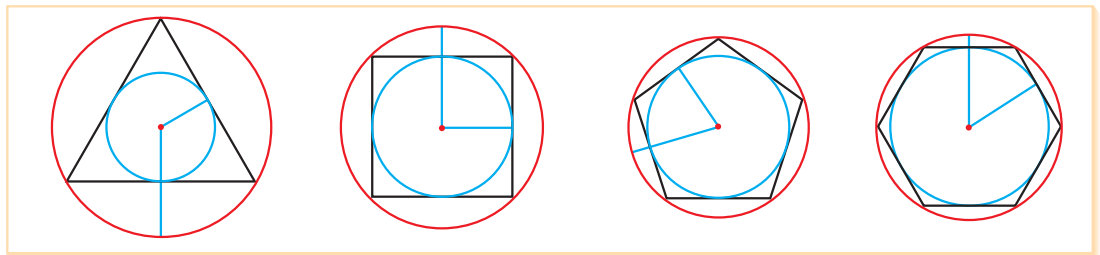
A polygon is called a **circumscribed polygon** if its sides are all tangent to the same circle. This circle is called the **inscribed circle** of the polygon.

Look at some examples of circumscribed polygons:



Property 1

A regular polygon is always an inscribed and circumscribed polygon, and its inscribed and circumscribed circles have the same center. Conversely, if the inscribed and circumscribed circles of a polygon have the same center then this polygon is a regular polygon.



Note that we may be able to draw both inscribed and circumscribed circles for other polygons, but if their centers are not at the same point then these polygons are not regular.

EXAMPLE

25

A square has sides 6 cm long. Find the radii of its inscribed and circumscribed circles.

Solution Look at the figure. The centers of the inscribed and circumscribed circles lie at the intersection of the diagonals of the square. The radius r of the inscribed circle is half the length of the side of the square, i.e. 3 cm.

The radius R of the circumscribed circle is half the length of the diagonal of the square.

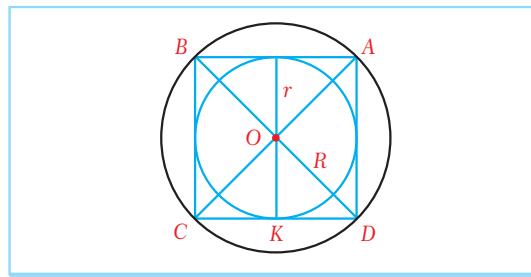
By the Pythagorean Theorem,

$$AD^2 + CD^2 = AC^2$$

$$6^2 + 6^2 = AC^2$$

$$AC = 6\sqrt{2}.$$

So the radius of the circumscribed circle is $R = 6\sqrt{2} \div 2 = 3\sqrt{2}$ cm.



EXAMPLE

26

A regular hexagon has sides of length 8 cm. Find the radii of its inscribed and circumscribed circles.

Solution

Look at the figure. The centers of the inscribed and circumscribed circles of the hexagon lie at point O . OH is the radius of the inscribed circle and OD is the radius of the circumscribed circle.

$\angle DOE$ is a central angle and measures $\frac{360^\circ}{6} = 60^\circ$.

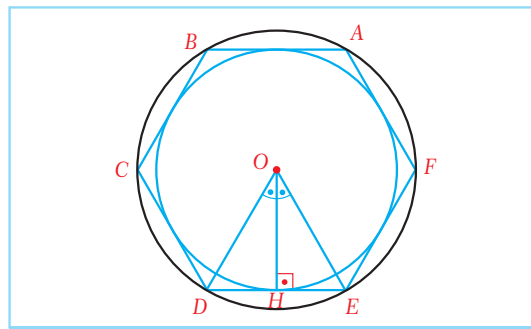
$\triangle DOE$ is an isosceles triangle with altitude OH .

In addition, $m(\angle HOE) = m(\angle DOE) \div 2 = 30^\circ$ and $HE = \frac{8}{2} = 4$ cm.

In the right triangle OHE , $\sin 30^\circ = \frac{HE}{OE}$ so $OE = \frac{HE}{\sin 30^\circ} = \frac{4}{0.5} = 8$ cm, and

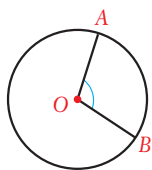
$$\tan 30^\circ = \frac{HE}{OH} \text{ so } OH = \frac{HE}{\tan 30^\circ} = \frac{4}{\frac{1}{\sqrt{3}}} = 4\sqrt{3} \text{ cm.}$$

So the radius of the circumscribed circle is 8 cm and the radius of the inscribed circle is $4\sqrt{3}$ cm.



In a circle, a central angle is an angle formed by two radii. In the figure below, $\angle AOB$ is a central angle and

$$m(\angle AOB) = m(\widehat{AB}).$$

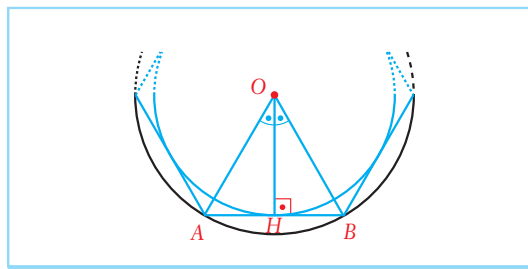




a. Formulas for the radii of inscribed and circumscribed circles

There is a relation between the side length of a regular polygon, the number of sides, and the radii of its inscribed and circumscribed circles.

Let a be the length of a side of a regular polygon, let n be the number of sides and let r and R be the radii of the polygon's inscribed and circumscribed circles, respectively.



Look at the figure. $\angle AOB$ is a central angle and $m(\angle AOB) = \frac{360^\circ}{n}$.

$\triangle AOB$ is isosceles and OH is its height. So $m(\angle HOB) = \frac{\angle AOB}{2} = \frac{180^\circ}{n}$ and $HB = \frac{AB}{2} = \frac{a}{2}$.

In the right triangle OHB , $\sin \angle HOB = \frac{HB}{OB}$; $OB = \frac{HB}{\sin \angle HOB}$; $R = \frac{a}{2 \sin \frac{180^\circ}{n}}$, and

$$\tan \angle HOB = \frac{HB}{OH}; OH = \frac{HB}{\tan \angle HOB}; r = \frac{a}{2 \tan \frac{180^\circ}{n}}.$$

So the radius of the circumscribed circle is $R = \frac{a}{2 \sin \frac{180^\circ}{n}}$ and the radius of the inscribed circle is $r = \frac{a}{2 \tan \frac{180^\circ}{n}}$.

The table at the right shows the relation between a , n , r and R for regular polygons with 3, 4 and 6 sides.

EXAMPLE

27

A 12-sided regular polygon has sides of length 10 cm. Find the radii of its inscribed and circumscribed circles.

Solution

We can use the formulas $R = \frac{a}{2 \sin \frac{180^\circ}{n}}$ and $r = \frac{a}{2 \tan \frac{180^\circ}{n}}$, where n is the number of



Use a trigonometric table or a calculator to find the trigonometric values of the angles.

sides and a is the length of one side. For $n = 12$ and $a = 10$ cm,

$$\begin{aligned} R &= \frac{10}{2 \sin \frac{180^\circ}{12}} = \frac{5}{\sin 15^\circ} & \text{and} & & r &= \frac{10}{2 \tan \frac{180^\circ}{12}} = \frac{5}{\tan 15^\circ} \\ &= \frac{5}{0.2558} & & & &= \frac{5}{0.2679} \\ R &= 19.55 \text{ cm} & & & r &= 18.66 \text{ cm.} \end{aligned}$$

EXAMPLE
28

The circumscribed circle of a regular hexagon with side length 10 cm is also the inscribed circle of an equilateral triangle. Find the length of one side of the triangle.

Solution

We will use the formulas $r = \frac{a}{2 \tan \frac{180^\circ}{n}}$ and

$$R = \frac{a}{2 \sin \frac{180^\circ}{n}}.$$

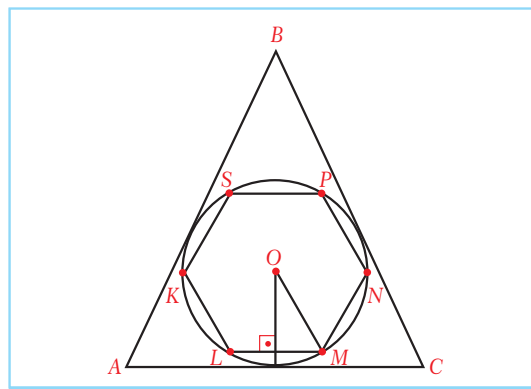
The radius of the inscribed circle of the triangle is equal to the radius of the circumscribed circle of the hexagon. In the triangle, $n = 3$ and a is unknown so

$$r = \frac{a}{2 \tan \frac{180^\circ}{n}} = \frac{a}{2 \tan \frac{180^\circ}{3}} = \frac{a}{2 \tan 60^\circ}.$$

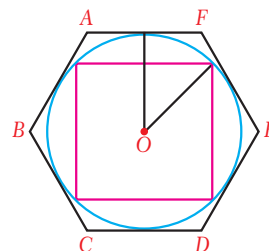
In the hexagon, $n = 6$ and $a = 10$ cm so

$$R = \frac{a}{2 \sin \frac{180^\circ}{n}} = \frac{10}{2 \sin \frac{180^\circ}{6}} = \frac{5}{\sin 30^\circ} = \frac{5}{0.5} = 10 \text{ cm}.$$

Since $r = R$ we have $\frac{a}{2 \tan 60^\circ} = 10$; $a = 20 \cdot \tan 60^\circ$; $a = 20\sqrt{3}$ cm. This is the length of one side of the triangle.


Check Yourself 4

- One side of an equilateral triangle measures $6\sqrt{3}$ cm. Find the radii of its inscribed and circumscribed circles.
- One side of a square measures $10\sqrt{3}$ cm. The inscribed circle of the square is also the circumscribed circle of an equilateral triangle. Find the length of one side of this triangle.
- The inscribed circle of a regular hexagon is the circumscribed circle of a square with side length $4\sqrt{2}$ cm. Find the length of one side of the hexagon.


Answers

- $r = 3$ cm, $R = 6$ cm
- 15 cm
- $\frac{8\sqrt{3}}{3}$ cm

Activity 1

The figure opposite shows a regular triangle, quadrilateral, pentagon and hexagon constructed on a common base which is 2 cm long. Use a ruler and a protractor to draw each of the following polygons on a base which is 4 cm long.

- a regular triangle
- a regular quadrilateral
- a regular pentagon
- a regular hexagon
- a regular octagon
- a regular decagon

(Hint: Calculate the size of the interior angles first.)

Activity 2

THE GAME OF SPROUTS

Sprouts is a pencil-and-paper game which has interesting mathematical properties. It was invented by the mathematicians John Horton Conway and Michael S. Paterson at England's Cambridge University in 1967. Here is a quote from Conway: 'The day after Sprouts sprouted, it seemed that everyone was playing it. At coffee or tea time there were little groups of people peering over ridiculous to fantastic sprout positions.'

The game is played by two players, starting with a few dots drawn on a sheet of paper. The game continues according to the following rules:

1. Players take turns drawing a line between two dots, or from a dot to itself.
2. The line may not cross any other line.
3. After drawing a line, the player adds a new dot on the same line.
4. No dot may have more than three lines coming out of it.
5. The player who makes the last move wins.

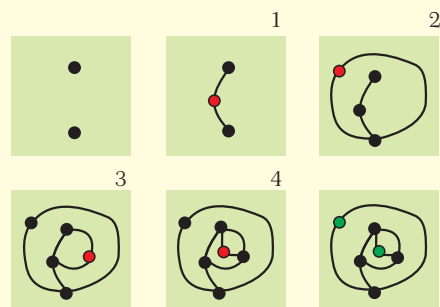
The diagram at the right shows a 2-dot game of Sprouts.

After the fourth move (picture 4), it is impossible to make another move, so the second player wins. The final figure shows that there are two dots (shown in green) that are still 'alive', i.e. they

are only connected to two lines. But since these two survivors are in separate regions, they cannot be joined together.

The secret to winning the game lies in using your lines to divide the paper up into parts that trap dots. If the 2-dot game gets too easy for you, start with more dots.

This game is a good way to improve your understanding of how space can be divided on a flat surface.

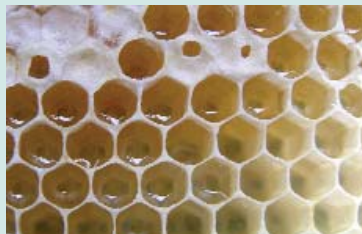


TESSELLATIONS

A **tessellation** (also called a **tiling**) is created when a shape is repeated over and over again to cover a plane without any gaps or overlaps. The word 'tessellate' means to form or arrange small squares in a checkered or mosaic pattern. It comes from the Greek word *tesseres*, which in English means 'four.' This is because the first tessellations were made from square (i.e. four-sided) tiles.

Tessellations are all around us in our natural and man-made environment. Whenever we want to cover or decorate a surface with shapes that neither overlap nor leave gaps, we work with tessellation.

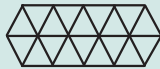
The honeycombs in a beehive are one of the best natural examples of tessellation. The hexagonal cells in a comb fit together perfectly without any gaps. Examples of man-made tessellations include floor tilings, brick walls, wallpaper patterns, textile patterns and some stained glass windows.



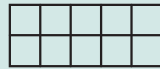
There are three types of tessellation: **regular**, **semi-regular** and **irregular**.

Regular Tessellations

A regular tessellation is a tessellation which is made up entirely of congruent regular polygons which all meet at their vertices. Only three regular polygons tessellate in the Euclidean plane: the triangle, the square and the hexagon.



a tessellation of triangles



a tessellation of squares

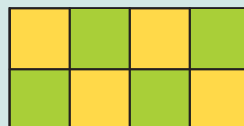


a tessellation of hexagons

How do we know that only three types of regular polygon form a regular tessellation? Since the regular polygons in a tessellation must fill the plane at each vertex, their interior angle must be an exact divisor of 360° . This only works for the triangle, square and hexagon. The interior angles of all other regular polygons are not exact divisors of 360° , and therefore these figures cannot fill the plane without leaving gaps. Look at some examples:



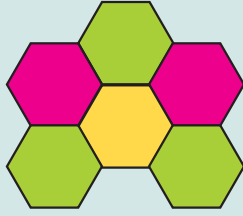
The interior angle of an equilateral triangle is 60° , and $6 \cdot 60^\circ = 360^\circ$.
So a triangle forms a regular tessellation



The interior angle of a square is 90° , and $4 \cdot 90^\circ = 360^\circ$.
So a square forms a regular tessellation.



The interior angle of a pentagon is 108° , and $4 \cdot 108^\circ = 324^\circ$.
So a pentagon does not form a regular tessellation.



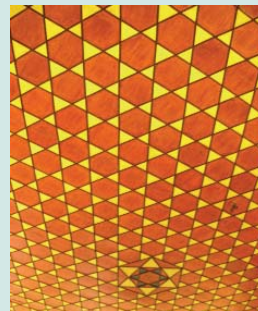
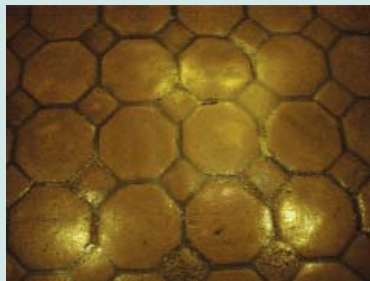
The interior angle of a hexagon is 120° ,
and $120^\circ + 120^\circ + 120^\circ = 360^\circ$.



For regular polygons with more than six sides, the polygons overlap.

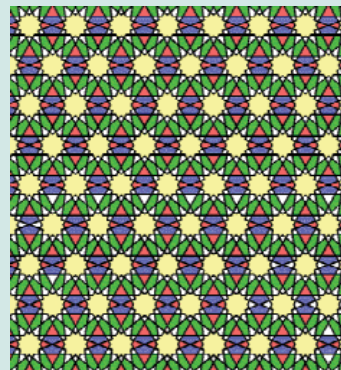
Semi-Regular Tessellations

Semi-regular tessellations are made up of two or more types of regular polygon which fit together in such a way that the arrangement of polygons at every vertex point is identical. There are eight possible semi-regular tessellations which are different combinations of equilateral triangles, squares, hexagons, octagons and dodecagons. The pictures show two examples of semi-regular tessellations.



Irregular Tessellations

Irregular tessellations place no restriction on the number of different polygons or the order of the polygons around the vertices. There are an infinite number of irregular tessellations. Here are two examples:



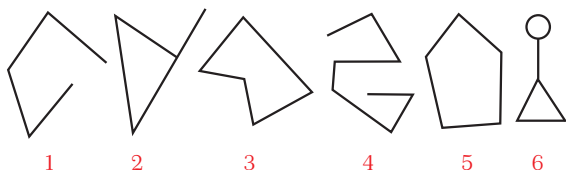
Can you see any other examples of tessellation in the world around you? Draw pictures of the patterns you find. How many different shapes make up your tessellations?

EXERCISES 5.1

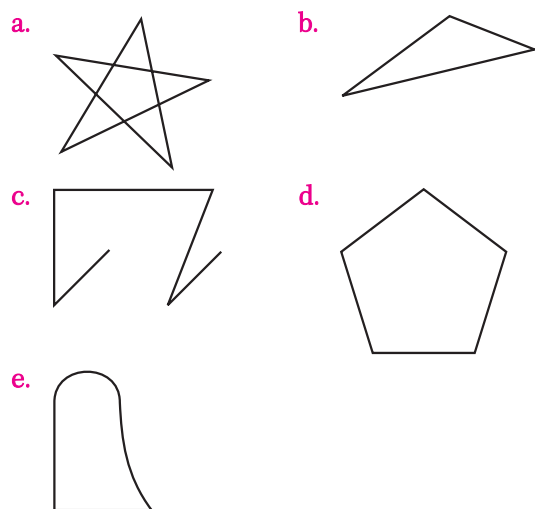
A. Basic Concepts

1. State which of curves 1-6 are

- a. simple. b. closed.
c. simple closed.

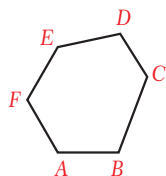


2. State whether each figure is a polygon. If it is not a polygon, explain why.

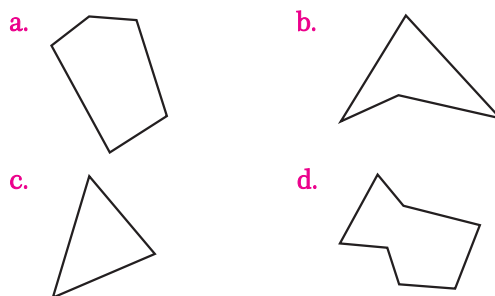


3. For the polygon opposite, list

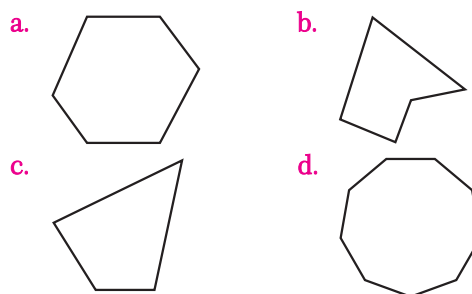
- a. its vertices.
b. its consecutive vertices and angles.



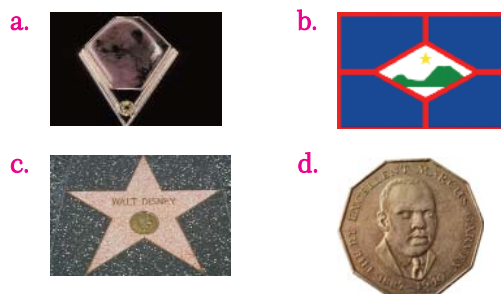
4. State whether each polygon is convex or concave.



5. Classify each polygon by counting its sides.

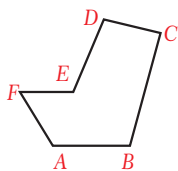


6. Find a polygon in each picture. Classify the polygon by counting its sides.

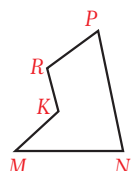


7. Name each polygon by listing its vertices.

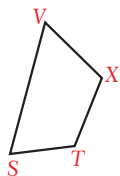
a.



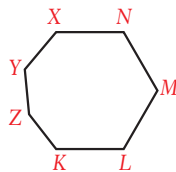
b.



c.

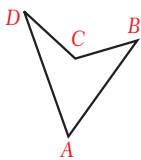


d.

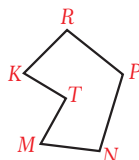


8. Name the sides of each polygon by referring to its vertices.

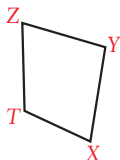
a.



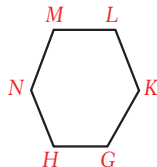
b.



c.

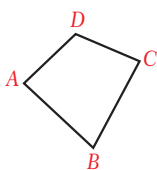


d.

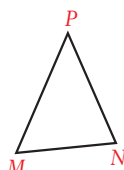


9. Name the diagonals in each polygon.

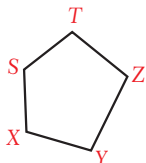
a.



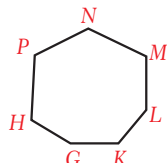
b.



c.



d.



B. Basic Properties of a Convex Polygon

10. Four polygons have 9, 12, 15 and 21 sides respectively. For each polygon, state

- the number of diagonals.
- the sum of the measures of its interior angles.

11. Find the number of sides of a convex polygon which has

- 14 diagonals.
- 35 diagonals.
- 77 diagonals.
- 170 diagonals.

12. How many sides does a polygon have if the sum of the measures of its interior angles is

- 540° ?
- 1440° ?
- 1800° ?
- 2880° ?

13. The number of diagonals in a given polygon is three times the number of sides of the polygon. Find the sum of the measures of the interior angles of this polygon.

14. The measures of the interior angles of a pentagon are in the ratio 3 : 4 : 5 : 7 : 8. Find the measure of each interior and exterior angle of this pentagon.

15. In the figure, $ABCDEF$ is a hexagon with

$$m(\angle E) = 105^\circ,$$

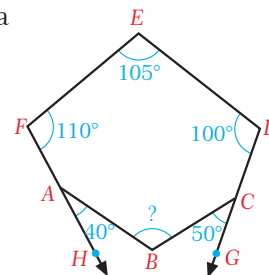
$$m(\angle D) = 100^\circ,$$

$$m(\angle F) = 110^\circ,$$

$$m(\angle HAB) = 40^\circ \text{ and}$$

$$m(\angle BCG) = 50^\circ.$$

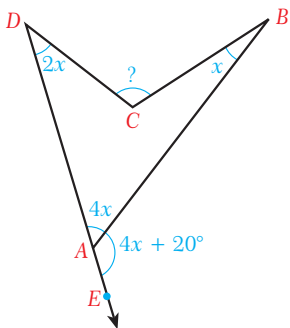
Find $m(\angle ABC)$.



16. In the figure, $ABCD$ is a concave quadrilateral with

$$\begin{aligned} m(\angle DAB) &= 4x, \\ m(\angle D) &= 2x, \\ m(\angle B) &= x \text{ and} \\ m(\angle BAE) &= 4x + 20^\circ. \end{aligned}$$

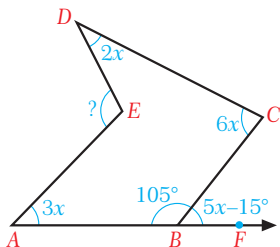
Find $m(\angle BCD)$.



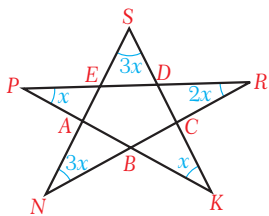
17. In the figure, $ABCD$ is a concave pentagon.

$$\begin{aligned} m(\angle A) &= 3x, \\ m(\angle D) &= 2x, \\ m(\angle C) &= 6x, \\ m(\angle ABC) &= 105^\circ \text{ and} \\ m(\angle CBF) &= 5x - 15^\circ \end{aligned}$$

are given. Find the measure of $\angle AED$.



18. In the figure,
 $m(\angle S) = m(\angle N) = 3x$,
 $m(\angle R) = 2x$ and
 $m(\angle P) = m(\angle K) = x$.
 Find the value of x .



19. Three interior angles of a polygon measure 110° , 125° and 135° respectively, and all the other interior angles measure 170° . How many sides does this polygon have?

20. Can the largest interior angle of a heptagon measure 128° ? Explain your answer.

21. What is the maximum possible number of right angles in a convex polygon with more than 4 sides?

C. Regular Polygons

22. State whether each shape is a regular polygon.

- | | |
|----------------------------|----------------|
| a. a rectangle | b. a trapezoid |
| c. a rhombus | d. a square |
| e. an equilateral triangle | f. a kite |

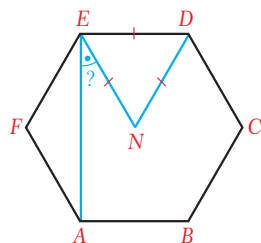
23. Three regular polygons have 9, 12 and 16 sides respectively. For each polygon state
- the measure of an interior angle.
 - the measure of an exterior angle.

24. The interior angle of a regular polygon measures 165° . How many sides does this polygon have?

25. The measure of an interior angle of a regular polygon is six times the measure of an exterior angle. How many sides does this polygon have?

26. $ABCDE$ is a regular polygon. Find the measure of the angle between the diagonals BE and BD .

27. In the figure, $ABCDEF$ is a regular hexagon and $\triangle END$ is an equilateral triangle. Find $m(\angle AEN)$.



28. The inscribed circle of a square has radius $2\sqrt{3}$ cm. Find

- the length of one side of the square.
- the radius of the circumscribed circle of the square.

29. One side of an equilateral triangle measures $4\sqrt{3}$ cm. Find the radii of the inscribed and circumscribed circles of this triangle.

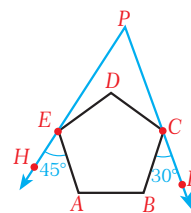
30. The circumscribed circle of a regular hexagon has radius 8 cm. Find

- the length of one side of the hexagon.
- the radius of the inscribed circle of the hexagon.

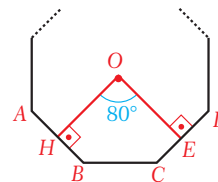
31. The circumscribed circle of a regular hexagon with side length 10 cm is also the inscribed circle of an equilateral triangle. Find the length of one side of this triangle.

32. The difference between the radii of the inscribed and circumscribed circles of an equilateral triangle is 4 cm. Find the length of one side of this

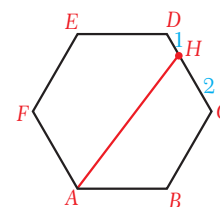
33. In the figure, $ABCDE$ is a regular pentagon. Points P , E and H are collinear and points P , C , F are also collinear. $m(\angle AEH) = 45^\circ$ and $m(\angle BCF) = 30^\circ$. Find the measure of $\angle EPF$.



34. In the figure, $ABCD\dots$ is a regular polygon with $OH \perp AB$ and $OE \perp CD$. O is at the center of the polygon and $m(\angle HOE) = 80^\circ$. How many sides does the polygon have?



35. In the figure, $ABCDEF$ is a regular hexagon with $DH = 1$ cm and $HC = 2$ cm. Find the length of AH .



36. The sum of the measures of five interior angles of a regular polygon is 270° more than the sum of the measures of all its other interior angles. How many sides does this polygon have?

37. Prove that any diagonal in a regular pentagon is parallel to one of its sides.

38. Prove that the intersection points of the diagonals of a regular pentagon form the vertices of another regular pentagon.

AREAS OF REGULAR POLYGONS AND CIRCLES

A. AREA OF A CIRCULAR REGION

Definition

circular region

The union of a circle and its interior region is called a **circular region**.

Theorem

area of a circular region

The area of a circular region is the product of π and the square of the radius of the circle:

$$A = \pi r^2 .$$

Proof

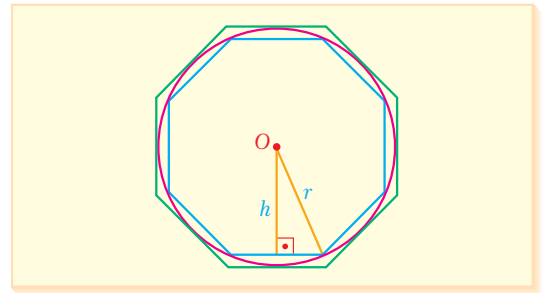
Let us draw two n -gons such that the circle is inscribed in one n -gon and circumscribes the other n -gon, as shown in the figure. Let the circle have area A and circumference C , and let us say that the area of the polygon inscribed in the circle is A_1 and its perimeter is P_1 , and the area of the polygon which circumscribes to the circle is A_2 and its perimeter is P_2 . Then $A_1 < A < A_2$.

Since $A_1 = \frac{1}{2} \cdot h \cdot P_1$ and $A_2 = \frac{1}{2} \cdot r \cdot P_2$, we get $\frac{1}{2} \cdot h \cdot P_1 < \frac{1}{2} \cdot r \cdot P_2$

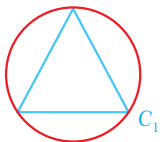
If we increase the number of sides of the polygons infinitely many times ($n \rightarrow \infty$) then h will get closer and closer to r and the areas A_1 and A_2 will both get closer and closer to the area of the circle.

Also, as $h \rightarrow r$ we get $P_1 \rightarrow P_2 \rightarrow C = 2\pi r$.

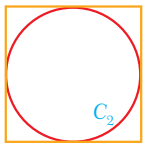
So $A = \frac{1}{2} \cdot r \cdot C = \frac{1}{2} \cdot r \cdot 2\pi r = \pi r^2$, as required.



Inscribed or circumscribed?



The triangle is inscribed in circle C_1 . Circle C_1 circumscribes the triangle.



Circle C_2 is inscribed in the square. The square circumscribes circle C_2 .

EXAMPLE**1**

Find the area of a circle with radius 3 cm.

Solution $r = 3$ cm means $A = \pi r^2 = \pi \cdot 3^2 = 9\pi$ cm².

EXAMPLE**2**

The circumference of a circle is 16π cm. Find its area.

Solution $C = 2\pi r = 16\pi$ gives us $r = 8$ cm.

So $A = \pi r^2 = \pi \cdot 8^2 = 64\pi$ cm².

EXAMPLE**3**

The area of a circle is 48 cm². Find its circumference.

Solution $A = \pi r^2 = 48$ gives us $r^2 = \frac{48}{\pi}$, $r = \sqrt{\frac{48}{\pi}} = \frac{4\sqrt{3}}{\sqrt{\pi}} = \frac{4\sqrt{3\pi}}{\pi}$ cm.

So $C = 2\pi r = 2\pi \frac{4\sqrt{3\pi}}{\pi} = 8\sqrt{3\pi}$ cm.

EXAMPLE

4

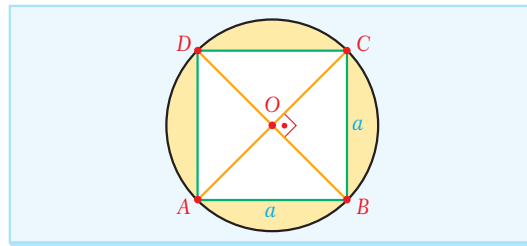
Within a circle with radius 8 cm we draw a square whose vertices lie on the circle. Find the area of the region between the square and the circle.

Solution A square is a regular polygon, so the center of the circle and the center of the square are the same point. So we can say that the diagonal of the square is equal to the diameter of the circle:

$$2r = d = a\sqrt{2}, \text{ i.e. } 2 \cdot 8 = a\sqrt{2} \text{ and}$$

$$a = \frac{16}{\sqrt{2}} = 8\sqrt{2} \text{ cm.}$$

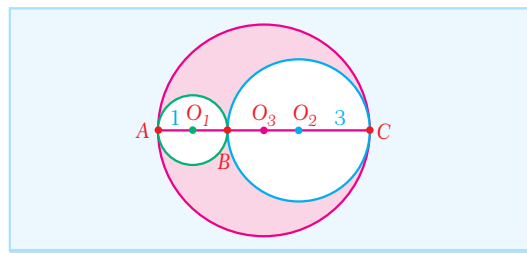
$$\begin{aligned} A_{\text{between the square and circle}} &= A_{\text{circle}} - A_{\text{square}} \\ &= \pi r^2 - a^2 \\ &= (\pi \cdot 8^2) - (8\sqrt{2})^2 \\ &= (64\pi - 128) \text{ cm}^2. \end{aligned}$$



EXAMPLE

5

In the figure, the given three circles are tangent to each other at the points A, B and C and the points O_1 , O_2 and O_3 on AC are the centers of these circles. The radii of the smallest and middle-sized circles are given as 1 cm and 3 cm respectively. Find the area of the shaded region.



Solution Let the radii of the circles with centers O_1 , O_2 and O_3 be r_1 , r_2 and r_3 and let their areas be A_1 , A_2 and A_3 respectively. We know that O_1 , O_2 and O_3 lie on the line AC.

Since $AB = 2r_1 = 2$ cm and $BC = 2r_2 = 2 \cdot 3 = 6$ cm, we have

$AC = AB + BC = 2 + 6 = 8$ cm. So $AC = 2r_3 = 8$ cm, i.e. $r_3 = 4$ cm.

So the area of the shaded region is

$$\begin{aligned} A_3 - A_2 - A_1 &= (\pi \cdot r_3^2) - (\pi \cdot r_2^2) - (\pi \cdot r_1^2) \\ &= \pi \cdot (4^2 - 3^2 - 1^2) \\ &= (16 - 9 - 1) \cdot \pi \\ &= 6\pi \text{ cm}^2. \end{aligned}$$

EXAMPLE

An equilateral triangle with side length 12 cm is inscribed in a circle and circumscribes another circle. Find the difference in area of these two circles.

Solution We know that the area of an equilateral triangle is $\frac{a^2\sqrt{3}}{4}$ or $u \cdot r$ or $\frac{a^3}{4R}$.

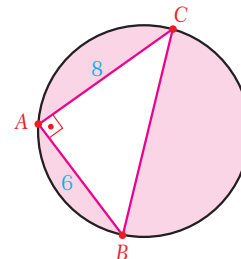
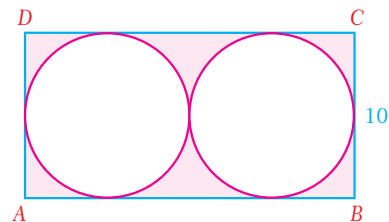
Equating these expressions gives us $\frac{12^2\sqrt{3}}{4} = \frac{12+12+12}{2} \cdot r = \frac{12 \cdot 12 \cdot 12}{4 \cdot R}$.

From this we get $r = 2\sqrt{3}$ cm and $R = 4\sqrt{3}$ cm. So

$$\begin{aligned} A_{\text{between the circles}} &= A_{\text{circumscribed circle}} - A_{\text{inscribed circle}} \\ &= \pi R^2 - \pi r^2 \\ &= \pi \cdot (4\sqrt{3})^2 - \pi \cdot (2\sqrt{3})^2 \\ &= 48\pi - 12\pi \\ &= 36\pi \text{ cm}^2. \end{aligned}$$

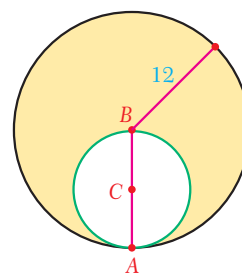
Check Yourself

1. A circle has diameter 14 units. Find its area.
2. The circumference of a circle is 4 units. Find the area of the circle.
3. A circle is inscribed in a square with side length 8 units. Find the area of the region between the square and the circle.
4. In the figure, $ABCD$ is a rectangle whose shorter side measures 10 cm. The two circles are internally tangent to this rectangle and externally tangent to each other. Find the area of the shaded region.
5. In the figure, $\triangle ABC$ is a right triangle and the circle circumscribes the triangle. Given that the legs of the triangle are 6 and 8 units long, find the area of the shaded region.





6. The two circles in the figure have centers at B and C . The circles are internally tangent to each other at point A . Given that the radius of the circle with center B is 12 cm, find the area of the shaded region.



Answers

1. 49π 2. $\frac{4}{\pi}$ 3. $64 - 16\pi$ 4. $(200 - 50\pi) \text{ cm}^2$ 5. $25\pi - 24$ 6. $108\pi \text{ cm}^2$

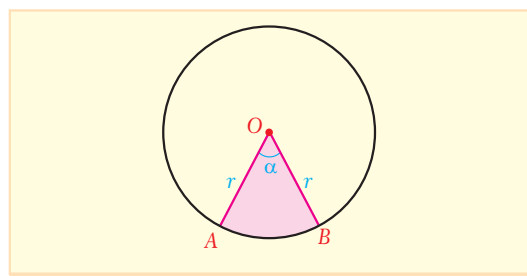
B. AREA OF A SECTOR

Definition

sector

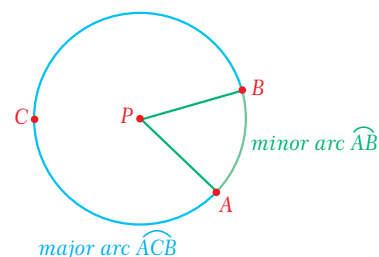
A **sector** of a circular region is a region bounded by two radii of the circle and the arc between the endpoints of these radii.

In the figure, OA and OB are radii of the circle, α is the angle between the radii, and $\widehat{AB}$ is the minor arc between the endpoints of these radii. So the shaded area is a sector of the circular region.



Note

In a circle with center P , if $m\angle APB < 180^\circ$ then the points A and B together with the points on the circle that lie in the exterior of $\angle APB$ form a **major arc** of the circle, denoted by three letters such as $\widehat{ACB}$. The points A and B together with the points on the circle that lie in the interior of $\angle APB$ form a **minor arc** of the circle, denoted by $\widehat{AB}$.



Theorem

area of a sector

The area of a sector is the product of the area of the circular region and the ratio of the angle α between the radii of the sector to a whole angle (360°):

$$A_{\text{sector}} = \frac{\alpha}{360^\circ} \pi r^2$$

Proof

Let us divide a whole circle into 360 congruent slices from its center.

Then the area of one slice is $\frac{\text{area of the circle}}{360} = \frac{\pi r^2}{360}$.

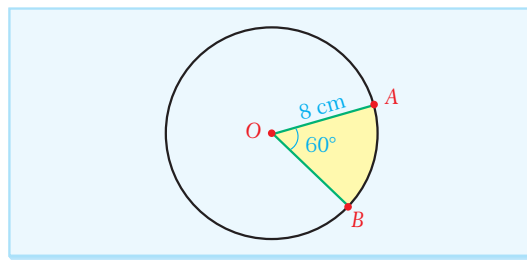
If a sector has angle α between its radii then we need α slices to match its area. So the area

of the sector is $\alpha \cdot \frac{\pi r^2}{360^\circ} = \frac{\alpha}{360^\circ} \cdot \pi r^2$.

EXAMPLE

7

The shaded region in the figure is a sector of the circle. Find the area of this sector.



Solution

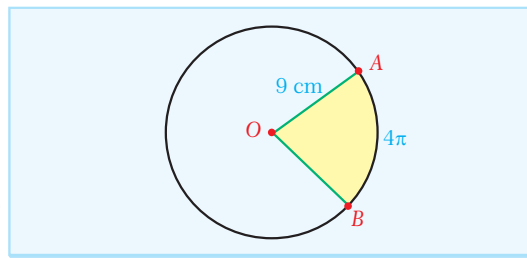
Given that the angle α is 60° and the radius of the circle is 8 cm, we have

$$A_{\text{sector}} = \frac{\alpha}{360^\circ} \pi r^2 = \frac{60}{360} \cdot \pi \cdot 8^2 = \frac{64\pi}{6} = \frac{32\pi}{3} \text{ cm}^2.$$

EXAMPLE

8

In the figure, AOB is a sector. Given that the length of the minor arc $\widehat{AB}$ is 4π cm, find the area of the sector.



Solution

We know that the length of an arc is $\frac{\alpha\pi r}{180^\circ}$. So

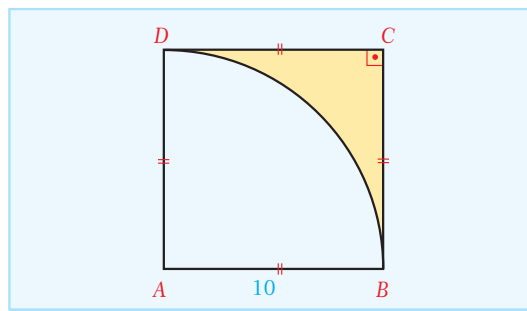
$$4\pi = \frac{\alpha\pi r}{180^\circ}, \text{ which means } 9\alpha = 720^\circ, \alpha = 80^\circ. \text{ So}$$

$$A_{\text{sector}} = \frac{\alpha}{360^\circ} \pi r^2 = \frac{80}{360} \cdot \pi \cdot 9^2 = 18\pi \text{ cm}^2.$$

EXAMPLE

9

In the figure, $ABCD$ is a square and A is the center of the quarter circle ABD . Given that one side of the square measures 10 cm, find the area of the shaded region.



Solution

We can see that one side of the square is the radius of the circle.

So $r = 10$ cm and $m(\angle DAB) = \alpha = 90^\circ$. So

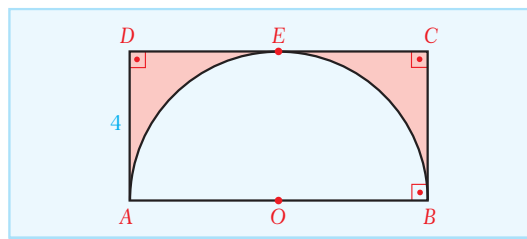
$$A_{\text{shaded region}} = A_{\text{square}} - A_{\text{sector}} = 10^2 - \left(\frac{90}{360} \cdot \pi \cdot 10^2\right) = (100 - 25\pi) \text{ cm}^2.$$

EXAMPLE

10

In the figure, $ABCD$ is a rectangle and the sector is part of a circle with center at point O which is tangent to the rectangle at point E .

Given that $AD = 4$ cm, find the sum of the areas of the shaded regions.



Solution Sector AEB is a semicircle and $m(\angle AOB) = \alpha = 180^\circ$. Since $AD = OE = r = 4$ cm, we have $AB = 2r = 8$ cm. So

$$A_{\text{shaded region}} = A_{\text{rectangle}} - A_{\text{semicircle}} = (4 \cdot 8) - \left(\frac{180}{360} \cdot \pi \cdot 4^2\right) = (32 - 8\pi) \text{ cm}^2.$$

EXAMPLE 11 A circle with radius 6 cm is given. AB is a diameter of this circle and BC is a chord. Given that $m(\angle ABC) = 45^\circ$, find the area of the region bounded by AB , BC and the minor arc $\widehat{AC}$.

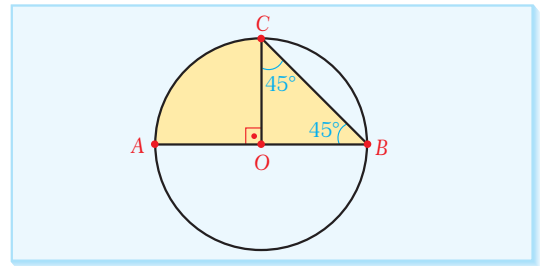
Solution Let us draw the figure and let O be the center of the circle. Drawing OC gives us $OB = OC = r$.

Since $m(\angle OBC) = 45^\circ$ and $\triangle OBC$ is isosceles, $m(\angle OCB) = 45^\circ$ and $m(\angle BOC) = \alpha = 90^\circ$.

AOC is a sector and $\triangle OBC$ is a right triangle, so

$$A_{\text{shaded region}} = A_{\text{sector}} + A(\triangle OBC) = \frac{90}{360} \cdot \pi \cdot 6^2 + \frac{6 \cdot 6}{2} = (9\pi + 18) \text{ cm}^2.$$

A chord is a straight line segment which connects two points on a circle.



EXAMPLE 12 A man has a rectangular garden with sides of length 4 m and 6 m surrounded by a fence. The man attaches a sheep to a ten-meter rope which he fixes to the middle of a longer side of the garden. What is the total area of grass outside the garden that the sheep can graze?

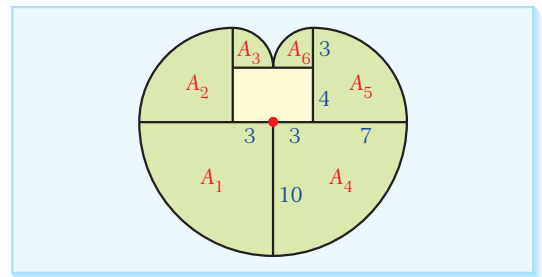
Solution Let us draw an appropriate figure (above right). In the figure, A_1, A_2, A_3, A_4, A_5 and A_6 are the quarter circles that the sheep can graze. As we can see in the figure, $A_1 = A_4, A_2 = A_5$ and $A_3 = A_6$. So

$$A_1 = A_4 = \frac{90}{360} \cdot \pi \cdot 10^2 = 25\pi$$

$$A_2 = A_5 = \frac{90}{360} \cdot \pi \cdot 7^2 = \frac{49\pi}{4}$$

$$A_3 = A_6 = \frac{90}{360} \cdot \pi \cdot 3^2 = \frac{9\pi}{4}$$

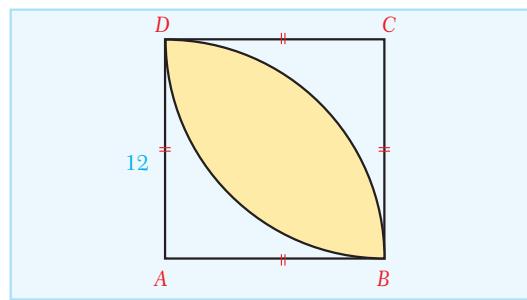
$$\begin{aligned} \text{total area} &= 2A_1 + 2A_2 + 2A_3 = 2 \cdot 25\pi + 2 \cdot \frac{49\pi}{4} + 2 \cdot \frac{9\pi}{4} \\ &= 50\pi + 29\pi = 79\pi \text{ m}^2. \end{aligned}$$



EXAMPLE

13

In the figure, $ABCD$ is a square with side length 12 cm and A and C are the centers of two quarter circles. Find the area of the shaded region.



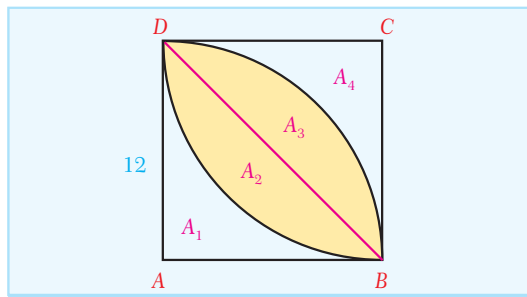
Solution

Let us draw the diagonal BD . We can easily conclude that $A_1 = A_4$ and $A_2 = A_3$. Also,

$$\begin{aligned} A_1 &= A_{\text{square}} - A_{\text{sectorBCD}} \\ &= 12^2 - \left(\frac{90}{360} \cdot \pi \cdot 12^2\right) \\ &= (144 - 36\pi) \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} A_2 &= A(\triangle ABD) - A_1 \\ &= \frac{12 \cdot 12}{2} - (144 - 36\pi) = (36\pi - 72) \text{ cm}^2. \end{aligned}$$

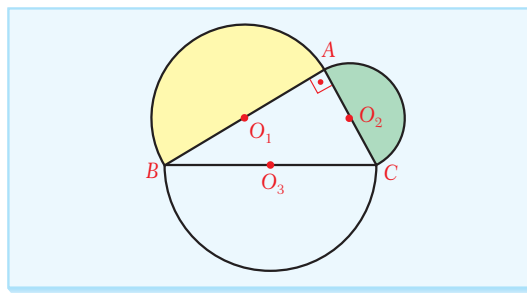
$$\text{So } A_{\text{shaded region}} = 2A_2 = 2 \cdot (36\pi - 72) = (72\pi - 144) \text{ cm}^2.$$



EXAMPLE

14

In the figure, $\triangle ABC$ is a right triangle with $m(\angle BAC) = 90^\circ$, and O_1 , O_2 , and O_3 are the centers of the three semicircles. Given that the area of the semicircle with center at O_3 is $20\pi \text{ cm}^2$, find the sum of the areas of the semicircles with centers at O_1 and O_2 .



Solution

Let the radii of the semicircles centered at O_1 , O_2 and O_3 be c , b and a respectively.

Then $AB = 2c$, $AC = 2b$ and $BC = 2a$.

By the Pythagorean Theorem we have $(2c)^2 + (2b)^2 = (2a)^2$, i.e. $c^2 + b^2 = a^2$.

Given that the area of the semicircle with center at point O_3 is 20π , we have

$$\frac{\pi a^2}{2} = 20\pi, \text{ i.e. } a^2 = 40 = c^2 + b^2.$$

The area of the semicircles with centers at O_1 and O_2 are $\frac{\pi c^2}{2}$ and $\frac{\pi b^2}{2}$ respectively.

$$\text{So the sum of their areas is } \frac{\pi c^2}{2} + \frac{\pi b^2}{2} = \frac{\pi}{2}(c^2 + b^2) = \frac{\pi}{2} \cdot 40 = 20\pi \text{ cm}^2.$$

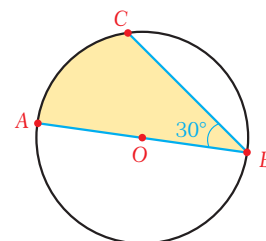


Note

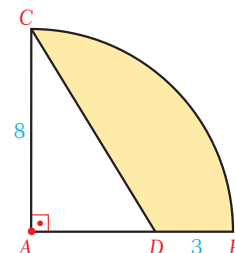
More generally, it can be shown that the sum of the areas of the circles whose diameters are the legs of a right triangle is equal to the area of the circle whose diameter is the hypotenuse.

Check Yourself

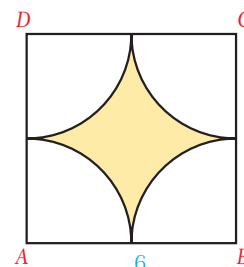
1. A sector of a circle with radius 12 units is given. The central angle of the sector is 120° . Find the area of this sector.
2. An arc of a circle measures 24 cm. Given that the radius of the circle is 8 cm, find the area of the sector created by this arc.
3. In the figure, AB is a diameter of the circle. Given that $AB = 16$ and $m(\angle ABC) = 30^\circ$, find the area of the shaded region.



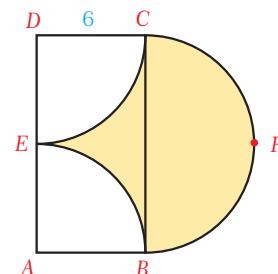
4. The figure shows a quarter circle with center at point A. Given that $AC = 8$ and $BD = 3$, find the area of the shaded region.



5. In the figure, $ABCD$ is a square and points A , B , C and D are the centers of four congruent and tangent quarter circles. Given that $AB = 6$, find the area of the shaded region.



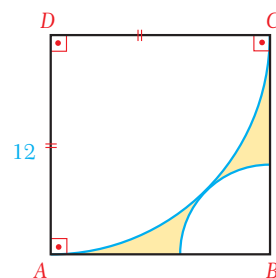
6. In the figure, A and D are the centers of two tangent quarter circles and $ABCD$ is a rectangle with shorter side length 6 units. BC is the diameter of the semicircle BFC . Find the area of the shaded region.



A central angle is an angle whose vertex lies at the center of a circle.



7. In the figure, $ABCD$ is a square with side length 12 units, and B and D are the centers of two tangent quarter circles. Find the sum of the areas of the shaded regions.



Answers

1. 48π 2. 96 cm^2 3. $\frac{32\pi}{3} + 16\sqrt{3}$ 4. $16\pi - 20$ 5. $36 - 9\pi$ 6. 72 7. $144 - 144\pi + 72\pi\sqrt{2}$

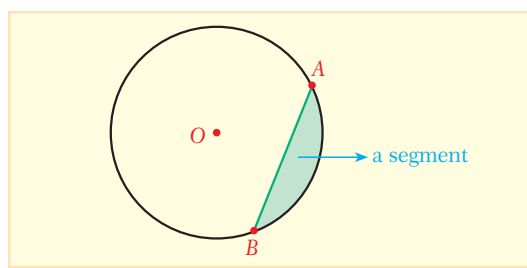
C. AREA OF A SEGMENT

Definition

segment

A **segment** is a part of a circular region which is bounded by a chord AB and the arc $\widehat{AB}$.

In the figure, AB is a chord and $\widehat{AB}$ is an arc. The shaded area shows the segment of the circle that they form.



Theorem

area of a segment

The area of a segment created by a chord AB is the difference of the areas of the sector including the arc $\widehat{AB}$ and the triangle whose vertices are A , B and the center of the circle:

$$A_{\text{segment}} = \frac{\alpha}{360^\circ} \pi r^2 - \frac{1}{2} r^2 \sin \alpha$$

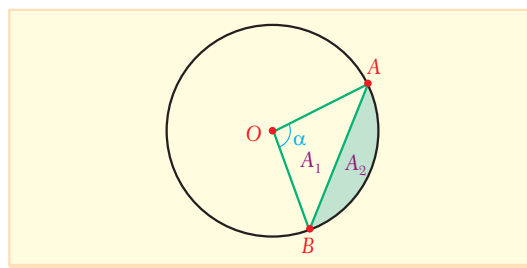
Proof

Look at the figure. $A_1 + A_2$ is the area of the sector which is bounded by the minor arc $\widehat{AB}$. Also, A_1 is an area of an isosceles triangle with sides r and vertex angle α .

So the area of the segment is

$A_2 = A_1 + A_2 - A_1$, which is the difference of the area of the sector and the area of the isosceles triangle:

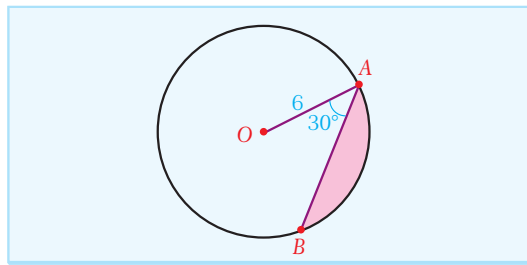
$$A_{\text{segment}} = \frac{\alpha}{360^\circ} \pi r^2 - \frac{1}{2} r^2 \sin \alpha$$



EXAMPLE

15

In the figure, O is the center of a circle with radius 6 cm. Given that $m(\angle OAB) = 30^\circ$, find the area of the segment bounded by the chord AB and the minor arc $\widehat{AB}$.



Solution Let us draw OB . Then

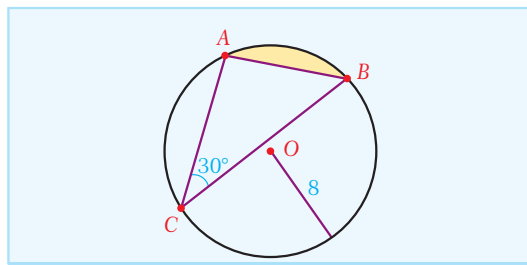
$OA = OB = 6$ cm and $m(\angle OBA) = m(\angle OAB) = 30^\circ$, which means $m(\angle AOB) = 120^\circ$. So

$$\begin{aligned} A_{\text{segment}} &= \frac{\alpha}{360^\circ} \pi r^2 - \frac{1}{2} r^2 \sin \alpha = \left(\frac{120}{360} \cdot \pi \cdot 6^2 \right) - \left(\frac{1}{2} \cdot 6^2 \cdot \sin 120^\circ \right) = 12\pi - \left(18 \cdot \frac{\sqrt{3}}{2} \right) \\ &= (12\pi - 9\sqrt{3}) \text{ cm}^2. \end{aligned}$$

EXAMPLE

16

The figure shows a circle with radius 8 cm. Given that $m(\angle ACB) = 30^\circ$, find the area of the shaded region.



Solution First of all let us find the central angle for the arc AB . $\angle ACB$ is an inscribed angle and $m(\angle ACB) = 30^\circ$, so

$m(\angle AOB) = \alpha = 2 \cdot 30^\circ = 60^\circ$. So

$$\begin{aligned} A_{\text{segment}} &= \frac{\alpha}{360} \pi r^2 - \frac{1}{2} r^2 \sin \alpha \\ &= \left(\frac{60}{360} \cdot \pi \cdot 8^2 \right) - \left(\frac{1}{2} \cdot 8^2 \cdot \sin 60^\circ \right) \\ &= \left(\frac{32\pi}{3} - 16\sqrt{3} \right) \text{ cm}^2. \end{aligned}$$



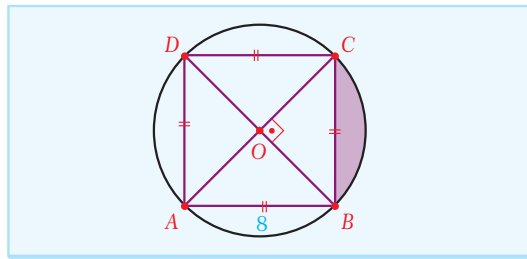
An angle ABC formed by three points A , B and C on the circumference of a circle is called an inscribed angle.



EXAMPLE

17

A circle circumscribes a square with side length 8 cm. Find the area of the segment created by one side of the square and its corresponding minor arc.



Solution 1 The diagonal of the square is equal to the diameter of the circle. So $2r = 8\sqrt{2}$, i.e. $r = 4\sqrt{2}$ cm. Since the sides of the square are the same length, the areas of the four segments are all equal. So

$$A_{\text{shaded region}} = \frac{A_{\text{circle}} - A_{\text{square}}}{4} = \frac{\pi \cdot (4\sqrt{2})^2 - 8^2}{4} = \frac{32\pi - 64}{4} = (8\pi - 16) \text{ cm}^2.$$

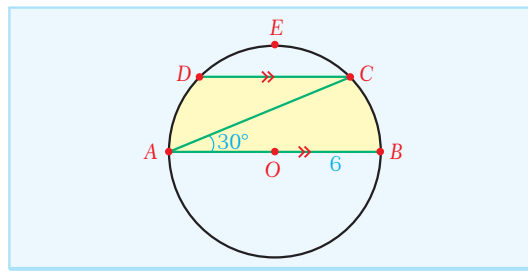
Solution 2 $r = 4\sqrt{2}$ cm (from above) and the central angle $\angle BOC$ measures 90° . So

$$A_{\text{segment}} = \frac{90}{360} \cdot \pi \cdot (4\sqrt{2})^2 - \frac{1}{2} \cdot (4\sqrt{2})^2 \cdot \sin 90^\circ = (8\pi - 16) \text{ cm}^2$$

EXAMPLE

18

The figure shows a circle with center O and radius 6 cm. Given that $AB \parallel DC$ and $m(\angle BAC) = 30^\circ$, find the area of the shaded region.

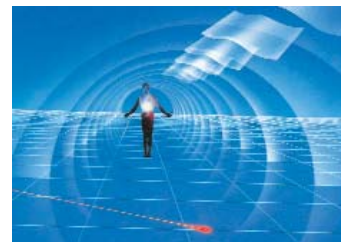


Solution $m(\angle BAC) = 30^\circ$ so the measure of the central angle of arc BC is

$$m(\widehat{BC}) = 2 \cdot m(\angle BAC) = 2 \cdot 30^\circ = 60^\circ$$

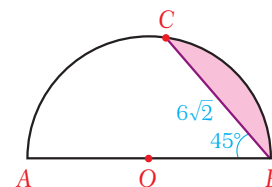
and $m(\widehat{AD}) = m(\widehat{BC}) = 2 \cdot 30^\circ = 60^\circ$, so $m(\widehat{DC}) = 60^\circ$. So

$$\begin{aligned} A_{\text{shaded region}} &= A_{\text{semicircle}} - A_{\text{segment DEC}} = \frac{\pi r^2}{2} - \left(\frac{60}{360} \pi r^2 - \frac{1}{2} r^2 \sin 60^\circ \right) \\ &= \frac{\pi \cdot 6^2}{2} - \left(\frac{60}{360} \pi \cdot 6^2 - \frac{1}{2} \cdot 6^2 \cdot \frac{\sqrt{3}}{2} \right) = 18\pi - (6\pi - 9\sqrt{3}) \\ &= (12\pi + 9\sqrt{3}) \text{ cm}^2. \end{aligned}$$

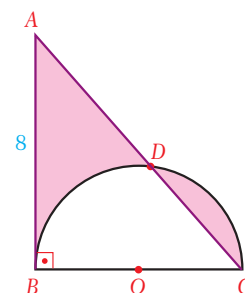


Check Yourself

1. A circle has radius 16 cm. Find the area of a segment of the circle if its central angle measures 30° .
2. The figure shows a semicircle with diameter AB . Given that $m(\angle ABC) = 45^\circ$ and $BC = 6\sqrt{2}$, find the area of the shaded region.
3. A regular hexagon is inscribed in a circle. Given that one side of the hexagon measures 12 units, find the area of the segment bounded by a side of the hexagon and its corresponding minor arc.



4. In the figure, $\triangle ABC$ is an isosceles right triangle and BC is the diameter of the semicircle. Given that $AB = 8$, find the sum of the areas of the shaded regions.



Answers

1. $(\frac{64\pi}{3} - 64) \text{ cm}^2$ 2. $9\pi - 18$ 3. $24\pi - 36\sqrt{3}$ 4. 16

D. AREA OF A RING

Definition

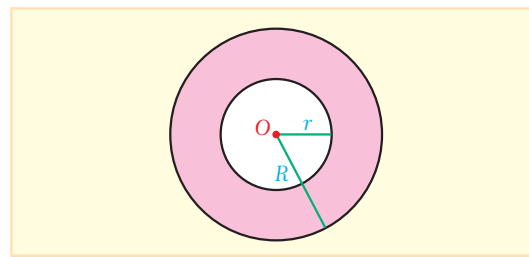
Concentric circles are circles which have a common center.



ring

The area between two concentric circles is called a **ring**.

In the figure, O is the common center of the circles and r and R are the respective radii of the smaller and bigger circles. So the shaded region is a ring.



We can easily see that the area of a ring is the difference of the areas of the larger and smaller circles:

$$A_{\text{ring}} = \pi R^2 - \pi r^2.$$

EXAMPLE

19

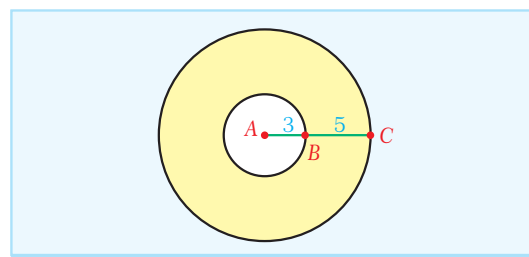
The figure shows two concentric circles with common center at A . $AB = 3$ and $BC = 5$ are given. Find the area of the shaded region.

Solution The shaded region is a ring.

We are given $AB = r = 3$ and

$AC = R = 3 + 5 = 8$.

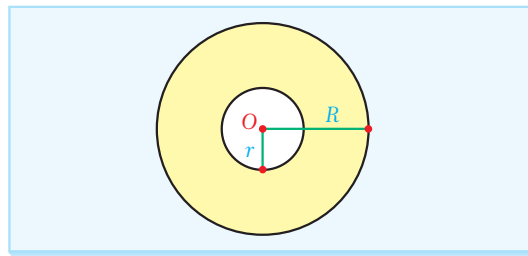
So $A_{\text{ring}} = \pi R^2 - \pi r^2 = (\pi \cdot 8^2) - (\pi \cdot 3^2) = 64\pi - 9\pi = 55\pi$.



EXAMPLE

20

The figure shows two concentric circles with center O . The difference between their circumferences is 8π cm and the area of the shaded region is 48π cm². Find the radius of each circle.



Solution

The difference of the circumferences of the circles is $2\pi R - 2\pi r = 8\pi$, which gives us

$R - r = 4$. Also, the shaded area is $\pi R^2 - \pi r^2 = 48\pi$, so $R^2 - r^2 = 48$.

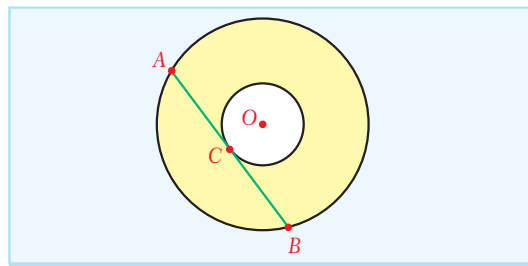
Writing $R^2 - r^2 = (R + r) \cdot (R - r) = 48$ and using $R - r = 4$ gives us $(R + r) \cdot 4 = 48$, i.e. $R + r = 12$.

Solving $R - r = 4$ and $R + r = 12$ simultaneously gives us $R = 8$ cm and $r = 4$ cm.

EXAMPLE

21

The figure shows two concentric circles. AB is a chord of the larger circle and is tangent to the smaller circle at point C . Given that $AB = 12$ cm, find the area of the shaded region.



Solution

Let the radius of the larger circle be $OB = R$, and the radius of the smaller circle be $OC = r$.

Let us draw OC . By the properties of chords we have $OC \perp AB$ and $AC = BC = 6$ cm. So

$$OB^2 = OC^2 + BC^2, \text{ i.e. } BC^2 = OB^2 - OC^2 = R^2 - r^2 = 6^2 = 36.$$

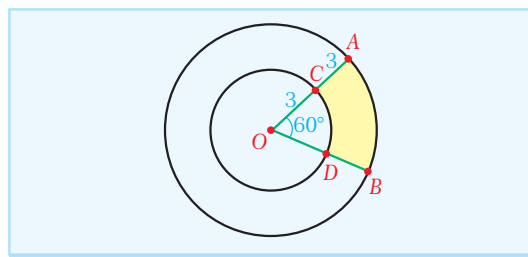
So the area of the shaded region is $\pi R^2 - \pi r^2 = (R^2 - r^2) = 36\pi$ cm².



EXAMPLE

22

The figure shows two concentric circles centered at point O . A sector AOB with a 60° central angle is drawn in these circles. Given that $OC = 3$ cm and $CA = 3$ cm, find the area of the shaded region.



Solution

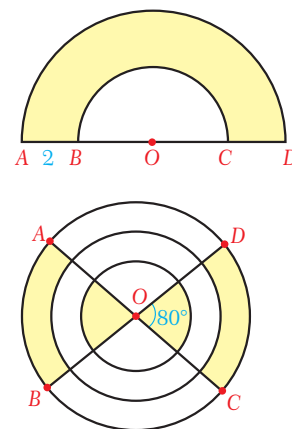
We have $\alpha = 60^\circ$, $r = 3$ cm and $R = 6$ cm.

Combining the formulas for the area of a ring and the area of a sector gives us the answer:

$$A_{\text{shaded region}} = \frac{\alpha}{360} (\pi R^2 - \pi r^2) = \frac{60}{360} (\pi \cdot 6^2 - \pi \cdot 3^2) = \frac{9\pi}{2} \text{ cm}^2.$$

Check Yourself

- Two concentric circles have radii 12 cm and 9 cm. Find the area of the region between these two circles.
- The figure shows two concentric semicircles with common center at the point O . Given that $AB = 2$ and $AD = 12$, find the area of the shaded region.
- The figure shows three concentric circles centered at point O . The radii of the circles are 3 cm, 4 cm and 5 cm, $m(\angle DOC) = 80^\circ$ and AC and BD are diameters of the biggest circle. Find the sum of the areas of the shaded regions.



Answers

- $63\pi \text{ cm}^2$
- 10π
- $8\pi \text{ cm}^2$

E. RATIOS IN CIRCLES

We can easily see that all circles are similar to each other. Their ratio of similarity is the same as the ratio of their radii.

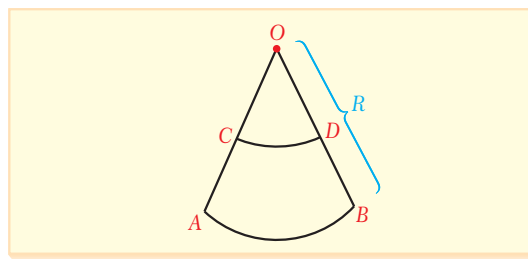
Properties 12

Let r and R be the radii of two circles. Then

- the ratio of similarity is $k = \frac{r}{R}$.
- the ratio of their circumferences is $k = \frac{r}{R}$.
- the ratio of their areas is $k^2 = \left(\frac{r}{R}\right)^2$.

Furthermore, if AOB and COD are sectors of two concentric circles as shown in the figure then

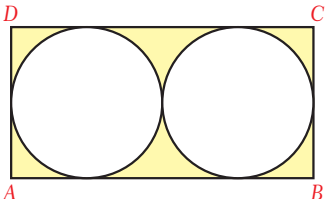
$$4a. \frac{\widehat{CD}}{\widehat{AB}} = \frac{r}{R}. \quad 4b. \frac{A(COD)}{A(AOB)} = \left(\frac{r}{R}\right)^2.$$



EXERCISES 6.1

A. Area of a Circular Region

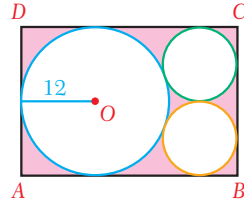
1. A circle has radius $\sqrt{7}$ units. Find its area.
2. A circle has area 1 square unit. Find its radius.
3. A circle has circumference 20π . Find its area.
4. Find the area of a circle whose diameter is 5.2 units.
5. A circle has circumference 10. Find its area.

6. In the figure, $ABCD$ is a rectangle and the two circles are tangent to the sides of the rectangle. Given that the circles are tangent to each other and both have radius 6 units, find the sum of the areas of the shaded regions.
 

7. Two circles are internally tangent to each other. The distance between their centers is 12 units and the diameter of the larger circle is 34 units. Find the difference between the areas of these two circles.

8. The area of a circle is twice the sum of the areas of three circles with radii 3 cm, 4 cm and 5 cm. Find the radius of this circle.

9. A man is connected to a boat with a rope which is 10 m long. The water is 6 m deep. What is the area of the region that the man can walk in?

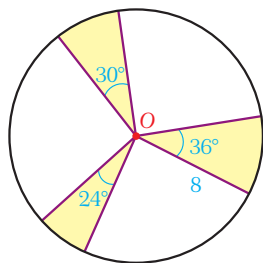
10. In the figure, $ABCD$ is a rectangle and the three circles are tangent to each other. The two small circles are also congruent to each other. Given that the radius of the largest circle is 12 units, find the sum of the areas of the shaded regions.
 

11. From one point on a circle we draw two chords with lengths 14 units and 48 units. The length of the line segment that connects the midpoints of these chords is 25 units. Find the area of the circle.

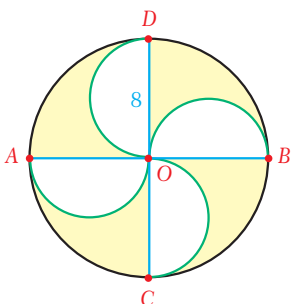
B. Area of a Sector

12. A circle has radius 12 units. Find the area of the sector of this circle which is bounded by an arc with central angle 120° .
13. A circle has radius 6 units. Find the area of the sector of this circle which is bounded by an arc with a central angle of 1.5 radians.

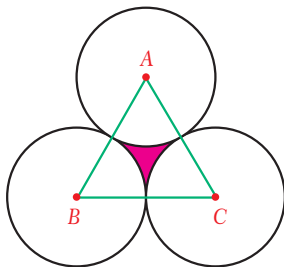
14. The figure shows, three central angles with measures 36° , 24° and 30° . Given that the radius of the circle is 8 units, find the sum of the areas of the shaded regions.



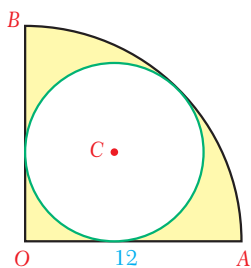
15. The figure shows a circle and four semicircles. The radius of the larger circle is 8 cm. Find the sum of the areas of the shaded regions.



16. In the figure, $\triangle ABC$ is an equilateral triangle with side length 8. Given that the three congruent circles have centers at points A, B and C, find the area of the shaded region.



17. The figure shows a quarter circle centered at O with radius 12 units. The circle with center at point C is tangent to OA, OB and the arc $\widehat{AB}$. Find the sum of the areas of the shaded regions.

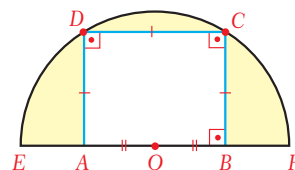


C. Area of a Segment

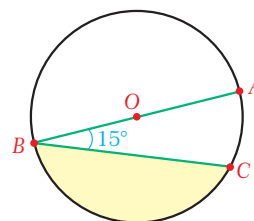
18. A regular octagon is inscribed in a circle with radius 8 units. Find the area of the segment bounded by one side of the octagon and the arc corresponding to this side.

19. An equilateral triangle $\triangle ABC$ has side length 8 units. Vertex A is at the center of a circle and vertices B and C are on the circle. Find the area of the segment which is between side BC and the minor arc $\widehat{BC}$.

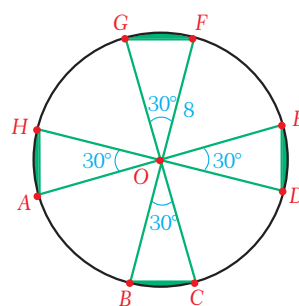
20. In the figure, ABCD is a square with side length 6. Given that O is the center of the semicircle and $AO = OB$, find the sum of the areas of the shaded regions.



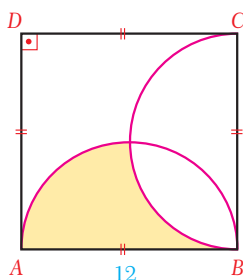
21. In the figure, O is the center of the given circle. Given that $m(\angle ABC) = 15^\circ$ and the radius of the circle is 12 units, find the area of the shaded region.



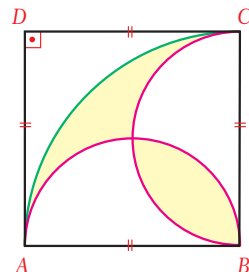
22. In the figure, O is the center of a circle with radius 8 units. Given that the measures of $\angle AOH$, $\angle BOC$, $\angle DOE$ and $\angle FOG$ are all 30° , find the sum of the areas of the shaded regions.



23. In the figure, $ABCD$ is a square with side length 12 units. Given that AB and BC are the diameters of the two semicircles, find the area of the shaded region.



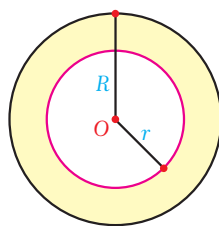
24. In the figure, $ABCD$ is a square with side length 8 units. Given that B is the center of the quarter circle and AB and BC are the diameters of the two semicircles, find the sum of the areas of the shaded regions.



D. Area of a Ring

25. Two concentric circles have radii 3 cm and 9 cm. Find the area of the region between these two circles.

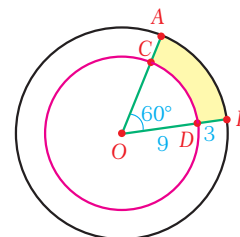
26. In the figure, O is the common center of two circles with radii R and r . Given that $R + r = 10$ and $R - r = m$, find the area of the shaded region in terms of m .



27. Two concentric circles have radii R and r . Given that the sum of the circumferences of these circles is 18π and the area of the region between them is 9π , find the radii of these circles.

28. A regular hexagon with side length 8 units is given. Find the area of the region between its inscribed and circumscribed circles.

29. In the figure, O is the center of both of the given circles. Given that $OD = 9$, $DB = 3$ and $m(\angle AOB) = 60^\circ$, find the area of the shaded region.



INTRODUCTION TO SPACE GEOMETRY

The geometry you have studied thus far has been largely plane geometry. In plane geometry you have learned definitions and properties of the figures in plane.

However we live in three dimensional world. Therefore it is necessary to extend your study of geometry to include figures having three dimensions, that is, figures having thickness as well as length and width.

The geometry of three dimensional figures is called Three-dimensional geometry or Space geometry.

In this chapter we will study lines and planes in space. We will state axioms, definitions, theorems about them. In proofs of theorems sometimes we will use theorems from plane geometry and we will not prove them.

Axioms of Space Geometry

In plane geometry you learned that two points determine a line. In space two points determine a line, too. That means, there can be drawn one and only one line passing through two points. The points which are on the same line are called as collinear points and the points which are not on the same line are called as non-collinear points. Now let us state three basic axioms of space geometry which will be helpful to prove theorems.

1. Axioms

Three non-collinear points determine a plane.

That means if we have three non-collinear points there can be drawn one and only one plane containing these points. Note that the points need to be non-collinear. If the points are collinear there can be drawn infinitely many planes containing these points.

2. Axioms

A plane having two points in common with a line contains this line.

If two points of a line belong to a plane, every point on this line is in the plane. So, to show that a line lies in a plane it is enough to find two points on the line belonging to the given plane.

3. Axioms

The intersection of two intersecting planes is a line.

Therefore if two planes have one point in common then they have a line in common and any point belonging to both planes is on this line.

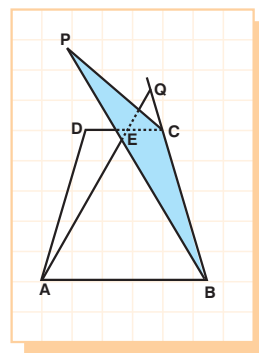
EXAMPLE

1 ABCD is a trapezoid so that $AB \parallel CD$, E is a point on the line segment DC, and P is a point not in plane ABC. Q is the intersection point of line AE and plane PBC. If $\angle CBA = 90^\circ$, $AB = 9$ cm, $EC = 3$ cm, and $BC = 8$ cm, find EQ.

Solution Since points A and E are in plane ABCD, all points on line AE are in that plane. Since the intersection of planes ABC and PCB is line BC, intersection of line AE and plane PCB is on BC. Since both AE and BC are in plane ABCD, they intersect. This intersection is given as Q. (Figure 1.1) From plane geometry we know that $\triangle DECQ$ and $\triangle DQAB$ are similar.

$$\text{So } \frac{QC}{QB} = \frac{EC}{AB} \Rightarrow \frac{QC}{QC+BC} = \frac{3}{9} \Rightarrow \frac{QC}{QC+8} = \frac{1}{3} \Rightarrow QC = 4 \text{ cm.}$$

$$\angle ECQ = \angle CBA = 90^\circ.$$



(Figure 1.1)

Hence, by applying the Pythagorean theorem in $\triangle ECQ$, we get $EQ = 5$ cm.

A. Lines and Planes in Space

1. Determination of a Plane

In Axiom 1.1 we stated that three non-collinear points determine a plane. We have other possibilities to determine a plane.

a. Parallel Lines

From the definition of parallel lines we know that parallel lines lie in a plane. So two parallel lines are coplanar.

Theorem

There can be drawn only one plane containing two parallel lines.

Proof

Let m and d be two parallel lines in space. Let a and b be two different planes containing both m and d . Since a and b have common points, which are the points on m and d , they are intersecting planes. By Axiom 1.3 this intersection must be a line. But here the intersection is the union of two lines m and n . There is a contradiction. So a and b are coincident

b. A Line and a Point

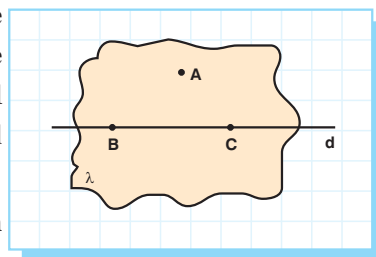
Theorem

A line and a point not on this line determine a plane.

Proof

Let d be a line and A be a point not on line d . Let us take points B and C on d . (Figure 1.2) Since points A, B, C are three non-collinear points they determine a plane l . Then l contains two points of line d . So d lies in l . Therefore line d and point A determine plane l .

Namely, there is only one plane, which is l , containing both point A and line d



c. Two Intersecting Lines

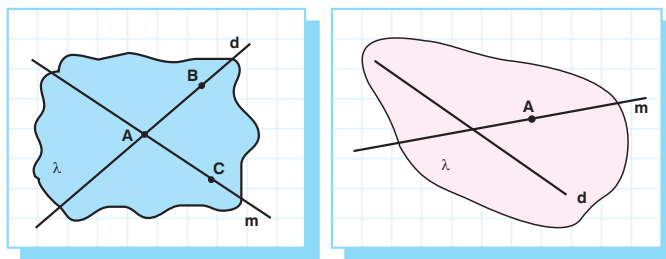
Theorem

Two intersecting lines determine a plane.

Proof

Let d and m be two intersecting lines and A be their intersection point. (Figure 1.3) Besides A , let us take two points B and C , on lines d and m , respectively. Since A, B, C are three non-collinear points, they determine a plane l . Then l will contain both lines since it has two points in common with each of the lines.

Proof Let d and m be two intersecting lines and A be a point on m but not on d . (Figure 1.4) By the previous theorem d and A determine a plane. Since d lies in this plane, the plane contains the intersection point of the lines. So line m lies in the plane, because the plane contains two points of m

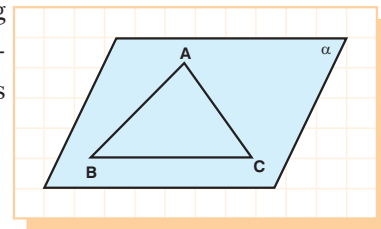


EXAMPLE

2

Show that all sides of a triangle are in the same plane.

Solution Let ABC be a triangle. Since AB and AC are intersecting lines they determine a plane a . Since two points of line segment BC are in a , a contains all points on BC . So, all sides of a triangle are in the same plane.



EXAMPLE

3

A, B, C, D are four non-coplanar points. Can three of these points be on the same line?

Solution No. Because if three of these points are on a line then for the fourth point we have two cases:

- a) It can be on this line. Then there can be drawn infinitely many planes containing these four points.
- b) If it is not on this line then the line and the point not on this line determine a plane, and this plane contains all the given points.

In both cases the points will be coplanar. However it is given that points are non-coplanar. So three of them can not be collinear.

2. Mutual Positions of Two Lines in Space

In space two lines may have infinitely many common points, one common point or no common point.

a. Infinitely Many Common Points

Two lines have infinitely many common points if they are coincident. In this case all of the points of one line will be on the other line. So their intersection is the set of the points on any of these lines. Actually, if two lines have two common points, they are coincident.

b. One Common Point

From plane geometry we know that if two lines are intersecting and not coincident then they intersect each other at a unique point. That means they have only one point in common. By intersecting lines we mean the lines having one common point.

We proved that two intersecting lines determine a plane. So intersecting lines are always coplanar lines.

c. No Common Point

i. Parallel Lines

Parallel lines are defined as coplanar lines having no points in common. So, by the definition, parallel lines have no common point.

From plane geometry we know that in a plane through a point not on a line there can be drawn one and only one line parallel to the given line. We proved that a line and a point not on this line determine a plane. Since this plane is unique and in this plane, through the given point, there can be drawn one and only one line parallel to the given line it can be concluded that:

In space, through a point not on a line, a line parallel to the given line can be drawn and this line is unique.

ii. Skew Lines

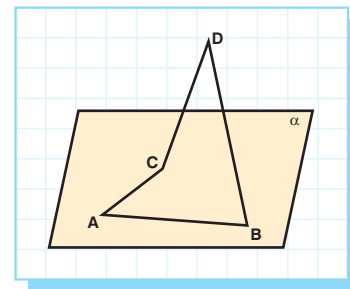
Two lines are skew to each other if there is not any plane containing both lines. So, these lines can not have any intersection. As it is proven, intersecting lines are always coplanar. By definition skew lines cannot be coplanar. So, skew lines have no common point.

Skew Quadrilateral

Let A, B, C be three non-collinear points. Then they determine a plane α . Let D be a point not in α . (Figure 1.6) Then we have four non-coplanar points. When we take the union of line segments AB, BD, CD and CA we get a quadrilateral called as a skew quadrilateral. Here is its formal definition.

Definition

A skew quadrilateral is a quadrilateral whose vertices are four non-coplanar points.



3. Mutual Positions of a Plane and a Line

There are three possible cases for mutual positions of a line and a plane.

a. Line Lying in the Plane

As mentioned in Axiom 1.2 if two points of a line belong to a plane, then the line lies in the plane. So, in order to show that a line lies in a plane we must find two points on the line belonging to the plane. For this case, the intersection of line and plane is the line itself. In other words, the set of the points on the line is a subset of the set of the points in the plane.

b. Line Intersecting the Plane

In this case they have one common point and their intersection is this point.

c. Line Parallel to the Plane

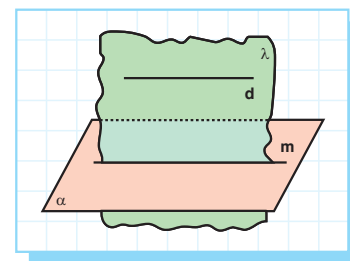
A line and a plane are said to be parallel if they have no point in common. After this definition let us state and prove some theorems about parallelity of a plane and a line.

Theorem

If a line is parallel to another line lying in a plane, the line will be parallel to the plane.

Proof

Let d be a line parallel to another line m lying in plane α . Since d and m are parallel lines they lie in a plane λ . α and λ are intersecting planes along line m . So, if d and α intersect each other, the intersection point must be on m . This contradicts with the parallelity of m and d . So d and α have no common point, in other words they are parallel

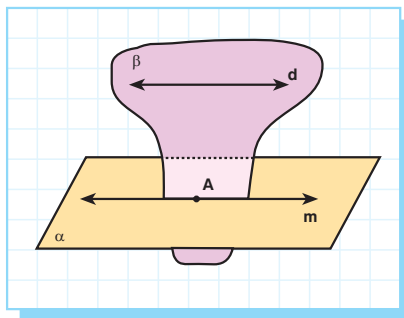


Theorem

If a line is parallel to a plane, in this plane there are lines parallel to the given line.

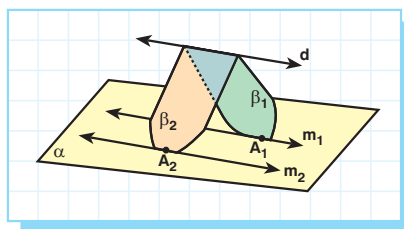
Proof

Let d be a line parallel to a given plane α and A be any point in α (Figure 1.8). Then d and A determine a plane β . β and α have a common point, that is A . So, they have a common line. Let us name this line as m . Both d and m are in β . Since d has no common point with α , it can not intersect m . So, d and m are parallel lines.



Therefore, a line is parallel to a plane if and only if it does not lie in the plane and it is parallel to a line lying in that plane.

Let d be a line parallel to a plane α and A_1 be a point in α . Then d and A_1 determine a plane β_1 . Let m_1 be the intersection of α and β_1 . We proved that $m_1 \parallel d$. Let A_2 be a point in α but not on m_1 . Line d and A_2 determine another plane β_2 . Let m_2 be the intersection of α and β_2 . Then $m_2 \parallel d$. Now let us think about m_1 and m_2 . They are both in α . If they have a common point, this point will be a common point of planes β_1 and β_2 . Since d is the intersection of β_1 and β_2 this common point must be on d . This is impossible. Because $d \parallel m_1$ and $d \parallel m_2$. So m_1 and m_2 can not have any common point. Therefore, they are parallel.



As a conclusion, if a line is parallel to a plane then in this plane there are infinitely many lines parallel to the given line and these lines are parallel to each other

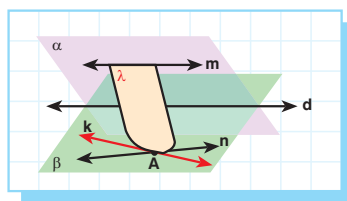
Theorem

Two lines parallel to the same line are parallel.

Proof

Let m , n and d be three lines in space so that $m \parallel d$ and $n \parallel d$

Since m and d are parallel they determine a plane α , and since n and d are parallel they determine another plane β . Let A be a point on n . Line m and point A determine a plane λ . Since β and λ have a common point A they have a common line k . We know that $m \parallel d$. So, m is parallel to k . Then lines d and k are two lines in plane β parallel to line m . So, k and d are parallel. Through point A there can be drawn only one line parallel to d . So, n and k should be coincident lines. Therefore, m and n are parallel lines



Conclusion

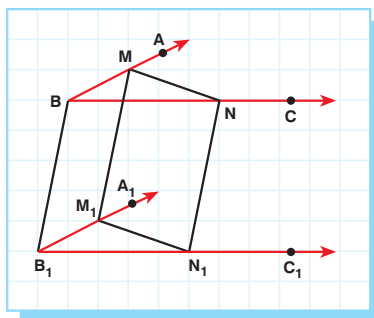
If one of two parallel lines is parallel to a plane the other is in the plane or parallel to the plane..

Theorem

Two angles with respectively parallel arms in the same direction are congruent.

Proof

Let $\angle ABC$ and $\angle A_1B_1C_1$ be two angles with respectively parallel arms in the same direction. Let M and N be any two points on arms BA and BC respectively. On B_1A_1 and B_1C_1 take two points M_1 and N_1 so that $M_1B_1 = MB$ and $N_1B_1 = NB$.



Since $BA \parallel B_1A_1$, BMM_1B_1 is a parallelogram.

So $BB_1 \parallel MM_1$ and $BB_1 = MM_1$. (1)

Similarly $BC \parallel B_1C_1$ and BNN_1B_1 is a parallelogram.

So $BB_1 \parallel NN_1$ and $BB_1 = NN_1$. (2)

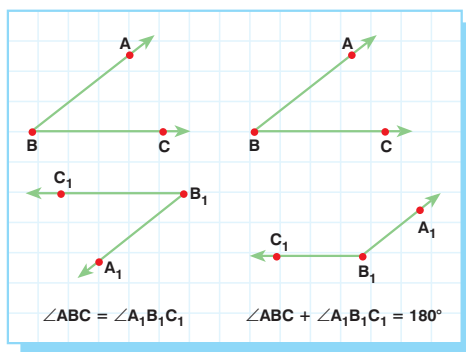
From (1) and (2) we get $NN_1 \parallel MM_1$ and $NN_1 = MM_1$.

So MNN_1M_1 is a parallelogram and $MN = M_1N_1$.

Then by S.S.S, $\triangle MBN$ and $\triangle M_1B_1N_1$ are congruent. That means $\angle MBN = \angle M_1B_1N_1$

Conclusion

1. If the corresponding arms of two angles are parallel and are in opposite directions, the angles are equal.
2. If the corresponding arms of two angles are parallel and if one of corresponding arms is in the same direction while the other is in opposite then the sum of the angles is 180° .



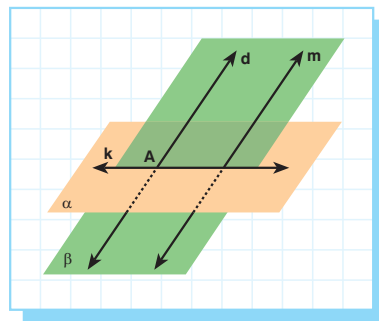
Theorem

If one of two parallel lines intersects a plane, the other intersects too.

Proof Let a be a plane and d, m be two parallel lines. Let d intersect plane a at a point A . We need to show that m also intersects a .

Since d and m are parallel they determine a plane b . Planes a and b have a common point. So they have a common line k .

Lines k, d and m are in the same plane, $d \parallel m$ and k intersects d . So k intersects m too. Since m intersects k and k is in a , m intersects a .



Proof Let a be a plane, d and m be two parallel lines and d intersect a . For m there are three positions:

It lies in a or it is parallel to a or it intersects a .

If m is in a then d will be parallel to a line in a . So d is parallel to a . This is a contradiction.

If m is parallel to a then in a there will be a line (for example n) parallel to m . Since $d \parallel m$ and $m \parallel n$, it can be concluded that $d \parallel n$. For this case again d will be parallel to a .

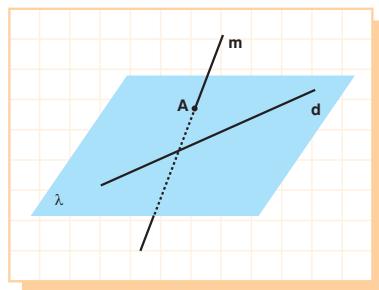
Hence m intersects a .

EXAMPLE

4

Show that if one of two lines lies in a plane and the other intersects this plane at a point not on the first line then these lines are skew to each other.

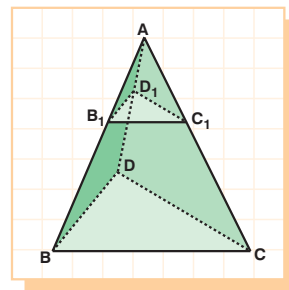
Solution Let d be a line in a plane l and m intersect l at a point A . Assume that there is a plane b containing both m and d . Then d and A will be in b . So b and l will be coincident. However l does not contain m . So b can not contain m either. Hence there is no plane containing both d and m . It means that they are skew.



EXAMPLE

5 A is not in (BCD), $B_1C_1 \parallel BC$,
and $C_1D_1 \parallel CD$.

If $\frac{AB_1}{B_1B} = \frac{1}{2}$ and the perimeter of DBCD is 24 cm, find the
perimeter of $DB_1C_1D_1$



Solution Since $B_1C_1 \parallel BC$, triangles AB_1C_1 and ABC are similar.

$$\text{So } \frac{B_1C_1}{BC} = \frac{AC_1}{AC} = \frac{AB_1}{AB}.$$

$$\text{Since } \frac{AB_1}{B_1B} = \frac{1}{2}, \quad \frac{AB_1}{AB} = \frac{1}{3}.$$

$$\text{So } \frac{B_1C_1}{BC} = \frac{AC_1}{AC} = \frac{1}{3}, \quad BC = 3B_1C_1$$

Moreover, since $C_1D_1 \parallel CD$, $\triangle C_1D_1A \sim \triangle CDA$.

$$\text{So } \frac{C_1D_1}{CD} = \frac{AD_1}{AD} = \frac{AC_1}{AC} = \frac{1}{3}, \quad CD = 3C_1D_1.$$

$$\text{Since } \frac{AB_1}{AB} = \frac{AD_1}{AD} = \frac{1}{3},$$

$$\triangle AB_1D_1 \sim \triangle ABD.$$

$$\text{Therefore, } \frac{B_1D_1}{BD} = \frac{AB_1}{AB} = \frac{1}{3}, \quad BD = 3B_1D_1.$$

$$P_{BCD} = 24 \text{ cm}, \quad BC + BD + CD = 24 \text{ cm},$$

$$3(B_1C_1 + B_1D_1 + C_1D_1) = 24 \text{ cm}, \quad P_{B_1C_1D_1} = 8 \text{ cm}.$$

Hence the perimeter of $DB_1C_1D_1$ is 8 cm.

EXAMPLE



Show that the midpoints of the sides of a skew quadrilateral are the vertices of a parallelogram.

Solution Let ABCD be a skew quadrilateral and M, N, P, Q be the midpoints of sides AB, BC, CD, DA, respectively.

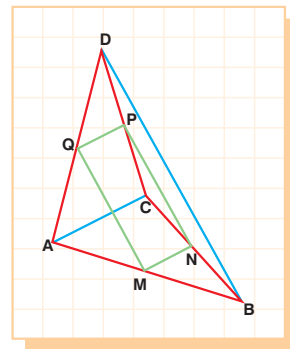
In $\triangle DAC$ and $\triangle BAC$, $QP \parallel AC$, $MN \parallel AC$, $QP = \frac{AC}{2}$, and

$$MN = \frac{AC}{2}.$$

So $QP \parallel MN$ and $QP = MN$.

If we use the same logic in $\triangle DAB$ and $\triangle DCB$ we will obtain that $QM \parallel PN$ and

$$QM = PN = \frac{BD}{2}. \text{ Hence } QMNP \text{ is a parallelogram.}$$



EXAMPLE

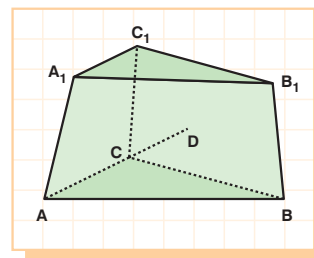


$A_1C_1 \parallel AC$,
 $C_1B_1 \parallel CB$, and $A_1B_1 \parallel AB$.

If $A_1C_1 = 5$ cm,

$A_1B_1 = 39$ cm and

$B_1C_1 = 7$ cm, find the
measure of angle DCB.



Solution Since $\angle A_1C_1B_1$ and $\angle ACB$ are two angles with respectively parallel arms in the same direction, they are equal. In $\triangle A_1B_1C_1$ by cosine theorem we get

$$A_1B_1^2 = A_1C_1^2 + B_1C_1^2 - 2A_1C_1 \times B_1C_1 \times \cos \angle C_1,$$

$$39^2 = 5^2 + 7^2 - 2 \times 5 \times 7 \times \cos \angle C_1, \quad \cos \angle C_1 = \frac{1}{2}.$$

So $\angle A_1C_1B_1 = 60^\circ$. Then $\angle ACB = 60^\circ$ and $\angle DCB = 120^\circ$.

EXAMPLE



One side of a rhombus ABCD is 4 cm. Sides AB and AD intersect a plane α at points P and Q respectively. $AP = 1$ cm and $AQ = 3$ cm are given.

- a) Show that lines CB and CD intersect α .
- b) If CB and CD intersect α at P₁ and Q₁, respectively, find the lengths of CP₁ and CQ₁

Solution a) In a rhombus opposite sides are parallel. So $AB \parallel CD$ and $AD \parallel BC$ (Figure 1.18). If one of two parallel lines intersects a plane, the other intersects too.

It is given that AB and AD intersect plane α . Hence CD and CB intersect α too.

- b) Points P₁, Q₁, P and Q are all intersections of planes α and ABC. So they are collinear.

Since $AQ \parallel P_1B$, triangles APQ and BPP₁ are similar.

$$\frac{BP_1}{BP} = \frac{AQ}{AP}, \quad \frac{BP_1}{3} = \frac{3}{1}, \quad BP_1 = 9 \text{ cm.}$$

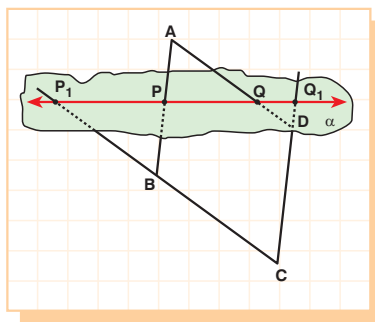
$$\text{So } CP_1 = CB + BP_1 = 13 \text{ cm.}$$

On the other hand, since $AP \parallel DQ_1$, we obtain

$\triangle APQ \sim \triangle DQ_1Q$. Then

$$\frac{DQ_1}{DQ} = \frac{AP}{AQ}, \quad \frac{DQ_1}{1} = \frac{1}{3}, \quad DQ_1 = \frac{1}{3} \text{ cm.}$$

$$\text{So } CQ_1 = CD + DQ_1 = 4 + \frac{1}{3} = \frac{13}{3} \text{ cm.}$$



EXAMPLE**9**

Show that when two parallel lines are intersected by a line, all these three lines lie in the same plane.

Solution

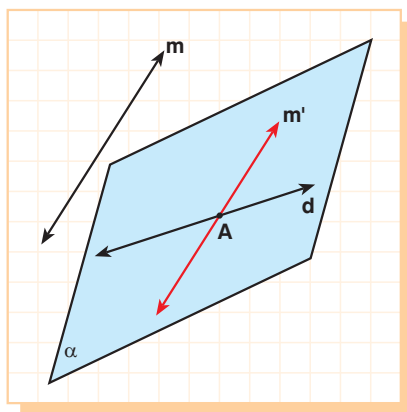
Two parallel lines determine a plane. The line intersecting these parallel lines has two points in common with this plane which are the intersection points. So, it lies in this plane too.

EXAMPLE**10**

Show that through one of two skew lines, there can be drawn a plane parallel to the other.

Solution

Let m and d be two skew lines and A be a point on d . Through A let us draw line m' parallel to m . d and m' determine a plane α . Since $m \parallel m'$ and m' is in α , $m \parallel \alpha$.



4. Mutual Positions Of Two Planes

Two planes can be coincident, intersecting or parallel.

a. Coincident Planes

If two planes have three non-collinear common points then these two planes are coincident. So all points of these two planes are common.

b. Intersecting Planes

Two planes may intersect each other. According to Axiom 1.3 the intersection of two intersecting planes is a line.

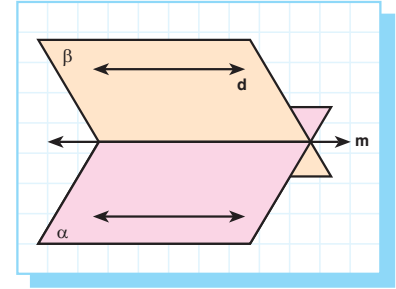
Theorem

If a plane passes through a line parallel to another plane and intersects that plane then the line of intersection of two planes is parallel to the given line.

Proof

Let d be a line parallel to a plane α , and β be a plane containing d and intersecting α along line m .

Then, d and m lie in β . Since m is in α and $d \parallel \alpha$, d and m can not intersect each other. Therefore they are parallel.

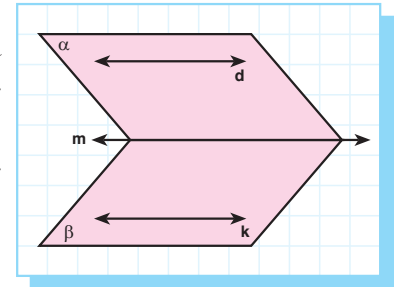


Theorem

If two parallel lines lie in two intersecting planes, the intersection of the planes is parallel to the given lines.

Proof

Let d and k be two parallel lines lying in two intersecting planes α and β , respectively. Let m be the intersection of α and β . Lines d and m are in the same plane. Since d is parallel to one line lying in β (line k) it is parallel to plane β . Since m is in β , d can not intersect m . So d and m are parallel. By analogy, k and m are parallel too.



Theorem

If a line is parallel to two intersecting planes, it is parallel to the intersection of these planes.

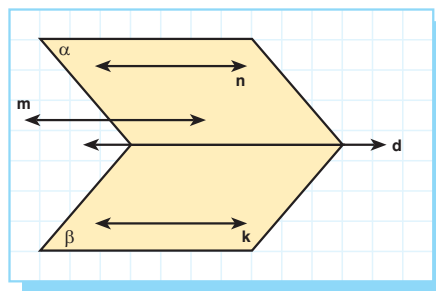
Proof Let α and β be two planes intersecting along line d and m be a line parallel to both α and β .

Since m is parallel to α and β , in α and β there can be found lines n and k parallel to m .

Since $n \parallel m$ and $k \parallel m$, $n \parallel k$.

Then by the previous theorem, $d \parallel n$ and $d \parallel k$.

Since $d \parallel n$ and $n \parallel m$, $d \parallel m$.



c. Parallel Planes

If two planes have no common point, they are called as parallel planes.

Theorem

If two intersecting lines in a plane are respectively parallel to two intersecting lines in another plane, the planes are parallel.

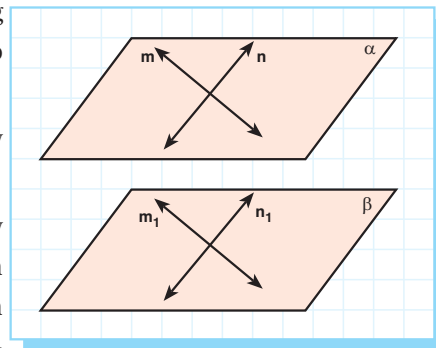
Proof Let α and β be two planes. Let m, n be two intersecting lines in α and let m_1, n_1 be two intersecting lines in β so that $m \parallel m_1$ and $n \parallel n_1$.

We need to prove that α and β are parallel, namely they do not have any common point.

Assume that they have a common point. Then they will have a common line. Let d be this line. Since m and n are parallel to m_1 and n_1 respectively both m and n are parallel to d . So none of m and n can intersect line d .

Since m, n and d are in the same plane, $m \parallel d$ and $n \parallel d$. But in this case, m and n must be coincident or parallel lines. However it is given that they are intersecting lines. So there is a contradiction.

Hence α and β can not have any common point. That means they are parallel planes.



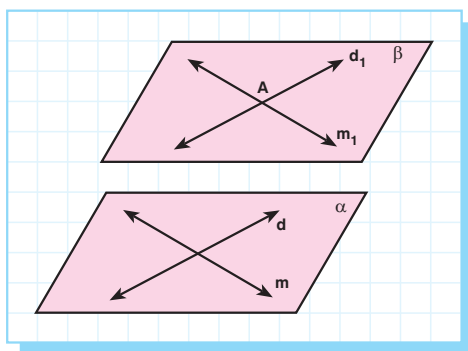
Theorem

Through a point not in a plane there can be drawn one and only one plane parallel to the given plane.

Proof

Let a be a plane and A be a point not in a . We need to prove that

- a. Through A , there can be drawn a plane parallel to a .
- b. This plane is unique.



a. The plane exists

Let d and m be two intersecting lines in a . Through A there can be drawn a line parallel to d and another line parallel to m . Let us name these lines as d_1 and m_1 . Lines d_1 and m_1 are intersecting lines. So they determine a plane b . By previous theorem a and b are parallel.

b. The plane is unique

Assume that there is another plane b' , containing A and parallel to a . b' can not contain both d_1 and m_1 . So at least one of d_1 and m_1 intersects b' .

Let d_1 be this line. Since d and d_1 are parallel, d also intersects plane b' .

This contradicts with the parallelity of b' and a . So plane b is unique

Conclusion

1. The lines parallel to a given plane and passing through a given point not in the given plane lie in the plane parallel to the given plane, and containing the given point.
2. Through a line parallel to a given plane there can be drawn a unique plane parallel to the given plane.
3. Any line in any of two parallel planes is parallel to the other plane.

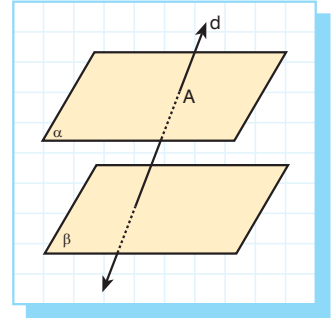
Theorem

If a line intersects one of two parallel planes, it intersects the other too.

Proof

Let α and β be two parallel planes and d be a line intersecting α at a point A . We need to prove that d intersects β . Line d can not lie in β , because d intersects α , and $\alpha \parallel \beta$. Any line drawn through A and parallel to β must lie in α . So if d is parallel to β it lies in α . However we know that d is not in α .

Hence there is only one possibility: Line d intersects β .



Conclusion

If a line is parallel to one of two parallel planes it is either in the second plane or parallel to the second plane.

5. Mutual Positions of Three Planes

Three planes may have no common point, one common point or one common line.

a. No Common Point

Three planes have no common point in three cases:

i. If three planes are parallel, they have no common point.

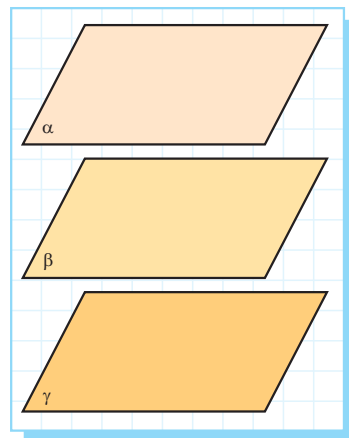
Theorem

The planes parallel to the same plane are parallel.

Proof Let a , b , and g be three planes such that $a \parallel g$ and $b \parallel g$. We need to prove that $a \parallel b$.

Assume that a and b are not parallel. Then they have a common line. Let m be this line. Through m we can draw only one plane parallel to g . This is a contradiction.

Hence $a \parallel b$



Conclusion

If a plane intersects one of two parallel planes, it intersects the other too.

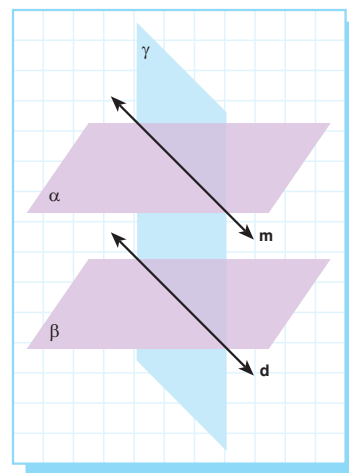
ii. When two planes are parallel and the third plane intersects these planes, the planes will have no common point.

Theorem

If two parallel planes are intersected by a third plane, the lines of intersection are parallel.

Proof Let a and b be two parallel planes and g be a plane intersecting both a and b . Let m and d be the intersections of a and g , and b and g , respectively.

Lines m and d are in the same plane. If m and d have a common point, this point will be a common point of a and b . However a and b are given as parallel planes. That means they can not have a common point. So, m and d can not have a common point. Hence, they are parallel



Theorem

The parallel line segments whose end points are on two parallel planes are equal.

Proof

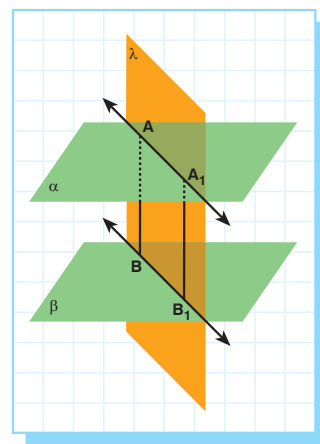
Let α and β be two parallel planes. Let AB and A_1B_1 be two parallel line segments, such that points A, A_1 in α and B, B_1 are in β .

Since AB and A_1B_1 are parallel lines, they determine a plane l . A and A_1 are two common points of l and α . So the line passing through A and A_1 is the intersection of l and α .

By the same logic the intersection of l and β is the line passing through B and B_1 .

Since $\alpha \parallel \beta$, $AA_1 \parallel BB_1$.

Additionally, it is given that $AB \parallel A_1B_1$. Therefore, AA_1B_1B is a parallelogram and $AB = A_1B_1$.



iii. If three planes intersect each other two by two and the lines of intersection are all parallel then the planes will have no common point.

Theorem

When three planes intersect each other if two of intersection lines are parallel then the third intersection line is parallel to these two lines.

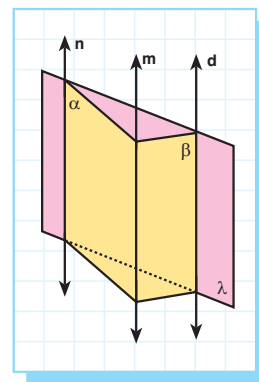
Proof

Let α, β and γ be three planes, and m, n and d be the intersections of $\alpha - \beta$, $\alpha - \gamma$ and $\beta - \gamma$, respectively. Assume that m and n are parallel lines. Let us prove that $d \parallel m$.

Lines d and m are in the same plane. If d intersects m at a point A then A will be a common point of planes γ and α . Because m is in α and d is in γ .

So, A must be on line n . Then m and n will have a common point. However m and n are given as parallel lines. So there cannot be such a point.

Therefore $m \parallel d \parallel n$.



b. One Common Point

Three planes may intersect each other in such a way that their intersection lines are concurrent.

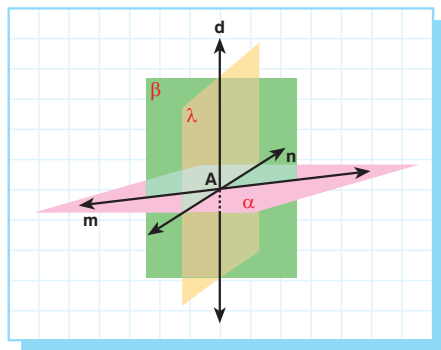
Theorem

If two of intersection lines formed by three intersecting planes intersect each other, the third intersects these lines at the same point.

Proof

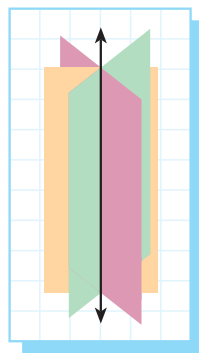
Let α , β and γ be three planes intersecting each other. Let m be the intersection of α and β , n be the intersection of α and γ , d be the intersection of β and γ . Let m and n intersect each other at a point A . We need to prove that d passes through A .

Since m is the intersection of α and β , A is in β , and since n is the intersection of α and γ , A is also in γ . So A is on d which is the intersection of β and γ . So d intersects m and n at A .



c. One Common Line

Three or more planes can intersect each other along a common line. Then these planes have a common line. Then such a figure is called as **bunch of planes**.



6. The Thales Theorem in Space

In plane geometry, the Thales theorem is stated with parallel lines and lines intersecting them. Now, we will write the same theorem with parallel planes and lines intersecting them.

Theorem

If two lines are intersected by parallel planes, the line segments between the planes are proportional.

Proof

Let α, β, γ be three planes. Let d be a line intersecting α at A , β at B and γ at C , and m be another line intersecting α, β, γ at points D, E and F respectively. We need to prove,

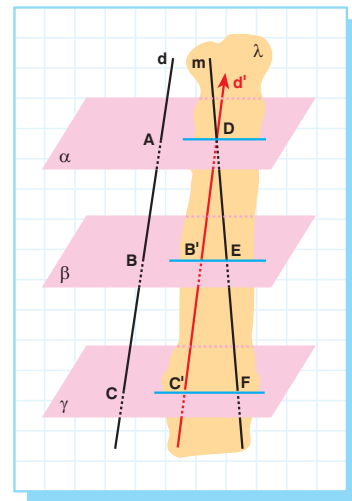
$$\frac{AB}{BC} = \frac{DE}{EF}.$$

Through point D there can be drawn line d' parallel to line d . Since d intersects β and γ , d' also intersects these planes. Let B' and C' be the intersection points. Since α, β and γ are parallel, and d and d' are parallel, $AB = DB'$, $BC = B'C'$. (I)

Lines m and d' are intersecting lines. So they determine a plane λ . Since β and γ are parallel planes, the intersection of λ and β is parallel to the intersection of λ and γ . So $B'E \parallel C'F$, and $\triangle DB'E$ and $\triangle DC'F$ are similar triangles.

$$\text{So } \frac{DB'}{B'C'} = \frac{DE}{EF}. \quad (\text{II})$$

From (I) and (II),



EXAMPLE

1 Show that if two intersecting lines in a plane a are parallel to a plane b then a and b are parallel planes.

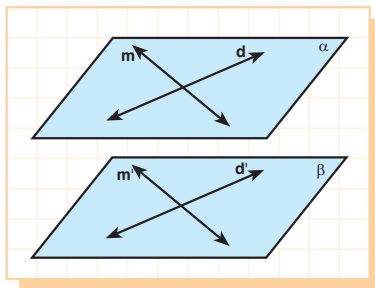
Solution

Let m and d be two intersecting lines in a parallel to plane b . Then in b there can be found two lines m' and d' parallel to m and d , respectively.

d' and m' can not be coincident lines. Because in this case m and d will be parallel to the same line which implies their parallelity.

If $d' \parallel m'$ then since $d \parallel d'$, d and m' will be parallel. Then d and m will be parallel to a common line, that is m' . So $d \parallel m$. However we know that they are intersecting lines. So d' and m' are not parallel.

Hence d' and m' are intersecting lines in b . Therefore $a \parallel b$.



EXAMPLE

12 ABC is a triangle and a is a plane. Show that if sides AB and BC are parallel to a then AC is also parallel to a .

Solution AB and BC are two intersecting lines in plane ABC. Since they are parallel to a , planes ABC and a are parallel. So AC is parallel to a .

EXAMPLE

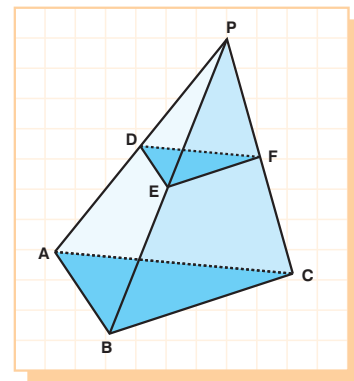
13 In planes DEF and ABC are parallel.

If $PF = 2$ cm,

$FC = 4$ cm, and

$S_{DEF} = 3$ cm²,

find the area of DABC.



Solution Lines PB and PC are intersecting lines. So they determine a plane. Since DEF and ABC are parallel planes which are intersected by plane BPC, the intersections will be parallel. So $EF \parallel BC$, and $\triangle DEF$ and $\triangle BPC$ are similar.

$$\text{So } \frac{EF}{BC} = \frac{PF}{PC} = \frac{2}{6} = \frac{1}{3}. \quad (I)$$

Similarly it can be obtained that $\triangle DEF \sim \triangle BPC$ and $\triangle DEF \sim \triangle BPC$.

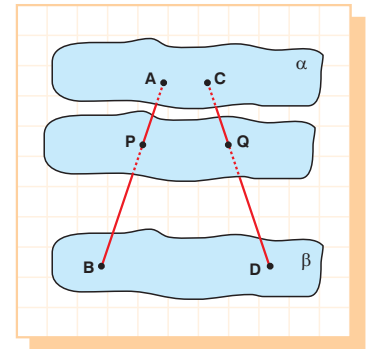
$$\text{So } \frac{DF}{AC} = \frac{PD}{PA} = \frac{PF}{PC} = \frac{1}{3} \quad (II) \quad \text{and} \quad \frac{DE}{AB} = \frac{PD}{PA} = \frac{1}{3}. \quad (III)$$

$$\text{From (I), (II), and (III), } \frac{EF}{BC} = \frac{DF}{AC} = \frac{DE}{AB} = \frac{1}{3},$$

$\triangle DEF$ and $\triangle ABC$ are similar triangles. Hence $\frac{S_{DEF}}{S_{ABC}} = \left(\frac{1}{3}\right)^2 = \frac{1}{9}$. S_{DEF} is given as 3 cm^2 . So, $S_{ABC} = 27 \text{ cm}^2$.

EXAMPLE

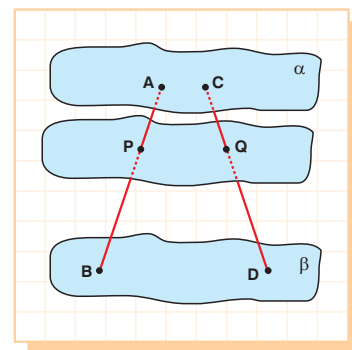
14 In, α and β are parallel planes. A plane parallel to α is intersecting AB at P and CD at Q. If $AP = 2 \text{ cm}$, $AB = 8 \text{ cm}$, and $QD = 9 \text{ cm}$, find CQ.



Solution By the Thales theorem, $\frac{AP}{PB} = \frac{CQ}{QD}$, $\frac{2}{6} = \frac{CQ}{9}$, so $CQ = 3 \text{ cm}$.

EXAMPLE

15 In, α and β are parallel planes. A plane parallel to α is intersecting AB at P and CD at Q . If $AP = 2$ cm, $AB = 8$ cm, and $QD = 9$ cm, find CQ .



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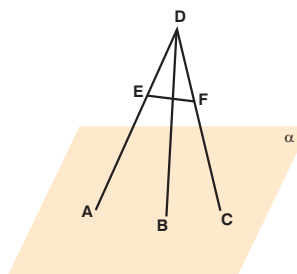
EXERCISES 7.1

A. Axioms of Space Geometry

1. State the followings as true or false
 - a) Three points determine a plane
 - b) If the endpoints of a line segment are in a plane its midsegment is in the same plane
 - c) There can be drawn two planes whose intersection is a unique point

2. There are n non-coplanar points ($n \geq 4$). Show that $n - 1$ of them can not be collinear.

3. In figure, A, B, C are non collinear points, E and F are on DA and DC respectively. Can EF intersect DB?



4. a and b are two intersecting planes. In a there is a line m intersecting b at point A and in b there is a line n intersecting a at point B where A and B are distinct points. Show that a and b intersect along line AB.

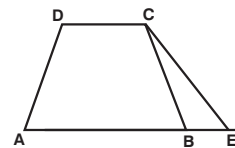
5. Given that points $A_1, A_2, A_3, \dots, A_{25}$ determine only one plane. At most how many of these points can be collinear?

B. Determination of a Plane

6. How many planes can be determined by three parallel lines? (Write all possible cases)

7. How many planes can be determined by three intersecting lines? (Write all possible cases)

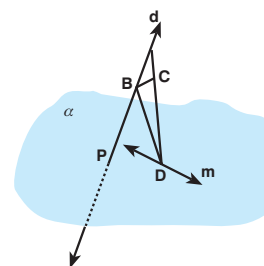
8. In figure, $AB \parallel CD$, points A, B and E are collinear. Show that all points in the figure are coplanar.



9. m, n and d are three lines in space such that m and n are intersecting at point A, and n and d are intersecting at point B. Do m, n, d always lie in the same plane?

C. Mutual Positions of Two Lines in Space

10. In the adjacent figure, m lies in a and d intersects a at point P which is not on m . Show that BC and m are skew lines.

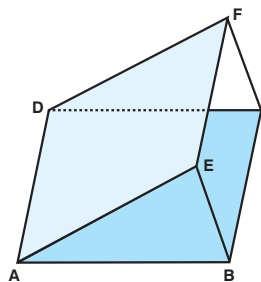


11. State the followings as true or false.
 - a) If lines m and n pass through points M and N, then m and n are coincident.
 - b) If two lines have no common point, they are parallel.
 - c) Skew lines can be coplanar or non-coplanar lines.

D. Mutual Positions of A Line and A Plane

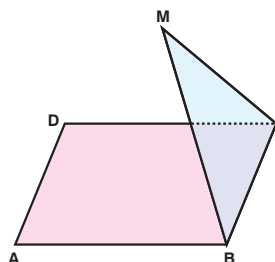
11. ABC is a triangle and P is a point not in (ABC). Show that the line joining the midpoints of PC and AB and the line joining the midpoints of PA and CB are intersecting lines.

13. In the adjacent figure, ABCD and AEFD are parallelograms. Show that DEAB and DFDC are congruent triangles.



14. ABCD is a trapezoid ($AB \parallel CD$). AB is in a plane α and CD is in a plane β . α and β are intersecting along the line m . Show that $MN \parallel m$ where M and N are the midpoints of AD and BC respectively.

15. In the adjacent figure, ABCD is a parallelogram. M is not in (ABC). Show that $AD \parallel (MBC)$.



16. State the followings as true or false

a) Two lines parallel to the same plane are always parallel.

b) If two lines m and n are parallel to a given line d then $m \parallel n$

17. m and n are two parallel lines not lying in a plane α . Show that if $m \parallel \alpha$, line n is parallel to α , too.

E. Mutual Positions of Two Planes

18. α and β are two intersecting planes and A is a point not in α or β . Show that through A there can be drawn one and only one line parallel to both α and β .

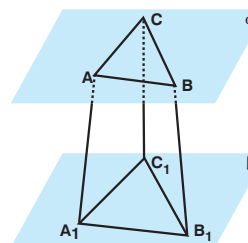
19. α and β are two intersecting planes and m is a line intersecting α and β at distinct points.

a) Show that through m there can be drawn a plane γ whose intersections with planes α and β are parallel.

b) Show that γ is unique.

20. Line m is parallel to α and intersects plane β . Show that α and β are intersecting planes.

21. In the adjacent figure, DABC lies in α and DA₁B₁C₁ lies in β . If $AC \parallel A_1C_1$, $AB \parallel A_1B_1$ and



$$\frac{AC}{A_1C_1} = \frac{AB}{A_1B_1} \quad \text{then show that}$$

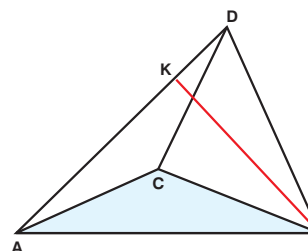
- a) DABC and DA₁B₁C₁ are similar.
b) $BC \parallel B_1C_1$.

22. m and n are skew lines. Show that there can be drawn two parallel planes α and β such that α contains m and β contains n , and show that these α and β are unique.

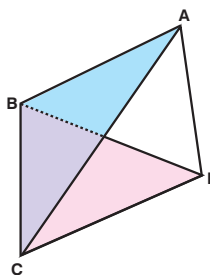
23. In the adjacent figure, D is not in (ABC).

$AC = AB$, $DC = DB$, and

$BK \perp DA$. If $KD = 3$ cm and $DB = 6$ cm, find KC .



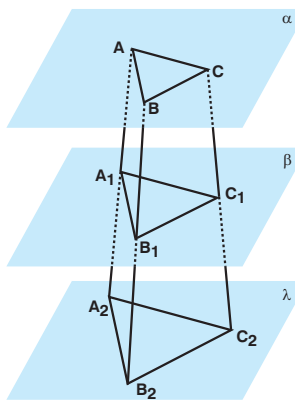
24. In the adjacent figure, $\triangle ABC$ and $\triangle BCD$ are equilateral.
If $\angle ACD = 20^\circ$ find $\angle BDA$.



25. α and β are two parallel planes. Plane l intersects α and β along lines m and n , and plane g intersects planes α and β along lines p and q , respectively. Show that $n \parallel q$.

26. $\triangle ABC$ and $\triangle A_1B_1C_1$ are two triangles in two parallel planes. If $AA_1 \parallel BB_1 \parallel CC_1$ then show that $\triangle ABC$ and $\triangle A_1B_1C_1$ are congruent triangles.

27. In the adjacent figure, $\alpha \parallel \beta \parallel \gamma$. Moreover $AB \parallel A_1B_1$, $BC \parallel B_1C_1$, $AC \parallel A_1C_1$. Find the perimeter of $\triangle A_2B_2C_2$ if perimeters of $\triangle ABC$ and $\triangle A_1B_1C_1$ are 6 cm and 9 cm, respectively, and $\frac{BB_1}{B_1B_2} = \frac{3}{5}$.



A. Perpendicular Lines

Definition

Two lines a and b are perpendicular to each other if the angle between them is 90° .

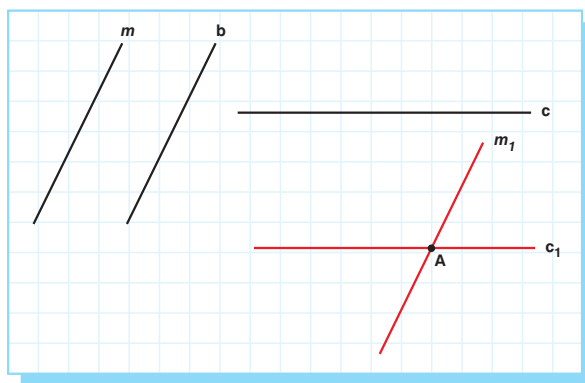
If the lines are intersecting, it is very easy to determine whether they are perpendicular or not. If they are skew to each other, we take any point on one of the lines, and through this point, we draw a line parallel to the other one. If the angle between these two intersecting lines is 90° then the given skew lines are said to be perpendicular.

Theorem

If one of two parallel lines is perpendicular to a third line, the other is perpendicular too.

Proof

Let m and b be two parallel lines and m be perpendicular to c (Figure 1.36). Through any point A , let us draw lines m_1 and c_1 so that $m_1 \parallel m$ and $c_1 \parallel c$. Since $m \perp c$, the angle between m_1 and c_1 is 90° . On the other hand, since $m_1 \parallel m$ and $m \parallel b$, we get $m_1 \parallel b$. So the angle between b and c_1 is also 90° . That means b and c_1 are perpendicular lines.



B. Line Perpendicular to a Plane

Definition

A line is said to be perpendicular to a plane if it is perpendicular to every line in this plane.

If a line m is perpendicular to every line in a plane a then m is perpendicular to a and it is shown by $m \perp a$. So if it is given that $m \perp a$ then m is perpendicular to any line in a .

If $m \perp a$ then m intersects a . To prove this statement let us assume that m does not intersect a . In this case there are two possibilities for m and a :

1. m is in a . Then since it is not perpendicular to a line in a , that is itself, m is not perpendicular to a .
2. m is parallel to a . In this case in a there can be found a line parallel to m . So m can not be perpendicular to a . In both possibilities m is not perpendicular to a . Therefore, m intersects a .

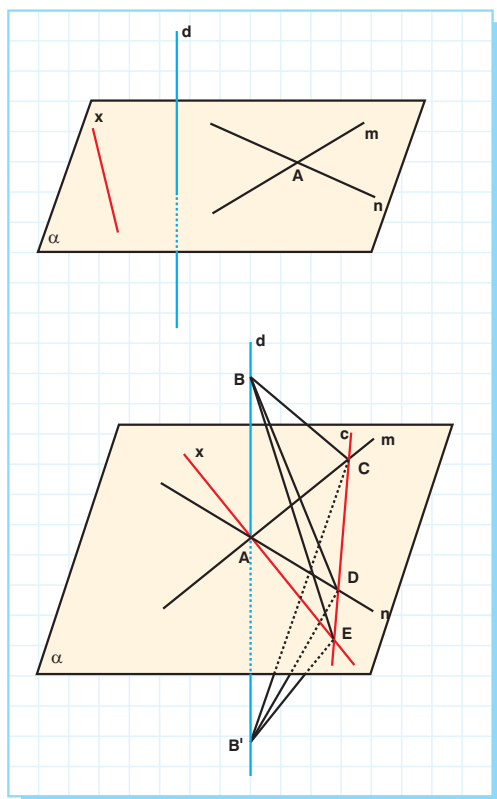
Definition

If a line intersects a plane but not perpendicular to the plane it is called an **inclined line**.

Theorem

If a line is perpendicular to two intersecting lines lying in a plane then it is perpendicular to the plane.

Proof



We need to prove that if a line is perpendicular to two intersecting lines in a plane it is perpendicular to any line in this plane. Let d be a line perpendicular to two lines m and n lying in a . Let A be the intersection point of m and n . It is obvious that d is perpendicular to every line in a which is parallel to either one of m or n .

So we should check for the lines which are not parallel to neither m nor n .

Let x be any line intersecting both m and n . We have to prove that d is perpendicular to x too. Let us shift lines d and x so that A is on d and x . Let c be any line in a intersecting m , n , x at points C , D , E respectively.

On line d let us take two points B and B' so that $BA = B'A$. Then triangles BAC and $B'AC$ are congruent, similarly triangles BAD and $B'AD$ are congruent (S.A.S.). So $BD = B'D$ and $BC = B'C$. Then triangles BDC and $B'DC$ are congruent (S.S.S.). That means $\angle BDC = \angle B'DC$. Then triangles BDE and $B'DE$ are congruent triangles (S.A.S.). So $BE = B'E$ and triangles BAE and $B'AE$ are congruent (S.S.S.).

Hence $\angle BAE = \angle B'AE = 90^\circ$. So d is perpendicular to x . Therefore d is perpendicular to any line in a . So $d \perp a$.

Theorem

Through any given point in space, there can be drawn one and only one plane perpendicular to a given line.

Proof

We have two cases:

1. The point is on the line.

Let d be a line and A be a point on d . Let us take two distinct planes α and β containing d .

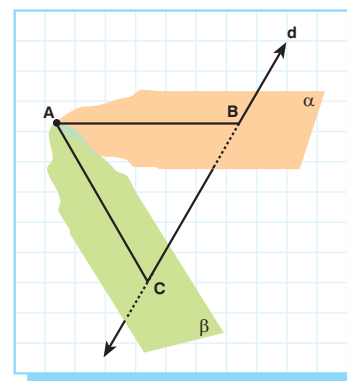
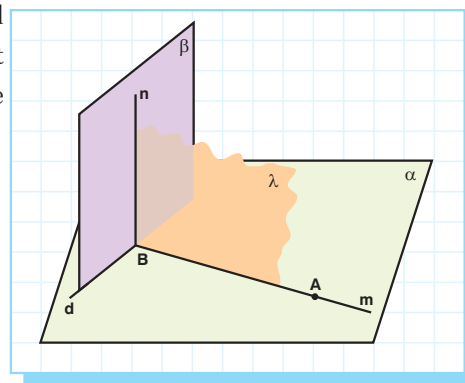
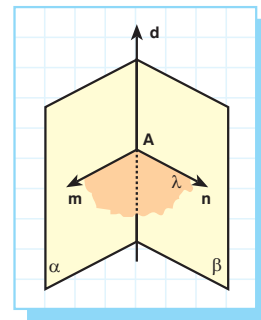
Let m be the line lying in α and perpendicular to d at A and n be the line lying in β and perpendicular to d at A . Then m and n determine a unique plane λ perpendicular to d .

2. The point is not on the line.

Let d be a line and A be a point in space not on d . Line d and point A determine a plane α . In α there can be drawn a line m passing through A and perpendicular to d at a point B . Let β be another plane containing line d . In β , through point B , let us draw a line n perpendicular to d . Since m and n are intersecting lines, they determine a plane λ . Since d is perpendicular to two intersecting lines in λ (those are m and n), d is perpendicular to λ .

Now let us prove that this plane is unique. Let d be a line and A be a point not on d . Assume that through A there can be drawn more than one plane perpendicular to d .

Let α and β be two of these planes. Let B be the intersection of d and α , and C be the intersection of d and β . Then $d \perp AB$ and $d \perp AC$ which is impossible



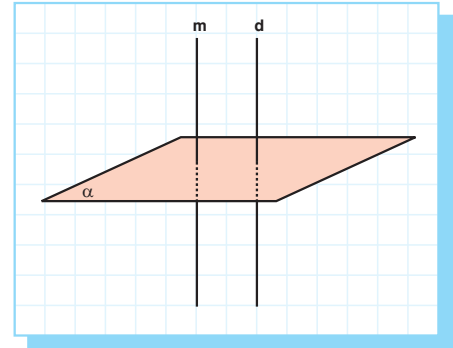
Theorem

If one of two parallel lines is perpendicular to a plane then the other line is also perpendicular to the same plane.

Proof

Let m and d be two parallel lines and α be a plane such that $m \perp \alpha$. We need to show that $d \perp \alpha$.

Since $m \perp \alpha$, m is perpendicular to every line in α . It is proven that if one of two parallel lines is perpendicular to a third line then the other one is perpendicular to the same line. So d is perpendicular to every line in α too. Therefore $d \perp \alpha$.

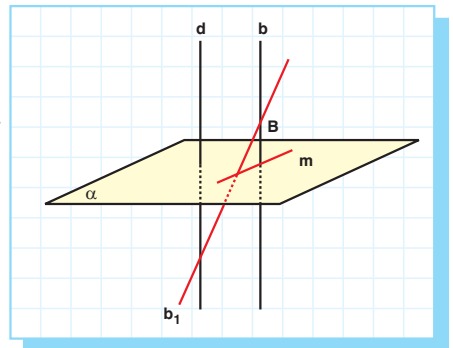


Theorem

Two lines perpendicular to the same plane are parallel.

Proof

Let lines d and b be perpendicular to a plane α . On b , take a point B which is not in α and through B , draw a line b_1 parallel to d . If b and b_1 are not coincident they determine a plane β . Let m be the intersection of β and α . Since $b_1 \parallel d$ and $d \perp \alpha$, $b_1 \perp \alpha$. So $b_1 \perp m$. Since b is perpendicular to α , b is also perpendicular to m . However through B there can be drawn only one line perpendicular to α . Hence b_1 and b are coincident. That means lines d and b are parallel.



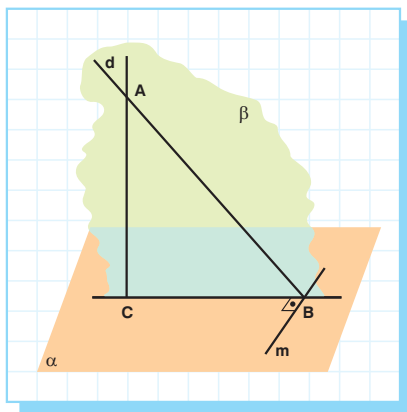
Theorem

A line drawn in a plane through the foot of an inclined line is perpendicular to the projection of the inclined line if and only if it is perpendicular to the inclined line itself.

Proof

In this theorem we need to prove that

1. If a line drawn in a plane through the foot of an inclined line is perpendicular to the projection of the inclined line, it is perpendicular to the inclined line.
2. If a line drawn in a plane through the foot of an inclined line is perpendicular to the inclined line, it is perpendicular to the projection of the inclined line.



1. Let m be the line lying in a plane α perpendicular to the projection of an inclined line d onto a at point B which is the intersection of d and a . Let A be any point on d and let the perpendicular drawn through A intersect a at C .

Point C is on the projection of d . CB and d determine a plane b . Since $AC \perp a$, $AC \perp m$. Given that $m \perp CB$, so $m \perp b$ and $m \perp d$.

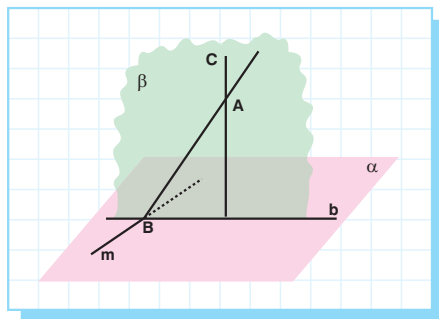
2. Referring to the same figure, we need to prove that if $m \perp d$ then $m \perp CB$. Since m is perpendicular to two intersecting lines in b which are d and AC , it is perpendicular to b . So it is perpendicular to CB .

Theorem

Through a point in space, there can be drawn a line perpendicular to a given plane.

Proof

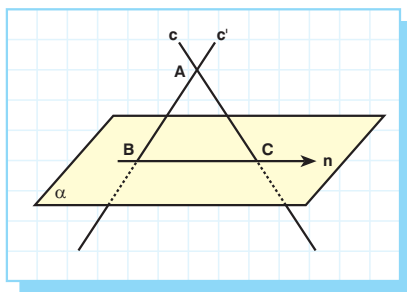
Let a be a plane and A be a point. In a take a line m (if A is in a take m in such a way that A is not on m). Through A there can be drawn a plane b perpendicular to m . a and b are intersecting planes because m intersects b at a point B and m is in a . So a and b have a common line b . In b through A there can be drawn line c perpendicular to b . Then since $c \perp b$ and $c \perp m$, it can be concluded that $c \perp a$.



Now let us prove that this line is unique.

Let a be a plane and A be a point not in a . Assume that through A there can be drawn more than one line perpendicular to a . Let c and c' be two of these lines. Since c and c' are perpendicular to a , they intersect a . Let B and C be the intersection points. B and C determine a line n .

Since $c \perp a$, $c \perp n$ and since $c' \perp a$, $c' \perp n$. Then $\triangle ABC$ is a triangle with two right angles which is impossible. So through A we can draw only one line perpendicular to a .



Theorem

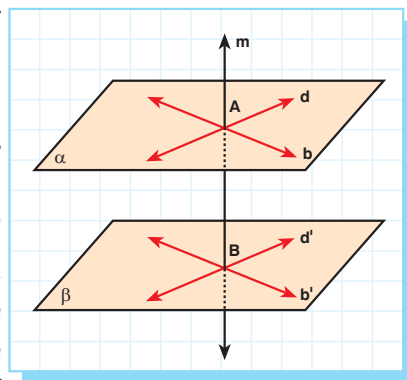
If a line is perpendicular to one of two parallel planes, it is perpendicular to the other.

Proof

Let α and β be two parallel planes and m be a line perpendicular to α .

Since m is perpendicular to α , it intersects α at a point A . If a line intersects one of two parallel planes, it intersects the other too. So m intersects β at a point B .

In α , let us take two lines d and b intersecting at A . Let d' be the intersection of b and the plane determined by d and m , and let b' be the intersection of b and the plane determined by m and b . When two parallel planes are intersected by a plane the intersections are parallel. $d' \parallel d$ and $b' \parallel b$. Since $m \perp \alpha$, $m \perp d$ and $m \perp b$. So $m \perp d'$ and $m \perp b'$. Since m is perpendicular to two intersecting lines in β , m is perpendicular to β .



EXAMPLE

16 DABC is an isosceles right triangle such that $AB = BC = 4$ cm. P is a point not in plane ACB and M is the midpoint of AC . If $PB \perp AB$, $PB \perp BC$ and $PB = 2\sqrt{2}$ cm, find PM .

Solution

AC is the hypotenuse of

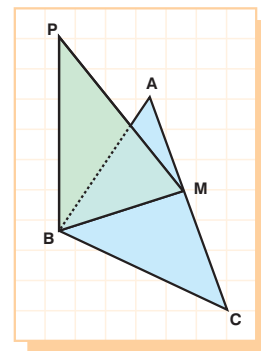
$\triangle ABC$, $AC = 4\sqrt{2}$ cm. Since BM is the median of the hypotenuse,

$BM = 2\sqrt{2}$ cm. Since $PB \perp BC$ and $PB \perp AB$, $PB \perp (ABC)$.

So $PB \perp BM$. Then

$$PM^2 = PB^2 + BM^2 = 8 + 8 = 16,$$

$$PM = 4 \text{ cm.}$$



Show that if two planes a and b are perpendicular to a line m then they are parallel.

Solution Since m is perpendicular to a and b , it intersects both planes. Let A and B be the intersections.

Assume that a and b are not parallel. Then they will have a common point P . Since AP is in a , $m \perp AP$ and since BP is in b , $m \perp BP$. So $DAPM$ is a triangle with two right angles which is impossible. Therefore a and b can not have any common point which means $a \parallel b$.

C. Perpendicular Planes

If a plane contains a line perpendicular to another plane then it is perpendicular to that plane. Every line on any of two parallel planes is parallel to the other plane. So, if a plane is perpendicular to another plane, it intersects the plane.

Let a and b be two planes so that $a \perp b$. Then in a there can be found a line m perpendicular to b . Then line m is perpendicular to the intersection of a and b because this line lies in b . Moreover in a through every point there can be drawn a line parallel to m . Since m is perpendicular to b all these lines are perpendicular to b . Therefore, it can be concluded that in a through every point there can be drawn a line perpendicular to b .

Theorem

If one of two planes is perpendicular to the other then the other is perpendicular to the first one too.

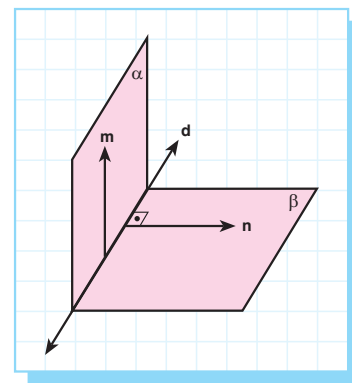
Proof

Let a be a plane perpendicular to a plane b . Then in a there is a line m so that $m \perp b$

Let d be the intersection of a and b . Since $m \perp b$, m intersects b . Since m is in a , m intersects b at a point which is on d . So m and d are two intersecting lines. Let n be any line in b which is perpendicular to d .

Since $m \perp b$, $m \perp n$. So n is perpendicular to two intersecting lines in b which are m and d . So $n \perp a$.

Hence $b \perp a$



Theorem

Any plane perpendicular to one of two parallel planes is perpendicular to the other.

Proof

Let a and b be two parallel planes and l be a plane perpendicular to a . Then in l there is a line perpendicular to a . This line will be also perpendicular to b . So $l \perp b$.

Theorem

If a line is perpendicular to a plane and parallel to another plane, these planes are perpendicular.

Proof

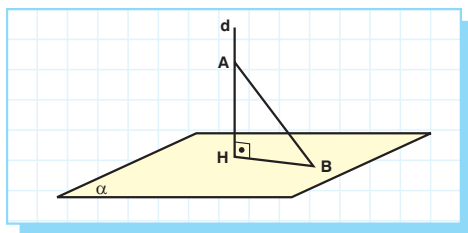
In the plane parallel to the line, there will be a line parallel to the given line and this line will be perpendicular to the other plane. So the planes are perpendicular.

D. Distance

1. Distance Between a Point and a Plane

Let A be a point and a be a plane. Through A let us draw line d perpendicular to a . Let d intersect plane a at point H . Then the length of line segment AH will be the distance between A and a . Let us show that the length of AH is indeed the shortest distance between A and a .

Let B be another point in a . Then B and H determine a line in a (Figure 1.49). Since $d \perp a$, $d \perp BH$. So triangle AHB is a right triangle with hypotenuse AB . So $AH < AB$.

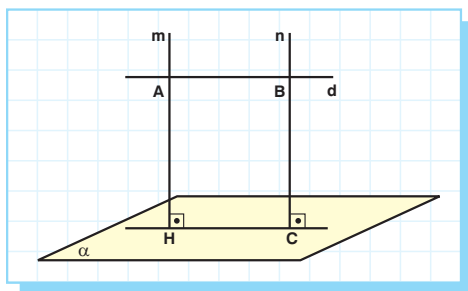


If point A is in a , the distance between A and a is zero.

2. Distance Between a Plane and a Line Parallel to the Plane

Let d be a line parallel to a plane a . Let A and B be two distinct points on d , and m and n be the lines perpendicular to plane a drawn through A and B . Let m and n intersect plane a at points H and C . So the lengths of line segments AH and BC are the distances from points A and B to a .

Since m and n are parallel, they are coplanar. So A, B, C, H are coplanar points and quadrilateral $ABCH$ is a rectangle. Therefore $AH = BC$.



So the distance from every point on line d to plane a is constant.

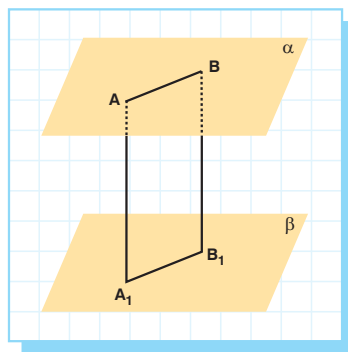
Hence to find the distance between a plane and a line parallel to the plane, we simply find the distance between any point on the line and the plane.

3. Distance Between Two Parallel Planes

Let a and b be two parallel planes. Let A and B be two points in a , and A_1 and B_1 be two points in b so that $AA_1 \perp b$ and $BB_1 \perp b$. Since AA_1 and BB_1 are perpendicular to the same plane they are parallel and determine a plane. When two parallel planes are intersected by a plane the intersections are parallel. So ABB_1A_1 is a rectangle and

$AA_1 = BB_1$. In other words the distance from any point in one of a or b to the other plane is constant.

To find the distance between two parallel planes, we simply find the distance from any point in any plane to the other plane.



4. Distance Between Skew Lines

To find the distance between two skew lines, we find the distance from one of these lines to the plane containing the other and parallel to the first line. This distance is equal to the length of the line segment perpendicular to both lines. Now let us show this.

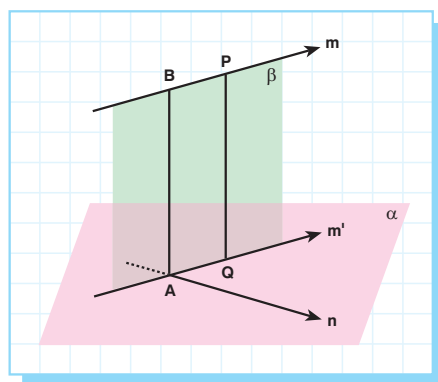
Let m and n be two skew lines. We proved that we can draw a unique plane parallel to m and containing n . Let a be this plane. Through any point P on m , there can be drawn a unique line perpendicular to a and let Q be the intersection of this line and a .

Through Q let us draw line m' parallel to m . m' is in a .

Since $m \parallel m'$, they determine a plane b . Let A be the intersection of m' and n . Through A there can be drawn a unique line parallel to PQ . This line will be in plane b . So it intersects line m at a point B .

Since $PQ \perp a$, $PQ \perp m'$. Since $m \parallel m'$, $PQ \perp m$. So $BA \perp m$. Since $PQ \perp a$ and $BA \parallel PQ$, $BA \perp a$. So $BA \perp n$. Therefore, BA is perpendicular to both m and n .

Hence BA is called as common perpendicular of m and n .



Definition

The line segment perpendicular to two given skew lines is called as the **common perpendicular of the given skew lines**.

Theorem

The common perpendicular of two skew lines is unique.

Proof

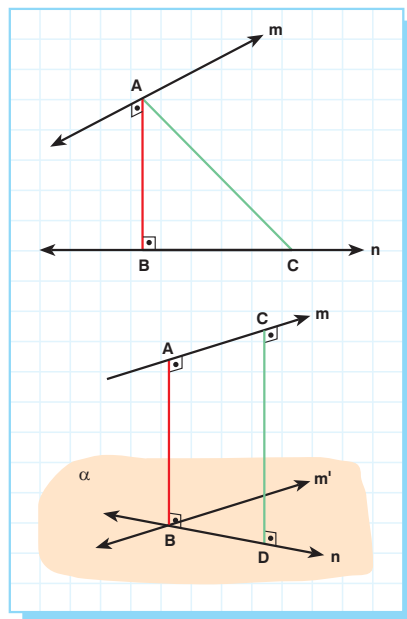
Let m and n be two skew lines and AB be their common perpendicular. Let C be a point on.

Since $\angle ABC = 90^\circ$, AC can not be perpendicular to n . So there can not be drawn any other common perpendicular containing A or B .

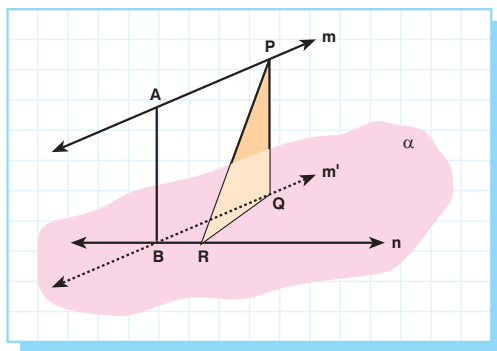
Now let us show that there is not any other common perpendicular except AB .

Assume that CD is a line segment perpendicular to both m and n . Let m' be the line parallel to m and passing through B . Since n and m' are intersecting lines, they determine a plane α . So $AB \perp \alpha$. Since $CD \perp m$, $CD \perp m'$. So $CD \perp \alpha$ too. Since $AB \perp \alpha$ and $CD \perp \alpha$, $AB \parallel CD$. Then points A, B, C, D will be coplanar which is impossible.

Hence the common perpendicular is unique



Now, let us show that **the length of the common perpendicular is the smallest distance between two skew lines**.



Let m, n be two skew lines and a be the plane containing n and parallel to m . Let AB be the common perpendicular of m and n , and m' be the line passing through B and parallel to m . Let P and R be any two other points on m and n .

Through P let us draw a line parallel to AB and let Q be the intersection of a and this line. Since $AB \perp a$ and $PQ \parallel AB$, $PQ \perp a$.

So, we have that $PQBA$ is a rectangle and $AB = PQ$.

Since $PQ \perp a$, $PQ \perp QR$. Therefore $PR > PQ$ and since $PQ = AB$, $PR > AB$.

So we can conclude that any line segment drawn between two skew lines is longer than their common perpendicular.

(To summarize: the length between two skew lines is the length of the common perpendicular of these lines. This length is equal to the distance from one of them to the plane containing the other and parallel to the first line.)

EXAMPLE

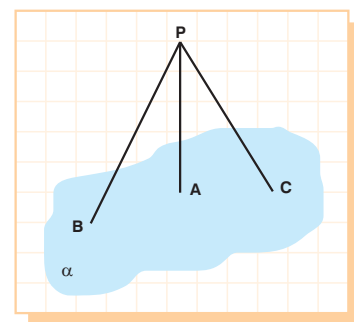
18 From a point P to a plane a , perpendicular OPA and two inclined line segments PB and PC are drawn.

If $PB = PC$,

$\angle BPA = 45^\circ$,

$\angle BPC = 60^\circ$, and

$PA = 2\sqrt{2}$ cm, find BC .



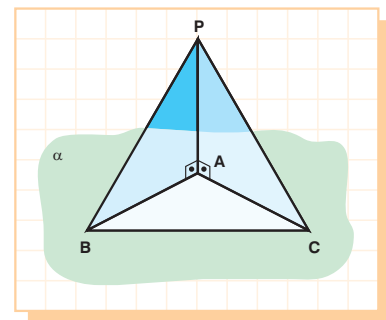
Solution Since $PA \perp a$, $PA \perp AB$, and since $\angle BPA = 45^\circ$,

$PB = PA \times \sqrt{2} = 2\sqrt{2} \times \sqrt{2}$ $PB = 4$ cm.

Since $PB = PC$ and $\angle BPC = 60^\circ$,

$\triangle PBC$ is an equilateral triangle.

So $BC = PB = 4$ cm.



EXAMPLE

19

A line segment AB intersects a plane α at a point C . If $\frac{AC}{AB} = \frac{5}{6}$ and the distance between B and α is 2 cm, find the distance between A and α .

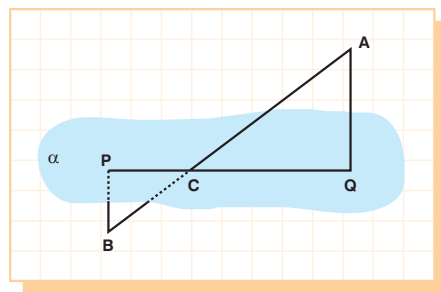
Solution Let $BP \perp \alpha$ and $AQ \perp \alpha$ where P, Q are in α . Then $BP \parallel AQ$. They determine a plane b and the intersection of this plane and α is line PQ .

Since A and B are in plane b , line AB is in b . So AB and PQ intersect and this intersection is in α . Since AB intersects α at C , the intersection point is C . So P, C and Q are collinear.

Since $BP \parallel AQ$, $\triangle BPC \sim \triangle AQC$.

So we have $\frac{BP}{AQ} = \frac{BC}{AC} = \frac{1}{5}$.

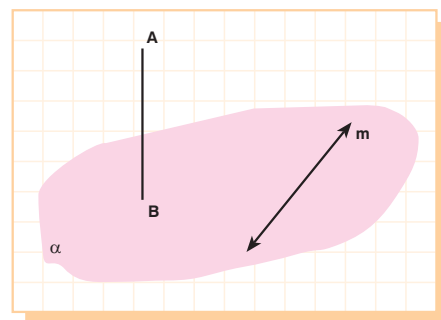
$AQ = 5 \times BP = 5 \times 2 = 10$ cm.



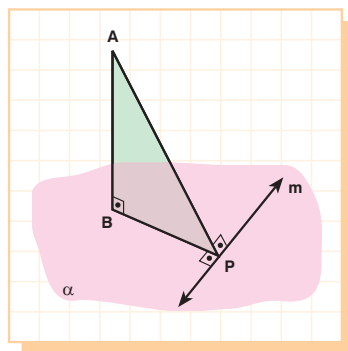
EXAMPLE

20

From a point A to a plane α , perpendicular AB is drawn. If the distance from B to m is 6 cm and $AB = 8$ cm, find the distance from A to m . (m is a line in α)



Solution Let $BP \perp m$ (Figure 1.59). Then $BP = 6$ cm. Since $AB \perp a$ and $BP \perp m$, by the three perpendiculars theorem, $AP \perp m$. So the distance from A to m is AP . Since $AB \perp a$, $AB \perp BP$. So $AP = \sqrt{AB^2 + BP^2} = \sqrt{8^2 + 6^2} = 10$ cm.



EXERCISES 7.2

A. Line Perpendicular to a Plane

1. In the adjacent figure,

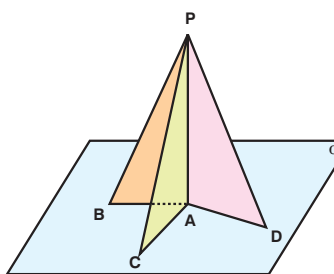
$$PA \perp a,$$

$$\angle BPA = 20^\circ,$$

$$\angle CPA = 10^\circ, \text{ and}$$

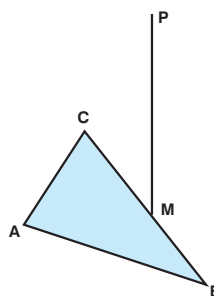
$$\angle APD = 30^\circ.$$

Write PB, PC, PD in ascending order.

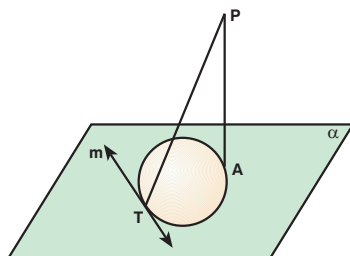


2. In the adjacent figure, M is the midpoint of CB and $PM \perp (ABC)$.

If $MC = CA$ and $PA = PB = PC$, find $\angle ABC$.



3.



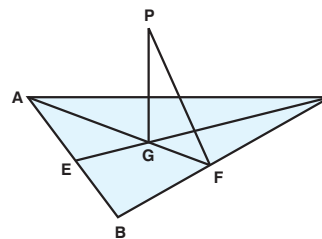
In the adjacent figure, in plane a there is a circle and a line m tangent to the circle at a point T . A is a point on circle such that $PA \perp a$ where P is not in a . If $PT \perp m$, $PT = 10$ cm, and $PA = 8$ cm, find the radius of the circle.

4. Lines m, b, c are coplanar and line d is not in the plane containing m, b, c . It is given that d is perpendicular to m and b but it is not perpendicular to c . What can be concluded about m and b ?

5. In the adjacent figure, PG is perpendicular to the plane of equilateral triangle ABC at its centroid G .

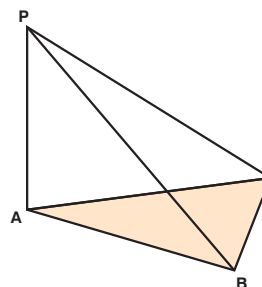
If $\angle GAF = 45^\circ$, what is

$$\frac{S_{ABC}}{S_{PBC}} = ?$$



6. In the adjacent figure, $PA \perp (ABC)$.

If $\angle APB = \angle APC = 45^\circ$ and $\angle BPC = 60^\circ$, find $\angle BAC$.



7. Triangle ABC , right angled at A , lies in a plane a and $PA \perp a$ where P is a point not in a . If A is equidistant from points B, C and P , find $\angle BPC$.

8. Show that if two planes are perpendicular to the same line then these planes are parallel.

9. State the followings as true or false

- If two lines are perpendicular to the same line, they are parallel.
- If two lines are perpendicular to the same plane, they are parallel.
- If a line is perpendicular to two lines lying in a plane, it is perpendicular to the plane.

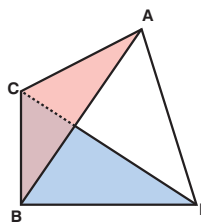
B. Perpendicular Planes

10. Show that through a given point A, there can be drawn infinitely many planes perpendicular to a given plane α .

11. Line m is parallel to plane α . Show that there can be drawn one and only one plane containing m and perpendicular to α .

12. In the adjacent figure, $DABC$ is an equilateral triangle and $DBDC$ is an isosceles triangle.

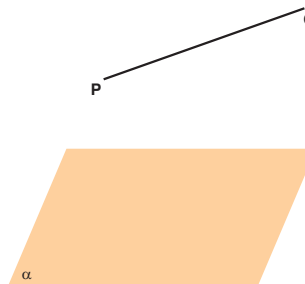
$AB = 6$ cm and $BD = DC = 5$ cm are given. Find AD , if $(ABC) \perp (BDC)$.



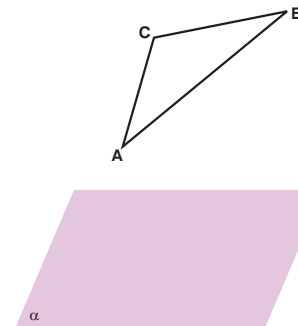
C. Distance

13. A and B are two points equidistant from a plane α and they are on the same side of α . Show that $AB \parallel \alpha$.

14. In the adjacent figure, PQ is a line segment on one side of α . If the distances from P and Q to α are 4 cm and 6 cm respectively, find the distance from the midpoint of PQ to α .



15. $DABC$ is a triangle on one side of a plane α . If the distances from vertices A, B, and C to α are 6 cm, 8 cm, and 10 cm respectively, find the distance from the centroid of $DABC$ to plane α .



16. $ABCD$ is a rhombus and $PA \perp (ABC)$. If $\angle BAD = 60^\circ$ and $AC = 2\sqrt{3}$, find the distance between lines PA and BC .

17. m and n are two lines parallel to a plane α and they are on the same side of α . If the distances from m and n to α are equal, can m and n be skew lines?

SOLIDS WITH CURVED SURFACES

A. Some Important Polyhedrons

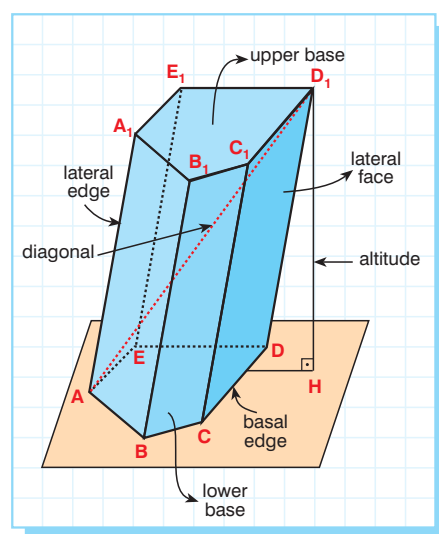
1. Prisms

Definition

A prism is a polyhedron formed by a closed prismatic surface and two parallel planes cutting all its elements.

The various parts of a prism have names consistent with the general meanings of the words.

- The polygons made by two parallel planes are the bases.
- The faces of a prism are the portions of planes which are enclosed by the polygons which form the prism. Two of the faces of a prism are enclosed by the bases. The remaining faces of the prism are called lateral faces.
- Intersections of the lateral faces are called the lateral edges.
- The sides of the polygons which form the bases are called the basal edges.
- An altitude of a prism is a line segment joining the two base planes and perpendicular to both. The length of an altitude is called the height of the prism.
- A line segment which joins two vertices not in the same face is called a diagonal of the prism.



- $ABCDE$ and $A_1B_1C_1D_1E_1$ are the bases of the prism.
- ABB_1A_1 and BCC_1B_1 are some of the lateral faces.
- AA_1 , BB_1 , and CC_1 are some of the lateral edges.
- AB , BC , A_1B_1 , and E_1D_1 are some of the basal edges.
- D_1H is an altitude of the prism.
- AD_1 is a diagonal of the prism.

Upper and lower bases correspond to each other.

- A and A_1 , B and B_1 are some of the corresponding vertices.
- AB and A_1B_1 , BC and B_1C_1 are some of the corresponding basal edges.
- We can represent a prism by using the end points of its any diagonal or the vertices of its lower and upper bases. For example, the prism in Figure 3.30 is represented as prism AD_1 or prism $ABCDEA_1B_1C_1D_1E_1$.

Conclusion

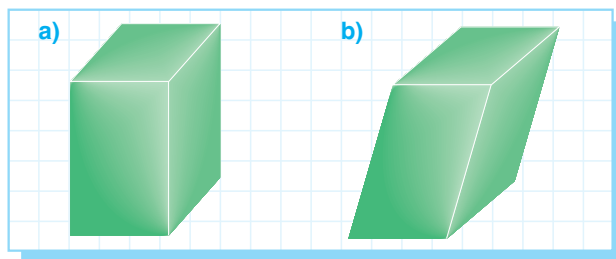
1. The bases of a prism are congruent polygons.
2. Every section of a prism made by a plane parallel to the bases is congruent to the bases.
3. All right sections of a prism are congruent.
4. Lateral faces of a prism are all parallelograms.

a. Classification of Prisms

A prism is either a right prism or an oblique prism.

Definition

A prism whose lateral edges are perpendicular to its bases is called a **right prism**, otherwise it is an **oblique prism**.



In, the prism in part (a) is a right prism and that in part (b) is an oblique prism.

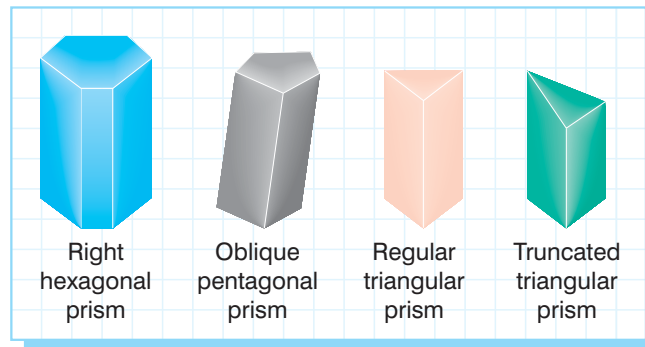
Conclusion

1. The lateral faces of a right prism are rectangles.
 2. The lateral edges of a right prism are altitudes.
- Prisms are also classified by the shapes of their bases. They are said to be triangular, quadrangular and so on, according to their bases which are triangles, quadrilaterals and so on.

Definition

A right prism whose bases are regular polygons is called a **regular prism**.

- If a plane not parallel to the bases of a prism and not intersecting the bases cuts all the lateral edges, it divides the given prism into two solids, either of which is called a truncated prism.



In, you may see how some of the prisms are named.

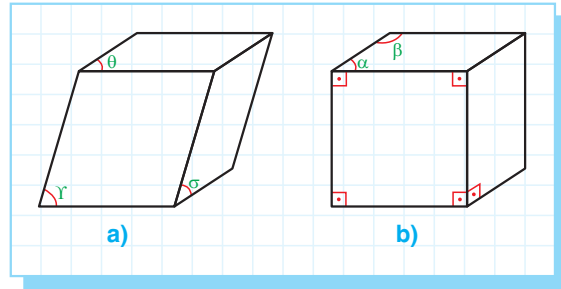
Here is another important definition that will be used in proving some theorems.

b. Some Special Prisms

i. Parallelepiped

Definition

A prism whose bases are parallelograms is called a **parallelepiped**. If the lateral edges are perpendicular to the bases then the parallelepiped is called a **right parallelepiped**.



As can be seen in a, the face angles at a vertex of a parallelepiped may be all different. On the other hand, even though the lateral edges of a right parallelepiped are perpendicular to its bases, the face angles on the bases may not be right angles as in part b) .

Theorem

The opposite faces of a parallelepiped are congruent and parallel.

Proof

Refer to the parallelepiped in

Since the bases are parallelograms, we can write

$AB \parallel DC$, $AB = DC$ and

$A_1B_1 \parallel D_1C_1$, $A_1B_1 = D_1C_1$...(1)

By the definition of a prism,

$AA_1 \parallel BB_1 \parallel CC_1 \parallel DD_1$ and

$AA_1 = BB_1 = CC_1 = DD_1$...(2)

By (1) and (2), ABB_1A_1 , BCC_1B_1 , CDD_1C_1 , and DAA_1D_1 are all parallelograms. (Why?)

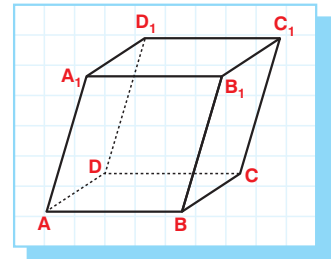
Moreover, if two angles not in the same plane have their sides respectively parallel and lying on the same side of the straight line joining their vertices, the angles are equal, and their planes are parallel. Therefore,

$\angle BAA_1 \cong \angle CDD_1$ and $ABB_1A_1 \parallel DCC_1D_1$,

$\angle CBB_1 \cong \angle DAA_1$ and $BCC_1B_1 \parallel ADD_1A_1$.

Since two parallelograms are said to be congruent if two sides and the included angle of the one are respectively equal to two sides and the included angle of the other, we can conclude that

$ABB_1A_1 \cong DCC_1D_1$ and $BCC_1B_1 \cong ADD_1A_1$ n



Conclusion

1. All faces of a parallelepiped are parallelograms.
2. Any two opposite faces of a parallelepiped can be taken as bases..

Definition

A right parallelepiped whose bases are rectangles is called a **rectangular parallelepiped** or more commonly a **rectangular solid**.

It can be asserted from the definition that a rectangular solid is a special parallelepiped. So, all properties of a parallelepiped hold for a rectangular solid.

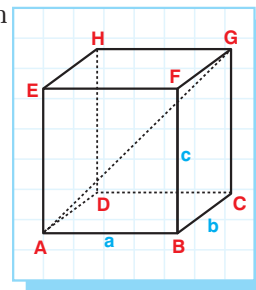
All faces of a rectangular solid are rectangles. Additionally, its diagonals are equal in length and bisect each other at the center of the solid.

- The lengths of the three edges of a rectangular parallelepiped (solid) which meet at a common vertex are called its dimensions.

Theorem

If the length of a diagonal of a rectangular solid is d and its dimensions are a , b , c then

$$d = \sqrt{a^2 + b^2 + c^2}$$



Proof

Let us draw AC as in. Since AC is the hypotenuse of right triangle ABC, we obtain

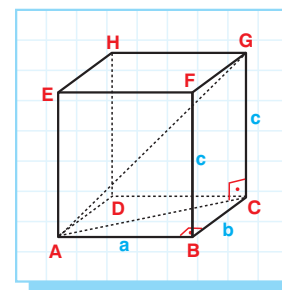
$$AB^2 + BC^2 = AC^2 \Rightarrow AC^2 = a^2 + b^2.$$

On the other hand, GC is perpendicular to AC, because of the fact that if a line is perpendicular to a plane then it is perpendicular to any line on that plane. Therefore, $AG^2 = AC^2 + GC^2 \Rightarrow$

$$d^2 = AC^2 + c^2$$

$$\Rightarrow d^2 = a^2 + b^2 + c^2 \quad (AC^2 = a^2 + b^2)$$

$$\Rightarrow d = \sqrt{a^2 + b^2 + c^2}$$



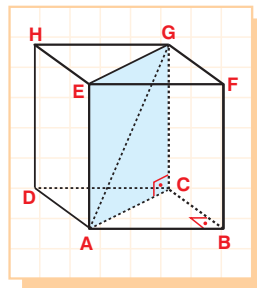
EXAMPLE

21

In the rectangular solid in the adjacent figure, $AB = 12$ cm, $BF = 4$ cm, and $BC = 5$ cm.

Accordingly,

- find AG .
- find the area of section $ACGE$.



Solution a) In right triangle ABC ,

$$AC^2 = AB^2 + BC^2 \Rightarrow AC^2 = 12^2 + 5^2$$

$$\Rightarrow AC^2 = 169$$

$$\Rightarrow AC = 13 \text{ cm}.$$

$$AG^2 = AC^2 + GC^2 \Rightarrow AG^2 = 13^2 + 4^2 \quad (GC = BF = 4 \text{ cm})$$

$$\Rightarrow AG^2 = 185$$

$$\Rightarrow AG = \sqrt{185} \text{ cm}$$

- As can be seen in, $ACGE$ is a rectangle. So, its area is equal to

$$S_{ACGE} = AC \times CG$$

$$= 13 \times 4 = 52 \text{ cm}^2.$$

iii. Cube

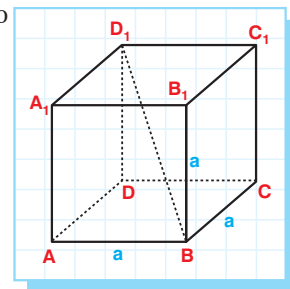
Definition

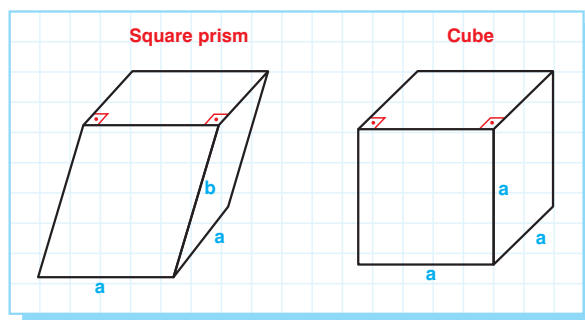
A parallelepiped whose six faces are all squares is called a **cube**.

Since a cube is a rectangular solid whose edges are all equal in length, the following conclusion can be written.

Conclusion

If the length of one edge of a cube is a then the length of its diagonal is $d = BD_1 = \sqrt{3} \times a$





Do not confuse a square prism whose bases are squares but lateral faces are any parallelograms with a cube whose six faces are squares.

EXAMPLE

22

The length of the diagonal of a face of a cube is equal to $5\sqrt{2}$ cm. Accordingly, find the length of a diagonal of the cube.

Solution Suppose that the length of one edge of the cube is a . So, the length of the diagonal of a face is $a \times \sqrt{2}$. Thus,

$$a \times \sqrt{2} = 5\sqrt{2} \Rightarrow a = 5 \text{ cm}.$$

By the conclusion above, we get the length of a diagonal of the cube as $d = \sqrt{3} \times 5 \Rightarrow d = 5\sqrt{3}$ cm

c. Areas of Prisms

In order to describe the measure of a polygonal region, we have used the term area. Similarly, we will use the same term to describe the measure of the surface of a prism (any polyhedron).

The area of a prism is the sum of the areas of its faces. The lateral area (S_L) of a prism is the sum of the areas of its lateral faces and the total area (S_T) of a prism is the sum of its lateral area and the areas of its two bases. Namely, if the area of a base of the prism is S_B then its total area is

$$S_T = S_L + 2 \cdot S_B$$

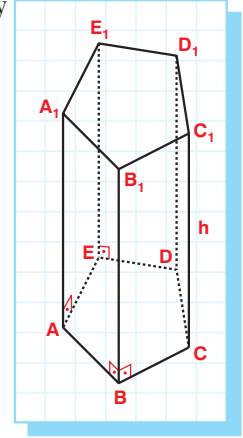
Because upper and lower bases of the prism are congruent.

Conclusion

The lateral area of a right prism is equal to the product of its height by the perimeter of its base.

For such a right prism in, if the perimeter of the base is p then

$$S_L = h \cdot p$$



EXAMPLE

23

The length of each edge of a regular hexagonal prism is 6 cm. Accordingly find the total area of the prism.

Solution

As stated in the previous conclusion, the lateral area of a right prism can be found by multiplying the perimeter of the base with the height of the prism. Therefore, the lateral area of the regular hexagonal prism in is equal to

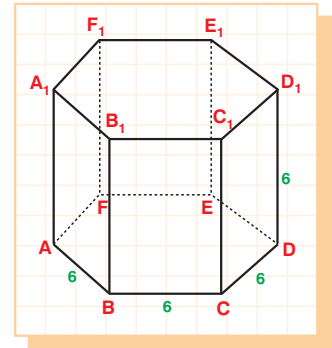
$$\begin{aligned}\Sigma L &= PAB CDE F \times DD_1 \\ &= 6 \times BC \times 6 \\ &= 36 \times 6 \\ &= 216 \text{ cm}^2.\end{aligned}$$

On the other hand, the area of the base is

$$\begin{aligned}\Sigma B &= S_{AB CDE F} = 3 \cdot \frac{BC^2 \cdot \sqrt{3}}{2} \\ &= 3 \cdot \frac{6^2 \cdot \sqrt{3}}{2} \\ &= 54\sqrt{3} \text{ cm}^2.\end{aligned}$$

Thus, the total area of the prism can be found as

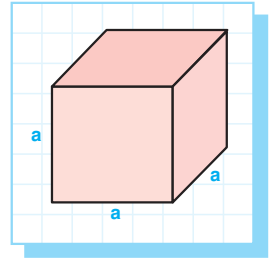
$$\begin{aligned}\Sigma T &= \Sigma L + 2 \times \Sigma B \\ \Sigma T &= 216 + 2 \times 54\sqrt{3} \\ \Sigma T &= (216 + 108\sqrt{3}) \text{ cm}^2 \\ \Sigma T &= 108 \times (2 + \sqrt{3}) \text{ cm}^2.\end{aligned}$$



Conclusion

If the dimensions of a rectangular solid are a , b and c then the total area of the prism equals

$$S_T = 2 \times (a \times b + b \times c + a \times c)$$



EXAMPLE

24

If the total area of a cube is 18 cm^2 , find the length of a diagonal of the cube.

Solution Let the length of one edge of the cube be a and the length of a diagonal d .

By the conclusion above,

$$ST = 6 \times a^2 \Rightarrow 18 = 6 \times a^2$$

$$\Rightarrow a^2 = 3$$

$$\Rightarrow a = \sqrt{3} \text{ cm.}$$

Since the length of a diagonal of a cube is equal to $\sqrt{3}$ times the length of an edge of the cube, we can find d as

$$d = a \times \sqrt{3} \Rightarrow d = \sqrt{3} \times \sqrt{3}$$

$$\Rightarrow d = 3 \text{ cm.}$$

Definition

A pyramid is a solid formed by one nappe of a pyramidal surface and a plane cutting all its elements.

- The section of the pyramidal surface made by the plane is the base of the pyramid.
- The triangular faces having a common vertex are called the lateral faces.
- The intersections of the lateral faces are called the lateral edges.
- The sides of the polygon which forms the base are called the basal edges.
- The common vertex is called the vertex of the pyramid. It is also named as apex.
- The segment drawn from the vertex and perpendicular to the base is the altitude of the pyramid. The length of the altitude is the height of the pyramid.
- Vertex angle at each face is called a plane angle at the vertex.

- The dihedral angles between adjacent lateral faces are called dihedral angles at the lateral edges.
- The dihedral angles between the lateral faces and the base are called the dihedral angles at the base.

As shown in, the parts of the pyramid are

Vertex : point V

Base : polygon ABCDE

Lateral faces : triangles VAB, VBC, etc.

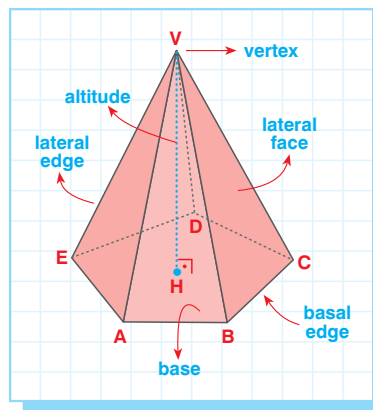
Altitude: perpendicular VH

Basal edges : AB, BC, CD, etc.

Lateral edges : VA, VB, VC, etc.

Plane angles at the vertex: $\angle AVB$, $\angle BVC$, $\angle CVD$, etc.

We can represent a pyramid by using its vertex and the vertices of its base. For example, the pyramid in is represented as VABCDE.



Theorem

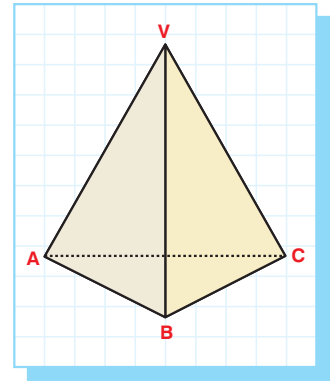
If a plane parallel to the base of a pyramid cuts all the lateral edges,

1. this plane divides the altitude and the lateral edges proportionally.
2. the section formed is similar to the base of the pyramid.

a. Classification of Pyramids

Pyramids are classified as to their bases. Namely, pyramids are said to be triangular, quadrangular, pentagonal, etc. , according to their bases that are triangles, quadrilaterals, pentagons, etc.

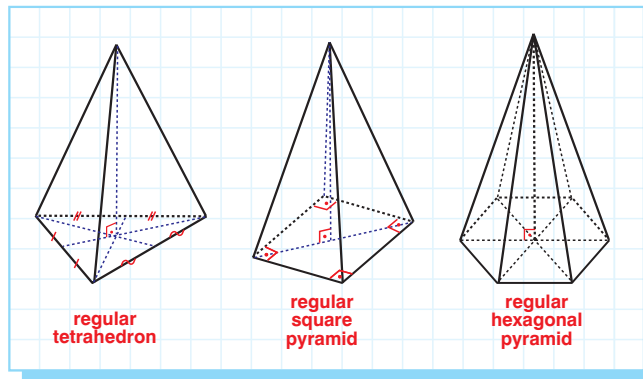
- A triangular pyramid has four triangular faces and is often called a tetrahedron. Any face of a tetrahedron may be used as the base.



i. Regular Pyramids

Definition

If the base of a pyramid is a regular polygon and if the projection of its vertex on its base is positioned at the center of the base then the pyramid is called as a **regular pyramid**.



In Figure 3.62, some types of regular pyramids are shown. As can be seen, if the base is an equilateral triangle then its center is the intersection point of its medians or if it is a square then the center is the intersection of its diagonals. Moreover, if the base is a regular hexagon then the center of the base is the center of its circumscribed circle.

- It can be generalized that center of the base of a regular pyramid is either the center of its circumscribed or inscribed circle.

Properties of a Regular Pyramid

1. The lateral edges of a regular pyramid are equal.

Explanation

The pyramid in Figure 3.63 is a regular hexagonal pyramid. Since the base is a regular hexagon, it can be divided into six congruent equilateral triangles. So, in triangle ABH,

$$AB = AH = BH .$$

On the other hand, since VH is the altitude of the pyramid,

$$\angle VHA = \angle VHB = 90^\circ .$$

By SAS congruence postulate, we can write

$$DVHA \cong DVHB .$$

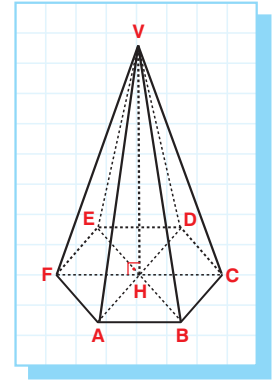
Therefore,

$$VA = VB$$

and by analogy, we can

$$VA = VB = VC = VD = VE = VF$$

2. The lateral faces of a regular pyramid are enclosed by congruent isosceles triangles.



Explanation

By Figure 3.63, this property claims that lateral faces $DVAB$, $DVBC$, $DVCD$, $DVDE$, $DVEF$, and $DVFA$ are all congruent isosceles triangles.

In the previous property, we have obtained that

$$VA = VB = VC \text{ and } AB = BC .$$

Therefore, by SSS congruence theorem,

$$DVAB \cong DVBC .$$

Similarly, we can conclude that

$$DVAB \cong DVBC \cong DVCD \cong DVDE \cong DVEF \cong DVFA .$$

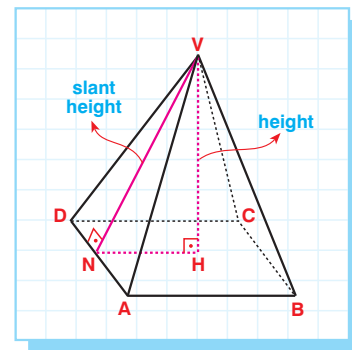
3. The altitudes of the triangular faces of a regular pyramid are equal.

Explanation

In the previous property, we have obtained that the lateral faces of a regular pyramid are all congruent isosceles triangles. So, their altitudes must be equal.

- The altitude of any of the lateral faces drawn from the vertex of a regular pyramid is called the slant height of the regular pyramid.

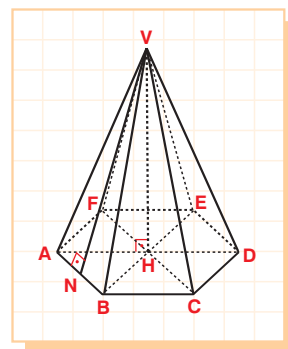
In, VN is the slant height of the regular square pyramid while VH is the height of the pyramid.



Note

Only a regular pyramid can have a slant height.

EXAMPLE 25 If the length of a basal edge of a regular hexagonal pyramid is 10 cm and the height of the pyramid is $\sqrt{69}$ cm then find the slant height of the pyramid.



Solution Let us construct regular hexagonal pyramid VABCDEF, its altitude VH and its slant height VN as in

Given that

$$AB = BC = 10 \text{ cm and } VH = \sqrt{69} \text{ cm .}$$

Since triangle ABH is an equilateral triangle,

$$AB = AH = BH = 10 \text{ cm .}$$

Therefore, in right triangle VHB,

$$VB^2 = VH^2 + BH^2 \Rightarrow VB^2 = (\sqrt{69})^2 + (10)^2$$

$$\Rightarrow VB^2 = 169$$

$$\Rightarrow VB = 13 \text{ cm .}$$

On the other hand, in isosceles triangle VAB, the slant height VN is also a median of the triangle. So,

$$AN = BN = \frac{AB}{2} = \frac{10}{2} = 5 \text{ cm.}$$

In right triangle VNB, we obtain

$$VB^2 = VN^2 + BN^2 \Rightarrow 132 = VN^2 + 5^2 \Rightarrow VN^2 = 144 \Rightarrow VN = 12 \text{ cm.}$$

EXAMPLE

26

VABCD is a regular pyramid whose base is square ABCD with a side length of 2. If the distance between AC and BV is 1 then find

- the measures of angles AVB and AVC.
- the dihedral angle formed by two adjacent lateral faces.
- the dihedral angle formed by two opposite lateral faces.

Solution a) If we draw altitude VH and diagonals AC and BD of the base, we have $VH \perp AC$, $VH \perp BD$, and $AC \perp BD$.

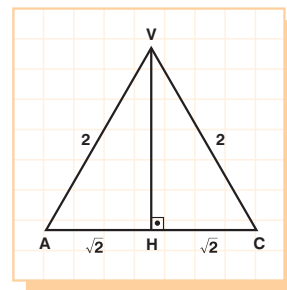
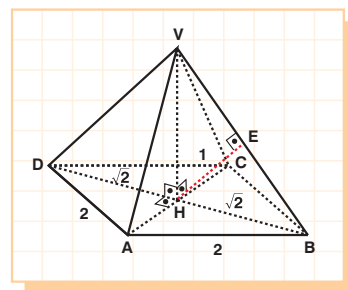
Since AC is perpendicular to two intersecting lines VH and BD, it is perpendicular to all lines lying on (VBD). In right triangle VHB, if we draw altitude HE to VB then AC will be perpendicular to HE and HE will be the common perpendicular of skew lines AC and VB ($HE \perp AC$ and $HE \perp VB$). So the distance between AC and VB is the length of common perpendicular HE. Therefore, we have

$$HE = 1, \quad DH = HB = AH = HC = \frac{AB \cdot \sqrt{2}}{2} = \frac{2 \cdot \sqrt{2}}{2} = \sqrt{2}.$$

If we apply the Pythagorean theorem in right triangle EHB, we obtain $EB = 1$. Then, it can be concluded that

$VB = VA = VC = VD = 2$. So, the lateral faces are all equilateral triangles which means that $\angle AVB = 60^\circ$. On the other hand, triangle AVC will be an isosceles triangle as in Figure 3.67.

Since $VA^2 + VC^2 = AC^2$, we can conclude that $\angle AVC = 90^\circ$



- b) In order to find the di-hedral angle between adjacent lateral faces, let us construct altitudes AE and CE on faces VAB and VBC, respectively (DVAB and DVBC are equilateral triangles, and E is the midpoint of VB). Angle AEC will be the plane angle of the dihedral angle between (VAB) and (VBC) and we need to find its measure.

Since DVAB and DVBC are equilateral, we have

$$AE = CE = \frac{AB \cdot \sqrt{3}}{2} = \frac{2\sqrt{3}}{2} = \sqrt{3}.$$

In triangle AEC, if we apply the Cosine theorem,

$$AC^2 = AE^2 + CE^2 - 2 \times AE \times CE \times \cos \angle AEC$$

$$\Rightarrow (2\sqrt{2})^2 = (\sqrt{3})^2 + (\sqrt{3})^2 - 2 \times \sqrt{3} \times \sqrt{3} \times \cos \angle AEC$$

$$\Rightarrow 8 = 6 - 6 \times \cos \angle AEC$$

$$\Rightarrow \cos \angle AEC = -\frac{1}{3}$$

$$\Rightarrow \angle AEC = \text{Arc cos}\left(-\frac{1}{3}\right) \approx 110^\circ$$

- c) If we draw a planethrough vertex V which is perpendicular to both faces (VAD) and (VBC), it will contain altitude VH of the pyramid and altitudes VK and VL of faces VAD and VBC respectively. So, we have

$$KL = AB = 2 \text{ and } VK = VL = \frac{VA\sqrt{3}}{2} = \frac{2\sqrt{3}}{2} = \sqrt{3}.$$

In triangle VKL, we may apply the Cosine theorem:

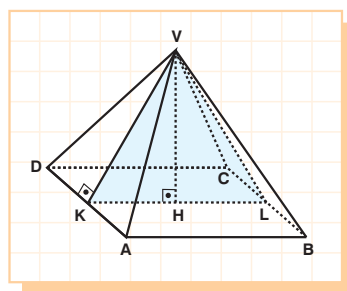
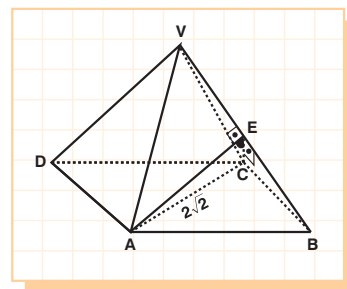
$$KL^2 = VK^2 + VL^2 - 2 \times VK \times VL \times \cos \angle KVL$$

$$22 = (\sqrt{3})^2 + (\sqrt{3})^2 - 2 \times \sqrt{3} \times \sqrt{3} \times \cos \angle KVL$$

$$4 = 6 - 6 \times \cos \angle KVL$$

$$\cos \angle KVL = \frac{1}{3} \Rightarrow \angle KVL = \text{Arc cos}\left(\frac{1}{3}\right) \approx 70^\circ.$$

Since angle KVL is the plane angle of the dihedral angle between faces VAD and VBC, we can conclude that the measure of the dihedral angle between opposite faces is approximately equal to 70° .

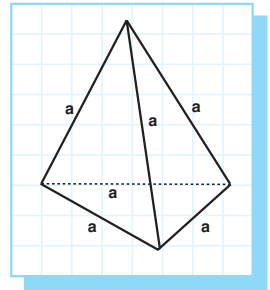


ii. Regular Tetrahedron

Definition

A triangular pyramid whose all edges are equal is called a **regular tetrahedron**.

As in, all four faces of a regular tetrahedron are equilateral triangles. The altitudes drawn to the faces of the regular tetrahedron are all equal and they are called as medians of the tetrahedron.



Note

A regular tetrahedron is a special regular triangular pyramid such that basal and lateral edges have the same length.

EXERCISES 7.3

A. Some Important Polyhedrons

Prism

1. A prism has 53 faces. Find the number of vertices and the number of edges.
2. A prism has 69 edges. Find the number of vertices and the number of faces.
3. The sum of the lengths of all edges of a rectangular parallelepiped is 28 cm and a diagonal is 6 cm. Calculate the surface area.
4. The length of diagonal of a cube is 1 greater than the length of diagonal of a face. Calculate the total area of the cube.
5. The sum of the dimensions of a rectangular parallelepiped is 9 cm and the surface area is 45 cm. Calculate the length of its diagonal.
6. A rectangular parallelepiped has a diagonal equal to $4\sqrt{3}$ cm and the surface area equal to 96 cm². Prove that this parallelepiped is a cube.
7. ABCA1B1C1 is a regular triangular prism. What is the distance from C1 to the center of base ABC, if $A1B1 = 6$ cm and $B1B = 4$ cm.
8. ABCDA1B1C1D1 is a parallelepiped. Prove that the midpoints of edges AB, BC, CC1, C1D1, D1A1, and AA1 are coplanar.
9. A right parallelepiped ABCDA1B1C1D1 has parallelogram ABCD as its base. If $AB = a$, $BC = b$, $AA1 = c$ and the distance between lines AB1 and CD1 is d, calculate:
 - a) the surface area of the parallelepiped
 - b) the lengths of the diagonals of the parallelepiped.
10. ABCDA1B1C1D1 is a cube with an edge of a. If M and N are the midpoints of CD and C1D1, respectively, find the distance between line segments
 - a) AC and B1D1
 - b) MN and D1A
 - c) MN and AC
11. ABCDA1B1C1D1 is a cube with an edge of a in length. We denote the midpoints of edges CD and C1D1 by M and N, respectively. Find the distance between lines:
 - a) AC and DD1
 - b) MN and BD
 - c) MN and BC1
12. ABCDA1B1C1D1 is a parallelepiped such that each face is a rhombus of edge length a, $\angle BCD = 60^\circ$, and $\angle BCC1 = \angle DCC1 = 120^\circ$.
 - a) Prove that planes ACC1A1 and ABCD are perpendicular.
 - b) Calculate the area of quadrilateral BB1D1D.
 - c) Prove that the projection of point A1 on plane ABCD is the center of the circumscribed circle of DABD.
 - d) Calculate the area of the parallelepiped.

Pyramid

13. A pyramid has 7 vertices. Find the number of faces and the number of edges of the pyramid.
14. A pyramid has 100 edges. Find the number of vertices and the number of faces of the pyramid.
15. ABCD is a regular tetrahedron of edge a . Calculate:
- the angle formed by edge AD and plane ABC.
 - the angle formed by planes ABC and ABD.
 - the distance between two opposite edges.
16. A quadrangular regular pyramid has equilateral triangles of side 2 cm for lateral faces. Calculate
- the surface area of the pyramid.
 - the angle between a lateral edge and the base plane.
 - the angle formed by two opposite lateral edges.
 - the dihedral angle formed by a lateral face and the base plane.
 - the dihedral angle formed by two adjacent lateral faces.
17. Prove that if a pyramid has congruent lateral edges then around the polygon which forms the pyramid's base, a circle can be circumscribed.
18. SABCD is a pyramid having parallelogram ABCD as its base. Prove that ABCD is a rectangle if and only if $SA^2 + SC^2 = SB^2 + SD^2$.
19. ABCD is a tetrahedron with $BC = CD$. The bisectors of angles ACB, ACD, BCD intersect AB, AD and BD in M, N and P, respectively. Prove that $CP \perp MN$.
20. SABCD is a regular pyramid with square ABCD of edge 2 cm as its base. The distance between lines AC and BS is 1 cm. Find:
- the lateral area.
 - the measures of angles ASB and ASC.
 - the cosine of the dihedral angle formed by faces ASB and DSC.

21. VABCD is a regular pyramid with base ABCD.
- Prove that the lateral faces are acute triangles.
 - If we denote the projection of B on VA by M and the projection of C on VD by N, prove that lines BM and CN are concurrent.
 - If P is the intersection point of BM and CN, prove that $PV \parallel (ABCD)$.
22. A regular pyramid has its base area equal to S and the lateral faces form 60° angles with the base. Calculate the surface area.
23. ABCD is a tetrahedron. The bisectors of angles BAC, CAD and DAB intersect edges BC, CD and DB in M, N and P, respectively. Prove that planes ANB, ACP and ADM share a common line.
24. VABCD is a pyramid whose base is parallelogram ABCD. We denote the midpoints of edges VA, VB, VC and VD by Q, M, N, P. Prove that:
- AP and BN are concurrent.
 - CM and DQ are concurrent.
25. $VA_1A_2\dots A_n$ is a regular pyramid having the polygon $A_1A_2\dots A_n$ as its base.
- Prove that $\angle A_1VA_2 < \frac{2\pi}{n}$.

Frustum of Pyramid

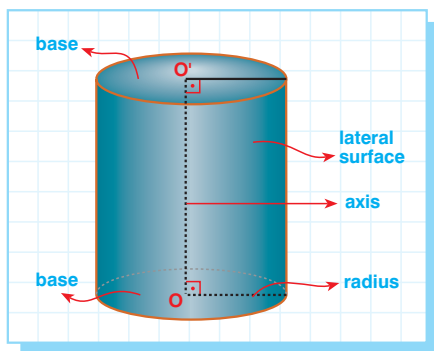
26. ABCDA₁B₁C₁D₁ is a frustum of a regular quadrangular pyramid. Prove that diagonals AC₁, BD₁, CA₁, and DB₁ are concurrent.
27. A frustum of a regular quadrangular pyramid ABCDA₁B₁C₁D₁ has the base edges equal to 5 cm and 3 cm, respectively. Lateral edge of frustum is $2\sqrt{2}$ cm. Find:
- the distance between lines AD and MN where M and N are the midpoints of edges BC and B₁C₁;
 - the distance from M to plane ADD₁.
28. Areas of bases of a frustum of a regular hexagonal pyramid are $54\sqrt{3}$ cm² and $24\sqrt{3}$ cm², and the height is $\sqrt{22}$ cm. What is the lateral area of this frustum?
29. ABCA₁B₁C₁ is a frustum of a regular triangular pyramid with bases DABC and DA₁B₁C₁. What is the distance between AB and B₁C₁ if AC = 6 cm, A₁B₁ = 4 cm and slant height of the frustum is $2\sqrt{2}$ cm?

A. Circular Cylinder

Definition

A **circular cylinder** is a cylinder whose bases are circles.

- Since the circle is the only plane curve studied in former Geometry courses, most of the postulates, theorems, and exercises which follow will refer only to circular cylinders and hereafter, unless otherwise indicated, the word “cylinder” will be used to mean “right circular cylinder”.
- The line segment joining the centers of the bases is the axis of the cylinder.
- An axial section of a circular cylinder is a plane section which contains the axis.
- The radius of a cylinder is the radius of any base. Some parts of a right circular cylinder can be seen in



Theorem

The axis of any cylinder is equal and parallel to its elements.

Proof Let A be any point on the circumference and K be the center of the base of the cylinder in .

If we draw, through A and K, the diameter AB of the lower base then there exists only one element BC containing B.

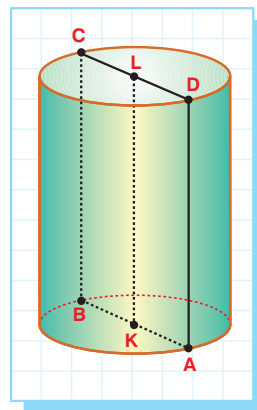
If we pass a plane through AB and BC, the formed section ABCD is a parallelogram by. Therefore,

$$AB = CD .$$

This implies that CD is a diameter of the upper base and passes through the center L .

Hence, $AK = DL$ and therefore AKLD is a parallelogram. So,

$$KL \parallel AD \text{ and } KL = AD$$



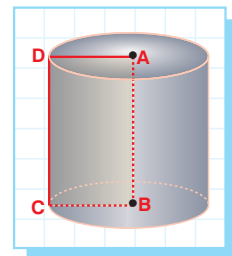
Conclusion

1. The axis of a cylinder passing through the centers of all sections parallel to its bases,
2. A cylinder may be generated by the revolution of a rectangle about one of its sides as an axis.

For this reason, a cylinder is also called a cylinder of revolution.

In, ABCD is given as a rectangle.

If AD , DC and BC are permitted to revolve about AB as an axis then AD and BC will generate parallel circular bases and CD will generate a cylindrical surface.



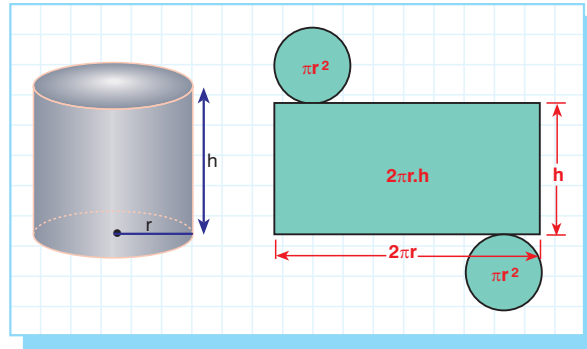
Conclusion

If the radius of a cylinder is r and its height is h then

1. its lateral area is equal to $SL = 2\pi r \times h$
2. its total area is equal to $ST = 2\pi r \times h + 2\pi r^2$

Explanation

The cylinder and its unfolded form can be seen in



The width of the rectangular region is equal to the circumference of the base of the cylinder and its height is equal to that of the cylinder.

At this point, it is easy to derive the given formulas for the lateral area and the total area of the cylinder.

EXAMPLE

27

The radius of a cylinder is 8 cm and its height is 5 cm. Accordingly, find the lateral area and the total area of the cylinder.

Solution

By the given conclusion, the lateral area of the cylinder is equal to

$$SL = 2\pi r \times h \Rightarrow SL = 2\pi \times 8 \times 5$$

$$\Rightarrow SL = 80\pi \text{ cm}^2$$

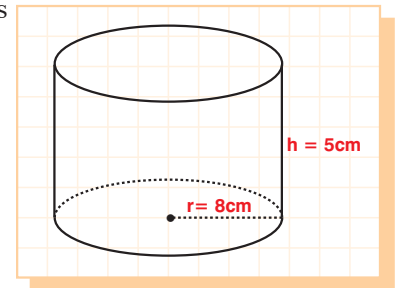
and its total area is equal to

$$ST = SL + 2\pi r^2$$

$$\Rightarrow ST = 80\pi + 2\pi \times (8^2)$$

$$\Rightarrow ST = 80\pi + 128\pi$$

$$\Rightarrow ST = 208\pi \text{ cm}^2.$$



Definition

A solid formed by one nappe of a conical surface and a plane cutting all the elements is called a **cone**.

- The base of the cone is the section of the conical surface made by the plane.
- The lateral surface is the curved part of the surface.
- The vertex of the conical surface is called the vertex of the cone.
- The altitude of a cone is the perpendicular from the vertex to the plane of the base. The length of the altitude is the height of the cone.
- An element of a cone is a line segment connecting the vertex with a point in the circumference of the base.

As can be seen in Figure 3.113, the parts of the cone are;

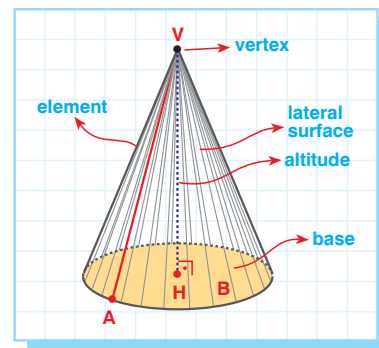
Vertex : point V

Base : section B

Lateral surface : the curved surface

Altitude : perpendicular VH

An element : VA .



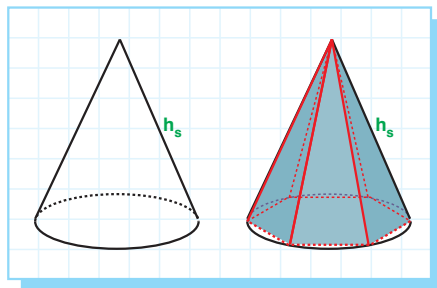
Every point of the lateral surface of a cone except the vertex is contained in exactly one element.

B. Areas of Cones

Theorem

The lateral area of a right cone is equal to half of the product of the slant height and the circumference of the base.

Proof



Let the lateral area of the cone be SL , the circumference of its base C and its slant height h_s .

Suppose a regular pyramid to be inscribed in the cone, the perimeter of its base being P and its lateral area SL' . Then, we can write

$$SL' = \frac{1}{2} \times h_s \times P .$$

If the number of lateral faces of the inscribed pyramid becomes infinite,

$$SL' \rightarrow SL \text{ and } P \rightarrow C .$$

Since h_s remains constant, we can write

$$P \rightarrow C \Rightarrow \frac{1}{2} \times h_s \times P \rightarrow \frac{1}{2} \times h_s \times C .$$

⏟

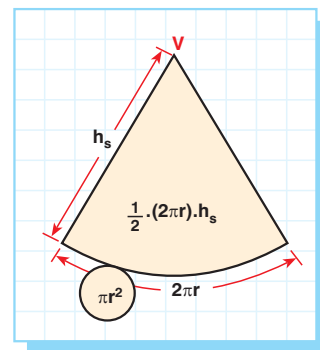
SL'

But, $SL' \rightarrow SL$. Hence, the right side of the above limit expression is equal to SL . Namely,

$$S_L = \frac{1}{2} \cdot h_s \cdot C$$

Explanation

Examine and try to get the above conclusion on your own.



EXAMPLE

28

Find the lateral area of a right cone with radius 6 cm and height 8 cm.

Solution In the right cone in Figure 3.134, $VO = h = 8$ cm,
 $OA = r = 6$ cm and $VA = h_s$.

In right triangle VOA , by the Pythagorean theorem, we obtain

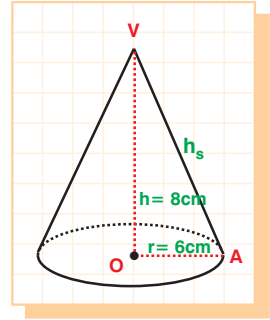
$$VA^2 = VO^2 + OA^2$$

$$h_s^2 = h^2 + r^2 \Rightarrow h_s^2 = 8^2 + 6^2$$

$$h_s = 10 \text{ cm} .$$

By the previous conclusion, the lateral area of the cone is equal to

$$SL = p \times r \times h_s \Rightarrow SL = p \times 6 \times 10 \Rightarrow SL = 60p \text{ cm}^2 .$$



C. Spheres

1. Fundamental Definitions

In this part, we will see that many properties of a sphere are similar to the ones of a circle. When we are defining the terms used for spheres, you may change the definition of the corresponding terms of circles, just by replacing the word circle by the word sphere and omitting the restriction in a plane.

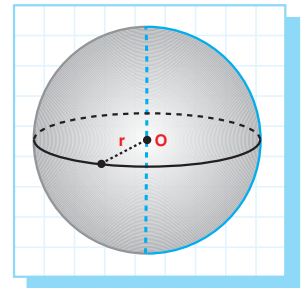
Definition

The set of all points in space which are equidistant from a fixed point is called a spherical surface. The fixed point is called the center and the fixed distance is called the radius.

In, O is the center and r is the radius of the spherical surface.

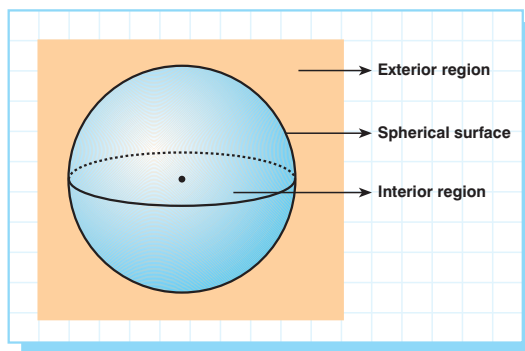
A spherical surface can be obtained by rotating a circle (or a semi-circle) about one of its diameters.

A spherical surface separates the whole space into three subsets; an exterior region, the spherical surface itself and an interior region.



Definition

A sphere is a solid formed by a spherical surface and its interior.



- A line which is intersecting the sphere is a secant of the sphere.
- A line segment whose endpoints are on the surface of a sphere is a chord of the sphere.
- Any chord which contains the center of a sphere is a diameter of the sphere.

EXERCISES 7.4

A. Cylinder

1. Through an element of a cylinder drawn two cross - sections. One of them is axial cross - section with area S . Two cross - sections make an angle of 45° . Find the area of second cross - section.
2. Diagonal of open form of lateral face of a cylinder makes an angle of α degrees with the base of this open form. Find the tangent of the angle between the diagonal of axial cross - section of the cylinder and the base of the cylinder.
3. Diagonal of axial cross section of a cylinder is 20 cm and makes 60° with the base of the cylinder. Find total surface area of the cylinder.
4. The lateral surface area of a cylinder is half of its total surface area. Find the cosine of the angle between the diagonal of an axial section and the base of the cylinder.

B. Cone and Frustum of Cone

5. An element of a right cone is 10 cm and forms 60° with the base of the cone. Find total area of the cone.
6. A right cone is cut through an element and opened. The central angle of this sector is 60° and surface area of cone is $\frac{3\pi}{2} \text{ cm}^2$. Find the area of axial cross section of cone.
7. Given a frustum of a right cone with circumferences of the bases 4π cm and 16π cm. Height of the frustum of the cone is 8 cm. Find its total surface area.
8. Find the radii of bases of a frustum of a right cone if its lateral area is $323\pi \text{ cm}^2$, its element is 17 cm, and its height is 8 cm.

9. A right triangle with legs 6 cm and 8 cm is rotated 360° about its hypotenuse. Find the surface area of the formed solid.

10. ABCD is a right trapezoid ($AD \parallel BC$ and $\angle BAD = 90^\circ$).

$AB = 5$ cm, $BC = 9$ cm, and $\angle CDA = 135^\circ$.

The trapezoid is rotated about AD. What will the surface area of the formed solid be?

C. Sphere

11. A sphere is cut by a plane and the distance from the center of the sphere to this plane is 8 cm. The area of the formed small circle is 36π cm². Find the surface area of the sphere.
12. Circumference of the intersection of a sphere with a plane is 16π cm and this plane is 7 cm away from the center of the sphere. Find the surface area of the sphere.

13. A sphere is intersected by two parallel planes. Areas of the formed small circles are 25π cm² and 144π cm². Center of the sphere lies between these two parallel planes. If the radius of the sphere is 13 cm, find the distance between the planes.

14. A, B, C are three points on a great circle of a sphere, and Δ is another point on the surface of the sphere which is not in (ABC). What is the maximum distance from Δ to (ABC) if $AB = 16$ cm, $AC = 30$ cm, and $BC = 34$ cm?

15. The area of a wedge of a sphere with an angle of 45° is 18π cm². Find the area of a spherical cone of this sphere if the angle of this cone is 120° .

16. A sphere is cut by a plane. The distance from the center of the sphere to the plane is half of radius of the sphere. Find the surface area of the sphere if the area of the formed spherical cap (small one) is 75π cm².

17. O is the center of a sphere and A is a point on its surface. Planes a and b are perpendicular to line

segment OA at B and C such that $\frac{OB}{BA} = \frac{OC}{CA} = \frac{1}{3}$.

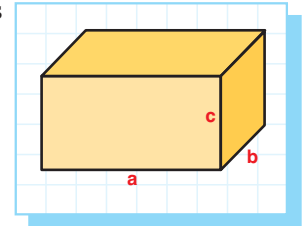
What is the ratio of the surface area of the segment of two bases whose bases are in a and b to the surface area of the segment of one base whose base is in b?

SECTIONS AND COMBINATIONS OF SOLIDS

Theorem

The volume of a rectangular solid is the product of the area of its base by its height or the product of its three dimensions.

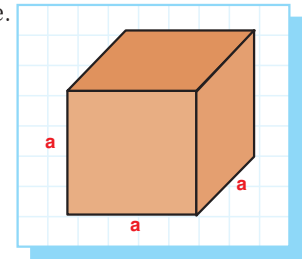
$$V = a \cdot b \cdot c$$



Conclusion

The volume of a cube is equal to the cube of the length of one edge.

$$V = a^3$$



EXAMPLE

29

If the total area of a cube is 24 cm^2 , find the volume of the cube.

Solution Let the length of an edge of the cube be a . So, the total area of the cube is equal to

$$S_T = 6 \cdot a^2 \Rightarrow 24 = 6 \cdot a^2 \Rightarrow a^2 = 4 \Rightarrow a = 2 \text{ cm}.$$

Therefore, by the conclusion above, the volume of the cube is obtained as

$$V = a^3 \Rightarrow V = 2^3 \Rightarrow V = 8 \text{ cm}^3.$$

1. Volume of a Right Prism

Theorem

The volume of a right prism is the product of the area of its base and its height.

2. Volume of a Pyramid

You learned that the volume of a prism is equal to the product of the area of its base and its height.

EXAMPLE 30 Find the volume of a square pyramid if its height is 9 cm and each basal edge is 8 cm.

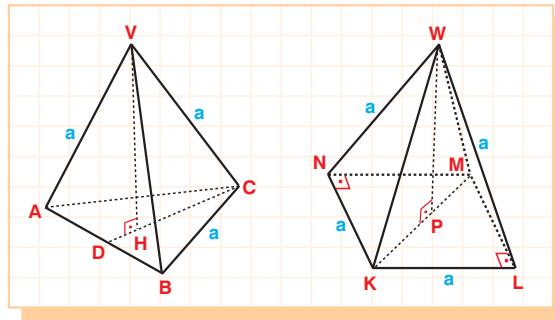
Solution The volume of the pyramid is

$$V = \frac{1}{3} \cdot S \cdot h = \frac{1}{3} \cdot 8^2 \cdot 9 = 192 \text{ cm}^3.$$

EXAMPLE 31 Find the ratio of the volumes of a regular quadrangular pyramid and a regular triangular pyramid if the lengths of their all edges are equal to a .

Solution First of all, let us try to get the height of each pyramid.

The height of regular tetrahedron VABC is $VH = \frac{a \cdot \sqrt{6}}{3}$.



Let P be the midpoint of KM which is a diagonal of square KLMN. Then,

$$KM = a \times \sqrt{2} \quad \text{and} \quad KP = PM = \frac{KM}{2} = \frac{a \cdot \sqrt{2}}{2}$$

In right triangle WPK, by the Pythagorean theorem,

$$WK^2 = WP^2 + KP^2 ,$$

$$a^2 = WP^2 + \left(\frac{a \cdot \sqrt{2}}{2}\right)^2 ,$$

$$WP^2 = \frac{a^2}{2} ,$$

$$WP = \frac{a \cdot \sqrt{2}}{2} .$$

Then, the ratio of the volumes of the pyramids is

$$\frac{V_{VABC}}{V_{WKLMN}} = \frac{\frac{1}{3} \cdot S_{ABC} \cdot VH}{\frac{1}{3} \cdot S_{KLMN} \cdot WP} = \frac{\frac{a^2 \cdot \sqrt{3}}{4} \cdot \frac{a \cdot \sqrt{6}}{3}}{a^2 \cdot \frac{a \cdot \sqrt{2}}{2}} = \frac{1}{2} .$$

Conclusion

If the length of each edge is a then

1. the volume of a regular tetrahedron is equal to

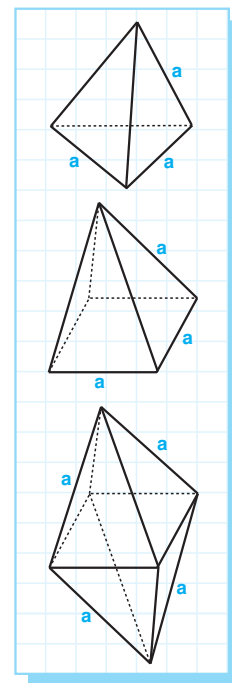
$$V = \frac{a^3 \cdot \sqrt{2}}{12}$$

2. the volume of a regular quadrangular pyramid is equal to

$$V = \frac{a^3 \cdot \sqrt{2}}{6}$$

3. the volume of a regular octahedron is equal to

$$V = \frac{a^3 \cdot \sqrt{2}}{3}$$



EXAMPLE

32

Find the volume of a regular octahedron if the length of one edge is $a = 6$ cm.

Solution By the above conclusion, the volume of the regular octahedron is equal to $V = \frac{a^3 \cdot \sqrt{2}}{3} = \frac{6^3 \cdot \sqrt{2}}{3} = 72\sqrt{2} \text{ cm}^3$.

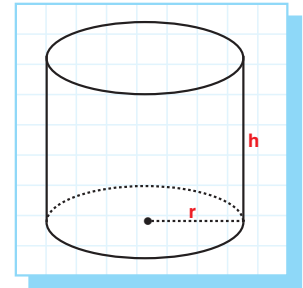
Theorem

The volume of a cylinder is equal to the product of its base area and its height.

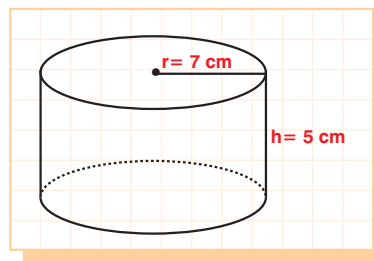
Conclusion

The volume of a cylinder with radius r and height h is equal to

$$V = \pi r^2 \cdot h$$



EXAMPLE 33 Find the volume of the cylinder according to the given values in.



Solution By the figure, $r = 7 \text{ cm}$ and $h = 5 \text{ cm}$. Therefore, the volume of the cylinder is

$$V = p \times r^2 \times h = p \times (7)^2 \times 5 = 245 p \text{ cm}^3$$

EXAMPLE

34 The altitude of a cylinder equals its diameter. If its volume is $128\pi \text{ cm}^3$, find the height and the lateral area of the cylinder.

Solution Let h be the height, r the radius and V the volume of the cylinder. So,

$$h = 2 \times r \quad (\text{height equals diameter})$$

$$V = \pi \times r^2 \times h,$$

$$128\pi = \pi \times r^2 \times 2r,$$

$$64 = r^3, \quad r = 4 \text{ cm}.$$

Hence, the height is

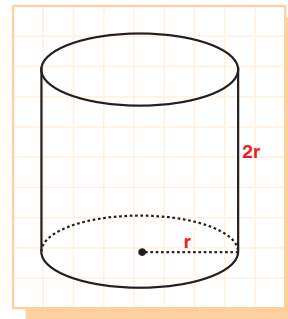
$$h = 2 \times r, \quad h = 2 \times 4 = 8 \text{ cm}$$

and the lateral area is equal to

$$SL = 2\pi r \times h,$$

$$SL = 2\pi \times 4 \times 8,$$

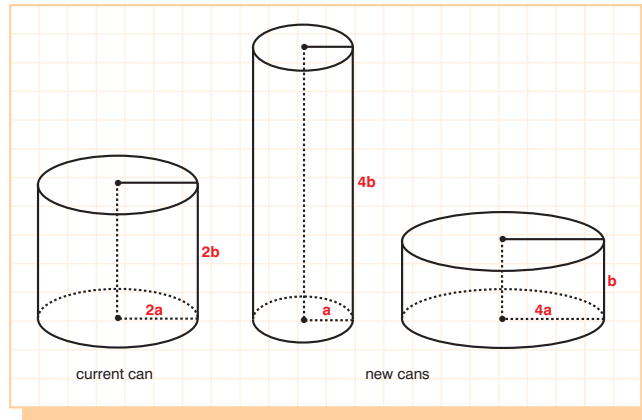
$$SL = 64\pi \text{ cm}^2.$$

**EXAMPLE**

35 A food-processing company is trying to test customer reaction to two new sizes of can. The first is half the radius and twice the height of the current can. The second is twice the radius and half the height of the current can.

- If each can sells for the same price, which is better to buy? Explain.
- If we are asked to design a can that has half the radius of the current can and has the same volume, how tall should the can be?
- We are asked to design a jumbo-sized can that has twice the radius and twice the height of the current can. If the current can sells for 5 \$, what would be reasonable price for the jumbo-sized can?

Solution **a)** If the radius of the current can is $r = 2a$ and the height is $h = 2b$ then the corresponding measures of the new cans are $r_1 = a$, $h_1 = 4b$, $r_2 = 4a$, and $h_2 = b$.



If we say V_1 and V_2 to the volumes of new cans, we get that

$$V_1 = \pi r_1^2 h_1 = \pi a^2 \times 4b = 4\pi a^2 \times b \quad \text{and}$$

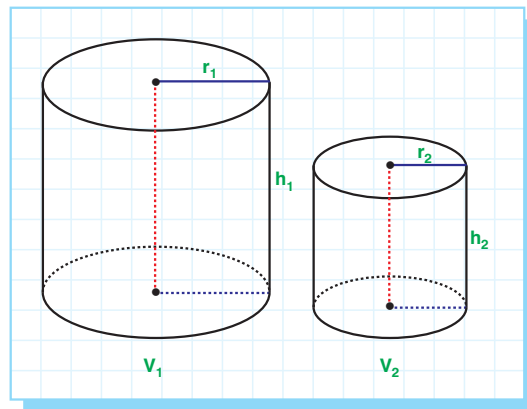
$$V_2 = \pi r_2^2 \times h_2 = \pi \times (4a)^2 \times b = 16\pi a^2 \times b.$$

Therefore, the volume of the second can is four times the volume of the first one. It is reasonable to buy the second one.

b) and c) are left to the student as an exercise.

Conclusion

If the radii and heights of two similar cylinders are r_1, r_2, h_1, h_2 then $\frac{V_1}{V_2} = \frac{r_1^3}{r_2^3} = \frac{h_1^3}{h_2^3}$.

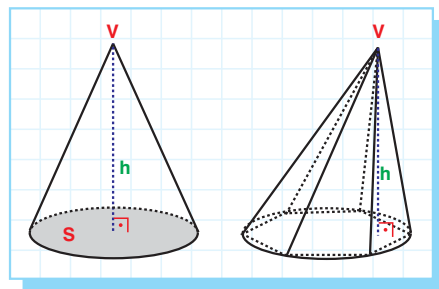


Volumes of Cones

Theorem

The volume of a cone is equal to one third the product of its base area and its height.

Proof



In other words, the volume of the cone is

$$V = \frac{1}{3} \cdot h \cdot S$$

Conclusion

If the radius of a cone is r and its height is h then its volume is

$$V = \frac{1}{3} \cdot \pi \cdot r^2 \cdot h$$

EXAMPLE

36 Find the volume of a cone if its height is $h = 10$ cm and its base is a circular region with radius $r = 3$ cm .

Solution

By the above conclusion, the volume of the cone is equal to

$$V = \frac{1}{3} \cdot \pi \cdot r^2 \cdot h = \frac{1}{3} \cdot \pi \cdot (3)^2 \cdot 10 = 30\pi \text{ cm}^3.$$

EXAMPLE

37 Find the volume of a right cone if its height is 9 cm and its slant height (the length of an element) is 15 cm .

Solution Let triangle VOA be the generator of the cone in.

In right triangle VOA,

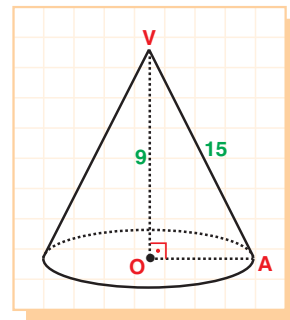
$$OA = 12 \text{ cm} .$$

Hence, the volume of the cone is equal to

$$V = \frac{1}{3} \cdot \pi \cdot OA^2 \cdot VO ,$$

$$V = \frac{1}{3} \cdot \pi \cdot 12^2 \cdot 9 ,$$

$$V = 432\pi \text{ cm}^3 .$$



EXERCISES 7.5

A. Volume of Rectangular Solid and Right

1. Dimensions of a rectangular solid are in the ratio of 3 : 4 : 5. A diagonal of the solid is $\sqrt{50}$ cm. Find its volume.
2. Base of a rectangular solid is a square. A diagonal of the rectangular solid is d units and makes 60° with a lateral edge. Find its volume.
3. Base of a right prism is an isosceles right triangle. A diagonal of the greatest face is 8 cm and makes 45° with the plane of base. Find the volume of the prism.
4. Base of a right prism ABCA₁B₁C₁ is a right triangle ABC with $\angle C = 90^\circ$. $AC = 10$ cm. Plane AB₁C makes 45° with the plane of base. Distance from point B to plane AB₁C is $2\sqrt{2}$ cm. Find the volume of the prism.
5. In a regular hexagonal prism ABCDEFA₁B₁C₁D₁E₁F₁, diagonals B₁F and B₁E are 12 cm and 13 cm, respectively. Find the volume of the prism.

B. Volume of an Oblique Prism

6. In an oblique triangular prism ABCA₁B₁C₁, lateral edge is 12 cm, distances from lateral edge AA₁ to edges CC₁ and BB₁ are 13 cm, distance from AA₁ to opposite lateral face is 12 cm. Find the volume of the prism.
7. Base of an oblique prism ABCA₁B₁C₁ is an equilateral triangle ABC with one side 4 cm. $\angle A_1AC = \angle A_1AB = 60^\circ$. One lateral edge is 5 cm. Find the volume of the prism.
8. In an oblique triangular prism, the height is $5\sqrt{2}$ cm and one lateral edge makes 45° with the plane of the base. Areas of two lateral faces are 50 cm^2 and 100 cm^2 and the angle between them is 120° . Find the volume of the prism.

C. Volume of a Cylinder

9. Diagonal of an axial cross section of a cylinder is 20 cm and makes 60° angle with the plane of the base of the cylinder. Find the volume of cylinder.
10. Given a cylinder and one of its cross - sections parallel to the axis of cylinder. Distance between the cross - section and the axis of cylinder is 15 cm, a diagonal of the cross - section is $2\sqrt{89}$ cm. Radius of the base is 17 cm. Find the volume of the cylinder.

11. Cross section which is parallel to the axis of a cylinder forms an arc of 120° in the base of the cylinder. Find the ratio of the volumes of parts of the cylinder divided by the cross - section.

D. Volume of a Pyramid

12. In a regular triangular pyramid, the height of the base is 3 cm and lateral edge makes 45° with the base of the pyramid. Find the volume of the pyramid.
13. In a regular quadrangular pyramid, a diagonal of the base is a units. Lateral face of the pyramid makes α degrees angle with the base of the pyramid. Find the volume of the pyramid.
14. Base of a pyramid is an isosceles trapezoid with acute angle 30° . Lateral faces of the pyramid make 60° angle with the base of the pyramid. Height of the pyramid is $6\sqrt{3}$ cm. Find the volume of the pyramid.

15. ABCD is a tetrahedron and O is a point inside it. We denote the heights of the tetrahedron which pass through A, B, C and D by h_1, h_2, h_3, h_4 and the distances from O to planes BCD, ACD, ABD and ABC by d_1, d_2, d_3, d_4 . AO, BO, CO, DO intersect planes BCD, ACD, ABD and ABC at points A_1, B_1, C_1, D_1 . Prove that:

a) $\frac{d_1}{h_1} + \frac{d_2}{h_2} + \frac{d_3}{h_3} + \frac{d_4}{h_4} = 1$

b) $\frac{OA_1}{AA_1} + \frac{OB_1}{BB_1} + \frac{OC_1}{CC_1} + \frac{OD_1}{DD_1} = 1$

E. Volume of a Cone

18. One element of a right cone is 6 cm and makes 60° angle with the base. Find the volume of the cone.
19. When a right cone is cut through an element and opened, the central angle of the formed sector is 60° and surface area of the cone is $\frac{3\pi}{2} \text{ cm}^2$. Find the volume of the cone.
20. Height of a right cone is twice that of another cone and radius of the base of the first cone is half of that of the second cone. What is the ratio of their volumes?

F. Volume of a Frustum of a Cone

21. Radii of the bases of a frustum of a right cone are in the ratio of 1 : 3. An element of the frustum is 8 cm and makes 60° with the base of the cone. Find the volume of the frustum.

22. Height of a frustum of a right cone is 12 cm and a diagonal of its axial section is 15 cm. Radii of the bases of this frustum are in the ratio of 1 : 2. Find the volume of the frustum.

G. Volume of Sphere

23. Surface area of a hemisphere is $75\pi \text{ cm}^2$. Find its volume.

24. Volume of a sphere is $36\pi \text{ cm}^3$. Find the surface area of the hemisphere.

25. AB is a diameter of a sphere. Two planes a and b are perpendicular to line segment AB at C and D such that $\frac{AC}{AB} = \frac{BD}{AB} = \frac{1}{3}$. What is the ratio of the volume of the formed spherical segment of two bases to the volume of the sphere?

26. The volume of a spherical wedge with 60° angle of a sphere is $48\pi \text{ cm}^3$. Then, what is the volume of a spherical cone of this sphere if the axial section of this cone is a circular sector with an angle of 90° ?